Cisco IoT Connectivity Management Platform

Application Programming Interface User Guide

SERVICE PROVIDER

An introduction to the Control Center platform APIs for service providers with Advantage, Essential, and Lite accounts.

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API OVERVIEW

The Control Center platform provides an application programming interface (API) that allows you to access and edit data in Control Center from an external application. Using the API, you can perform many of the same tasks you can handle through Control Center's web interface, including:

- Managing the device lifecycle
- Monitoring usage
- Sending messages to a device
- Editing device attributes

Control Center offers multiple types of APIs:

SOAP APIs

The SOAP APIs have been available for many years and provide a wide range of functions. Basic access to these APIs is available to all accounts. Learn more ›

REST APIs

The REST APIs provide device attributes, usage information, session details, and more. Over time this set of APIs will grow. Learn more ›

Push APIs

Control Center uses push APIs to notify you when an automation rule triggers. You must implement a push receiver to interpret the messages. Learn more ›

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Real Time Data. REST and SOAP APIs are not intended to provide real-time monitoring of service usage. For immediate feedback when certain conditions occur, use automation rules.

Backward Compatibility. Follow these guidelines to ensure that your application code functions properly with any API framework changes.

Essential/Lite Differences. For information about differences in Essential and Lite accounts, see the *Control Center User Guide*.

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API Policies and Best Practices

Cisco has established an API fair use policy to ensure an optimal experience for all Control Center users. Any client-side code that uses Control Center APIs must adhere to the limitations described below.

Concurrent Connections

Cisco advises enterprises to avoid concurrent processing because it has the potential to slow down response time considerably. We recommend using APIs in single-threaded applications or routines only. Accounts using APIs in multi-threaded scenarios may experience throttling or other restrictions, depending on the circumstances.

***Policy*:** Use single-threaded code—one API call at a time.

API TPS

Cisco ensures effective API performance by tracking the number of API calls/requests an enterprise makes per second. This metric is knows as TPS (Transactions Per Second).

When an enterprise exceeds a certain TPS limit, Control Center throttles (rejects) their API requests. Service providers do not have a TPS limit.

Control Center only counts succesfully authenticated API requests toward the enterprise TPS limit. Likewise, Control Center only considers successfully authenticated APIs when throttling for TPS limits. However, Control Center will reject APIs before authentication because of incorrect credentials.

Example

An enterprise has three users. Each user makes 3 API calls in the same second, resulting in a total of 9 API calls for that second. Control Center successfully authenticates 7 of the API calls and rejects 2 of the API calls due to incorrect credentials. Control Center calculates the usage for that second as 7 TPS.

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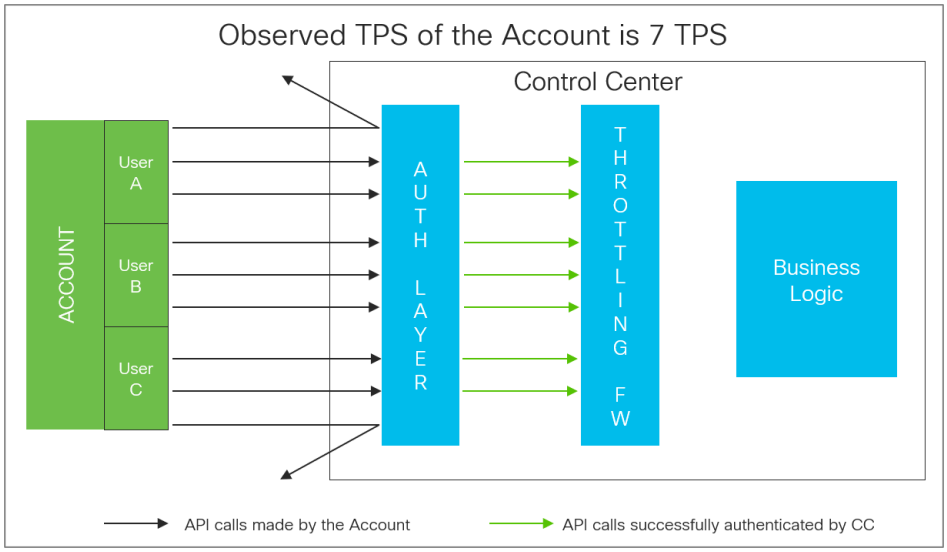
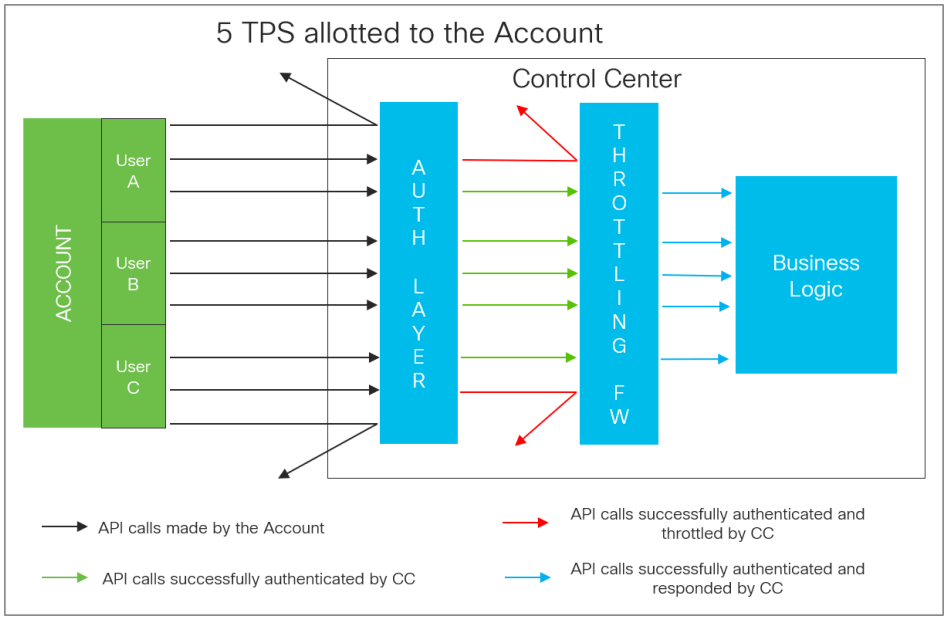


Example

An enterprise with three users has an allotment of 5 TPS. Each user makes 3 API calls in the same second, resulting in a total of 9 API calls for that second. Control Center successfully authenticates 7 API calls and rejects 2 API calls due to incorrect credentials. The TPS limit for the account is 5, so Control Center accepts 5 of the successfully authenticated API calls and throttles 2 of them.



Base Contracted TPS

Control Center provides a standard amount of API TPS to enterprises. This TPS is known as Base Contracted TPS (BCTPS). The amount an enterprise receives depends on the account type.

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Account Type	Base Contracted TPS
Advantage	5 TPS
Essential	1 TPS
Lite	1 TPS

TPS Autoscaling

When an enterprise reaches 1 million devices, Control Center automatically increases the Base Contracted TPS by the Base Contracted TPS amount for no extra charge. Each subsequent addition of 500,000 devices results in another increase. Similarly, each reduction of 500,00 devices results in a decrease of the Base Contracted TPS. This adjusted TPS is known as Scaled-Base Contracted TPS (SBCTPS).

Autoscaling Activation. To activate this feature, the service provider must request the API TPS Service add-on from Cisco and implement Base Contracted TPS for all of their enterprises.

The table below shows the Scaled-Base Contracted TPS for each account type. Control Center increases or decreases the enterprise Base Contracted TPS by this amount when the device count for the enterprise increases or decreases by 500,000, respectively.

Account Type	Base Contracted TPS	Scaled-Base Contracted TPS
Advantage	5 TPS	5 TPS
Essential	1 TPS	1 TPS
Lite	1 TPS	1 TPS

Example

An enterprise increases their device count (DC) from 1 million devices to 5 million devices (10 blocks of 500,000 devices).

- For an Advantage account, the new SBCTPS is 50 (10 device blocks * 5 BCTPS).
- For an Essential and Lite account, the new SBCTPS is 10 (10 device blocks * 1 BCTPS).

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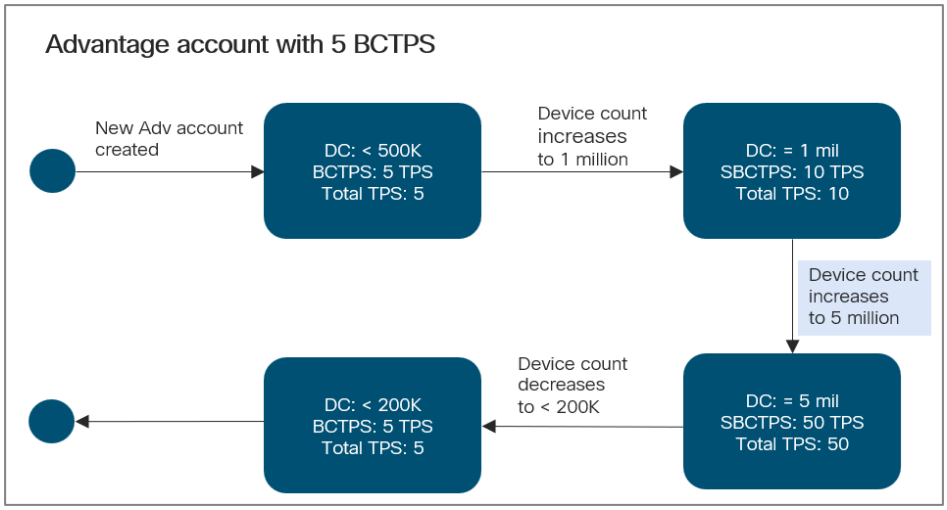

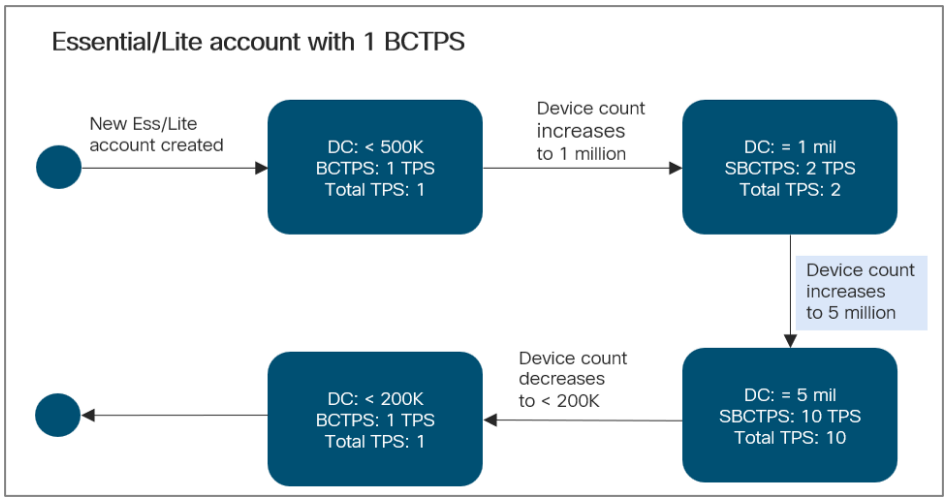

Example

An enterprise increases their device count from 1 million devices to 2.7 million devices (5.4 blocks of 500,000 devices). In such scenarios, Control Center always rounds the number of blocks up to the next whole number (6 blocks of 500,000 devices).

- For an Advantage account, the SBCTPS is 30 (6 device blocks * 5 BCTPS).
- For an Essential account, the SBCTPS is 6 (6 device blocks * 1 BCTPS).
- For an Lite account, the SBCTPS is 6 (6 device blocks * 1 BCTPS).

SBCTPS lower limit. Scaled-Base Contracted TPS never falls below the original Base Contracted TPS for the enterprise.

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Purchasing Additional TPS

A service provider can offer additional units of TPS to an enterprise through the Marketplace API TPS Service add-on. One TPS unit contains 5 TPS.

This add-on provides two types of additional TPS:

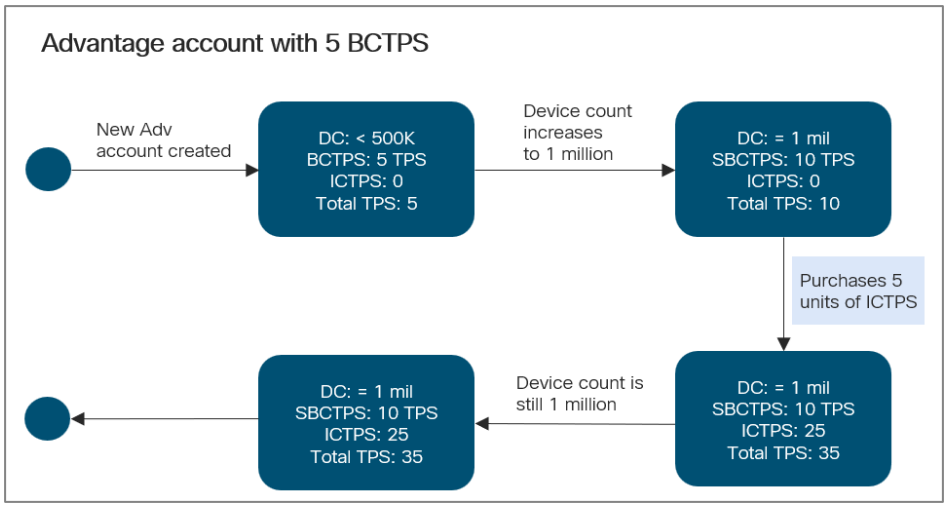
- Incremental Contracted TPS
- Overage TPS

Incremental Contracted TPS

With Incremental Contracted TPS (ICTPS), you pay a monthly fee for additional TPS whether you use it or not. If you exceed the Total Contracted TPS limit (Base/Scaled-Base Contracted TPS + Contracted TPS + Incremental Contracted TPS) and did not opt in for Overage TPS, Control Center throttles the API call and returns an error message. You will need to run the API again later.

Example

An enterprise with a device count of 1 million and a Scaled-Base Contracted TPS of 10, purchases 5 units (25 TPS) of Incremental Contracted TPS. As a result, the Total Contracted TPS for the enterprise increases from 10 to 35. When the enterprise uses more than 10 TPS, Control Center applies the Incremental Contracted TPS. Control Center throttles all API requests when TPS usage exceeds 35 TPS.



Overage TPS

With Overage TPS (OPTPS), you pay for additional TPS only when you use it. You can opt-in to Overage TPS without purchasing Incremental Contracted TPS, making it a flexible option for those who do not want to pay a monthly fee.

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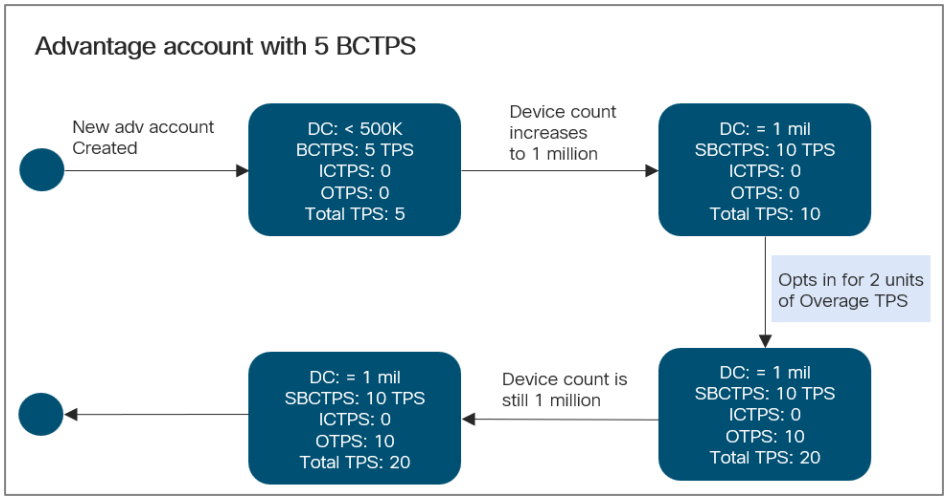
Control Center calculates Overage TPS usage based on the peak TPS level that the enterprise reaches during the billing cycle.

If the enterprise exceeds the Total Contracted TPS limit, the service provider applies the Overage TPS and charges the enterprise accordingly.

If the enterprse exceeds the Total Available TPS (Base/Scaled-Base Contracted TPS + Incremental Contracted TPS + Overage TPS), Control Center throttles the API call and returns an error message. You will need to run the API again later.

Example

An enterprise with a device count of 1 million and a Scaled-Base Contracted TPS of 10, opts in for 2 units (10 TPS) of Overage TPS. As a result, the Total Contracted TPS for the enteprise increases from 10 to 20. The service provider charges for Overage TPS when TPS usage exceeds 10 TPS. Control Center throttles all API requests when the TPS usage exceeds 20 TPS.



TPS Increments. Control Center offers both Incremental Contracted TPS and Overage TPS in units of 5, with a maximum combined total of 200 units. This excludes Scaled-Base Contracted TPS

Total Available TPS

Total Available TPS is the amount of TPS that an enterprise can use before Control Center starts rejecting API requests. Control Center calculates it as: Contracted TPS (Base/Scaled-Base Contracted TPS + Incremental Contracted TPS) + Overage TPS.

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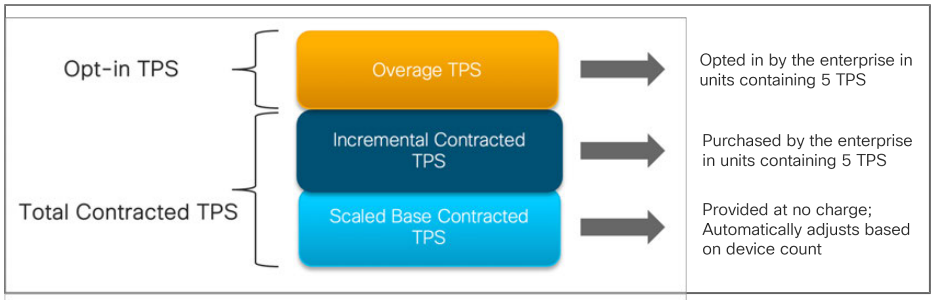
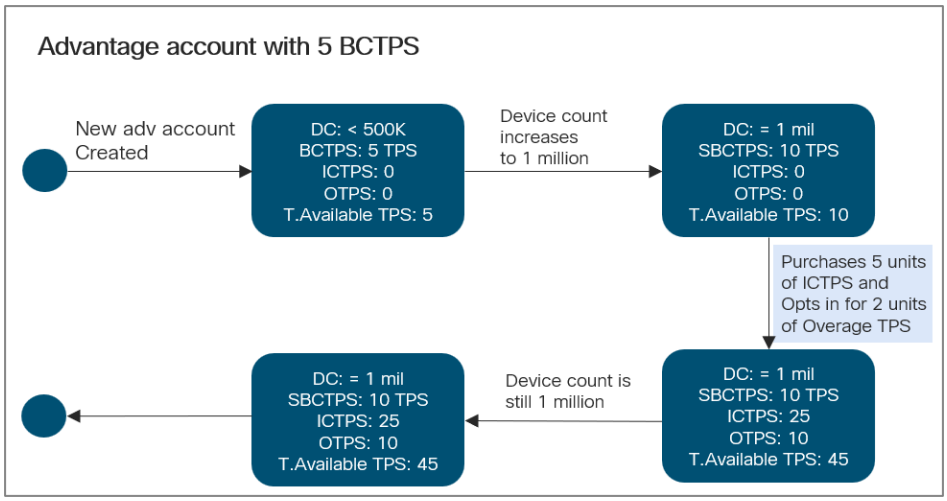


Example

An enterprise with a device count of 1 million and a Scaled-Base Contracted TPS of 10, purchases 5 units (25 TPS) of Incremental Contracted TPS and opts in for 2 units (10 TPS) of Overage TPS. As a result, the Total Available TPS for the enteprise increases from 10 to 45. When the enterprise uses more than 10 TPS, Control Center applies the Incremental Contracted TPS. The service provider charges for Overage TPS when TPS usage exceeds 35 TPS. Control Center throttles all API requests when the TPS usage exceeds 45 TPS.



TPS Limit for 'Write' Operations. Control Center caps APIs that trigger the 'write' operations towards the service provider's core network (e.g., changes to SIM state, Communication Profile changes) at either your maximum Total Available TPS (Total Contracted TPS + Overage TPS) or 25 TPS, whichever is lower.

Viewing API Usage

Control Center provides API Usage Dashboards that give service providers visibility into REST and SOAP API usage for their own account and their enterprise accounts. The API Usage Overview dashboard provides an at-a-

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glance overview of your API usage by transaction metrics. The API Usage Details Dashboard provides a graphical view of your API usage. See

If you offer the API TPS Service add-on to your enterprises, additional TPS-related information appears on the dashboards.

TPS Limits

To effectively manage network traffic, Control Center throttles (rejects) successfully authenticated API requests after an enterprise reaches certain TPS limits. These limits fall into two categories: soft limits and hard limits.

If you opt-in for Overage TPS, the soft limit is your Total Contracted TPS (Scaled-Base Contracted TPS + Incremental Contracted TPS). At this limit, Control Center continues to accept API requests and charges you for them accordingly.

The hard limit is your Total Available TPS (Total Contracted TPS + Overage TPS). At this limit, Control Center throttles the API requests and provides the following response codes:

API Type	Return Code	Message
REST	429	Rate Limit Exceeded
SOAP	500	License has exceeded the rate limit for API calls

Be aware that this limit includes API calls made by both the account and the account's customers. For example, if an account has two API users and they, along with five of their customers, made simultaneous API calls, Control Center would attribute a total of 7 calls per second to the account.

The limit also applies to the combination of REST and SOAP API calls. For example, suppose an account made 1 REST API call and 1 SOAP API call simultaneously and, at the same time, one of the account's customers made a single REST API call. Control Center would attribute 3 calls per second to the account.

General TPS Policies

As we continue to refine the platform for speed and reliability, Cisco may update the calls per second limit from time to time. These updates may occur without advance notice, however we will document them in the Release Notes.

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Be aware that this limit includes API calls made by both the account and the account's customers. For example, if an account has two API users and they, along with five of their customers, made simultaneous API calls, Control Center would attribute a total of 7 calls per second to the account. The initial API calls would succeed, but after the limit is hit, the rest of the API calls would return one of the messages shown above.

The limit also applies to the combination of REST and SOAP API calls. For example, suppose an account made 1 REST API call and 1 SOAP API call simultaneously and, at the same time, one of the account's customers made a single REST API call. Control Center would attribute 3 calls per second to the account.

Policy: Limit calls to 1 per second for Essential accounts and 5 per second for Advantage accounts.

Best Practices

The following tips can help you avoid throttling or other usage restrictions that may occur when your API use is excessive.

Add Rate Limiting Code

The best way to avoid exceeding TPS limits is to create single-threaded code. However, you may have a scenario where that's not possible. For example:

- Multiple API users are making calls.
- You need to use both REST and SOAP functions.
- You have a high volume of calls requiring multi-threaded processes.

In these cases, we recommend creating a rate limiting routine that monitors and regulates your API calls. For instance, if your account receives numerous error messages when processing API calls, you might introduce a wait time between API requests. You can increase the wait time until the errors stop.

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Pseudo Code Example. The following pseudo code illustrates how rate limiting code might work.

```
// DISCLAIMER: THIS EXAMPLE IS PROVIDED FOR ILLUSTRATION PURPOSES ONLY,
// AND WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED

WHILE ( <<list is not empty>> )
{
    // API_call attempts to perform an action to update the DEVICE status
    // it returns the API response code
    response = API_call( <<list element>> )

    // API call returns the "limit exceeded" error, increase loop delay
    IF ( response == ERROR_LIMIT_EXCEEDED )
    {
        delay += ADJUST_DELAY_INCR
    }
    ELSE
    {   // ensure delay never goes below 0
        IF ( delay > ADJUST_DELAY_DECR )
        {
            delay -= ADJUST_DELAY_DECR
        }
        ELSE
        {
            delay = 0
        }
    }

    sleep( delay )
}
```

Adjust API Request Frequency

When tracking device usage with APIs, it is important to align your API requests with the frequency of your service provider updates.

For example, if your service provider updates their usage information once every 6 hours, there's no need to request the information via API more often than once every 6 hours. Also, there is often a delay between the time when usage data is available and when Control Center receives that usage data from the service provider. In some cases, the delay can be as long as 24 hours.

Use Caching

Cisco encourages local caching. For example, instead of calling an API on every page load, create a simple routine to check the cache once a day.

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Use Filters to Limit API Calls

Avoid unnecessary API requests by checking for certain conditions before you execute a program. For example:

- If a device has been deactivated, do not check for usage.
- If an account has not requested usage on a device, do not check every hour.

Lengthen Times Between Searches

If you receive API responses that are empty or unchanged, increase your query time. For example, consider running your queries once a day rather than every two hours.

FTP Reports. For enterprises that require access to large amounts of data on a daily basis, FTP reports may be a better solution than APIs. See the reports chapter in the Control Center User Guide for more information.

Backward Compatibility

Cisco regularly updates our APIs to provide new or enhanced functionality. Our goal is to keep our API changes backward-compatible within the same API version (see *API Versions on the facing page*) so that you don't have to update your code whenever we make these changes.

Backward-compatible changes for our SOAP and REST APIs include:

- Adding new APIs
- Adding optional API request parameters
- Adding new fields to existing API responses
- Adding new values to existing API parameters that have enumerated sets of values
- Adding HTTP response headers or adding values to existing response headers

Backward-compatible changes for our push APIs include:

- Adding new Push API event types
- Adding HTTP POST form parameters
- Adding new fields to the “data” XML content
- Adding new values to existing API parameters that have enumerated sets of values
- Adding HTTP request headers or adding values to existing request headers

You should write your API clients (for SOAP and REST APIs) and push receivers (for push APIs) with the expectation that these API changes will happen, so that they can handle such changes gracefully. For example, do not hard-code to a fixed set of API response fields, since new response

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fields may be added in the future; instead, look for only the fields that you are interested in, and ignore any other fields.

While Cisco strives to ensure that the majority of API changes are backward-compatible, there are times when incompatible changes need to be made. Backward-incompatible changes include:

- Removing or renaming an API, request parameter, response field, or enum value
- Changing the visible behavior of existing APIs

Through the Release Notes, Cisco will provide at least six months' notice before backward-incompatible changes go into effect. This notice will include specific information about the change along with instructions for upgrading your code without service disruption.

API Versions

Every API function includes a version number that identifies the API structure and behavior. Cisco increments this version number under the following circumstances:

- The API receives a backward-incompatible change AND
- The previous version of the API continues to be supported.

Refer to the detailed documentation to learn more about version numbers for each type of API.

API Sandbox

For API development purposes, Control Center maintains a sandbox environment, completely separate from the production environment. In this protected environment, you can familiarize yourself with the APIs and try your applications on test data.

If you decide to develop in the sandbox environment, you must submit a change request in Control Center asking for sandbox access. Once you receive access, you can use this sandbox account to access Control Center through either the web interface or the APIs. You can also create sandbox test accounts for your enterprises.

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Chapter 1

REST APIS

The following table lists all the available REST API functions. For general information about how to use these APIs, see Getting Started.

API Function	Description
Accounts	
Get Account Details	Returns information about a given account.
Get All Accounts	Returns information about all accounts.
Create Account	Creates an enterprise account.
Edit Account Details	Modifies attributes for a given account.
Get Account Application Settings	Returns information about application settings for a given account.
Edit Account Application Settings	Modifies attributes about application settings for a given account.
Get Account Billing Settings	Returns information about billing settings for a given account.
Edit Account Billing Settings	Modifies attributes about billing settings for a given account.
Get Available Rate Plans for an Account	Returns information about available rate plan for a given account.
Edit Available Rate Plans for an Account	Modifies attributes about available rate plans for a given account.

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API Function	Description
API Key	
Create/Reset API Key for Given User	Creates or resets a given user's API key.
Get API Key	Returns the API key for a given user.
Reset API Key for Current User	Resets the API key for the current user.
Customers	
Get Customer Details	Returns information about a given customer.
Get All Customers	Returns information about all customers.
Create a Customer	Creates a customer.
Edit Customer Details	Modifies attributes for a given customer.
Devices	
Search Devices	Searches for devices based on various filters.
Get Device Details	Retrieves detailed information for a given device.
Get Device Details by EID	Retrieves device details for a specific EID.
Edit Device Details	Modifies any device attributes such as SIM status, rate plan, communication plan, custom fields and other identifiers.
Get Device Location	Returns location history information for a given device during a specified time period.
Get Device Audit History	Returns information about changes made to a given device.
Get Device Usage	Retrieves usage-related details for a given device.
Get Device Usage by Zone	Returns usage details for a given device, based on a specified zone and billing cycle.
Get Session Details	Retrieves information related to the current or most recent session for a given device.
Transfer Device to Account	Transfers up to 10,000 devices from one account to another.

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API Function	Description
Dynamic Reporting	
Get Aggregated Usage Details	Returns aggregated device usage details.
Get Device Subscriptions	Returns device subscription details.
Get Top K Usage	Returns usage details for the Top K devices.
Create Device Report	Creates a custom device report.
Create Aggregated Billing Report	Creates a custom billing report.
Create Aggregated Usage Report	Creates a custom usage report.
Get Report Status	Returns the status of a generated report.
Multi-Party Billing	
Get Partner Accounts for a Device	Get partner accounts for a device.
Assign Partner Account to Device	Assign partner account to device.
Change Partner Rate Plan for Device	Change partner rate plan for device.
Remove Partner Account from Device	Remove partner account from device.
Assign Partner Event Rate Plan to Device	Assigns a partner event rate plan to a device. You can specify one or more instances.
Get Event Details	Retrieve information on existing prepaid event rate plan associated with devices.
Rate Plans	
Get Rate Plans	Retrieves information about published rate plans.

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API Function	Description
Assign Event Rate Plan to Device	Assigns an event rate plan to a device. You can specify one or more instances.
Get Event Rate Plans Assigned to Device	Returns information about the event rate plans assigned to a device.
Edit Event Rate Plan/Device Assignment	Allows you to modify the assignment of an event rate plan to a device. For example, you can cancel future instances of the event rate plan.
SMS Messages	
Search SMS	Searches for messages that have been sent or received during a specified time frame.
Get SMS Details	Retrieves detailed information about a message.
Send SMS	Sends an SMS message to a device.
Usages	
Get Usage by Rate Plan	Returns usage information for all devices using a given rate plan.
Users	
Get User Details	Returns information about a given user.
Get All User Details	Returns information about all users.
Create User	Creates a user.
Edit User Details	Modifies attributes for a given user.
Reset User Password	Resets the web interface password for a given user.
Delete User	Deletes a given user.
General	
Echo	Returns the value passed.
QoS on Demand (Network APIs)	
Get Services for Account	Retrieves a list of available QoS profiles in Control Center.
Activate QoS Profile for Device	Activates a QoS profile for a device or application in real time.

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API Function	Description
Get QoS Job Status	Tracks the status of a QoS Profile activation request that was made using the Activate a QoS Profile for a Device API.
Revoke QoS Profile Activation	Deactivates a previously activated QoS profile for a device in real time.

Essential/Lite Differences. For information about Essential and Lite account differences, see the *Control Center User Guide*.

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Getting Started

This section provides general information to help users execute the Control Center APIs, from authentication and error handling to best practices and backward compatibility.

In This Section

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SMS Validity Period
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Roles with Access

Service provider, account, and customer users can use the APIs. However, user access is limited by the following:

- **Access Type**. The user must have an Access Type of *API Only* or *Both API and UI* in order to access the APIs. You can find that information on the User Details page: Admin > Users > *User Name* link > Access Type.
- **Role**. If a user with UI access can perform the function within the Control Center web interface, then a user with the same role and API access will be able to complete the task via an API call. There is no single role that will guarantee access for all function calls. See the API/Role matrix in the Knowledge Base for a complete list of roles and the APIs they can use.

Typically, an enterprise will create user accounts with access to both the UI and the APIs for their engineering team during development. For auditing purposes, Cisco recommends that production code use a dedicated Control Center user whose sole purpose is to execute API functions. When you create the user, you can specify that the user has API-only access to Control Center.

Field Restrictions. While multiple roles may have access to the same API, not every role will have access to all the fields listed in the documentation. You can verify field access for a particular role by logging into the web interface as a user with the same role who also has web interface access and then looking for the field.

AccountAPIOnly Role. This legacy role has access to Control Center through the API, but not through the web interface. The purpose of this role was to provide restricted access to Control Center for service provider partners using a different platform. We recommend that current users take advantage of the Access Type attribute to create users with API-only access to Control Center rather than creating users with the AccountAPIOnly role.

IP Address Restrictions . If an enterprise implements IP Address Restrictions, an API user might not have API access if their IP address falls within a restricted range.

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Single Sign-On Restriction. APIs are not available to users who access Control Center using the Single Sign-On (SSO) authentication process. SSO users are UI-only users by default.

Access by Account Type

Different account types have access to different REST APIs. The following table provides a side-by-side comparison of API access for each of the three account types: Advantage, Essential, and Lite.

API Function	Lite	Essential	Advantage
Accounts			
Get Account Details	For Service Providers Only		
Get All Accounts			
Create Account			
Edit Account Details			
Get Account Application Settings			
Edit Account Application Settings			
Get Account Billing Settings			
Edit Account Billing Settings			
Get Available Rate Plans for an Account			
Edit Available Rate Plans for an Account			
API Key			
Create/Reset API Key for Given User			✓
Get API Key			✓
Reset API Key for Current User			✓
Customers			
Get Customer Details			✓
Get All Customers			✓
Create a Customer			✓
Edit Customer Details			✓
Devices			
Search Devices	✓	✓	✓

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API Function	Lite	Essential	Advantage
Get Device Details	✓	✓	✓
Get Device Details by EID	✓	✓	✓
Edit Device Details	✓	✓	✓
Get Device Location		✓	✓
Get Device Audit History			✓
Get Device Usage		✓	✓
Get Device Usage by Zone			✓
Get Session Details	✓	✓	✓
Transfer Device to Account			✓
Dynamic Reporting APIs			
Get Aggregated Usage Details		✓	✓
Get Device Subscriptions		✓	✓
Get Top K Usage Details		✓	✓
Create Device Report		✓	✓
Create Aggregated Billing Report	For Service Providers Only		
Create Aggregated Usage Report		✓	✓
Get Report Status		✓	✓
Multi-Party Billing			
Get Partner Accounts for a Device	For Service Providers Only		
Assign Partner Account to Device			
Change Partner Rate Plan for Device			
Remove Partner Account from Device			
Activate Partner Prepaid Recurring Event			
Get Event Details			
QoS on Demand			
Get Services for an Account		✓	✓
Activate a QoS Profile for a Device		✓	✓
Get QoS Job Status		✓	✓
Revoke QoS Profile Activation		✓	✓

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API Function	Lite	Essential	Advantage
Rate Plans			
Assign Event Rate Plan to Device			✓
Edit Event Rate Plan/Device Assignment			✓
Get Event Rate Plans Assigned to Device			✓
Get Rate Plans			✓
SMS Messages			
Search SMS		✓	✓
Get SMS Details		✓	✓
Send SMS		✓	✓
Usages			
Get Usage by Rate Plan			✓
Users			
Get User Details			✓
Get All User Details			✓
Create User			✓
Edit User Details			✓
Reset User Password			✓
Delete User			✓
General			
Echo	✓	✓	✓

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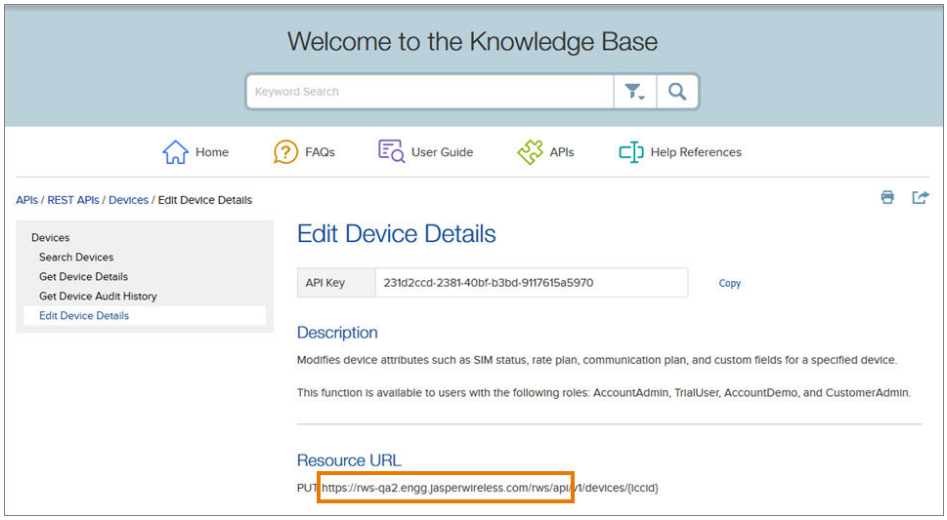
Base URL

The Control Center REST API is securely available over HTTPS. The Knowledge Base displays your base URL on the REST API Getting Started page. The base URL also appears as part of the resource URL on every REST API reference page.



API Sandbox

For API development purposes, Control Center maintains a sandbox environment, completely separate from the production environment, that you can use to test your code. If you decide to develop in the sandbox environment, you must submit a change request in Control Center asking for sandbox access. Once you receive access, you can create sandbox test accounts for your enterprises.

Base URL for Sandbox: https://rws-jpotest.jasper.com/rws/api/

If you use the sandbox account to test your code, you'll need to replace the credentials and URL in your calls when you're ready to put your code into production. Be aware that you cannot migrate any data from the sandbox environment to the production environment or vice versa.

Sandbox Support. Be aware that Control Center's API test environment, JPOTest, is not governed by Cisco's service-level agreement. Cisco does not guarantee continuous support for the sandbox environment, but we will do our best to keep the environment available.

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API Key

Each user with API access has a unique API key that they must use in the function calls. Be aware that you'll need different API keys for the production and sandbox environments.

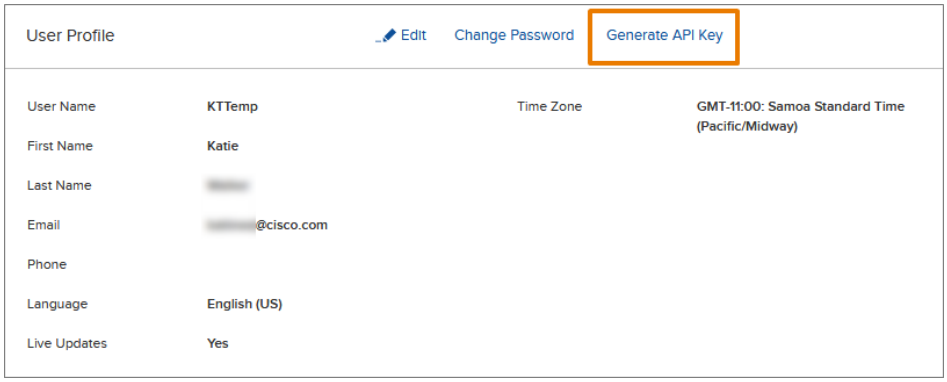
There are two ways to request a unique API Key for your user name:

From the Knowledge Base. Click the "Generate new key" link on any REST API reference page.

After you generate your API key, it appears at the top of every function page in the Knowledge Base.

A user with API-only access can also reset their API key using the *Create/Reset API Key for Given User* API, if needed.

From the User Profile page. Click the Generate API Key button on the User Profile or User Details page.



The new key appears as a series of asterisks. Click the Show API Key button to view and copy the actual values. Click the Reset API Key button to reset an existing key that has been compromised.

A user must have web interface access to Control Center in order to create or view their API key. Users with *API Only* access must rely on an administrative user with appropriate privileges to generate an API key for them in the User Details page. Although administrator users can generate and view API keys for users with the *API Only* access type, they cannot perform those functions for users with *Both API and UI* access types.

API Key Expiration. Be aware that API keys can expire, depending on the password policies associated with your account. Users with web interface access can reset the API key in their user profile. Otherwise, a service provider or account administrator must reset the API key.

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Authentication

All API calls must be authenticated through the HTTP Basic Authentication scheme. To do this, you create an authorization header that you send as part of the API request. This header contains encoded versions of a Control Center user name and its associated API key.

Follow these instructions to create an authorization header:

1. Combine the user name and API key into a single string with a colon separating the values. For example, if the user name is *starterkit* and the API key is *d703f52b-1200-4318-ae0d-0f6092b2e6ab*, the concatenated string would be:
   starterkit:d703f52b-1200-4318-ae0d-0f6092b2e6ab
2. Encode the concatenated string using Base64 (i.e. RFC2045-MIME):
   c3RhcnRlcmtpdDpkNzAzZjUyYi0xMjAwLTQzMTgtYWUwZC0wZjYw
   OTJiMmU2YWI=
3. Set the Authorization header value to *Basic* followed by the encoded string from step 2. Make sure there is a space between *Basic* and the encoded string:
   Basic c3RhcnRlcmtpdDpkNzAzZjUyYi0xMjAwLTQzMTgtYWUwZC0w
   ZjYwOTJiMmU2YWI=

For example, a Search Devices request with an authorization header might look like this:

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
c3RhcnRlcmtpdDpkNzAzZjUyYi0xMjAwLTQzMTgtYWUwZC0wZjYwOTJiMmU2YWI="
"https://<YOUR-BASE-
URL>/rws/api/v1/devices?accountId=100020620&modifiedSince=2016-04-
18T17%3A31%3A34%2B00%3A00&pageSize=50&pageNumber=1"
```

If you're using Curl, you can use the user parameter followed by your username and API key (--user <your-username>:<your-API-key>) instead of an authorization header. Curl automatically encrypts your credentials and generates the appropriate HTTP authorization header. For example:

```
curl -X GET --header "Accept: application/json" --user <your-username>:<your-
API-key>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices?accountId=100020620&modifiedSince=2016-04-
18T17%3A31%3A34%2B00%3A00&pageSize=50&pageNumber=1"
```

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Resources

Control Center supports the following resources and their associated functions:

Resource	Functions
Accounts	• Get Account Details • Get All Accounts • Create Account • Edit Account Details • Get Account Application Settings • Edit Account Application Settings • Get Account Billing Settings • Edit Account Billing Settings • Get Available Rate Plans for an Account • Edit Available Rate Plans for an Account
API Key	• Create/Reset API Key for Given User • Get API Key • Reset API Key for Current User
Customers	• Get Customer Details • Get All Customers • Create a Customer • Edit Customer Details
Devices	• Search Devices • Get Device Details • Get Device Details by EID • Get Device Location • Get Device Usage • Get Device Usage by Zone • Get Device Audit History • Get Session Details • Edit Device Details • Transfer Device to Account
Dynamic Reporting APIs	• Get Aggregated Usage Details • Get Device Subscriptions • Get Top K Usage Details • Create Device Report • Create Aggregated Billing Report • Create Aggregated Usage Report • Get Report Status
Echo	• Echo

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Resource	Functions
Multi-Party Billing	• Get Partner Accounts for Device • Assign Device to Partner Account • Change Partner Rate Plan for Device • Remove Device from Partner Account • Assign Partner Event Rate Plan to Device • Get Event Details
QoS on Demand (Network APIs)	• Get Services for an Account • Activate a QoS Profile for a Device • Get Job Status • Revoke QoS Profile Activation
Rate Plans	• Get Rate Plans • Assign Event Rate Plan to Device • Get Event Rate Plans Assigned to Device • Edit Event Rate Plan/Device Assignment
SMS Messages	• Search SMS Messages • Get SMS Details • Send an SMS
Usages	• Get Usage by Rate Plan
Users	• Get User Details • Get All User Details • Create User • Edit User Details • Reset User Password • Delete User

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Partial Responses

When you make a REST API request, Control Center returns a JSON-formatted response. The documentation for each API function lists the request parameters and contains a sample JSON response for that function.

For most GET functions, you can request a partial response to return just the data of interest to you. For example, if you want to get the rate plan for a particular device, you can instruct the Get Device Details function to return the rate plan name only and no other fields. You can request one or more fields, including subfields, subfields within an array, and subfields of any depth in a hierarchy.

Partial Response Guidelines
Partial Response Use Case #1: One field
Partial Response Use Case #2: Multiple fields
Partial Response Use Case #3: Subfields
Partial Response Use Case #4: Subfields in an array
Partial Response Use Case #5: All subfields
Partial Response Use Case #6: Subfields of any depth
Partial Response Code Samples

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Partial Response Guidelines

When requesting partial responses, keep in mind these guidelines. For use cases and sample code, see the following section.

Guideline	Description
Syntax	Use "fields=" followed by a comma-separated list of field names with no spaces between names. The GET function returns the fields in the order you request them. For example, "fields=smsMsgId,messageText,messageType,status" returns four attributes for a particular SMS message: ID, content, type (MO or MT), and status. You must specify the individual fields you want in the response. Otherwise, the function returns nothing.
Hierarchies	To indicate fields in a hierarchy, use a forward slash. For example, "field=contact/name,contact/email,billingAddress/country" returns the name and email of the contact person and the country associated with the billing address.
Wildcards	You can use an asterisk (*) to request all the subfields of a parent field. For example, "field=shippingAddress/* returns all the attributes of the shipping address. If you omit the asterisk and do not specify a subfield by name, the function does not return any subfields. For example, "field=shippingAddress/" returns no data.
Errors	If the function call incorrectly formats the request parameters or uses an incorrect field name, Control Center returns as much data as it can, but does not supply an error message. For example, "field=iccid,ratePlan,commPlan" returns the ICCID and the rate plan name only. The function does not return the communication plan because the parameter should be "communicationPlan" not "commPlan". Similarly, "field=contact/ name, contact/email" returns the contact email, but not the contact name. Control Center trims spaces before and after parameters, but not within them. Be aware that Control Center ignores duplicate parameters.

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Guideline	Description
API Functions	You can request partial responses for most GET functions. The "fields=" request parameter does not work with PUT, POST, or DELETE functions.
Try It Support	Currently, you cannot request partial responses in the Knowledge Base Try It tool.

Partial Response Use Case #1: One field

Syntax: fields=field1

Example: Get Device Details
https://<YOUR-BASE-URL>/rws/api/v1/devices/8988216716970004975?fields=status

Sample Response:

```
{
   "status": "ACTIVATED"
}
```

Partial Response Use Case #2: Multiple fields

Syntax: fields=field1,field2,field3

Notes: Do not use spaces between field names.

Example: Get Device Details
https://<YOUR-BASE-URL>/rws/api/v1/devices/8988216716970004975?fields=iccid,status,ratePlan

Sample Response:

```
{
   "iccid": "8988216716970004975",
   "status": "ACTIVATED",
   "ratePlan": "Monthly Flexible Pool 2GB"
}
```

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Partial Response Use Case #3: Subfields

Syntax: fields=field1/subfield1,field1/subfield2

Example: Get Customer Details
https://<YOUR-BASE-URL>/rws/api/v1/customers/16716970?fields=name,
billingAddress/addressLine1,
billingAddress/postalCode

Sample Response:

```
{
   "name": "Northwest Region",
   "billingAddress": {
      "addressLine1": "123 Main Street",
      "postalCode": "10155"
   }
}
```

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Partial Response Use Case #4: Subfields in an array

Syntax: fields=field1/subfield1,field1/subfield2

Example: Get Device Usage in a Zone
https://<YOUR-BASE-URL>/rws/api/v1/devices/89302720396916972826/usageInZone?fields=iccid, cycleStartDate, cycleEndDate, deviceCycleUsageInZones/ratePlan, deviceCycleUsageInZones/ratePlanVersion, deviceCycleUsageInZones/zone, deviceCycleUsageInZones/dataUsage, deviceCycleUsageInZones/dataUsageUnit

Sample Response:

```
{
   "iccid": "89302720396916972826",
   "cycleStartDate": "2019-04-18T00:00:00.000Z",
   "cycleEndDate": "2019-05-17T23:59:59.000Z",
   "deviceCycleUsageInZones": {
      "Acme Monthly Individual, 3": [
      {
         "ratePlan": "Acme Monthly Individual",
         "ratePlanVersion": "3",
         "zone": "Roaming",
         "dataUsage": 20971924,
         "dataUsageUnit": "bytes"
      },
      {
         "ratePlan": "Acme Monthly Individual",
         "ratePlanVersion": "3",
         "zone": "Other Usage",
         "dataUsage": 1375776,
         "dataUsageUnit": "bytes"
      }
   ]
}
```

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Partial Response Use Case #5: All subfields

Syntax: fields=field1/*

Notes: To include subfields in the response, you must specify the individual field names or use an asterisk (*) to return all fields. If you use a forward slash alone, the function returns no subfields.

Example: Get Customer Details
https://<YOUR-BASE-URL>/rws/api/v1/customers/16716970?fields=
name,
contacts/name,
contacts/email,
billingAddress/*

Sample Response:

```
{
   "name": "Northwest Region",
   "contacts": [
      {
         "name": "John Doe",
         "email": "jdoe@acme.com"
      },
      {
         "name": "Annette Wong",
         "email": "awong@acme.com"
      }
   ]
   "billingAddress": {
      "addressLine1": "593 Palm Dr.",
      "addressLine2": null,
      "city": "Los Angeles",
      "state": "CA",
      "country": "USA",
      "postalCode": "91201"
   }
}
```

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Partial Response Use Case #6: Subfields of any depth

Syntax: fields=field1/subfield1/subsubfield1

Example: Get Account Details
https://<YOUR-BASE-URL>/rws/api/v1/accounts/100020704?fields=accountName,
security/passwordPolicySettings/ui/passwordExpirationInDays,
security/passwordPolicySettings/ui/content/minLength

Sample Response:

```
{
   "accountName": "Acme Widgets",
   "security": {
      "passwordPolicySettings": {
         "ui": {
            "passwordExpirationInDays": "90",
            "content": {
               "minLength": "6"
            }
         }
      }
   }
}
```

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Partial Response Code Samples

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

Python

```
import requests
import json
import base64
import pprint

# Replace the URL and credentials with your own
apiUrl= 'https://<your-base-URL>/rws/api/v1/accounts/100000000'

parameters= {'fields': "accountId,shippingAddr/addr1"}

myResponse = requests.get(url=apiUrl, auth=(raw_input("username: "),raw_input
("api_key: ")), params=parameters)
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
   # Loading the response data into a dict variable
   # json.loads takes in only binary or string variables so using content to
fetch binary content
   # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
   pp = pprint.PrettyPrinter(indent=4)
   jData = json.loads(myResponse.content)
   pp.pprint(jData)
else:
   # If response code is not ok (200), print the resulting http error code
with description
   myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];

// Replace the URL and credentials with your own
request.get('https://<your-base-
URL>/rws/api/v1/accounts/100000000?fields=accountID,shippingAddr/addr1').auth
('user_name', 'password', false)
   .on('error', function(error){
      console.log('Error:', error);
   })
   .on('response', function(response) {
      console.log(response.statusCode); // return statusCode
      console.log(response.headers['content-type']); // return contentType
   })
   .on('data',function(chunk){
      body.push(chunk);
   })
   .on('end',function(){
      body = Buffer.concat(body).toString();
      console.log(body);
   });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

# Replace the URL and credentials with your own
url = 'https://<your-base-
URL>/rws/api/v1/accounts/100000000?fields=accountId,shippingAddr/addr1'

response = RestClient::Request.execute(
   method: :get,
   url: url,
   user: 'user_name',
   password: 'password',
   headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

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Asynchronous Responses

Some REST APIs support an optional asynchronous response in addition to a synchronous response. These APIs include a callback request parameter that allows users to specify a destination URL for the callback notification.

To receive the callback notification, you must implement a simple receiver (for example, a Java servlet) that understands HTTPS or HTTP POST requests, depending on which protocol you specify in the URL. For increased security, we recommend using HTTPS in the URL.

Many accounts already have a push API receiver for automation rule notifications that can be adapted for REST API callbacks. Be aware that REST API callbacks use a JSON format, rather than the XML format generally used in push APIs. For information about setting up a push receiver, see *Implementing a Push Receiver on page 440*.

Typical Event Sequence
Callback Notification Format
Sample Callback

Typical Event Sequence

Here's a typical event sequence for REST API callbacks:

1. **You specify a callback in the REST API**. If the URL you supply uses HTTPS protocol (recommended), your push receiver must have an SSL certificate signed by one of the common trusted Certificate Authorities (CA). For details, see *SSL Certificates on page 444*.
2. **Control Center sends a callback notification** when the API work is complete. If a time-out occurs or your server returns a status code other than 200, Control Center will wait a few minutes and resend the message. For more information, see *Retry Handling on page 441*.
3. **(Optional) You verify the signature**. You can add another level of security to the communication by implementing a Push API Shared Secret. For details, see *Signature Verification (Optional) on page 441*.
4. **The push receiver processes the returned data**. After successfully parsing the notification, your receiver should respond with the HTTP status code “200 OK”. If there are errors, the receiver should return an HTTP error code. For more information about the callback content, refer to the specific REST API reference page.

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Callback Notification Format

Each callback is an HTTPS/HTTP POST with the following URL-encoded form parameters:

Parameter	Description
eventId	A unique identifier for the event that remains the same even if the same event is re-sent multiple times due to network problems, server downtime, and so on. See *Asynchronous Responses on the previous page*.
eventType	The type of callback event, such as "TRANSFER_DEVICE_TO_ACCOUNT_API_RESPONSE."
timestamp	The request time in UTC, such as: 2009-12-10T01:16:20.026Z.
signature	An HMAC SHA2 hash of the timestamp using the API shared secret, which can be used to verify the identity of the caller. See *Signature Verification (Optional) on page 441*.
signature2	An HMAC SHA2 hash of the timestamp using the API shared secret, which can be used to verify the identity of the caller. See *Signature Verification (Optional) on page 441*.
data	The JSON content, specific to the event type. Refer to the REST API documentation for the corresponding callback formats.

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Sample Callback

Here is a sample callback message generated by the Transfer Device to Account API. All messages have similar headers. See the REST API documentation for the data format returned in each message.


```
{
   "eventId": "TRANSFER_DEVICE_TO_ACCOUNT_API_RESPONSE-123",
   "data": {
      "correlationId": "1",
      "apiCallbackNotificationId": "4205",
      "sourceApi": "TransferDevicesToAcctResource",
      "requestDate": "148413074253",
      "transferDeviceToAccountStatusList": {
         "transferDeviceToAccountStatus": [
         { "iccid": "89011986000009860012",
           "transferStatus": "success" }
         ]
      }
   },
   "signature": "s6WfKld6Ru8SIBHdrH+w1obm7Eg=",
   "eventType": "TRANSFER_DEVICE_TO_ACCOUNT_API_RESPONSE",
   "timestamp": "2022-03-21T16:36:45.471Z",
   "signature2": "oqb+ww+ihEAKEb9Rb20PaBBaqT6L5BS31gSAg3rT/F8="
}
```


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Pagination

For functions that typically return numerous records (such as Search Devices and Search SMS), the API provides pagination controls that allow you to process one set of results and then request more.

These request parameters manage the pagination process:

Request Parameter	Description
pageSize	Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50.
pageNumber	Specifies the number of the response page to return. This value defaults to 1.

These response values allow your code to process the returns properly:

Response Value	Description
pageNumber	An integer specifying the number of the current response page.
lastPage	A true or false value indicating whether the current response page is the last in the series.

For example, the first API request would define the number of records to include in the response (pageSize=50) and ask for the first page of records (pageNumber=1). The response would send an array containing the first 50 records and then let you know if there are any more pages to return (lastPage=false).

The second request would define the page size again (pageSize=50) and ask for the second page of results (pageNumber=2). The second response would return the next 50 records and alert you if there were more records to retrieve (lastPage=false). This process would continue until one of the responses indicates that the current page is the last page of results (lastPage=true).

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Date/Time Formats

The REST APIs use the following date and time formats for requests and responses.

Use Case	Format
Search Requests, for example: • Search Devices (modifiedSince) • Search SMS (fromDate)	yyyy-MM-ddTHH:mm:ssZ (ISO 8601 standard) Example: 2016-04-18T17:31:34+00:00
Edit Device Request (effectiveDate) Get Device Usage in a Zone Request (cycleStartDate)	yyyy-MM-ddZ (ISO 8601 standard; default time zone is UTC) Example: 2016-04-18Z or 2016-04-18+05:00
Responses, for example: • Get Device Details (dateShipped) • Get Session Info (lastSessionEndTime)	yyyy-MM-dd HH:mm:ss.SSSZ (ISO 8601 standard, based on UTC time) Example: 2016-04-18 17:31:34.121+0050
Get Device Usage by Zone Response (all dates)	yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601 standard) Example: 2016-12-06T15:58:06.466Z
Response timestamps (some)	UNIX Epoch clock format. This format counts the number of seconds since the start of the UNIX Epoch on January 1, 1970 (UTC). There are a number of converters available online. Example: 1571416432 (equivalent to October 18, 2019 at 4:33pm UTC)

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Error Handling

The REST APIs use standard HTTP status codes as well as business-specific error messages. If an error occurs, Control Center will return a JSON response containing an error code and an error message. For example:

```
{
   "errorMessage": "The API credentials are invalid.",
   "errorCode": "10000001"
}
```

The documentation for each API function contains a list of the errors that may occur for that function.

The standard HTTP status codes are shown below.

HTTP Status Code	Description
200	Success
202	Accepted. The request is scheduled for processing and the task should be complete within several seconds.
400	Bad request
401	Invalid credentials
404	Resource not found
500	Server error

SIM Status Values

Many API functions take the device SIM status as a parameter or return the information in the response. Following is a list of the values you may see, depending on your role.

- ACTIVATED
- ACTIVATION_READY
- DEACTIVATED
- INVENTORY
- ONHOLD (Start SIM state)
- PURGED
- REPLACED
- RETIRED
- TEST_READY

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Language Values

Several API functions take the language preference as a request parameter or return the information in the response. Here are the valid values:

- "English"
- "English (UK)"
- "Portuguese (BR)"
- "Portuguese (PT)"
- "Spanish (MX)"
- "Spanish (ES)"
- "German"
- "French"
- "Japanese"
- "Russian"
- "Dutch"
- "Chinese (Simple)"
- "Chinese (Traditional)"
- "Indonesian"
- "Italian"
- "Thai"
- "Swedish"
- "Korean"

Currency Values

Several API functions take currency as a request parameter or return the information in the response. Here are the valid values:

- AED
- ARS
- AUD
- BOB
- BRL
- CAD
- CHF
- CLP
- CNY
- COP
- CZK
- DKK
- EEK
- EUR
- GBP
- GTQ
- HKD
- HNL
- IDR
- INR
- ITL
- JPY
- KRW
- KZT
- LTL
- LVL
- MOP
- MXN
- NGN
- NOK
- NTD
- NZD
- PYG
- RUB
- SAR
- SEK
- SGD
- SVC
- THB
- USD
- ZAR

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User Role Values

For API functions that take user role as a request parameter or return the information in a response, here is a list of valid values:

- SERVICEPROVIDERACCOUNTREP
- SERVICEPROVIDERADMIN
- SERVICEPROVIDERCSRUSER
- SERVICEPROVIDERDATAHELPDESK
- SERVICEPROVIDERDEVICESORDERS
- SERVICEPROVIDERFINANCE
- SERVICEPROVIDERFINANCEAPPROVAL
- SERVICEPROVIDERFINANCEOPERATIONS
- SERVICEPROVIDERGLOBALADMIN
- SERVICEPROVIDERREADONLY
- SERVICEPROVIDERSERVICEADMIN
- SERVICEPROVIDERSUPPORTONLY
- SERVICEPROVIDERTECHSUPPORT
- SERVICEPROVIDERUSER
- SPSECURITYADMIN
- SPSECURITYADMINAPPROVER
- ACCOUNTADMIN
- ACCOUNTAPIONLY
- ACCOUNTDEMO
- ACCOUNTFINANCE
- ACCOUNTFINANCEAPPROVAL
- ACCOUNTFINANCEREADONLY
- ACCOUNTGLOBALADMIN
- ACCOUNTLBSUSER
- ACCOUNTORGADMIN
- ACCOUNTREADONLY
- ACCOUNTSECURITYADMIN
- ACCOUNTUSER
- TRIALUSER
- CUSTOMERADMIN
- CUSTOMERDEMO
- CUSTOMERREADONLY
- CUSTOMERSUPERVISOR
- CUSTOMERTERMINALEDIT
- CUSTOMERTRIALUSER
- CUSTOMERUSER

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Industry Vertical Values

For API functions that take vertical industry as a request parameter or return the information in a response, here is a list of the valid values:

- AGRICULTURE
- ASSET_MANAGEMENT
- AVIATION
- CONNECTED_CAR
- CONSTRUCTION
- CONSUMER_ELECTRONICS
- DIGITAL_SIGNAGE
- DRONES
- ECOMMERCE
- EDUCATION
- FINANCE
- FLEET_MANAGEMENT
- GOVT_AND_PUBLIC_SERVICE
- HOSPITALITY
- INDUSTRIAL
- INSURANCE
- MANUFACTURING
- MEDIA_AND_ENTERTAINMENT
- MEDICAL_DEVICES
- MOBILITY
- OIL_AND_GAS
- OTHER
- PAYMENT_SOLUTIONS
- RETAIL
- ROBOTICS
- SECURITY_AND_AUTOMATION
- SMART_BIKES
- SMART_METERS
- TECHNOLOGY
- TELEMATICS
- UTILITIES

For information on the available industry verticals, see the *Control Center User Guide* or the Knowledge Base (Reference > Admin > Account Wizard Page 1 > Industry Vertical Descriptions).

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SMS Validity Period

The Send SMS function uses a *tpvp* request parameter for time-sensitive messages. This value defines the length of time the message is available before expiring. The SMSC will attempt to deliver the SMS message to the mobile device within this validity period. If the message cannot be delivered to the device by the end of this period (because the device is off or out of coverage, for example), the message will be dropped.

For example, suppose you have an application that sends a Door Lock/Unlock command to a vehicle via SMS. If the car is out of coverage when the message is sent, you wouldn't necessarily want to deliver the message hours later when the car is back in coverage. In this scenario, you might choose to expire the message after a 5 minute validity period.

The tpvp parameter supports validity periods between 5 minutes and 63 weeks in length using the TP-VP relative format defined by the GSM 03.40 standards:

TPVP Value	Validity Period	Possible Validity Periods
0-143	(TP-VP + 1) x 5 minutes	5, 10, 15 minutes ... 11:55, 12:00 hours
144-167	(12 + (TP-VP - 143) / 2) hours	12:30, 13:00, ... 23:30, 24:00 hours
168-196	(TP-VP - 166) days	2, 3, 4, ... 30 days
197-255	(TP-VP - 192) weeks	5, 6, 7, ... 63 weeks

In the Door Lock/Unlock example, if you pass a value of 0 for *tpvp*, the message would expire in 5 minutes: (0 + 1) x 5 = 5 minutes.

The maximum validity period depends on the SMSC configuration. If Cisco hosts the SMSC, the maximum validity period is 5 days (tpvp = 171). If the service provider hosts the SMSC, the maximum validity period may be different. If the SMSC doesn't support the validity period you specify in the API call, the value defaults to the maximum validity period for that SMSC. If you do not specify a validity period, the API uses 3 days as the default.

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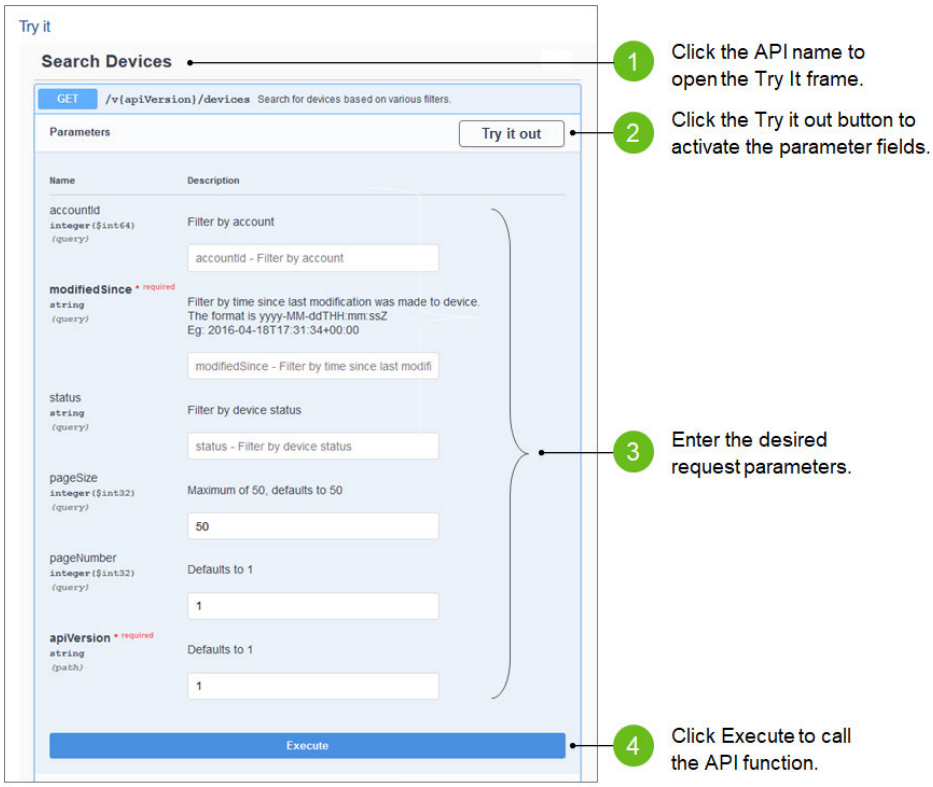
Try It Tool

From within the Knowledge Base, you can execute individual REST APIs without writing any code.

Location: Help > Find everything you need in the Knowledge Base > APIs > REST APIs > Devices > Search Devices



1. Scroll down to the bottom of any REST API reference page until you see the Try It section. Click the function name to open the Try It tool.
2. Click the *Try it out* button to activate the parameter fields.
3. Enter the desired request parameter values. For functions that require a date, copy and paste the sample date (which uses the correct date format) and then edit the date to suit your use case.
4. Click Execute to send the request.

Control Center displays both the request code and the JSON response. You can copy and paste the request code into a separate code file for future use.

No Partial Responses.Currently, you cannot request partial responses in the Knowledge Base Try It tool.

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Caution. The Try It tool acts on the environment you are currently logged into — production or test (sandbox). Be aware that any PUT or POST calls will update the current database. If you are interested in setting up a sandbox environment, submit a change request in Cisco IoT Connect Portal.

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Accounts

Get Account Details

Description

Returns the account details for a given account. See the Response Parameters section for a list of the specific account details returned.

You must have the ServiceProviderAdmin role to perform this function.

Resource URL

GET BaseURL/v{apiVersion}/accounts/{accountId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The ID of the account whose details you want to retrieve.

Response Parameters

Return Value	Description
accountName	The account name appears throughout Control Center as an identifier for the account.
accountId	A Control Center-generated identifier for the account.
defaultSIMState	This field defines the default SIM state that devices receive when the service provider transfers them from the inventory account into the enterprise account. For valid values, see *SIM Status Values on page 50*.

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Return Value	Description
type	The type of account. Here are some values you may see: • *STANDARD*. For Advantage enterprise accounts. • *ESSENTIAL*. For Essential enterprise accounts. • *OPERATOR*. For service provider accounts. Each service provider has only one of these accounts. This account should contain all service provider users, but no devices. • *INVENTORY*. For service provider inventory accounts. Every service provider has at least one inventory account. An inventory account contains devices that have not been assigned to an enterprise account. The devices in this account have a special status called *Start*.
currency	The currency value is the unit of measure used for the rate plan and billing information.
language	The default account language. The preference of the logged in user will always take precedence over the account language.
operatorAccountId	The account ID used by the service provider's internal IT systems. Control Center uses this ID in the invoice report feed that you receive for IT integration.
status	Indicates whether the account is active or inactive. When an account is inactive, account users cannot log into Control Center through the web interface or the APIs. However, account devices remain active in Control Center and on the network. Automation rules for the account also remain active and may trigger based on the defined conditions. Typically, service providers use this flag as a way to "delete" an account. Valid values are *Active* and *Inactive*.
taxId	The tax or VAT ID is an account identifier that may be used for tax purposes.
industryVertical	This flag indicates the industry vertical for the account. For valid values, see *Industry Vertical Values on page 53*.
commPlanDetails	See the Communication Plan Parameters table below.
defaultRatePlan	See the Rate Plan Parameters table below.
billingAddr	See the Address Parameters table below.

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Return Value	Description
shippingAddr	See the Address Parameters table below.
ppuAddress	See the Address Parameters table below.
primaryContact	See the Contact Parameters table below.
billingContact	See the Contact Parameters table below.
acctOnboardedBy	Indicates how the enterprise account was onboarded. For accounts onboarded before this field was implemented, this field is blank. • SP_OPERATIONS. For accounts onboarded by the service provider. • STARTERKIT. For accounts onboarded using the starter kit process.
accountSegment	Account segments are service provider-created categories that reflect the service provider's business requirements.
applicationSettings	The application settings are grouped in categories that mirror the organization of the Account Details > Application Settings subtab in the web interface. See the tables below.
applicationFeature	See the Application Settings > Application Feature Parameters table below.
simFeature	See the Application Settings > SIM Feature Parameters table below.
billingSettings	The GET function currently returns only the general billing settings. In the web interface, this information appears on the Account Details page in the Billing Settings > General subtab.
general	See the Billing Settings > General Parameters table below.
security	All the security settings appear in this section. See the Security Parameters table below.

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Communication Plan Parameters

Control Center uses the default communication plan for any account device that does not have a communication plan assigned to it.

Parameter	Description
defaultCommPlan	The name of the default communication plan.
defaultOnCommProfile	The name of the communication profile devices in the ON state will use.
defaultOffCommProfile	The name of the communication profile devices in the OFF state will use.
onCommProfileHlrTemplateId	A unique identifier for the HLR template containing the ON communication profile information.
offCommProfileHlrTemplateId	A unique identifier for the HLR template containing the OFF communication profile information.

Rate Plan Parameters

Control Center applies the default rate plan to any SIMs you transfer into the account unless you specify a different rate plan.

Parameter	Description
defaultRatePlanId	The Plan ID for the default rate plan.
defaultRatePlanName	The Plan Name for the default rate plan.

Address Parameters

These parameters are the same for all three types of addresses: billing, shipping, and primary place of use.

Parameter	Description
addr1	Street address, line 1. String, 1-100 characters
addr2	(Optional) Street address, line 2. String, 1-100 characters
city	The city name. String, 1-100 characters
region	The region name — a state or province, for example. String, 1-100 characters
countryCode	A two character country code. See ISO 3166 for two-letter codes (alpha-2).
postalCode	A postal code. String, 1-50 characters

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Contact Parameters

These parameters are the same for both the primary contact and the billing contact, with one exception. The primary contact can have a job title, but the billing contact does not.

Parameter	Description
firstName	A first name for the contact person. String, 1-49 characters
lastName	A last name for the contact person. String, 1-49 characters
email	An email for the contact person. This string must conform to standard email format. No more than 320 characters.
phone	A phone number for the contact person. String, 1-25 characters
fax	A fax number for the contact person. String, 1-25 characters
name	This string is the combination of the first and last name values. Control Center uses the string for the Primary Contact field on the Accounts list and the Account Details pages. String, up to 100 characters.
jobTitle	The primary contact can have a job title, but the billing contact does not. String, up to 100 characters.

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Application Settings > Application Feature Parameters

Parameter	Description
accessCommPlan	All account users can see the communication plan name in the device Edit form. This toggle determines whether the field is editable. If this toggle is set to *true*, then account users have the ability to change a device's communication plan. In addition, Control Center displays the communication plan name in the Device List and on the device Details page. If the toggle is set to *false*, account users cannot change a device's communication plan and will not see the communication plan name in the Device List and on the device Details page.
businessRulesEnabled	If this toggle is set to *true*, then the rules feature is enabled for this account. The rules functionality is visible in the Automation category to users associated with this account. If this toggle is set to *false*, then rules are disabled, and the rules functionality is not visible to users associated with this account.
simOrderingEnabled	If this toggle is set to *true*, then the SIM ordering feature is enabled for this account. This feature allows account users to place an order as soon as the need arises, and to assign a default rate plan and communication plan for the new SIMs at the time of the order. If this toggle is set to *false*, SIM ordering is disabled for this account.
smsAccess	If this toggle is set to *true*, then the SMS fields in the user interface are visible to account users. If this toggle is set to *false*, then the SMS fields are not visible.
voiceAccess	If this toggle is set to *true*, then the voice fields in the user interface are visible to account users. If this toggle is set to *false*, then the voice fields are not visible.

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Application Settings > SIM Feature Parameters

Parameter	Description
blockBackwardSimState	If this toggle is set to *true*, users cannot change a device status from *Activated* to the pre-activated status (*Activation Ready*, *Test Ready*, or *Inventory*). If the toggle is set to *false*, there are no restrictions on changing a device status to *Activation Ready*, *Test Ready*, or *Inventory*. This setting prevents an account from moving under-used devices to *Activation Ready* where they can avoid being charged a subscription fee until they attempt to make a connection.
displayDiagnosticWizard	The Diagnostics Wizard provides a very simple interface that users with little wireless knowledge can use to quickly see what is happening with a device on the network. If this toggle is set to *true*, then Control Center displays the Diagnostics Wizard to account users. If this toggle is set to *false*, then Control Center hides the Diagnostics Wizard from account users.
displaySpotlight	Spotlight is a troubleshooting tool that provides a detailed snapshot of device connectivity history over the past 30 days so you can observe long-term trends. This tool is designed for advanced support representatives, who have technical expertise in wireless connectivity. If this toggle is set to *true*, then Control Center displays Spotlight to account users. If this toggle is set to *false*, then Control Center hides Spotlight from account users.
fixedIpEnabled	If this toggle is set to *true*, Control Center displays the fixed IP address (if any) for devices in the Device List page. If this toggle is set to *false*, Control Center does not display the fixed IP addresses.
testReadyDataLimit	This field specifies the amount of data (in bytes) a device in the *Test Ready* state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it initiates traffic.

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Parameter	Description
testReadySmsLimit	This field specifies the number of messages a device in the *Test Ready* state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it sends or receives a message.
testReadyVoiceLimit	This field specifies the number of voice minutes a device in the *Test Ready* state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it places or receives a phone call.
testReadyDataState	Once the device reaches the configured limit in the *Test Ready* state, Control Center automatically moves the device to another SIM state. This field specifies the next SIM state. Most service providers use either *Activation Ready* or *Inventory*. For valid values, see *SIM Status Values on page 50*.

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Billing Settings > General Parameters

Parameter	Description
billable	This toggle indicates whether the account is billable. Valid values are *Yes* and *No*.
prorationRule	Indicates whether or not the billing engine will prorate subscription fees and included usage in the first billing cycle for devices using monthly rate plans. • *None*. The billing engine charges for the full month regardless of when the device was first activated. The device receives all the included usage. • *Daily*. The billing engine charges the account based on the number of days since the device was activated for the first time. The device receives a portion of the included usage, based on the number of days since the first activation. • *Semi-Monthly*. The billing engine charges for either a full or half month depending on whether the device was activated for the first time in the first half or second half of the month. The device receives all or half of the included usage, based on the first activation date. • *Unknown*. You may also see this value.
renewalProrationRule	Similar to prorationRule, this parameter covers the special case where a device ends a prepaid rate plan and moves onto a monthly plan. Depending on the value of prorationRule, Control Center will prorate the new subscription fee on a semi-monthly or daily basis – or not at all. Valid values include: *None*, *Semi-Monthly*, *Daily*, *Unknown*.
simFee	This fee indicates how much the account is charged for any SIMs they purchase from you.

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Parameter	Description
activationPlan	The default activation plan defines how Control Center will apply the activation fee to each device owned by the account. • *First Time Only*. The account pays the activation fee when the device is activated for the first time (typical). • *Each Time*. The account pays every time the device SIM status changes to *Activated*. • *Not First Time*. The account does not pay for the first device activation, but it does pay for every subsequent activation. • *None*. No charge. You can override the default activation fee and plan settings by defining this information at the rate plan level. In that case Control Center will charge an activation fee based on the settings associated with the current device rate plan.
activationFee	Control Center charges this fee based the activationPlan setting. You can override the default activation fee and plan settings by defining this information at the rate plan level. In that case Control Center charges an activation fee based on the settings associated with the current device rate plan.
vpnPlan	Control Center supports two types of VPN: GPRS and SMPP. The account can choose to purchase one, both, or neither. You must select one of these options before you can define a setup fee or monthly fees for the VPN. Valid values include: *GPRS Only*, *SMPP Only*, *GPRS + SMPP*, and *None*.
retailBilling	The retail billing feature is enabled if the value is *Retail Customer*.

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Parameter	Description
immediateRatePlanChange	Rate plan changes for devices with non-active SIM status values (such as *Test Ready* or *Deactivated*) always take effect immediately. For devices with an *Activated* SIM status, you can control the rate plan change behavior with this setting. • If the toggle is on, any mid-cycle rate plan changes for devices with an *Activated* status take effect immediately. Control Center will bill for the device as if it had been on the new rate plan during the entire billing cycle. • If the toggle is off, Control Center schedules any mid-cycle rate plan changes for devices with an *Activated* status to take effect in the next billing cycle. See the Scheduled Changes subtab on the Device Details page. This behavior applies to devices with monthly rate plans only. You cannot change the rate plan for a device on a prepaid rate plan until the plan expires. Essential accounts do not support immediate rate plan changes for devices with an *Activated* SIM status, regardless of this parameter value.
defaultTimeZone	The default time zone for the account. Unless otherwise specified, Control Center uses the time zone associated with the logged in user to display all dates. The information is stored as text. Sample values include: *UTC, Pacific/Midway, America/Costa_Rica, America/Nome, Pacific/Marquesas, America/Anchorage.*
eventPlanEnabled	If this toggle is set to *true*, then event plans are enabled for this account. An event plan is a prepaid rate plan that's designed for special circumstances or "events" that occur outside of a normal rate plan. If this toggle is set to *false*, then this account cannot use event plans.

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Parameter	Description
prepaidRenewalPolicy	When a prepaid rate plan expires, Control Center either deactivates the device or gives it a different rate plan. You can specify the default renewal behavior for all the account's prepaid rate plans. Valid options are: • *Deactivate*. Deactivate the device. • *AutoRenew*. Renew the current prepaid rate plan. • *NamedPlan*. Use the first rate plan in the device queue. If there is no device queue, Control Center uses the named rate plan (prepaidRenewalRatePlanId). • *None*. • *Unknown*.
prepaidRenewalRatePlanId	If prepaidRenewalPolicy is NamedPlan, then Control Center requires a rate plan. When a prepaid device rate plan expires, Control Center attaches the named rate plan to the device. If the device has a queue, Control Center uses the rate plan at the top of the queue instead of the plan named here.

Security Parameters

Parameter	Description
ipLoginRestrictionEnabled	Indicates whether the account can use IP address restrictions .
secureSimEnabled	Indicates whether the account can use the Secure SIM feature .
twoStepVerification	Indicates whether the account can use the 2-Step Verification feature.
passwordPolicySettings	See Password Parameters table below.
ui	These parameters define password policies for users who access the product through the web interface.
soap	These parameters define password policies for users who access the product through the SOAP API.

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Password Parameters

Parameter	Description
passwordExpirationInDays	The number of days before the user's password expires automatically. If the user does not reset their password before the current password expires, the user must ask an administrator for assistance. This parameter applies to all new users. For existing users, this parameter applies only if an administrator forces the user to change their password, or if the user initiates the password reset. If passwords do not expire, this field contains the string "Passwords do not expire." This policy is useful for SOAP API users who do not typically change passwords.
numAttemptsBefore TempLockout	The number of times a user can enter the wrong password or 2-Step Verification code before Control Center locks them out of the system. If Control Center never locks users out of the system, this field contains the string "User accounts never lock out."
tempLockoutPeriodInMinutes	If the numAttemptsBeforeTempLockout parameter is used, then the service provider can enforce a specified time (in minutes) for the lockout period.
numHistoryChecks	Control Center examines previous passwords assigned to a user ID and ensures that the newly created password is not identical to the specified past number of versions.
content	A group of fields that specify rules for the password string.
minLength	The minimum number of characters that a password must contain.
maxLength	The maximum number of characters a password can contain.
minUppercaseChars	The minimum number of uppercase characters that a password must contain.
minNumerics	The minimum number of numeric characters that a password must contain.
minSpecialChars	The minimum number of special characters that a password must contain.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/accounts/100020704"
```

Response Example


```
{
  "accountName": "Widgets Inc.",
  "accountId": "100020704",
  "defaultSIMState": "TEST_READY",
  "type": "STANDARD",
  "currency": "USD",
  "language": "English",
  "operatorAccountId":"123",
  "status": "Active",
  "taxId": "123",
  "industryVertical": "AVIATION",
  "commPlanDetails": {
    "defaultCommPlan": "QACP4",
    "defaultOnCommProfile": "QACPR3",
    "defaultOffCommProfile": "QACPR4",
    "onCommProfileHlrTemplateId": "999999102236001",
    "offCommProfileHlrTemplateId": "999999101669802"
    },
  "defaultRatePlan": {
    "defaultRatePlanId": 78138,
    "defaultRatePlanName": "29658_RPID_MonInd"
    },
  "billingAddr": {
    "addr1": "123 Main St",
    "addr2": null,
    "city": "Danville",
    "region": "CA",
    "countryCode": "US",
    "postalCode": "94526"
    },
  "shippingAddr": {
    "addr1": "123 Main St",
    "addr2": null,
    "city": "Danville",
    "region": "CA",
    "countryCode": "US",
    "postalCode": "94526"
    },
  "ppuAddress": {
    "addr1": "123 Main St",
    "addr2": null,
    "city": "Danville",
    "region": "CA",
    "countryCode": "US",
    "postalCode": "94526"
    },
  "primaryContact": {
    "firstName": "Katie",
```


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```
  "lastName": "Duncan",
  "email": "katiedu@widgets.com",
  "phone": "123-345-2345",
  "fax": null,
  "name": null,
  "jobTitle": null
  },
"billingContact": {
  "firstName": "Katie",
  "lastName": "Duncan",
  "email": "katiedu@widgets.com",
  "phone": "123-345-2345",
  "fax": null,
  "name": null,
  "jobTitle": null
  },
"acctOnboardedBy": "SP_OPERATIONS",
"acctSegment": null,
"applicationSettings": {
  "applicationFeature": {
  "accessCommPlan": true,
  "businessRulesEnabled": true,
  "simOrderingEnabled": true,
  "smsAccess": true,
  "voiceAccess": true
  },
"simFeature": {
  "blockBackwardSimState": true,
  "displayDiagnosticWizard": true,
  "displaySpotlight": true,
  "fixedIpEnabled": true,
  "testReadyDataLimit": 20480,
  "testReadySmsLimit": 2,
  "testReadyVoiceLimit": 120,
  "testReadyDataState": "ACTIVATION_READY"
      }
},
"billingSettings": {
  "general": {
    "billable": "No",
    "prorationRule": "None",
    "renewalProrationRule": "None",
    "simFee": null,
    "activationPlan": "First Time Only",
    "activationFee": null,
    "vpnPlan": "None",
    "retailBilling": "None",
    "immediateRatePlanChange": true,
    "defaultTimeZone": null,
    "eventPlanEnabled": false,
    "prepaidRenewalPolicy": "Deactivate",
    "prepaidRenewalRatePlanId": null
    }
 },
 "security": {
   "ipLoginRestrictionEnabled": true,
   "secureSimEnabled": false,
   "twoStepVerification": true,
   "passwordPolicySettings": {
```


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```
        "ui": {
          "passwordExpirationInDays": "Passwords do not expire.",
          "numAttemptsBeforeTempLockout": "User accounts never lock out.",
          "tempLockoutPeriodInMinutes": null,
          "numHistoryChecks": null,
          "content": {
            "minLength": "6",
            "maxLength": "25",
            "minUppercaseChars": null,
            "minNumerics": null,
            "minSpecialChars": null
            }
          },
        "soap": {
          "passwordExpirationInDays": "Passwords do not expire.",
          "numAttemptsBeforeTempLockout": "User accounts never lock out.",
          "tempLockoutPeriodInMinutes": null,
          "numHistoryChecks": null,
          "content": {
          "minLength": "6",
          "maxLength": "25",
          "minUppercaseChars": null,
          "minNumerics": null,
          "minSpecialChars": null
         }
       }
     }
   }
}
```


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Code Samples

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/accounts/'+input("Account ID: ")
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-URL>/rws/api/v1/accounts/147060908').auth
('username', 'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/accounts/147060908'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
20000003	404	Resource not found - Invalid accountId.
30000001	500	Unknown server error.

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Get All Accounts

Description

Returns all accounts associated with the service provider. The response body contains a subset of the details for each account. Use the Get Accounts Details API to get additional information for a particular account.

You must have the ServiceProviderAdmin role to perform this function.

Resource URL

GET BaseURL/v{apiVersion}/accounts

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.

Response Parameters

Return Value	Description
pageNumber	An integer specifying the number of the current response page. See *Pagination on page 48*.
accounts	An array of accounts associated with the service provider.
accountName	The account name appears throughout Control Center as an identifier for the account.
accountId	A Control Center-generated identifier for the account.

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Return Value	Description
type	The type of account. Here are some values you may see: • *STANDARD*. For Advantage enterprise accounts. • *ESSENTIAL*. For Essential enterprise accounts. • *OPERATOR*. For service provider accounts. Each service provider has only one of these accounts. This account should contain all service provider users, but no devices. • *INVENTORY*. For service provider inventory accounts. Every service provider has at least one inventory account. An inventory account contains devices that have not been assigned to an enterprise account. The devices in this account have a special status called *Start*.
status	Indicates whether the account is active or inactive. When an account is inactive, account users cannot log into Control Center through the web interface or the APIs. However, account devices remain active in Control Center and on the network. Automation rules for the account also remain active and may trigger based on the defined conditions. Typically, service providers use this flag as a way to "delete" an account. Valid values are *Active* and *Inactive*.
currency	The currency value is the unit of measure used for the rate plan and billing information.
operatorAccountId	The account ID used by the service provider's internal IT systems. Control Center uses this ID in the invoice report feed that you receive for IT integration.
taxId	The tax or VAT ID is an account identifier that may be used for tax purposes.
industryVertical	This flag indicates the industry vertical for the account. For valid values, see *Industry Vertical Values on page 53*.
commPlanDetails	See the Communication Plan Parameters table below.
defaultRatePlan	See the Rate Plan Parameters table below.

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Return Value	Description
acctOnboardedBy	Indicates how the enterprise account was onboarded. For accounts onboarded before this field was implemented, this field is blank. • SP_OPERATIONS. For accounts onboarded by the service provider. • STARTERKIT. For accounts onboarded using the starter kit process.
lastPage	A true or false value indicating whether the current response page is the last in the series. See *Pagination on page 48*.

Communication Plan Parameters

Control Center uses the default communication plan for any account device that does not have a communication plan assigned to it.

Parameter	Description
defaultCommPlan	The name of the default communication plan.
defaultOnCommProfile	The name of the communication profile devices in the ON state will use.
defaultOffCommProfile	The name of the communication profile devices in the OFF state will use.
onCommProfileHlrTemplateId	A unique identifier for the HLR template containing the ON communication profile information.
offCommProfileHlrTemplateId	A unique identifier for the HLR template containing the OFF communication profile information.

Rate Plan Parameters

Control Center applies the default rate plan to any SIMs you transfer into the account unless you specify a different rate plan.

Parameter	Description
defaultRatePlanId	The Plan ID for the default rate plan.
defaultRatePlanName	The Plan Name for the default rate plan.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/accounts"
```

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Response Example


```
{
  "pageNumber": 1,
  "accounts": [{
    "accountName": "Acme Corp",
    "accountId": "100595340",
    "type": "STANDARD",
    "status": "Active",
    "currency": "USD",
    "operatorAccountId": null,
    "taxId": "78593",
    "industryVertical": "AVIATION",
    "commPlanDetails": {
      "defaultCommPlan": "Acme Default",
      "defaultOnCommProfile": "Acme ON",
      "defaultOffCommProfile": "Acme OFF"
      "onCommProfileHlrTemplateId": "999999108256301",
      "offCommProfileHlrTemplateId": "999999109629702"
    },
    "defaultRatePlan": {
      "defaultRatePlanId": 1126395,
      "defaultRatePlanName": "Acme Flex Pool"
    },
    "acctOnboardedBy": "SP_OPERATIONS"
  }, {
    "accountName": "Widgets Inc",
    "accountId": "100613714",
    "type": "STANDARD",
    "status": "Active",
    "currency": "USD",
    "operatorAccountId": "39599",
    "taxId": "23090",
    "industryVertical": "SMART_METERS",
    "commPlanDetails": {
      "defaultCommPlan": "Widgets Default",
      "defaultOnCommProfile": "Widgets ON",
      "defaultOffCommProfile": "Widgets OFF"
      "onCommProfileHlrTemplateId": "999999105236806",
      "offCommProfileHlrTemplateId": "999999129729305"
    },
    "defaultRatePlan": {
      "defaultRatePlanId": 1993832,
      "defaultRatePlanName": "Widgets Fixed Pool"
    },
    "acctOnboardedBy": "SP_OPERATIONS"
  }],
  "lastPage": true
}
```


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Code Samples

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/accounts'
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-URL>/rws/api/v1/accounts').auth('username',
'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/accounts'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000062	400	Access Denied.
30000001	500	Unknown server error.

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Create Account

Description

Creates a standard enterprise account with the minimum information required for a new account. Using this REST API is equivalent to creating an account in the web interface with just the information on the first page of the account creation form.

You must have the ServiceProviderAdmin role to perform this function.

Resource URL

POST BaseURL/v{apiVersion}/accounts

Request Parameters

All parameters are required unless the description says otherwise.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountName	The account name appears throughout Control Center as an identifier for the account. We recommend using the company name. This name must be unique with at least 2 characters, but no more than 60. Commas (,) and asterisks (*) are not allowed.
type	The type of enterprise account: • *ESSENTIAL* for Essential accounts • *STANDARD* for Advantage accounts
operator	(Optional) The name of the service provider. This field is optional for service provider users.
currency	The currency value is the unit of measure used for the rate plan and billing information. Note that you can use only one currency type to bill an account. For valid values, see *Currency Values on page 51*.
language	The default account language. The preference of the logged in user will always take precedence over the account language. For valid values, see *Language Values on page 51*.
taxId	The tax or VAT ID for the account appears in the invoice report feeds that Control Center sends to the service provider. You can use this account identifier for tax purposes.

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Parameter	Description
industryVertical	(Optional) This flag indicates the industry vertical for the account. If the value is left blank, the system defaults to *OTHER*. For valid values, see *Industry Vertical Values on page 53*.
defaultSIMState	This field defines the default SIM state that devices receive when the service provider transfers them from the inventory account into the enterprise account. For valid values, see *SIM Status Values on page 50*.
defaultCommPlanName	Control Center uses the default communication plan for any account device that does not have a communication plan assigned to it. You must supply the name of an existing shared communication plan.
defaultRatePlanName	Control Center applies the default rate plan to any devices you transfer into the account. You must supply the name of an existing shared base rate plan. Control Center automatically assigns that rate plan to the account.
p5gCommercialStatus	(Optional) This field is only for accounts using the Cisco IoT Private 5G service.
primaryContact	Every account must have a primary contact with detailed information. See Contact Parameters table below for details.
billingContactSameAsPrimary	This flag indicates whether the billing contact is the same as the primary contact. If you choose *true*, you can omit the billing contact information. If you choose *false*, the billing contact fields are required.
billingContact	(Optional) Every account must have a billing contact. If the billingContactSameAsPrimary flag is *true*, you can omit the fields. Otherwise, you must provide values for these fields (see Contact Parameters table below).
billingAddr	The billing address is required. See the Address Parameters table below for details.
shippingAddrSameAs Billing	This flag indicates whether the shipping address is the same as the billing address. If you choose *true*, you can omit the shipping address information. If you choose *false*, the shipping address fields are required.

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Parameter	Description
shippingAddr	(Optional) Every account must have a shipping address. If the shippingAddrSameAsBilling flag is *true*, you can omit the fields. Otherwise, you must provide values for this list of fields (see Address Parameters table below).
ppuaddrSameAs Shipping	This flag indicates whether the Primary Place of Use (PPU) address is the same as the shipping address. If you choose *true*, you can omit the PPU address information. If you choose *false*, the PPU address fields are required.
ppuAddr	(Optional) Every account must have a Primary Place of Use (PPU) address. If the ppuaddrSameAsShippingflag is *true*, you can omit the fields. Otherwise, you must provide values for this list of fields (see Address Parameters table below).
accountSegment	(Optional) Account segments are service provider-created categories that reflect the service provider’s business requirements.
notes	(Optional) Provides additional information about the account. Maximum length is 2000 characters.
operatorAccountId	(Optional) The account ID used by the service provider's internal IT systems. Control Center uses this ID in the invoice report feed that you receive for IT integration.
customFields	(Optional) If you have already created custom account fields to record business-specific information, you can supply values for these fields when you create a new account. To view, edit, or create custom account fields and their values, go to Admin > Service Provider Profile > Account Custom Fields. See fieldValue1...10.
fieldValue1...10	(Optional) You can provide account-specific values for existing custom account fields. You must use the parameter names "fieldValue1", "fieldValue2", and so on, regardless of the field names that appear in the web interface.

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Contact Parameters

These parameters are the same for both the primary contact and the billing contact, with one exception. The primary contact can have a job title, but the billing contact does not.

Parameter	Description
firstName	A first name for the contact person. String, 1-49 characters
lastName	A last name for the contact person. String, 1-49 characters
email	An email for the contact person. This string must conform to standard email format. No more than 320 characters.
phone	A phone number for the contact person. String, 1-25 characters
fax	A fax number for the contact person. String, 1-25 characters
name	This string is the combination of the first and last name values. Control Center uses the string for the Primary Contact field on the Accounts list and the Account Details pages. String, up to 100 characters.
jobTitle	The primary contact can have a job title, but the billing contact does not. String, up to 100 characters.

Address Parameters

These parameters are the same for all three types of addresses: billing, shipping, and primary place of use.

Parameter	Description
addr1	Street address, line 1. String, 1-100 characters
addr2	(Optional) Street address, line 2. String, 1-100 characters
city	The city name. String, 1-100 characters
region	The region name — a state or province, for example. String, 1-100 characters
countryCode	A two character country code. See ISO 3166 for two-letter codes (alpha-2).
postalCode	A postal code. String, 1-50 characters

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Response Parameters

Control Center returns the account ID for the new account.

Return Value	Description
accountId	An unique identifier for the new account.
timestamp	The date and time when the account was created. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials.

```
{ curl -X POST --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"accountName\": \"ProDevice Inc.\",
  \"type\": \"STANDARD\",
  \"currency\": \"USD\",
  \"language\": \"English\",
  \"taxId\": \"34006\",
  \"industryVertical": \"AVIATION\",
  \"defaultSIMState\": \"TEST_READY\",
  \"defaultCommPlanName\": \"QACP4\",
  \"defaultRatePlanName\": \"29658_RPID_MonInd\",
  \"primaryContact\": {
    \"name\": \"John Doe\",
    \"firstName\": \"John\",
    \"lastName\": \"Doe\",
    \"email\": \"john@prodevice.com\",
    \"phone\": \"432-556-4545\" },
  \"billingContactSameAsPrimary\": true,
  \"billingAddr\": {
    \"addr1\": \"335 Elm St\",
    \"city\": \"Sacramento\",
    \"region\": \"CA\",
    \"countryCode\": \"US\",
    \"postalCode\": \"94632\" },
  \"shippingAddrSameAsBilling\": true,
  \"ppuaddrSameAsShipping\": true
}" "https://<YOUR-BASE-URL>/rws/api/v1/accounts"
```

Response Example

```
{
    "accountId": 105520704,
    "timestamp": 1527590173
}
```

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Errors

Error Code	HTTP Code	Error Message
1000591	400	Setting P5G attributes are not allowed since P5GaaS is not enabled for the operator Description: You cannot set the p5gCommercialStatus field because the service provider does not support the Cisco IoT Private 5G (P5G) service.
1000593	400	Setting p5gCommercialStatus access is not allowed for the user
1000594	400	Updating from commercial status is not allowed
1000621	400	Invalid Industry Vertical.
10000013	404	Invalid ratePlan.
10000014	400	Invalid communicationPlan.
10000023	400	The JSON in the request is not well formed. Please ensure that commas, colons, braces etc. are formatted properly.
10000024	400	Invalid apiVersion.
10000062	400	Access Denied.
10000500	400	Service provider not found.
10000501	400	Could not create account.
10000504	400	Account already exists.
10000505	400	Invalid region Id.
10000515	400	Invalid BillingCycleStart.
10000516	400	Not allow to set BillingCycleStart.
10000521	400	Invalid Country Code.
10000528	400	Partner Account creation not allowed.
10000548	400	Primary contact firstName and lastName are required.
30000001	500	Unknown server error.

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Error Code	HTTP Code	Error Message
50001225	400	Shipping Address cannot be null as it is not same as the billing address.
50001226	400	PPU Address cannot be null is it as not same as the shipping address.
50001224	400	Billing Contact cannot be null is it is not same as primary contact.
50001229	400	Currency code is invalid.
50001227	400	Service Provider name is invalid.
50001228	400	Account type is invalid.
50001233	400	JSON validation error.
50001230	400	Language is invalid.
50001231	400	Sim state name is invalid.

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Edit Account Details

Description

Changes one or more account settings for a single active account. You cannot make changes to an inactive account.

You must have the ServiceProviderAdmin role to perform this function.

Resource URL

PUT BaseURL/v{apiVersion}/accounts/{accountId}

Request Parameters

The following tables list all the parameters you can update using this API function. Typically, users will update one or two parameters at a time.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The ID of the account you want to change. You cannot change this value.
accountName	The account name appears throughout Control Center as an identifier for the account. We recommend using the company name. This name must be unique with at least 2 characters, but no more than 60. Commas (,) and asterisks (*) are not allowed.
status	Indicates whether the account is active (*A*) or inactive (*I*).
currency	The currency value is the unit of measure used for the rate plan and billing information. Note that you can use only one currency type to bill an account. For valid values, see *Currency Values on page 51*.
language	The default account language. The preference of the logged in user will always take precedence over the account language. For valid values, see *Language Values on page 51*.
taxId	The tax or VAT ID for the account appears in the invoice report feeds that Control Center sends to the service provider. You can use this account identifier for tax purposes.
industryVertical	This flag indicates the industry vertical for the account. If the value is left blank, the system defaults to *OTHER*. For valid values, see *Industry Vertical Values on page 53*.

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Parameter	Description
p5gCommercialStatus	This field is only for accounts using the Cisco IoT Private 5G service.
primaryContact	Every account must have a primary contact with detailed information. See Contact Parameters table below for details.
billingContact	Every account must have a billing contact with detailed information. See Contact Parameters table below for details.
billingAddr	A billing address. See the Address Parameters table below for details.
shippingAddr	A shipping address. See the Address Parameters table below for details.
ppuAddr	A primary place of use (PPU) address. See the Address Parameters table below for details.
accountSegment	Account segments are service provider-created categories that reflect the service provider’s business requirements. You can view, add, edit, or delete these values in the web interface: Admin > Service Provider Profile > Account Segments subtab.
notes	Provides additional information about the account. Maximum length is 2000 characters.
operatorAccountId	The account ID used by the service provider's internal IT systems. Control Center uses this ID in the invoice report feed that you receive for IT integration.
customFields	If you have already created custom account fields to record business-specific information, you can specify values for these fields. To view, edit, or create custom account fields and their values, go to Admin > Service Provider Profile > Account Custom Fields.
fieldValue1-10	You can provide account-specific values for existing custom account fields. You must use the parameter names "fieldValue1", "fieldValue2", and so on, regardless of the field names that appear in the web interface.

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Contact Parameters

These parameters are the same for both the primary contact and the billing contact, with one exception. The primary contact can have a job title, but the billing contact does not.

Parameter	Description
firstName	A first name for the contact person. String, 1-49 characters
lastName	A last name for the contact person. String, 1-49 characters
email	An email for the contact person. This string must conform to standard email format. No more than 320 characters.
phone	A phone number for the contact person. String, 1-25 characters
fax	A fax number for the contact person. String, 1-25 characters
name	This string is the combination of the first and last name values. Control Center uses the string for the Primary Contact field on the Accounts list and the Account Details pages. String, up to 100 characters.
jobTitle	The primary contact can have a job title, but the billing contact does not. String, up to 100 characters.

Address Parameters

These parameters are the same for all three types of addresses: billing, shipping, and primary place of use.

Parameter	Description
addr1	Street address, line 1. String, 1-100 characters
addr2	(Optional) Street address, line 2. String, 1-100 characters
city	The city name. String, 1-100 characters
region	The region name — a state or province, for example. String, 1-100 characters
countryCode	A two character country code. See ISO 3166 for two-letter codes (alpha-2).
postalCode	A postal code. String, 1-50 characters

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Response Parameters

Control Center returns the account ID for the updated account.

Return Value	Description
accountId	An unique identifier for the updated account.
timestamp	The date and time when the account was updated. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Account updates are typically brief, targeting one or two parameters at a time. Make sure to use your own URL and user credentials.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"currency\": \"USD\" }
}" "https://<YOUR-BASE-URL>/rws/api/v1/accounts/100020704"
```

Response Example

```
{
    "accountId": 100020704,
    "timestamp": 1527590173
}
```

Errors

Error Code	HTTP Code	Error Message
1000591	400	Setting P5G attributes are not allowed since P5GaaS is not enabled for the operator Description: You cannot set the p5gCommercialStatus field because the service provider does not support the Cisco IoT Private 5G (P5G) service.
1000593	400	Setting p5gCommercialStatus access is not allowed for the user
1000594	400	Updating from commercial status is not allowed
1000621	400	Invalid Industry Vertical.
1400109	404	User not found.

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Error Code	HTTP Code	Error Message
10000002	400	AccountId is required.
10000004	400	Invalid accountId.
10000005	400	Invalid status.
10000013	404	Invalid ratePlan.
10000014	400	Invalid communicationPlan.
10000024	400	Invalid apiVersion.
10000033	400	Invalid contactName.
10000521	400	Invalid Country Code.
20000003	404	Resource not found - Invalid accountId.
30000001	500	Unknown server error.
50001229	400	Currency code is invalid.
50001234	400	CommPlan not allowed for Partner Account.
50001233	400	JSON validation error.
50001230	400	Language is invalid.
50001235	400	Invalid AccountSegment.

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Get Account Application Settings

Description

Returns account application settings for a specified account. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

GET BaseURL/v{apiVersion}/accounts/{accountId}/applicationSettings

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The ID of the account whose application settings you want to retrieve.

Response Parameters

This function returns data organized into four different groups corresponding to the groups you'll see in the web interface (Admin > Account Details > Application Settings):

- Application Feature
- SIM Feature
- Site Configuration
- Support

Return Value	Description
applicationFeature	See the Application Feature table below.
simFeature	See the SIM Feature table below.
siteConfiguration	See the Site Configuration table below.
support	See the Support table below.

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Application Feature

Return Value	Description
accessCommPlan	All account users can see the communication plan name in the device Edit form. This toggle determines whether the field is editable. If this toggle is set to *true*, then account users have the ability to change a device's communication plan. In addition, Control Center displays the communication plan name in the Device List and on the device Details page. If the toggle is set to *false*, account users cannot change a device's communication plan and will not see the communication plan name in the Device List and on the device Details page.
csdAccess	A *true* or *false* indicates whether or not the application displays access to Circuit Switched Data (CSD) fields in the user interface are visible to account users. Not supported for Essential accounts.
businessRulesEnabled	A *true* or *false* indicates whether or not business rules are available for the account. The rules functionality is visible in the Automation category to users associated with the account. Be aware that any existing business rules associated with the account continue to function even it you disable rules for the account. Not supported for Essential accounts.
globalAccountEnabled	If set to *true*, the Global SIM feature is enabled for this account. This feature enables service providers to create partnerships with other service providers to serve multinational enterprise customers. Unlike a typical roaming agreement between service providers, a global alliance allows the roaming SIM to actually swap credentials (IMSIs, ICCIDs, and MSISDNs) from the original service provider to the partner service provider using over-the-air (OTA) swap technology. If set to *false*, the Global SIM feature is disabled for this account. Not supported for Essential accounts.
policiesEnabled	This legacy field is no longer used. Not supported for Essential accounts.

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Return Value	Description
simOrderingEnabled	A *true* or *false* indicates whether or not the SIM ordering feature is enabled for the account. If set to true, account users can place an order that includes the default rate plan and communication plan for the new SIMs.
smsAccess	If this toggle is set to *true*, then the SMS fields in the user interface are visible to account users. If this toggle is set to *false*, then the SMS fields are not visible.
subscriptionChannel	A *true* or *false* indicates whether or not the application has enabled a subscription channel. This capability allows a deactivated account to go to a specific URL and reactivate itself. Not supported for Essential accounts.
voiceAccess	If this toggle is set to *true*, then the voice fields in the user interface are visible to account users. If this toggle is set to *false*, then the voice fields are not visible.
smsEnabled	Indicates whether or not the account can use the SMS service. To use SMS, a device must also have a communication plan that allows SMS. Only for Essential accounts. If Simplified Packaging is enabled, this parameter is not available.
voiceEnabled	Indicates whether or not the account can use the voice service. To use voice, a device must also have a communication plan that allows voice. Only for Essential accounts. If Simplified Packaging is enabled, this parameter is not available.

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SIM Feature

Return Value	Description
blockBackwardSimState	If set to *true*, account users cannot change a device status from *Activated* to the pre-activated status (such as *Activation Ready*, *Inventory*, or *Test Ready*). If set to *false*, there are no restrictions on changing a device status to *Activation Ready*, *Inventory*, or *Test Ready*. This setting prevents an account from moving under-used devices to *Activation Ready* where they can avoid being charged a subscription fee until they attempt to make a connection.
defaultSimState	This field defines the default SIM state that devices receive when the service provider transfers them from the inventory account into the enterprise account. For a list of valid values, see *SIM Status Values on page 50*.
displayDiagnosticWizard	The Diagnostics tool provides a very simple interface that users with little wireless knowledge can use to quickly see what is happening with a device on the network. If set to *Yes*, then Control Center displays the Diagnostics tool to account users. If set to *No*, then Control Center hides the Diagnostics tool from account users.
displaySpotlight	Spotlight is a troubleshooting tool that provides a detailed snapshot of device connectivity history over the past 30 days so you can observe long-term trends. This tool is designed for advanced support representatives, who have technical expertise in wireless connectivity. If set to Yes, then Control Center displays Spotlight to account users. If set to No, then Control Center hides Spotlight from account users.
fixedIpEnabled	If set to *true*, Control Center displays the fixed IP address (if any) for devices in the Device List page. If set to *false*, Control Center does not display the fixed IP addresses.

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Return Value	Description
testReadyDataLimit	This field specifies the amount of data (in bytes) a device in the *Test Ready* state receives free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it initiates traffic. The upper values is 10.
testReadySmsLimit	This field specifies the number of messages a device in the *Test Ready* state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it sends or receives a message. The upper values is 10.
testReadyVoiceLimit	This field specifies the number of voice minutes a device in the *Test Ready* state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it places or receives a phone call. The upper values is 10.
testReadyCsdLimit	This field specifies the number of Circuit Switched Data in seconds (CSD) a device in the *Test Ready* state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it places or receives a phone call. The upper values is 10.
testReadyDataState	Once the device reaches the configured limit in the *Test Ready* state, Control Center automatically moves the device to another SIM state. This field specifies the next SIM state. Most service providers use either *Activation Ready* or *Inventory*. For a list of valid values, see *SIM Status Values on page 50*.

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Site Configuration

Return Value	Description
cobrandSite	The URL for the branded instance of Control Center that account users will use to launch Control Center.
fileExportMethod	This field determines the format of an export files the user creates. The options are *EXCEL* and *CSV*.
eulaEnabled	If set to *true*, Control Center prompts a new user to view and accept the End User License Agreement (EULA) the first time the user logs into this account. If set to false, Control Center does not prompt account users to view and accept the EULA when they log in for the first time.

Support

Return Value	Description
supportEmail	The support email address. If showAccountSupport is set to *yes*, then this support email displays in the Account Support section of the Help page.
supportPhone	The support phone number. If showAcctSupport is set to *yes*, then this support phone number displays in the Account Support section on the Help page.
showAcctSupport	If set to *yes*, then the support phone number and support email displays in the Account Support section on the Help page.
showSpSupport	If set to *yes*, Control Center displays Service Provider Support information on the Help page. On this page, the account will see a support site URL, an email, and a phone number. All this information is associated with the service provider profile located in Admin > Service Provider Profile > Support Access. If you select *false*, then users will not see the Service Provider Support information on the Help page.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/accounts/130265703/applicationSettings"
```

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Response Example


```
{
  "applicationFeature": {
     "accessCommPlan": false,
     "csdAccess": false,
     "businessRulesEnabled": true,
     "globalAccountEnabled": false,
     "policiesEnabled": false,
     "simOrderingEnabled": false,
     "smsAccess": true,
     "subscriptionChannel": false,
     "voiceAccess": false
  },
  "simFeature": {
     "blockBackwardSimState": false,
     "defaultSimState": "TEST_READY",
     "displayDiagnosticWizard": true,
     "displaySpotlight": true,
     "fixedIpEnabled": false,
     "testReadyDataLimit": 20480,
     "testReadySmsLimit": 2,
     "testReadyVoiceLimit": 120,
     "testReadyCsdLimit": 120,
     "testReadyDataState": "ACTIVATION_READY"
  },
  "siteConfiguration": {
     "cobrandSite": null,
     "fileExportMethod": "EXCEL",
     "eulaEnabled": true
  },
  "support": {
     "supportEmail": null,
     "supportPhone": null,
     "showAcctSupport": false,
     "showSpSupport": true
  }
}
```


Errors

Error Code	HTTP Code	Error Message
10000024	400	Invalid apiVersion.
20000003	404	Resource not found - Invalid accountId.
50001236	400	Unsupported Account Type.

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Edit Account Application Settings

Description

Edits the application settings for a specific account. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

PUT BaseURL/v{apiVersion}/accounts/{accountId}/applicationSettings

Request Parameters

This function organizes the application settings data into four different groups corresponding to the groups you'll see in the web interface (Admin > Account Details > Application Settings):

- Application Feature
- SIM Feature
- Site Configuration
- Support

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The ID of the account whose application settings you want to edit.
applicationFeature	See the Application Feature table below.
simFeature	See the SIM Feature table below.
siteConfiguration	See the Site Configuration table below.
support	See the Support table below.

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Application Feature

All parameters are required.

Parameter	Description
accessCommPlan	All account users can see the communication plan name in the device Edit form. This toggle determines whether the field is editable. If this toggle is set to *true*, then account users have the ability to change a device's communication plan. In addition, Control Center displays the communication plan name in the Device List and on the device Details page. If the toggle is set to *false*, account users cannot change a device's communication plan and will not see the communication plan name in the Device List and on the device Details page.
csdAccess	A *true* or *false* indicates whether or not the application displays access to Circuit Switched Data (CSD) fields in the user interface are visible to account users. Not supported for Essential accounts.
businessRulesEnabled	A *true* or *false* indicates whether or not the application has enabled business rules for the account. The rules functionality is visible in the Automation category to users associated with the account. Be aware that any existing business rules associated with the account continue to function even it you disable rules for the account. Not supported for Essential accounts.
globalAccountEnabled	If set to *true*, the Global SIM feature is enabled for this account. This feature enables service providers to create partnerships with other service providers to serve multinational enterprise customers. Unlike a typical roaming agreement between service providers, a global alliance allows the roaming SIM to actually swap credentials (IMSIs, ICCIDs, and MSISDNs) from the original service provider to the partner service provider using over-the-air (OTA) swap technology. If set to *false*, the Global SIM feature is disabled for this account. Not supported for Essential accounts.

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Parameter	Description
policiesEnabled	This legacy field is no longer used. Not supported for Essential accounts.
simOrderingEnabled	A *true* or *false* indicates whether or not the SIM ordering feature is enabled for the account. If set to true, account users can place an order that includes the default rate plan and communication plan for the new SIMs.
smsAccess	If this toggle is set to *true*, then the SMS fields in the user interface are visible to account users. If this toggle is set to *false*, then the SMS fields are not visible.
subscriptionChannel	A *true* or *false* indicates whether or not the application has enabled a subscription channel. This capability allows a deactivated account to go to a specific URL and reactivate itself. Not supported for Essential accounts.
voiceAccess	If this toggle is set to *true*, then the voice fields in the user interface are visible to account users. If this toggle is set to *false*, then the voice fields are not visible.
smsEnabled	Indicates whether or not the account can use the SMS service. To use SMS, a device must also have a communication plan that allows SMS. Only for Essential accounts. If Simplified Packaging is enabled, this parameter is not available.
voiceEnabled	Indicates whether or not the account can use the voice service. To use voice, a device must also have a communication plan that allows voice. Only for Essential accounts. If Simplified Packaging is enabled, this parameter is not available.

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SIM Feature

All parameters are required.

Parameter	Description
blockBackwardSimState	If set to *true*, account users cannot change a device status from *Activated* to the pre-activated status (such as *Activation Ready*, *Inventory*, or *Test Ready*). If set to *false*, there are no restrictions on changing a device status to *Activation Ready*, *Inventory*, or *Test Ready*. This setting prevents an account from moving under-used devices to *Activation Ready* where they can avoid being charged a subscription fee until they attempt to make a connection.
defaultSimState	This field defines the default SIM state that devices receive when the service provider transfers them from the inventory account into the enterprise account. For a list of valid values, see *SIM Status Values on page 50*.
displayDiagnosticWizard	The Diagnostics tool provides a very simple interface that users with little wireless knowledge can use to quickly see what is happening with a device on the network. If set to Yes, then Control Center displays the Diagnostics tool to account users. If set to No, then Control Center hides the Diagnostics tool from account users.
displaySpotlight	Spotlight is a troubleshooting tool that provides a detailed snapshot of device connectivity history over the past 30 days so you can observe long-term trends. This tool is designed for advanced support representatives, who have technical expertise in wireless connectivity. If set to Yes, then Control Center displays Spotlight to account users. If s set to No, then Control Center hides Spotlight from account users.
fixedIpEnabled	If set to *true*, Control Center displays the fixed IP address (if any) for devices in the Device List page. If set to *false*, Control Center does not display the fixed IP addresses.

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Parameter	Description
testReadyDataLimit	This field specifies the amount of data (in bytes) a device in the *Test Ready* state receives free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it initiates traffic. The upper values is 10.
testReadySmsLimit	This field specifies the number of messages a device in the Test Ready state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it sends or receives a message. The upper values is 10.
testReadyVoiceLimit	This field specifies the number of voice minutes a device in the Test Ready state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it places or receives a phone call. The upper values is 10.
testReadyCsdLimit	This field specifies the number of Circuit Switched Data in seconds (CSD) a device in the Test Ready state will receive free of charge. Once the configured limit is reached for any service (voice, SMS, data, or other), the device will transition to the target state, and no more free test traffic will be available for any service. A blank value indicates there is no limit. For a zero value, the device will reach the limit the first time it places or receives a phone call. The upper values is 10.
testReadyDataState	Once the device reaches the configured limit in the *Test Ready* state, Control Center automatically moves the device to another SIM state. This field specifies the next SIM state. Most service providers use either Activation Ready or Inventory. For a list of valid values, see *SIM Status Values on page 50*.

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Site Configuration

All parameters are required, but those marked as optional can take null values.

Parameter	Description
cobrandSite	(Optional). The URL for the branded instance of Control Center that account users will use to launch Control Center.
fileExportMethod	This field determines the format of an export files the user creates. The options are *Excel* or *CSV*.
eulaEnabled	If set to *true*, Control Center prompts a new user to view and accept the End User License Agreement (EULA) the first time the user logs into this account. If set to false, Control Center does not prompt account users to view and accept the EULA when they log in for the first time.

Support

All parameters are required, but those marked as optional can take null values.

Return Value	Description
supportEmail	(Optional) The support email address. If showAccountSupport is set to *yes*, then this support email displays in the Account Support section of the Help page.
supportPhone	(Optional) The support phone number. If showAcctSupport is set to *yes*, then this support phone number displays in the Account Support section on the Help page.
showAcctSupport	If set to *yes*, then the support phone number and support email displays in the Account Support section on the Help page.
showSpSupport	If set to *yes*, Control Center displays Service Provider Support information on the Help page. On this page, the account will see a support site URL, an email, and a phone number. All this information is associated with the service provider profile located in Admin > Service Provider Profile > Support Access. If you select *false*, then users will not see the Service Provider Support information on the Help page.

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Response Parameters

Control Center returns the account ID for the updated account.

Return Value	Description
accountId	An unique identifier for the updated account.
timestamp	The date and time when the account was updated. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{{
  \"applicationFeature\": {
    \"accessCommPlan\": false, "
    \"csdAccess\": false, "
    \"businessRulesEnabled\": false,
    \"globalAccountEnabled\": false,
    \"policiesEnabled\": false,
    \"simOrderingEnabled\": false,
    \"smsAccess\": true,
    \"subscriptionChannel\": false,
    \"voiceAccess\": false
    \"smsEnabled\": false,
    \"voiceEnabled\": false
    },
  \"simFeature\": {
    "blockBackwardSimState\": false,
    "defaultSimState\": "TEST_READY",
    "displayDiagnosticWizard\": true,
    "displaySpotlight\": true,
    "fixedIpEnabled\": false,
    "testReadyDataLimit\": 20480,
    "testReadySmsLimit\": 2,
    "testReadyVoiceLimit\": 120,
    "testReadyCsdLimit\": 120,
    "testReadyDataState\": "ACTIVATION_READY"
    },
  \"siteConfiguration\": {
    \"cobrandSite\": null,
    \"fileExportMethod\": "EXCEL",
    \"eulaEnabled\": true
    },
  \"support\": {
    \"supportEmail\": null,
    \ "supportPhone\": null,
    \"showAcctSupport\": false,
    \"showSpSupport\": true
    }
}" "https://<YOUR-BASE-
URL>/rws/api/v1/accounts/100020704/applicationSettings"
```

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Response Example

```
{
  "accountId": 100020704,
  "timestamp": 1527590173
}
```

Errors

Error Code	HTTP Code	Error Message
10000024	400	Invalid apiVersion.
10000301	400	Test Ready Data Limit length must be between 0 and 10.
10000302	400	Test Ready Sms Limit length must be between 0 and 10.
10000303	400	Test Ready Voice Limit length must be between 0 and 10.
10000304	400	Test Ready Csd Limit length must be between 0 and 10.
10000305	400	Test Ready Data Limit must be a positive number.
10000306	400	Test Ready Sms Limit must be a positive number.
10000307	400	Test Ready Voice Limit must be a positive number.
10000308	400	Test Ready Csd Limit must be a positive number.
10000309	400	Invalid Default Sim State.
10000310	400	Invalid Test ready data state.
10000311	400	Invalid File export method
10000312	400	No support information to show. Either Account Support Email and/or Account Support Phone must be specified.
10000313	400	Account Support Email must be a correctly formatted email address and cannot contain commas.

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Error Code	HTTP Code	Error Message
10000314	400	Account Support Phone cannot be more than 25 characters and cannot contain commas (,) or asterisks (*).
10000315	400	Invalid SMS Enabled value Only true/false allowed.
10000316	400	Invalid Voice Enabled value Only true/false allowed.
10000321	400	Application feature cannot be null.
10000322	400	Sim feature cannot be null.
10000323	400	Site configuration cannot be null.
10000324	400	Support cannot be null.
10000325	400	AccessCommPlan cannot be null.
10000326	400	CsdAccess cannot be null.
10000327	400	BusinessRulesEnabled cannot be null.
10000328	400	GlobalAccountEnabled cannot be null.
10000329	400	PoliciesEnabled cannot be null.
10000330	400	SimOrderingEnabled cannot be null.
10000331	400	SmsAccess cannot be null.
10000332	400	SubscriptionChannel cannot be null.
10000333	400	VoiceAccess cannot be null.
10000334	400	SmsEnabled cannot be null.
10000335	400	VoiceEnabled cannot be null.
10000336	400	BlockBackwardSimState cannot be null.
10000337	400	DefaultSimState cannot be null.
10000338	400	DisplayDiagnosticWizard cannot be null.
10000339	400	DisplaySpotlight cannot be null.
10000340	400	FixedIpEnabled cannot be null.
10000341	400	TestReadyDataLimit cannot be null.

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Error Code	HTTP Code	Error Message
10000342	400	TestReadySmsLimit cannot be null.
10000343	400	TestReadyVoiceLimit cannot be null.
10000344	400	TestReadyCsdLimit cannot be null.
10000345	400	TestReadyDataState cannot be null.
10000346	400	FileExportMethod cannot be null.
10000347	400	EulaEnabled cannot be null.
10000348	400	ShowAcctSupport cannot be null.
10000349	400	ShowSpSupport cannot be null.
20000003	404	Resource not found - Invalid accountId.
50001236	400	Unsupported Account Type.

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Get Account Billing Settings

Description

Retrieves the billing settings for a given account. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

GET BaseURL/v{apiVersion}/accounts/{accountId}/billingSettings

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The ID of the account.

Response Parameters

Parameter	Description
general	Contains the general billing setting information. See the General table for parameter definitions.
commitments	Contains information about device and account commitments. See the Commitments table for parameter definitions.

General Parameters

Return Value	Description
billable	If set to *Yes*, the account is billable. The default value is *No*.
currentBillingCycleStart	Indicates the first day of the account's current billing cycle.

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Return Value	Description
prorationRule	Indicates whether or not the billing engine will prorate subscription fees in the first billing cycle for devices using monthly rate plans. • *None*. The first month's subscription is charged for the full month regardless of when the device was first activated. • *Daily*. The first month's subscription is charged based on the number of days since the device was activated for the first time. • *Semi-Monthly*. The first month's subscription is charged either a full or half month depending on whether the device was activated for the first time in the first half or second half of the month.
renewalProrationRule	Similar to activation proration, this parameter covers the special case where a device ends a prepaid rate plan and moves onto a monthly plan. Control Center prorates the new subscription fee on a semi-monthly or daily basis – or not at all, depending on the value of this parameter. Valid values are: *Daily*, *Semi-Monthly*, and *None*.
simFee	This fee indicates how much the account is charged for any SIMs purchased. Not supported for Essential accounts.
activationPlan	The default activation plan defines how Control Center applies the activation fee to each device owned by the account. • *First Time Only*. The account pays the activation fee when the device is activated for the first time (typical). • *Each Time*. The account pays every time the device SIM status changes to *Activated*. • *Not First Time*. The account does not pay for the first device activation, but it does pay for every subsequent activation. • *None*. No charge. Not supported for Essential accounts.

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Return Value	Description
activationFee	Control Center charges this fee based the Activation Plan setting. This setting might be replaced if it is set at the rate plan level. In that case Control Center charges an activation fee based on the settings associated with the current device rate plan. (Admin > Accounts > Account Name link > Rate Plans subtab > Edit Settings button). Not supported for Essential accounts.
retailBilling	Indicates whether the account can use the retail billing feature: • *None*. The account does not support retail billing. • *Retail Customer*. The account supports retail billing. Not supported for Essential accounts.
immediateRatePlanChange	Rate plan changes for devices with non-active SIM status values (such as *Test Ready* or *Deactivated*) always take effect immediately. For devices with an *Activated* SIM status, you can control the rate plan change behavior with this setting. • If the toggle is on, any mid-cycle rate plan changes for devices with an *Activated* status take effect immediately. Control Center will bill for the device as if it had been on the new rate plan during the entire billing cycle. • If the toggle is off, Control Center schedules any mid-cycle rate plan changes for devices with an *Activated* status to take effect in the next billing cycle. See the Scheduled Changes subtab on the Device Details page. This behavior applies to devices with monthly rate plans only. You cannot change the rate plan for a device on a prepaid rate plan until the plan expires. Essential accounts do not support immediate rate plan changes for devices with an *Activated* SIM status, regardless of this parameter value.
defaultTimeZone	The default time zone to use for this account. Not supported for Essential accounts.

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Return Value	Description
eventPlanEnabled	If set to *true*, then event plans are enabled for this account. An event plan is a prepaid rate plan that's designed for special circumstances or "events" that occur outside of a normal rate plan. If set to *false*, then this account cannot use event plans. The default is *false*. Not supported for Essential accounts.
prepaidRenewalPolicy	When a prepaid rate plan expires, Control Center either deactivates the device or gives it a different rate plan. You can specify the default renewal behavior for all the account's prepaid rate plans. The options are as follows: • *Deactivate*. (Default) Deactivate the device. • *AutoRenew*. Renew the current prepaid rate plan. • *NamedPlan*. Assign a specific rate plan or assign the next rate plan in the device queue.
prepaidRenewalRatePlanId	When a prepaid device rate plan expires, Control Center attaches the named rate plan to the device. If the device has a queue, Control Center uses the rate plan at the top of the queue instead of the named plan here.

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Commitments Parameters

Return Value	Description
activationGracePeriod	This length of time (specified in days) is the grace period between the time a device is transferred to an account and Control Center starts billing for the device. If the device is not active (and billable) by the time this grace period ends, Control Center starts billing the subscription fee in the next billing cycle. To enable this feature, you must also set the Bill Activation Grace Period value to true. The grace period for devices using prepaid rate plans is indefinite. Control Center does not charge for these devices until they become active. This API does not return commitment data for Essential accounts.
billActivationGracePeriod	When this parameter is set to *true*, Control Center enables the activation grace period feature and bill the account for a device, regardless of its status, after the activation period ends. You also need to define the length of the grace period with the Activation Grace Period parameter. The default is *false*. This API does not return commitment data for Essential accounts.
monthlyDevicesMinimum Term	This parameter specifies the length of time the monthly device minimum will be in effect for each device in the account. The term begins on the device's activation date. The maximum term is 999 months. Control Center does not support Monthly Device Minimum commitments for Essential accounts.
monthlyDeviceMinimum	This parameter specifies the minimum revenue a device is expected to provide in each billing cycle. The maximum amount is 9999.99 (in the account's default currency). Control Center does not support Monthly Device Minimum commitments for Essential accounts.
billMonthlyDeviceMinimum	If set to *true*, Control Center enables the monthly device minimum commitment using the values in the monthlyDevicesMinimumTerm and monthlyDeviceMinimum parameters. Control Center does not support Monthly Device Minimum commitments for Essential accounts.

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Return Value	Description
minimumActivationTerm	The number of months each device must be active. To enable this feature, you must also set the Bill Minimum Activation Term parameter to *true*. This API does not return commitment data for Essential accounts.
billMinimumActivation Term	If set to *true*, Control Center enables the minimum activation term feature and bills the subscription fee for any devices that have not been active long enough to meet the minimum requirement. The default is *false*. This API does not return commitment data for Essential accounts.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/accounts/100015068/billingSettings"
```

Response Example


```
{
"general": {
  "billable": "No",
  "currentBillingCycleStart": "12th of month(Europe/Moscow)",
  "prorationRule": "None",
  "renewalProrationRule": "None",
  "simFee": null,
  "activationPlan": "None",
  "activationFee": null,
  "retailBilling": "None",
  "immediateRatePlanChange": true,
  "defaultTimeZone": null,
  "eventPlanEnabled": false,
  "prepaidRenewalPolicy": "Deactivate",
  "prepaidRenewalRatePlanId": null
}
"commitments": {
  "activationGracePeriod": 365,
  "billActivationGracePeriod": false,
  "monthlyDevicesMinimumTerm": null,
  "monthlyDeviceMinimum": null,
  "billMonthlyDeviceMinimum": false,
  "minimumActivationTerm": null,
  "billMinimumActivationTerm": false
}
}
```


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Errors

Error Code	HTTP Code	Error Message
10000004	400	Invalid accountId.
10000024	400	Invalid apiVersion.
20000003	404	Resource not found - Invalid accountId.
20000031	400	Invalid request to view account billing settings.
30000001	500	Unknown server error.

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Edit Account Billing Settings

Description

Changes the billing settings for a given account. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

PUT BaseURL/v{apiVersion}/accounts/{accountId}/billingSettings

Request Parameters

All parameters are required.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The identification number of an existing account.
general	Contains the general billing setting information. See the General table for parameter definitions.
commitments	Contains information about device and account commitments. See the Commitments table for parameter definitions.

General Parameters

Parameter	Description
taxable	If set to *true*, the account is taxable. The default value is *false*.
billable	If set to *true*, the account is billable. The default value is *false*.

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Parameter	Description
prorationRule	Indicates whether or not the billing engine will prorate subscription fees in the first billing cycle for devices using monthly rate plans. • *None*. The first month's subscription is charged for the full month regardless of when the device was first activated. • *Daily*. The first month's subscription is charged based on the number of days since the device was activated for the first time. • *Semi-Monthly*. The first month's subscription is charged either a full or half month depending on whether the device was activated for the first time in the first half or second half of the month.
renewalProrationRule	Similar to activation proration, this parameter covers the special case where a device ends a prepaid rate plan and moves onto a monthly plan. Control Center prorates the new subscription fee on a semi-monthly or daily basis – or not at all, depending on the value of this parameter. Valid values are: *Daily*, *Semi-Monthly*, and *None*.
simFee	This fee indicates how much the account is charged for any SIMs purchased. Not supported for Essential accounts.
activationPlan	The default activation plan defines how Control Center applies the activation fee to each device owned by the account. • *First Time Only*. The account pays the activation fee when the device is activated for the first time (typical). • *Each Time*. The account pays every time the device SIM status changes to *Activated*. • *Not First Time*. The account does not pay for the first device activation, but it does pay for every subsequent activation. • *None*. No charge. Not supported for Essential accounts.

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Parameter	Description
activationFee	Control Center charges this fee based the Activation Plan setting. This setting might be replaced if it is set at the rate plan level. In that case Control Center charges an activation fee based on the settings associated with the current device rate plan. (Admin > Accounts > Account Name link > Rate Plans subtab > Edit Settings button). Not supported for Essential accounts.
retailBilling	Not supported for service provider users. Control Center ignores any values you supply for this parameter.
immediateRatePlanChange	Rate plan changes for devices with non-active SIM status values (such as *Test Ready* or *Deactivated*) always take effect immediately. For devices with an *Activated* SIM status, you can control the rate plan change behavior with this setting. • If the toggle is on, any mid-cycle rate plan changes for devices with an *Activated* status take effect immediately. Control Center will bill for the device as if it had been on the new rate plan during the entire billing cycle. • If the toggle is off, Control Center schedules any mid-cycle rate plan changes for devices with an *Activated* status to take effect in the next billing cycle. See the Scheduled Changes subtab on the Device Details page. This behavior applies to devices with monthly rate plans only. You cannot change the rate plan for a device on a prepaid rate plan until the plan expires. Essential accounts do not support immediate rate plan changes for devices with an *Activated* SIM status, regardless of this parameter value.
defaultTimeZone	The default time zone to use for this account Not supported for Essential accounts.

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Parameter	Description
eventPlanEnabled	If set to *true*, then event plans are enabled for this account. An event plan is a prepaid rate plan that's designed for special circumstances or "events" that occur outside of a normal rate plan. If set to *false*, then this account cannot use event plans. The default is *false*. Not supported for Essential accounts.
prepaidRenewalPolicy	When a prepaid rate plan expires, Control Center either deactivates the device or gives it a different rate plan. You can specify the default renewal behavior for all the account's prepaid rate plans. The options are as follows: • *Deactivate*. (Default) Deactivate the device. • *AutoRenew*. Renew the current prepaid rate plan. • *NamedPlan*. Assign a specific rate plan or assign the next rate plan in the device queue.
prepaidRenewalRatePlanId	When a prepaid device rate plan expires, Control Center attaches the named rate plan to the device. If the device has a queue, Control Center uses the rate plan at the top of the queue instead of the named plan here.

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Commitments Parameters

Parameter	Description
activationGracePeriod	This length of time (specified in days) is the grace period between the time a device is transferred to an account and Control Center starts billing for the device. If the device is not active (and billable) by the time this grace period ends, Control Center starts billing the subscription fee in the next billing cycle. To enable this feature, you must also set the Bill Activation Grace Period value to true. The grace period for devices using prepaid rate plans is indefinite. Control Center does not charge for these devices until they become active. This API does not allow you to update commitment data for Essential accounts.
billActivationGracePeriod	When this parameter is set to *true*, Control Center enables the activation grace period feature and bill the account for a device, regardless of its status, after the activation period ends. You also need to define the length of the grace period with the Activation Grace Period parameter. The default is *false*. This API does not allow you to update commitment data for Essential accounts.
monthlyDevicesMinimum Term	This parameter specifies the length of time the monthly device minimum will be in effect for each device in the account. The term begins on the device's activation date. The maximum term is 999 months. Control Center does not support Monthly Device Minimum commitments for Essential accounts.
monthlyDeviceMinimum	This parameter specifies the minimum revenue a device is expected to provide in each billing cycle. The maximum amount is 9999.99 (in the account's default currency). Control Center does not support Monthly Device Minimum commitments for Essential accounts.
billMonthlyDeviceMinimum	If set to *true*, Control Center enables the monthly device minimum commitment using the values in the monthlyDevicesMinimumTerm and monthlyDeviceMinimum parameters. Control Center does not support Monthly Device Minimum commitments for Essential accounts.

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Parameter	Description
minimumActivationTerm	The number of months each device must be active. To enable this feature, you must also set the Bill Minimum Activation Term parameter to true. This API does not allow you to update commitment data for Essential accounts.
billMinimumActivation Term	If set to *true*, Control Center enables the minimum activation term feature and bills the subscription fee for any devices that have not been active long enough to meet the minimum requirement. The default is *false*. This APII does not allow you to update commitment data for Essential accounts.

Response Parameters

Return Value	Description
accountID	The ID of the account.
timestamp	The date and time when the account was updated. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  {
  \"general\":
    {
    \"taxable\": false,
    \"billable\": false,
    \"prorationRule\": \"None\",
    \"renewalProrationRule\": \"None\",
    \"simFee\": 0,
    \"activationPlan\": \"None\",
    \"activationFee\": 0,
    \"retailBilling\": \"None\",
    \"immediateRatePlanChange\": true,
    \"defaultTimeZone\": null,
    \"eventPlanEnabled\": false,
    \"prepaidRenewalPolicy\": \"Deactivate\",
    \"prepaidRenewalRatePlanId\": null
    },
  \"commitments\":
    {
    \"activationGracePeriod\": 180,
    \"billActivationGracePeriod\": true,
    \"monthlyDevicesMinimumTerm\": 0,
    \"monthlyDeviceMinimum\": 0,
    \"billMonthlyDeviceMinimum\": false,
    \"minimumActivationTerm\": 0,
    \"billMinimumActivationTerm\": false
    }
}" "https://<YOUR-BASE-URL>/rws/api/v1/accounts/100015068/billingSettings"
```

Response Example

```
  {
    "accountId": 100015068,
    "timestamp": 1527590173
  }
```

Errors

Error Code	HTTP Code	Error Message
10000004	400	Invalid accountId.
10000024	400	Invalid apiVersion.
20000003	404	Resource not found - Invalid accountId.
20000013	400	Account billing settings cannot be edited for service provider account.

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Error Code	HTTP Code	Error Message
20000014	400	Empty monthly devices minimum term (months).
20000016	400	Invalid activation proration.
20000015	400	Empty monthly device minimum.
20000017	400	Invalid renewal proration.
20000018	400	Invalid sim fee scale.
20000021	400	Invalid default activation fee scale.
20000020	400	Invalid default activation plan.
20000019	400	Sim fee should be between [0-9999999].
20000022	400	Default activation fee should be between [0-9999999].
20000023	400	Invalid enable retail billing.
20000025	400	Invalid prepaid renewal policy.
20000024	400	Invalid default timezone.
20000026	400	Invalid prepaid renewal rate plan.
20000029	400	Monthly device minimum should be between [0-999.99].
20000028	400	Monthly devices minimum term should be between [0-999].
20000027	400	Invalid activation grace period.
20000030	400	Invalid minimum activation term.
20000033	400	Invalid request to perform edit operation.
20000032	400	Billing commitment settings cannot be edited for Essential/Partner accounts.
20000031	400	Invalid request to view account billing settings.
20000034	400	Account billing settings cannot be edited for NBIOT account.
20000041	400	Taxable cannot be null.

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Error Code	HTTP Code	Error Message
20000042	400	Billable cannot be null.
20000043	400	ProrationRule cannot be null.
20000044	400	RenewalProration cannot be null.
20000045	400	ActivationPlan cannot be null.
20000046	400	RetailBilling cannot be null.
20000047	400	ImmediateRatePlanChange cannot be null.
20000049	400	PrepaidRenewalPolicy cannot be null.
20000048	400	EventPlanEnabled cannot be null.
20000050	400	PrepaidRenewalRatePlanId cannot be null.
20000051	400	BillActivationGracePeriod cannot be null.
20000052	400	MonthlyDevicesMinimumTerm cannot be null.
20000053	400	MonthlyDeviceMinimum cannot be null.
20000054	400	BillMonthlyDeviceMinimum cannot be null.
20000056	400	Commitments settings cannot be null.
20000055	400	BillMinimumActivationTerm cannot be null.
30000001	500	Unknown server error.

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Get Available Rate Plans for an Account

Description

Returns a list of rate plans available for an account. These rate plans include custom rate plans associated with the account as well as shared rate plans that the service provider assigned to the account. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

GET BaseURL/v{apiVersion}/accounts/{accountId}/ratePlans

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The ID of the account whose rate plan(s) you want to retrieve.
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.

Response Parameters

Return Value	Description
pageNumber	An integer specifying the number of the current response page. This value defaults to 1. See *Pagination on page 48*.
ratePlans	An array of rate plans.
ratePlanId	A rate plan identifier, that is not necessarily unique. A rate plan can have multiple versions that share the same ratePlan ID. See versionId.
ratePlanName	The name of the rate plan.
versionId	Uniquely identifies a particular version of a rate plan in relation to every other rate plan version in the system. Because a rate plan may have multiple active versions, the versionId, and not the ratePlanId is the only truly unique rate plan identifier.

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Return Value	Description
shared	If set to *true*, it indicates that this rate plan is available to multiple accounts. If set to *false*, the rate plan is assigned to a specific account.
lastPage	A true or false value indicating whether the current response page is the last in the series. See *Pagination on page 48*.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>
" "https://<YOUR-BASE-
URL>/rws/api/v1/accounts/100015068/ratePlans?pageSize=50&pageNumber=1"
```

Response Example

```
{
"pageNumber": 1,
 "ratePlans": [
  {
   "ratePlanId": 39374,
   "ratePlanName": "QASPAWTRP",
   "versionId": 28060,
  "shared": true
  }
],
"lastPage": true
 }
```

Errors

Error Code	HTTP Code	Error Message
10000004	400	Invalid accountId.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000024	400	Invalid apiVersion.
20000003	404	Resource not found - Invalid accountId.
20000035	400	Shared rate plans can only be viewed by Advantage/Essential accounts.
30000001	500	Unknown server error.

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Edit Available Rate Plans for an Account

Description

Changes the shared rate plans associated with a specified account. The function allows a service provider to add or remove a shared rate plan from an account. You must have a ServiceProvideFinance or ServiceProviderCSRUser role to perform this function.

Resource URL

PUT BaseURL/v{apiVersion}/accounts/{accountId}/ratePlans

Request Parameters

This function specifies all the shared rate plans associated with the specified account. If you remove an existing shared rate plan from your request parameters, you will delete the shared rate plan from the account. All request parameters are required.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	The ID of the account whose rate plans you want to change.
sharedRatePlans	A comma-separated list of shared rate plans that you want to assign to the account. The named rate plans will overwrite any shared rate plans already associated with the account. You must include a minimum of one shared rate plan.

Response Parameters

Return Value	Description
accountId	The ID of the account you want to change.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"sharedRatePlans\": [
    \"100MB Pool\",
    \"200MB Pool\",
    \"300MB Pool\"
    ]
}" "https://<YOUR-BASE-URL>/rws/api/v1/accounts/100015068/ratePlans"
```

Response Example

```
{
   "accountId": 100015068
}
```

Errors

Error Code	HTTP Code	Error Message
10000004	400	Invalid accountId.
10000024	400	Invalid apiVersion.
20000003	404	Resource not found - Invalid accountId.
20000037	400	Operation cannot be performed as there are no rate plans available for this account.
20000036	400	Invalid request to assign shared rate plans to the account.
20000038	400	Cannot perform operation as one or more shared rate plans not available for this account.
20000039	500	Error adding shared rate plan id mappings to the account.
20000040	400	Shared rate plans can only be added by Advantage/Essential accounts.
20000057	400	Failed to update shared plans. Cannot remove the account default rate plan.
30000001	500	Unknown server error.

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API Key

Create/Reset API Key for Given User

Description

Resets the API key for a given user and returns the new value. If the user does not have an API key, the function creates a new key for the user. You must have access to this user's information in order to change or create an API key. In addition, you cannot reset or create an API key for a user if they have web interface access and can perform these tasks for themselves.

To reset your own API key, you must use a different API (see *Reset API Key for Current User on page 135*). You cannot create an API key for yourself.

This function is available to service provider and account administrators, but not customer users. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Warning. Resetting the API key immediately invalidates any API key in use. We recommend you save the response containing the new key and then use it to update any programs using the previous key.

Resource URL

PUT BaseURL/v{apiVersion}/apikey/{userId}

Request Parameters

Return Value	Description
apiVersion	The version number for this API. The current version for all functions is 1.
userId	The ID of the user whose API key you want to reset or create. If there is no user ID, the function resets the API key for the user making the function call. See *Reset API Key for Current User on page 135*.

Response Parameters

Return Value	Description
userId	The ID of the requested user. If you reset your own API key, the function returns your own user ID.
apiKey	The new API key value for the requested user.

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Return Value	Description
timestamp	The date and time when the API key was created or updated. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/apikey/401038105"
```

Response Example

```
{
    "userId": 401038105,
    "apiKey": "84b5c96e-9bfb-4565-b4b5-e73fd41b998f",
    "timestamp": 1549369025
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000438	404	Resource not found - Invalid username.
10000439	400	Unauthorized access to the user.
30000001	500	Unknown server error.

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Get API Key

Description

Returns the API key for a given user. If the user doesn't have an API key or if the key has expired, the function returns an empty value. You must have access to this user's information in order to get the API key. Be aware that you cannot get a user's API key if the user has web interface access.

This function is available to service provider and account administrators, but not customer administrators. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

GET BaseURL/v{apiVersion}/apikey/{userId}

Request Parameters

Return Value	Description
apiVersion	The version number for this API. The current version for all functions is 1.
userId	The ID of the user whose API key you want to get. The user must have the API-only access type.

Response Parameters

Return Value	Description
userId	The ID of the requested user.
apiKey	The API key value for the requested user. If the user doesn't have an API key or if the key has expired, the function returns an empty value.
timestamp	The date and time when the API key was returned. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/apikey/401056705"
```

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Response Example

```
{
    "userId": 401056705,
    "apiKey": "84b5c96e-9bfb-4565-b4b5-e73fd41b998f",
    "timestamp": 1549365650
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000438	404	Resource not found - Invalid username.
10000439	400	Unauthorized access to the user.
10001550	400	You cannot get an API key for a user with access to the web interface.
30000001	500	Unknown server error.

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Reset API Key for Current User

Description

Resets the API key for the user making the function call. For security reasons, you may want to reset the key on a regular basis—or whenever you have reason to believe the key has been compromised.

All service provider and account users have access to this function. Customer users do not have access to this API and must request a key reset from an account administrator. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Warning. Resetting the API key immediately invalidates the API key in use. We recommend you save the response containing the new key and then use it to update any programs using the previous key.

Resource URL

PUT BaseURL/v{apiVersion}/apikey

Request Parameters

Return Value	Description
apiVersion	The version number for this API. The current version for all functions is 1.

Response Parameters

Return Value	Description
apiKey	New API key value.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/v1/apikey"
```

Response Example

```
{
    "apiKey": "7a8af469-fc46-40da-903a-00acaea9940d"
}
```

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Code Samples

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'
require 'base64'
$auth = 'Basic' + Base64.encode64( 'apiuser:469bfd67-62a2-4ab6-9df6-
d5cf5527e7a8' ).chomp
$url = 'https://<your-base-URL>/rws/api/v1/apikey'
@resource = RestClient::Resource.new( $url )
@response = @resource.put( :Authorization => $auth )
puts JSON.parse(response)
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
10000440	400	You cannot reset an API key for a user with access to the web interface.
30000001	500	Unknown server error.

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Customers

Get Customer Details

Description

Returns the customer details for a given customer. Users can view detailed information for any customers to which they have access. In general, service provider users can view all customers, account users can view all customers within their account, and customer users can view their own customer information. If a user is part of an account group or a customer group, however, the group may restrict or expand the user's access to information. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

GET BaseURL/v{apiVersion}/customers/{customerId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
customerId	The identification number of the customer.

Response Parameters

Return Value	Description
name	The name of the customer.
accountName	The name of the account to which the customer belongs.
securityQuestion	The security question.
securityAnswer	The answer to the security question.
shipToBillAddress	If true, the customer's billing address is the same as the shipping address.
contacts	An array of contacts. See the Contacts table below.
billingAddress	See the Address table below.
shippingAddress	See the Address table below.
customerId	The identification number of the customer.

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Contacts Parameters

The function returns an array of up to four contacts.

Return Value	Description
name	The name of the customer contact.
title	The title of the customer contact.
phone	The phone number of the customer contact.
mobile	The mobile phone number of the customer contact.
email	The email address of the customer contact.

Address Parameters

The shipping and billing address use the same response parameters.

Return Value	Description
addressLine1	The first line of the address.
addressLine2	The second line of the address.
city	The city.
state	The region, for example, state or province.
country	The country.
postalCode	Postal code.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/customers/10008314"
```

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Response Example

```
{
   "name": "Acme Corporation",
   "accountName": "QASPAWT_Act003",
   "securityQuestion": null,
   "securityAnswer": null,
   "shipToBillAddress": true,
   "contacts": [
      {
      "name": "Jules Bron",
      "title": "Director",
      "phone": "323-948-2312",
      "mobile": "408-796-9432",
      "email": "julesbron@acme.com"
      }
   ],
   "billingAddress": {
      "addressLine1": "123 Main Street",
      "addressLine2": null,
      "city": "Buckley",
      "state": "Tennessee",
      "country": "United States",
      "postalCode": "42308"
      },
   "shippingAddress": {
      "addressLine1": "123 Main Street",
      "addressLine2": null,
      "city": "Buckley",
      "state": "Tennessee",
      "country": "United States",
      "postalCode": "42308"
   },
   "customerId": 100021774
 }
```

Errors

Error Code	HTTP Code	Error Message
10000024	400	Invalid apiVersion.
10000603	400	Invalid account. Account erasure request is in progress for the account.
10000604	400	Invalid Customer Id.
10000605	404	Customer not found with given Id.

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Get All Customers

Description

Returns details for all active customers associated with a specific account.

Resource URL

GET BaseURL/v{apiVersion}/customers/acct/{acctId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
acctId	The ID for the account whose customers you want to retrieve.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.
limit	(Optional) The maximum number of customers (up to 50) to include in the response. For example, if you enter "25", Control Center returns details for 25 customers. If you leave this parameter blank, Control Center uses the default value of 50.
sortedBy	(Optional) The attribute to sort the customer list by. Possible values are: • *name* • *customerId* • *dateAdded* • *dateModified*
sortDirection	(Optional) The sort order for the customer list. Enter *ASC* (ascending) or *DESC* (descending). The default value is *ASC*.

Response Parameters

Return Value	Description
customerId	The identification number of the customer.
name	The name of the customer.
accountName	The name of the account to which the customer belongs.

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Return Value	Description
shipToBillAddress	If true, the customer's billing address is the same as the shipping address.
contacts	This section includes the contact information for the customer. See the Contact Parameters table below.
billingAddress	See the Address Parameters table below.
shippingAddress	See the Address Parameters table below.

Contact Parameters

Each customer can have a maximum of four contacts. A customer contact is optional.

Parameter	Description
name	The name of the contact. The string allows 0 to 40 characters.
title	The title of the contact. The string allows 0 to 40 characters.
phone	The phone number of the contact. The string allows 0 to 25 characters.
mobile	The mobile number of the contact. The string allows 0 to 25 characters.
email	The email address of the contact. The string allows 0 to 320 characters.

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Address Parameters

The following parameters are used for both billingAddress and shippingAddress. Billing addresses are optional.

Parameter	Description
addressLine1	The first line of the contact's billing/shipping address. This is typically a street number and a street name. The string allows 0 to 40 characters.
addressLine2	The second line of the contact's billing/shipping address. The string allows 0 to 40 characters.
city	The city of the contact. The string allows 0 to 40 characters.
state	The state or province of the contact. The string allows 0 to 40 characters.
country	The country of the contact. The string allows 0 to 150 characters.
postalCode	The postal code of the contact. The string allows 0 to 15 characters.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/customers/acct/12345"
```

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Response Example


```
{
  "customers": [{
    "name": "Alex Demo",
    "accountName": "Company10",
    "shipToBillAddress": true,
    "contacts": [{
      "name": "Sam Sample",
      "title": null,
      "phone": "123-548-6372",
      "mobile": "305-674-7427",
      "email": "sam.sample@example.com"
    },
    {
      "name": "Jane Doe",
      "title": "Ms",
      "phone": "292-628-7232",
      "mobile": "301-181-2971",
      "email": "jane.doe@example.com"
    },
    {
      "name": "Jamie Test",
      "title": "Mrs",
      "phone": "171-242-4218",
      "mobile": "505-342-9074",
      "email": "jamie.test@example.com"
    }
    ],
    "billingAddress": {
      "addressLine1": "Address Line 1",
      "addressLine2": "Address Line 2",
      "city": "Sample City",
      "state": "Sample State",
      "country": "India",
      "postalCode": "123123"
    },
    "shippingAddress": {
      "addressLine1": "Address Line 1",
      "addressLine2": "Address Line 2",
      "city": "Sample City",
      "state": "Sample State",
      "country": "India",
      "postalCode": "123123"
    },
    "customerId": 131914403
       },
       {
       "name": "Richard Roe",
       "accountName": "Company10",
    "shipToBillAddress": true,
    "contacts": [
    {
       "name": "Pat Example",
       "title": null,
       "phone": "181-292-5201",
       "mobile": "308-627-2953",
       "email": "pat.example@example.com"
    }
    ],
```


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```
"billingAddress": {
   "addressLine1": "dsfdsf",
   "addressLine2": "rwedsa",
   "city": "Sample City",
   "state": "Sample State",
   "country": "India",
   "postalCode": "123123"
},
"shippingAddress": {
   "addressLine1": "dsfdsf",
   "addressLine2": "rwedsa",
   "city": "Sample City",
   "state": "Sample State",
   "country": "India",
   "postalCode": "123123"
},
"customerId": 131908403
}
],
"pageNumber": 1,
"totalCount": 3,
"lastPage": false
}
```


Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000002	400	AccountId is required.
10000004	400	Invalid accountId.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000024	400	Invalid apiVersion.
10000603	400	Invalid account. Account erasure request is in progress for the account.
15000015	400	Invalid element provided in sort_by parameter.
20000003	404	Resource not found - Invalid accountId.
30000001	500	Unknown server error.

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Edit Customer Details

Description

Updates the detailed information for any customers to which users have access. In general, service provider users can view all customers, account users can view all customers within their account, and customer users can view their own customer information. If a user is part of an account group or a customer group, however, the group may restrict or expand the user's access to information. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

PUT BaseURL/v{apiVersion}/customers/{customerId}

Request Parameters

Caution. This function overwrites existing customer information using parameter values you supply. Control Center uses default values for any parameters that are missing from the call. To avoid losing data, make sure to provide existing values for any unchanged parameters.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
name	The name of the customer that you want to update.
accountName	The account name for the customer.
securityAnswer	The answer to the security question. If a securityAnswer is used, the securityQuestion is required. The string allows a maximum of 100 characters. A null value is acceptable
securityQuestion	A security question. If a securityAnswer is used, the securityQuestion is required. The string allows a maximum of 100 characters. A null value is acceptable.
shipToBillAddress	Set this value to true if you want to to use the customer's billing address as the shipping address.
contacts	See the Contact Parameters table below. A null value is acceptable.
billingAddress	See the Address Parameters table below.
shippingAddress	See the Address Parameters table below.

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Contact Parameters

Each customer can have a maximum of four contacts.

Parameter	Description
name	The name of the contact. The string allows 0 to 40 characters.
title	The title of the contact. The string allows 0 to 40 characters.
phone	The phone number of the contact. The string allows 0 to 25 characters.
mobile	The mobile number of the contact. The string allows 0 to 25 characters.
email	The email address of the contact. The string allows 0 to 320 characters.

Address Parameters

The following parameters are used for both billingAddress and shippingAddress.

Parameter	Description
addressLine1	The first line of the contact's billing/shipping address. This is typically a street number and a street name. The string allows 0 to 40 characters.
addressLine2	The second line of the contact's billing/shipping address. The string allows 0 to 40 characters.
city	The city of the contact. The string allows 0 to 40 characters.
state	The state or province of the contact. The string allows 0 to 40 characters.
country	The country of the contact. The string allows 0 to 150 characters.
postalCode	The postal code of the contact. The string allows 0 to 15 characters.

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Response Parameters

Return Value	Description
customerId	The customer identification number associated with the customer name.
timestamp	The date and time when the customer was updated. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>"
-d "{
   \"name\": \"Acme Customer\",
   \"accountName\": \"Acme\",
   \"securityQuestion\": null,
   \"securityAnswer\": null,
   \"shipToBillAddress\": true,
   \"contacts\": [
      {
      \"name\": \"Jules Bron\",
      \"title\": \"Director\",
      \"phone\": \"323-948-2312\",
      \"mobile\": \"408-796-9432\",
      \"email\": \"julesbron@acme.com\"
      },
      {
      \"name\": \"Guillaume Avicci\",
      \"title\": \"VP\",
      \"phone\": \"323-948-5722\",
      \"email\": \"guillaumea@acme.com\"
      }
   ],
   \"billingAddress\":
      {
      \"addressLine1\": \"123 Main Street\",
      \"addressLine2\": \"string\",
      \"city\": \"Buckley\",
      \"state\": \"Tennessee\",
      \"country\": \"United States\",
      \"postalCode\": \"42308\"
      }
   \"shippingAddress\":
      {
      \"addressLine1\": \"123 Main Street\",
      \"addressLine2\": \"string\",
      \"city\": \"Buckley\",
      \"state\": \"Tennessee\",
      \"country\": \"United States\",
      \"postalCode\": \"42308\"
      }
}" "https://<YOUR-BASE-URL>/rws/api/v1/customers/10008314/"
```

Response Example

```
{
  "customerId": 10008314,
  "timestamp": 1527590173
}
```

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Errors

Error Code	HTTP Code	Error Message
10000004	400	Invalid accountId.
10000024	400	Invalid apiVersion.
10000580	400	Customer details required.
10000581	400	Customer name is required.
10000582	400	Invalid customer name. Number of characters allowed 1 - 60.
10000583	400	Account Id is required.
10000584	400	Security answer missing.
10000585	400	Security question missing.
10000586	400	Invalid security question. Number of characters allowed 0 - 100.
10000587	400	Invalid security answer. Number of characters allowed 0 - 100.
10000588	400	Maximum 4 contacts can be associated with a customer.
10000589	400	Invalid contact name. Number of characters allowed 0 - 40.
10000590	400	Invalid contact title. Number of characters allowed 0 - 40.
10000591	400	Invalid contact phone. Number of characters allowed 0 - 25.
10000592	400	Invalid contact mobile. Number of characters allowed 0 - 40.
10000593	400	Invalid contact email. Number of characters allowed 0 - 320.
10000594	400	Invalid address line1. Number of characters allowed 0 - 40.
10000595	400	Invalid address line2. Number of characters allowed 0 - 40.
10000596	400	Invalid city. Number of characters allowed 0 - 40.
10000597	400	Invalid state/region. Number of characters allowed 0 - 40.

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Error Code	HTTP Code	Error Message
10000598	400	Invalid country. Number of characters allowed 0 - 100.
10000599	400	Invalid postal code. Number of characters allowed 0 - 15.
10000602	400	Duplicate customer name.
10000603	400	Invalid account. Account erasure request is in progress for the account.
20000003	404	Resource not found - Invalid accountId.

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Create Customer

Description

Creates a new customer. Both service provider and account users can create customers. In general, service provider users can create customers for any account and account users can create customers within their own account. If a user is part of an account group or a customer group, however, the group may restrict or expand the user's ability to create customers within the given account. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

POST BaseURL/v{apiVersion}/customers

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
name	The name of the customer. The string must contain at least one character with a maximum of 60 characters. You can include spaces and special characters in the customer name.
accountName	The name of the parent account containing the customer.
securityAnswer	(Optional) The answer to the security question. This field is required if there is a securityQuestion. The string allows a maximum of 100 characters.
securityQuestion	(Optional) A security question. If a securityAnswer is used, the securityQuestion is required. The string allows a maximum of 100 characters.
shipToBillAddress	(Optional) Set this value to true if you want to use the customer's billing address as the shipping address. The default is *false*.
contacts	(Optional) See the Contact Parameters table below.
billingAddress	(Optional) See the Address Parameters table below.
shippingAddress	(Optional) See the Address Parameters table below. This address is required if shipToBillAddress is *false*.

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Contact Parameters

Each customer can have a maximum of four contacts. A customer contact is optional.

Parameter	Description
name	The name of the contact. The string allows 0 to 40 characters.
title	The title of the contact. The string allows 0 to 40 characters.
phone	The phone number of the contact. The string allows 0 to 25 characters.
mobile	The mobile number of the contact. The string allows 0 to 25 characters.
email	The email address of the contact. The string allows 0 to 320 characters.

Address Parameters

The following parameters are used for both billingAddress and shippingAddress. Billing addresses are optional.

Parameter	Description
addressLine1	The first line of the contact's billing/shipping address. This is typically a street number and a street name. The string allows 0 to 40 characters.
addressLine2	The second line of the contact's billing/shipping address. The string allows 0 to 40 characters.
city	The city of the contact. The string allows 0 to 40 characters.
state	The state or province of the contact. The string allows 0 to 40 characters.
country	The country of the contact. The string allows 0 to 150 characters.
postalCode	The postal code of the contact. The string allows 0 to 15 characters.

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Response Parameters

Return Value	Description
customerId	The identification number of the customer that was created.
timestamp	The date and time when the customer was created. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X POST --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>"
-d "{
   \"name\": \"Acme Customer\",
   \"accountName\": \"Acme\",
   \"securityQuestion\": \"FirstConcert\",
   \"securityAnswer\": \"Journey\",
   \"shipToBillAddress\": true,
   \"billingAddress\":
      {
      \"addressLine1\": \"123 Main\",
      \"addressLine2\": \"string\",
      \"city\": \"Anytown\",
      \"state\": \"FLA\",
      \"country\": \"USA\",
      \"postalCode\": \"36301\"
      }
}" "https://<YOUR-BASE-URL>/rws/api/v1/customers"
```

Response Example

```
{
  "customerId": 1234,
  "timestamp": 1527590173
}
```

Errors

Error Code	HTTP Code	Error Message
10000004	400	Invalid accountId.
10000024	400	Invalid apiVersion.
10000580	400	Customer details required.
10000581	400	Customer name is required.

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Error Code	HTTP Code	Error Message
10000582	400	Invalid customer name. Number of characters allowed 1 - 60.
10000583	400	Account Id is required.
10000584	400	Security answer missing.
10000585	400	Security question missing.
10000586	400	Invalid security question. Number of characters allowed 0 - 100.
10000587	400	Invalid security answer. Number of characters allowed 0 - 100.
10000588	400	Maximum 4 contacts can be associated with a customer.
10000589	400	Invalid contact name. Number of characters allowed 0 - 40.
10000590	400	Invalid contact title. Number of characters allowed 0 - 40.
10000591	400	Invalid contact phone. Number of characters allowed 0 - 25.
10000592	400	Invalid contact mobile. Number of characters allowed 0 - 40.
10000593	400	Invalid contact email. Number of characters allowed 0 - 320.
10000594	400	Invalid address line1. Number of characters allowed 0 - 40.
10000595	400	Invalid address line2. Number of characters allowed 0 - 40.
10000596	400	Invalid city. Number of characters allowed 0 - 40.
10000597	400	Invalid state/region. Number of characters allowed 0 - 40.
10000598	400	Invalid country. Number of characters allowed 0 - 100.
10000599	400	Invalid postal code. Number of characters allowed 0 - 15.
10000602	400	Duplicate customer name.

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Error Code	HTTP Code	Error Message
10000603	400	Invalid account. Account erasure request is in progress for the account.
20000003	404	Resource not found - Invalid accountId.

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Dynamic Reporting

Get Aggregated Usage Details

Description

Returns aggregated device usage details for a service provider. You can filter the information by billing cycle or date range.

With this function, you have the option to specify a date range. You must include both a start date and end date, which are no more than 30 days apart.

The API response limit is 15,000 records per request.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

GET BaseURL/v{apiVersion}/dynareport/operator/usage

Request Parameters

Parameter	Description
apiVersion (Path)	The version number for this API. The current version for all functions is 1.
accountIds (Query)	(Optional) The enterprise accounts you want to retrieve usage details for. Separate multiple entries with a comma.
startDate (Query)	(Optional) The start date for the time period you want to retrieve usage details for. The date format is YYYYMMDD.
endDate (Query)	(Optional) The end date for the time period you want to retrieve usage details for. The date format is YYYYMMDD.
billingCycle (Query)	(Optional) The billing cycle date that you want to retrieve usage details for. The date format is YYYYMM.
groupBy (Query)	(Optional) The attribute(s) to group the usage details by. Possible values are: • account • carrier • country • rate plan • rating zone Separate multiple entries with a comma.

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Parameter	Description
metrics (Query)	(Optional) The metric type(s) that you want to retrieve usage details for. Possible values are: • voice • data • sms • vmo • smo • vmt • smt Separate multiple entries with a comma.
sortedBy (Query)	(Optional) The attribute to sort the usage details by. Possible values are: • account • voice • data • sms • vmo • smo • vmt • smt
sortDirection (Query)	(Optional) The sort order for the usage data. Enter asc (ascending) or desc (descending). The value defaults to desc.
pageNumber (Query)	(Optional) Specifies the number of the response page to return. The value defaults to 1. See *Pagination on page 48*.
pageSize (Query)	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.

Response Parameters

Return Value	Description
Metadata	
startDate	The start date for the usage details.
endDate	The end date for the usage details.
billingCycle	The billing cycle for the usage details.
metrics	The metric type for the usage details.
groupBy	The attribute(s) the usage details are grouped by.
accountIds	The ID of the enterprise account(s).

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Return Value	Description
sortedBy	The sort attributes for the usage details.
sortDirection	The sort order for the usage details.
Pagination	
pageNumber	The current page number.
pageSize	The number of records returned in each response page. The maximum value is 50.
totalPages	Total number of pages in the response.
totalItems	Total number of items per page in the response.
Group Elements	
operatorId	The ID for the service provider.
accountId	The ID for the enterprise account.
accountName	Name of the enterprise account.
carrierName	Name of the carrier.
ratePlanName	Name of the rate plan.
ratingZoneName	Name of the rating zone.
countryName	Name of the country.
Metric Data	
metricType	The metric type.
usage	The usage count for the metric type.
count	The total number of devices. The count of devices depends on the group by attribute.
unit	The metric attributes for the usage details.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET "Content-Type:application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/
v1/dynareport/operator/usage?pageNumber=1&pageSize=2&startDate=20240402&endDa
te=20240428&groupBy=account,carrier,country,rateplan,ratingzone&metrics=data,
sms,voice&sortDirection=asc&accountIds=131085003,130827103
```

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Response Example


```
{
       "meta": {
       "startDate": "20240402",
       "endDate": "20240428",
       "metrics": "[data, sms, voice]",
       "groupBy": "[account, carrier, country, rateplan, ratingzone]",
       "accountIds": "[131085003, 130827103]",
       "sortedBy": "account",
       "sortDirection": "asc",
               "pagination": {
                       "pageNumber": 1,
                       "pageSize": 2,
                       "totalPages": 3,
                       "totalItems": 5
                }
               },
               "data": [
               {
                 "groups": [
                  {
                        "groupId": "accountId",
                        "value": "131085003"
                       },
                       {
                        "groupId": "accountName",
                        "value": "InvMod"
                       },
                       {
                        "groupId": "operatorId",
                        "value": "30131703"
                       },
                       {
                        "groupId": "carrierName",
                        "value": ""
                       },
                       {
                        "groupId": "countryName",
                        "value": "United State of America"
                       },
                       {
                        "groupId": "ratePlanName",
                        "value": "InvMod-RP-1"
                       },
                       {
                        "groupId": "ratingZoneName",
                        "value": "default zone"
                       }
               ],
               "metrics": [
               {
                 "metricType": "data",
                 "usage": 0,
                 "count": 3,
                 "unit": "bytes"
                },
                {
                  "metricType": "sms",
                  "usage": 0,
```


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```
                "count": 3,
                "unit": "messages"
              },
              {
                "metricType": "voice",
                "usage": 290400,
                "count": 3,
                "unit": "minutes"
              }
          ]
        },
             {
              "groups": [
               {
                "groupId": "accountId",
                "value": "130827103"
               },
               {
                "groupId": "accountName",
                "value": "QaAWTBillOpr0051"
               },
               {
                "groupId": "operatorId",
                "value": "30131703"
                },
                {
                     "groupId": "carrierName",
                     "value": ""
                     },
                     {
                      "groupId": "countryName",
                      "value": "United State of America"
                     },
                     {
                      "groupId": "ratePlanName",
                      "value": "Default Pricing - 30131703"
                     },
                     {
                      "groupId": "ratingZoneName",
                      "value": "default zone"
                     }
                ],
                     "metrics": [
                      {
                        "metricType": "data",
                        "usage": 0,
                        "count": 1,
                        "unit": "bytes"
                       },
                       {
                        "metricType": "sms",
                        "usage": 2,
                        "count": 1,
                        "unit": "messages"
                       },
                       {
                        "metricType": "voice",
                        "usage": 0,
                        "count": 1,
```


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```
                        "unit": "minutes"
                      }
                    ]
                }
        ]
}
```


Errors

Error Code	HTTP Code	Error Message
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000012	400	Invalid date format.
10000062	400	Access Denied.
15000001	400	Start date should be paired with end date.
15000002	400	Either provide start and end date or billing cycle.
15000003	400	End date should be greater than start date.
15000004	400	Maximum duration between start date and end date is allowed 30 days.
15000005	400	Records count exceeded the maximum permissible limit in response(15k).
15000006	400	Provided login session is not valid.
15000007	400	Not a valid group by element.
15000008	400	Not a valid metric element.
30000001	500	Unknown server error.

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Get Device Subscriptions

Description

Returns device subscription details for the service provider. You can filter the information by billing cycle or date range.

With this function, you have the option to specify a date range. You must include both a start and end date, which are no more than 30 days apart.

The API response limit is 15,000 records per request.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

GET BaseURL/v{apiVersion}/dynareport/operator/device/subscriptions

Request Parameters

Parameter	Description
apiVersion (Path)	The version number for this API. The current version for all functions is 1.
startDate (Query)	(Optional) The start date for the time period you want to retrieve subscription details for. The date format is YYYYMMDD.
endDate (Query)	(Optional) The end date for the time period you want to retrieve subscription details for. The date format is YYYYMMDD.
accountIds (Query)	(Optional) The enterprise accounts you want to retrieve usage details for. Separate multiple entries with a comma.
billingCycle (Query)	(Optional) The billing cycle date that you want to retrieve subscription details for. The date format is YYYYMM.
pageNumber (Query)	(Optional) Specifies the number of the response page to return. The value defaults to 1. See *Pagination on page 48*.
pageSize (Query)	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.
billable (Query)	(Optional) Specifies whether the enterprise account is billable or not.

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Response Parameters

Return Value	Description
Meta	
startDate	The start date for the subscription details.
endDate	The end date for the subscription details.
billingCycle	The billing cycle date for the subscription details.
billable	Indicates if the account is billable.
accountIds	The ID of the enterprise accounts.
Pagination	
pageNumber	The current page number.
pageSize	The number of records returned in a response page.
totalPages	Total number of pages in the response.
totalItems	Total number of items per page in the response.
Data	
accountId	The ID of the enterprise account.
accountName	Name of the enterprise account.
periodEnding	The end date of the billing cycle or date range.
beginingOfPeriod	Number of device subscriptions at the beginning of the billing cycle or date range.
grossAdds	Number of activated device subscriptions during the billing cycle or date range.
deactivations	Number of deactivated device subscriptions during the billing cycle or date range.
netSubscriptions	The net device subscriptions during the billing cycle or date range. This is the number of activated device subscriptions minus the number of deactivated device subscriptions.
endOfPeriod	Number of device subscriptions at the end of the billing cycle or date range.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

Date Range

```
curl -X GET "Content-Type:application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/
v1/dynareport/operator/device/subscriptions?startDate=20250101&endDate=202501
31&pageNumber=1&pageSize=3&billable=N&accountIds=100000022,100000086
```

Billing Cycle

```
curl -X GET "Content-Type:application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/
v1/dynareport/operator/device/subscriptions?billingCycle=202404&pageNumber=1&
pageSize=50"
```

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Response Example


```
{
        "meta": {
        "startDate": "20250101",
        "endDate": "20250131",
        "billable": "N",
        "accountIds": "[100000022, 100000086]",
                "pagination": {
                        "pageNumber": 1,
                        "pageSize": 3,
                        "totalPages": 16,
                        "totalItems": 48
                        }
                },
                "data": [
                {
                  "accountId": 100000022,
                  "accountName": "",
                  "billable": "N",
                  "periodEnding": "20250131",
                  "beginningOfPeriod": 5,
                  "grossAdds": 0,
                  "deactivations": 0,
                  "netSubscriptions": 0,
                  "endOfPeriod": 5
                },
                {
                  "accountId": 100000086,
                  "accountName": "",
                  "billable": "N",
                  "periodEnding": "20250131",
                  "beginningOfPeriod": 2,
                  "grossAdds": 0,
                  "deactivations": 0,
                  "netSubscriptions": 0,
                  "endOfPeriod": 2
                },
                {
                  "accountId": 100000022,
                  "accountName": "",
                  "billable": "N",
                  "periodEnding": "20250130",
                  "beginningOfPeriod": 5,
                  "grossAdds": 0,
                  "deactivations": 0,
                  "netSubscriptions": 0,
                   "endOfPeriod": 5
                 }
        ]
}
```


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Errors

Error Code	HTTP Code	Error Message
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000012	400	Invalid date format.
10000062	400	Access Denied.
15000001	400	Start date should be paired with end date.
15000002	400	Either provide start and end date or billing cycle.
15000003	400	End date should be greater than start date.
15000004	400	Maximum duration between start date and end date is allowed 30 days.
15000005	400	Records count exceeded the maximum permissible limit in response(15k).
15000006	400	Provided login session is not valid.
15000007	400	Not a valid group by element.
15000008	400	Not a valid metric element.
30000001	500	Unknown server error.

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Get Top K Usage Details

Description

Returns the voice, data, or SMS usage data for devices with the highest usage. When running the API, you must specify a value for k, which determines how many devices Control Center includes in the return. For example, if you enter "50", Control Center returns details for the 50 devices with the highest usage. You can filter Top K usage data by billing cycle or date range.

With this function, you have the option to specify a date range. You must include both a start and end date, which are no more than 30 days apart.

The API response limit is 100-records per request.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

GET BaseURL/v{apiVersion}/dynareport/operator/topk/usage

Request Parameters

Parameter	Description
apiVersion (Path)	The version number for this API. The current version for all functions is 1.
k (Query)	The value for k. This is the number of devices that you want to include in the return. For example, if you enter "25", Control Center returns details for the 25 devices with the highest usage. The default value is 10. The max limit is 100.
metrics (Query)	The metric type that you want to retrieve usage details for. Possible values are: • voice • data • sms • vmo • smo • vmt • smt
accountIds (Query)	(Optional) The enterprise accounts you want to retrieve usage details for. Separate multiple entries with a comma.
startDate (Query)	(Optional) The start date for the time period you want to retrieve usage details for. The date format is YYYYMMDD.

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Parameter	Description
endDate (Query)	(Optional) The end date for the time period you want to retrieve usage details for. The date format is YYYYMMDD.
billingCycle (Query)	(Optional) The billing cycle date that you want to retrieve usage details for. The date format is YYYYMM.
groupBy (Query)	(Optional) The attribute(s) to group the usage details by. Possible values are: • account • carrier • country • rate plan • rating zone Separate multiple entries with a comma.
sortedBy (Query)	(Optional) The attributes to sort the usage details by. Possible values are: • account • voice • data • sms • vmo • smo • vmt • smt
sortDirection (Query)	(Optional) The sort order for the usage data. Enter asc (ascending) or desc (descending). The value defaults to desc.
pageNumber (Query)	(Optional) Specifies the number of the response page to return. The value defaults to 1. See *Pagination on page 48*.
pageSize (Query)	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.

Response Parameters

Return Value	Description
Metadata	
topk	The value for k, which is the number of devices that Control Center returns.
startDate	The start date for the usage details.

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Return Value	Description
endDate	The end date for the usage details.
metrics	The metric type.
groupBy	The attribute(s) usage details are grouped by.
accountIds	The ID of the enterprise account(s).
sortedBy	The sort attributes for the usage details.
sortDirection	The sort order for the usage details.
Pagination	
pageNumber	The current page number.
pageSize	The number of records returned in a response page.
totalPages	Total number of pages in the response.
totalItems	Total number of items per page in the response.
Group Elements	
operatorId	The ID for the service provider.
iccid	The ICCID of the device.
eid	The EID (eSIM ID) is a 32-digit serial number that uniquely identifies an eSIM installed in a device.
accountId	The ID for the enterprise account.
accountName	Name of the enterprise account.
carrierName	Name of the carrier.
ratePlanName	Name of the rate plan.
ratingZoneName	Name of the rating zone.
countryName	Name of the country.
Metric Data	
metricType	The metric type.
usage	The usage count for the metric type.
count	The total number of devices.
unit	The metric attributes for the usage details.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

Date Range

```
curl -X GET "Content-Type:application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/
v1/dynareport/operator/topk/usage?k=2&startDate=20240401&endDate=20240430&pag
eNumber=1&pageSize=10&sortDirection=desc&metrics=sms&groupBy=account,carrier,
rateplan&sortedBy=sms&accountIds=131085003
```

Response Example


```
{
   "meta": {
                "topK": 2,
                "startDate": "20240401",
                "endDate": "20240430",
                "metrics": "[sms]",
                "groupBy": "[account, carrier, rateplan]",
                "accountIds": "[131085003]",
                "sortedBy": "sms",
                "sortDirection": "desc",
                        "pagination": {
                           "pageNumber": 1,
                           "pageSize": 10,
                           "totalPages": 1,
                           "totalItems": 7
                         }
                  },
                "data": [
                {
                "groups": [
                {
                 "groupId": "operatorId",
                 "value": "30131703"
                },
                {
                 "groupId": "accountId",
                 "value": "131085003"
                },
                {
                 "groupId": "accountName",
                 "value": "InvMod"
                },
                {
                 "groupId": "iccid",
                 "value": "8900215100000005021"
                },
                {
                 "groupId": "eid",
                 "value": ""
                },
                {
                 "groupId": "carrierName",
                 "value": ""
```


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```
                },
                {
                 "groupId": "ratePlanName",
                 "value": "InvMod-RP-1"
                }
            ],
            "metrics": [
            {
                  "metricType": "sms",
                  "usage": 117,
                  "count": 1,
                  "unit": "messages"
                }
            ]
        },
         {
          "groups": [
                {
                  "groupId": "operatorId",
                  "value": "30131703"
                },
                {
                  "groupId": "accountId",
                  "value": "131085003"
                 },
                 {
                   "groupId": "accountName",
                   "value": "InvMod"
                 },
                 {
                   "groupId": "iccid",
                   "value": "8900215100000018997"
                 },
                 {
                   "groupId": "eid",
                   "value": ""
                 },
                 {
                   "groupId": "carrierName",
                   "value": ""
                 },
                 {
                   "groupId": "ratePlanName",
                   "value": "InvMod-RP-1"
                 }
                ],
                 "metrics": [
                 {
                   "metricType": "sms",
                   "usage": 113,
                   "count": 1,
                   "unit": "messages"
                  }
                ]
        }
    ]
}
```


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Errors

Error Code	HTTP Code	Error Message
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000012	400	Invalid date format.
15000001	400	Start date should be paired with end date.
15000002	400	Either provide start and end date or billing cycle.
15000003	400	End date should be greater than start date.
15000004	400	Maximum duration between start date and end date is allowed 30 days.
15000005	400	Records count exceeded the maximum permissible limit in response(15k).
15000006	400	Provided login session is not valid.
15000007	400	Not a valid group by element.
15000008	400	Not a valid metric element.
15000009	400	Provide one element in metrics parameter.
15000010	400	Max limit for Top K is 100.
30000001	500	Unknown server error.

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Create Device Report

Description

Creates a Dynamic Reporting device report. When you run this API, Control Center starts the report generation process, which takes some time to complete. The API returns a request ID and timestamp as an acknowledgement for the request.

The service provider or enterprise can use the Get Report Status API to check the current status of the report. When the report is available (it has a status of *Completed*), you can access it in a few ways:

- Use the link from the Get Report Status API response to download the report.
- Open the Dynamic Reporting page in Control Center to download the report. See *Downloading a Dynamic Report* in the *Control Center User Guide*.
- Click the link in the body of the email notification to download the report. You must enable email delivery through this API to use this option.
- Access the report through an sFTP server. The service provider must have sFTP enabled to use this feature.

Control Center can generate the device report on a one-time, daily, weekly, or monthly basis.

You must have a ServiceProviderAdmin or AccountAdmin role to run this API.

Resource URL

POST BaseURL/v{apiVersion}/dynareport/requests

Request Parameters

Parameter	Description
apiVersion (Path)	The version number for this API. The current version for all functions is 1.
reportName (Body)	Specify the name of the report, for example, *Custom Device Report*.
reporttype (Body)	The report type. The value is always *device.*

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Schedule Parameters

Use these parameters to specify when Control Center generates the report, and whether it creates the report once, or on a recurring basis.

Parameter	Description
frequency	This is how often the report generates. Possible values include: • *One Time* • *Daily* • *Weekly* • *Monthly*
daysOfWeek	If Control Center generates the report weekly, specify a day of the week for the report. For example, if you enter *Monday*, the date range for the report is the previous Monday to Sunday. If Control Center generates the report monthly, leave this parameter blank.
daysOfMonth	If Control Center generates the report monthly, specify a day of the month for the report. Possible values range from *1* to *31*. For example, if you enter *15* for this parameter, the report will generate on the 15th of each month. If Control Center generates the report weekly, leave this parameter blank.
generationStartDate	The start date for the time period you want to generate the custom device report for. The date format is YYYYMMDD.
generationEndDate	The end date for the time period you want to generate the custom device report for. The date format is YYYYMMDD.

Settings Parameters

Use these parameters to specify the report file format. In addition, you can specify whether Control Center sends an email notification when the report is ready and/or delivers the report through sFTP.

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Parameter	Description
reportFormat	The file format for the report. Possible values include: • *CSV* • *DAT*
notificationEmail	
enabled	To receive notifications about generated reports via email, set this value to *true.* When the report is available, Control Center sends an email containing a download link for the report.
emailIds	The email address that Control Center sends the notification email to. Separate multiple entries with a comma.
sftp	
enabled	To receive the report through sFTP, set this value to *true.* The service provider must have sFTP enabled to use this feature.

Field Parameters

Use this parameter to specify the fields for the report.

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Parameter	Description
fields	Specify the fields to include on the report. Separate entries with a comma. The following are required fields that you must include on the report: • *opAcctId* • *acctId* • *acctName* • *operatorId* • *ICCID* • *IMSI* The following are optional fields that you can choose to include or exclude from the report: • *activationDate* • *imei* • *ratePlanName* • *commPlan* • *simProfileID* • *staticAPN* • *msisdn* • *customer* • *deviceId* • *linePayStatus* • *countryCode* • *dateShipped* • *ratePlanId* • *staticIPAddress*

Filter Parameters

Use these parameters to filter the report by specific fields on the report (only some fields are available). For example, if you want to filter the report by an ICCID range, enter *ICCID* for the field parameter, enter *Range* for the operator parameter, and enter the lower and upper values of the ICCID (separated by a comma) for the values parameter. You can add multiple filters to the report. For each filter, you must specify the field, operator, and values. See the *Request Example on page 178*.

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Parameter	Description
operator	Always set this field to *AND*.
items	
field	The fields to apply the filters to. You can filter by any one of these fields as long as you include it on the report: • *opAcctId* • *acctId* • *acctName* • *operatorId* • *ICCID* • *IMSI* • *activationDate* • *imei* • *ratePlanName* • *commPlan* • *simProfileID* • *staticAPN* • *msisdn* • *customer* • *deviceId* • *linePayStatus* • *countryCode* • *dateShipped* • *ratePlanId* • *staticIPAddress*
operator	The filter options include operators. Possible values are: • *Range - Specify lower and upper values.* • = • > • <
values	The value to filter the report by.

Response Parameters

Parameter	Description
requestId	A unique ID that Control Center generates when it accepts the report request.
reportName	The name of the report.
dateCreated	Date the report was created.

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Request Example

```
curl -X POST \
  --header "Content-Type: application/json" \
  --header "Accept: application/json" \
  --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>" \
  -d '{
    "reportName": "custom_device_report",
       "reporttype": "Device",
       "schedule": {
         "frequency": "monthly",
         "daysOfMonth": ["24"],
         "generationStartDate": "2025-06-10",
         "generationEndDate": "2025-06-30"
       },
       "settings": {
         "reportFormat": "csv",
         "notificationEmail": {
           "enabled": "true",
           "emailIds": [
             "someone@somedomain.com",
             "someone2@somedomain.com"
           ]
         },
         "sftp": {
           "enabled": "false"
         }
       },
       "fields": [
         "opAcctId",
         "acctId",
         "acctName",
      "operatorId",
         "ICCID",
         "IMSI"
       ],
       "filters": {
         "operator": "AND",
         "items": [
           {
             "field": "IMSI",
             "operator": ">",
             "values": [
                   "100567"
             ]
           },
           {
             "field": "ICCID",
             "operator": "Range",
             "values": [
                   "1230768789876789873",
                   "1089787655677777773"
             ]
           }
         ]
       }
  }'
```

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Response Example

```
{
  "requestId": 473720,
  "reportName": "custom_device_report",
  "dateCreated": "2025-04-16 08:03:36:503+0530"
}
```

Errors

Error Code	HTTP Code	Error Message
15000020	400	Requested report type is not valid.
15000022	400	Requested field name is not valid.
15000026	400	OperatorId and acctId are mandatory fields.
15000027	400	Requested accountId(s) in the filters section do not belong to the user.
15000028	400	Both operatorId and accountId can not be null or empty.
15000029	400	Incorrect filter value(s) provided for {fieldNames}.
15000033	400	Internal system error.
15000034	400	Requested filter is not present in the fields.
15000036	400	GenerationEndDate cannot be before GenerationStartDate.
15000037	400	Mandatory Fields are missing for the report.
15000038	400	Past dates are not allowed. Please choose a valid date.
15000039	400	Invalid date format.
15000040	400	Invalid report format. Only CSV and DAT format are supported.
15000041	400	Invalid Frequency.
15000042	400	Please contact Cisco Support to enable SFTP.
15000043	400	Email Id(s) must be provided.

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Error Code	HTTP Code	Error Message
15000044	400	Invalid email provided for email notification.
15000045	400	DaysOfMonth must be specified for a monthly request.
15000046	400	DaysOfWeek must be specified for a weekly request.
15000047	400	Generation start date and end date are required for daily, weekly, and monthly requests.
15000048	400	Filtering data from other accounts is not permitted.
15000049	400	For the filter section allowed options for the operator are Range, <, >, =.

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Create Aggregated Billing Report

Description

Creates a Dynamic Reporting aggregated billing report. When you run this API, Control Center starts the report generation process, which takes some time to complete. The API returns a request ID and timestamp as an acknowledgement for the request.

The service provider can use the Get Report Status API to check the current status of the report. When the report is available (it has a status of *Completed*), you can access it in a few ways:

- Use the link from the Get Report Status API response to download the report.
- Open the Dynamic Reporting page in Control Center to download the report. See *Downloading a Dynamic Report* in the *Control Center User Guide*.
- Click the link in the body of the email notification to download the report. You must enable email delivery through this API to use this option.
- Access the report through an sFTP server. The service provider must have sFTP enabled to use this feature.

Control Center can generate the billing report on a one-time or monthly basis.

To run this API, you must have a ServiceProviderAdmin role. This API is not available to enterprise accounts.

Resource URL

POST BaseURL/v{apiVersion}/dynareport/requests

Request Parameters

Parameter	Description
apiVersion (Path)	The version number for this API. The current version for all functions is 1.
reportName (Body)	Specify the name for the report. For example, *Custom Billing Report*.
reporttype (Body)	The report type. The value is always *billing.*

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Parameter	Description
category	The billing report category. Choose from one of the possible values: • *account* • *device* • *ratePlan* • *ratePlanZone*
subCategory	Enter a charge type for the category you selected. • ***account.*** You must enter at least one charge type for this category. Possible values are: *Pooled Charges* and *Non-Pooled Charges*. If desired, you can enter both *Pooled Charges* and *Non-Pooled Charges* for this category. • ***device***. Control Center automatically assigns *Non-Pooled Charges* as the default for this category. • ***ratePlan***. Charge types are not applicable for this category. • ***ratePlanZone***. Charge types are not applicable for this category.
metrics	The metric type(s) that you want to retrieve billing details for. Possible values are: • *voice* • *data* • *sms* Separate multiple entries with a comma.

Schedule Parameters

Use these parameters to specify when Control Center generates the report, and whether it creates the report once, or on a recurring basis.

Parameter	Description
frequency	This is how often the report generates. Possible values include: • ***One Time***. You can specify a billing cycle to determine the generation date of the report. • ***Monthly***. You can specify a generation start date and generation end date for the report's billing cycle.
billingCycle	The billing cycle date that you want to retrieve billing details for. The date format is MMYYYY.

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Parameter	Description
generationStartDate	The start date for the time period you want to generate the custom billing report for. The date format is YYYYMMDD.
generationEndDate	The end date for the time period you want to generate the custom billing report for. The date format is YYYYMMDD.

Settings Parameters

Use these parameters to specify the report file format. In addition, you can specify whether Control Center sends an email notification when the report is ready and/or delivers the report through sFTP.

Parameter	Description
reportFormat	The file format for the report. Possible values include: • *CSV* • *DAT*
notificationEmail	
enabled	To receive notifications about generated reports via email, set this value to *true.* When the report is available, Control Center sends an email containing a download link for the report.
emailIds	The email address that Control Center sends the notification email to. Separate multiple entries with a comma.
sftp	
enabled	To receive the report through sFTP, set this value to *true.* The service provider must have sFTP enabled to use this feature.

Field Parameters

Use this parameter to specify the fields for the report.

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Parameter	Description
fields	Specify the fields to include on the report. Separate entries with a comma. Available fields vary based on the chosen category and metrics. The following are required fields that you must include on the report: • *acctName* • *ratePlanId* • *ratingZoneId* • *operatorId* • *cycleId* • *acctId* • *dataCharge* • *smsCharge* • *voiceCharge* • *ICCID* • *IMSI* The following are optional fields that you can choose to include or exclude from the report: • *ratePlanType* • *ratePlanName* • *invoiceId* • *billingCycleStatus* • *currency* • *billingAddress* • *avgdatausage* • *DATACDRCOUNT* • *subscriptionCharge* • *overageDataCharge* • *actualDataUsage* • *roamingDataCharge* • *overageDataUsage* • *roamingDataUsage* • *smsUsage* • *SMSMOCharge* • *SMSMTCharge* • *actualSMSUsage* • *voiceMOCharge* • *voiceMTCharge* • *voiceUsage* • *actualVoiceUsage* • *totalUsageCharge* • *terminalCount* • *ratingZoneName* • *ratingZoneId* • *msisdn* • *includedDataUsage* • *includedSMSUsage*

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Parameter	Description
	• *includedVoiceUsage* • *termStartDate* • *termEndDate* • *opAcctId* • *orderId*

Filter Parameters

Use these parameters to filter the report by specific fields on the report (only some fields are available). For example, if you want to filter the report by an ICCID range, enter *ICCID* for the field parameter, enter *Range* for the operator parameter, and enter the lower and upper values of the ICCID (separated by a comma) for the values parameter. You can add multiple filters to the report. For each filter, you must specify the field, operator, and values. See the *Request Example on page 187*.

Parameter	Description
operator	Always set this field to *AND*.
items	
field	The fields to apply the filters to. You can filter by any one of these fields as long as you include it on the report: • *acctName* • *ratePlanId* • *ratingZoneId* • *operatorId* • *cycleId* • *acctId* • *dataCharge* • *smsCharge* • *voiceCharge* • *ICCID* • *IMSI*
field (cont.)	• *ratePlanType* • *ratePlanName* • *invoiceId* • *billingCycleStatus* • *currency* • *billingAddress* • *avgdatausage* • *DATACDRCOUNT* • *subscriptionCharge* • *overageDataCharge* • *actualDataUsage*

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Parameter	Description
field (cont.)	• *roamingDataCharge* • *overageDataUsage* • *roamingDataUsage* • *smsUsage* • *SMSMOCharge* • *SMSMTCharge* • *actualSMSUsage* • *voiceMOCharge* • *voiceMTCharge* • *voiceUsage* • *actualVoiceUsage* • *totalUsageCharge*
field (cont.)	• *terminalCount* • *ratingZoneName* • *ratingZoneId* • *msisdn* • *includedDataUsage* • *includedSMSUsage* • *includedVoiceUsage* • *termStartDate* • *termEndDate* • *opAcctId* • *orderId*
operator	The filter options include operators. Possible values are: • *Range - Specify lower and upper values.* • *=* • *>* • *<*
values	The value to filter the report by.

Response Parameters

Parameter	Description
requestId	A unique ID that Control Center generates when it accepts the report request.
reportName	The name of the report.
dateCreated	Date the report was created.

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Request Example

```
curl -X POST \
  --header "Content-Type: application/json" \
  --header "Accept: application/json" \
  --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>" \
  -d '{
       "reportName": "billing_Monthly_Report",
       "reportType": "Billing",
       "category": "account",
       "subCategory":["pooled","non-pooled"],
       "metrics": ["data"],
       "schedule": {
       "frequency": "Monthly",
       "billingCycle":"Aug 2025",
       "generationStartDate": "2025-10-25",
       "generationEndDate": "2025-10-30"
       },
       "settings": {
               "reportFormat": "csv",
               "notificationEmail": {
               "enabled": "true",
               "emailIds": [
               "someone@somedomain.com",
               "someone2@somedomain.com",
               ]
       },
       "sftp": {
       "enabled": "false"
       }
},
"fields": ["acctId","acctName","operatorId","cycleId","dataCharge"],
"filters": {
       "operator": "AND",
       "items": [
       {
        "field": "dataCharge",
        "operator": ">",
        "values": [
        "120456"

       ]
   }
  ]
  }
  }'
```

Response Example

```
{
  "requestId": 613720,
  "reportName": "billing_Monthly_Report",
  "dateCreated": "2025-10-07 03:54:04:399-0700"
}
```

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Errors

Error Code	HTTP Code	Error Message
15000020	400	Requested report type is not valid.
15000022	400	Requested field name is not valid.
15000026	400	OperatorId and acctId are mandatory fields.
15000027	400	Requested accountId(s) in the filters section do not belong to the user.
15000028	400	Both operatorId and accountId can not be null or empty.
15000029	400	Incorrect filter value(s) provided for {fieldNames}.
15000030	400	Requested category is not valid.
15000031	400	Category is empty.
15000032	400	Requested metrics is not valid.
15000033	400	Internal system error.
15000034	400	Requested filter is not present in the fields.
15000035	400	Metric is empty.
15000036	400	GenerationEndDate cannot be before GenerationStartDate.
15000037	400	Mandatory Fields are missing for the report.
15000038	400	Past dates are not allowed. Please choose a valid date.
15000039	400	Invalid date format.
15000040	400	Invalid report format. Only CSV and DAT format are supported.
15000041	400	Invalid Frequency.
15000042	400	Please contact Cisco Support to enable SFTP.
15000043	400	Email Id(s) must be provided.
15000044	400	Invalid email provided for email notification.

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Error Code	HTTP Code	Error Message
15000047	400	Generation start date and end date are required for daily, weekly, and monthly requests.
15000048	400	Filtering data from other accounts is not permitted.
15000049	400	For the filter section allowed options for the operator are Range, <, >, =.
15000051	400	The requested Sub-Category is not valid.
15000052	400	Sub-Category is only applicable for Billing account level report.
15000053	400	'Subscription Charges' and 'Rating Zone ID' cannot be provided together.
15000054	400	'Rating Zone ID' cannot be provided with any Prepaid Term fields.
15000055	400	Requested billing cycle is not valid.
15000056	400	Sub-category must be provided for billing account level report.
15000057	400	Filter section value can not be empty.
15000058	400	Only one value is allowed when using the <, > or = operator.
15000059	400	The Range operator requires exactly two values.
15000060	400	Billing cycle is not allowed for this report type.
15000061	400	Billing cycle is empty.

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Create Aggregated Usage Report

Description

Creates a Dynamic Reporting aggregated usage report. When you run this API, Control Center starts the report generation process, which takes some time to complete. The API returns a request ID and timestamp as an acknowledgement for the request.

The service provider or enterprise can use the Get Report Status API to check the current status of the report. When the report is available (it has a status of *Completed*), you can access it in a few ways:

- Use the link from the Get Report Status API response to download the report.
- Open the Dynamic Reporting page in Control Center to download the report. See *Downloading a Dynamic Report* in the *Control Center User Guide*.
- Click the link in the body of the email notification to download the report. You must enable email delivery through this API to use this option.
- Access the report through an sFTP server. The service provider must have sFTP enabled to use this feature.

Control Center can generate the usage report on a one-time, daily, weekly, or monthly basis.

You must have a ServiceProviderAdmin or AccountAdmin role to run this API.

Resource URL

POST BaseURL/v{apiVersion}/dynareport/requests

Request Parameters

Parameter	Description
apiVersion (Path)	The version number for this API. The current version for all functions is 1.
reportName (Body)	Specify the name for the report. For example, *Custom Usage Report*.
reporttype (Body)	The report type. The value is always *usage*.

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Parameter	Description
category	The usage report category. Choose from one of the possible values: • *account* • *device*
metrics	The service type(s) that you want to retrieve usage details for. Possible values are: • *voice* • *data* • *sms*

Schedule Parameters

Use these parameters to specify when Control Center generates the report, and whether it creates the report once, or on a recurring basis.

Parameter	Description
frequency	This is how often the report generates. Possible values include: • *One Time* • *Daily* • *Weekly* • *Monthly*
daysOfWeek	If Control Center generates the report weekly, specify a day of the week for the report. For example, if you enter *Monday*, the date range for the report is the previous Monday to Sunday. "frequency": "Weekly", "daysOfWeek": ["Monday"], "generationStartDate": "2025-06-30", "generationEndDate": "2025-07-01" If Control Center generates the report monthly, leave this parameter blank.

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Parameter	Description
daysOfMonth	If Control Center generates the report monthly, specify a day of the month for the report. Possible values range from *1* to *31*. For example, if you enter *15* for this parameter, the report will generate on the 15th of each month. "frequency": "Monthly", "daysOfMonth": ["15"], "generationStartDate": "2025-06-30", "generationEndDate": "2025-07-01" If Control Center generates the report weekly, leave this parameter blank.
generationStartDate	The start date for the time period you want to generate the custom usage report for. The date format is YYYYMMDD. For example: "frequency": "Daily", "generationStartDate": "2025-06-30", "generationEndDate": "2025-07-01" If Control Center generates the report one time, leave this parameter blank.
generationEndDate	The end date for the time period you want to generate the custom usage report for. The date format is YYYYMMDD. If Control Center generates the report one time, leave this parameter blank.
billingCycle	The billing cycle date that you want to retrieve usage details for. The date format is MMYYYY. If Control Center generates the report daily, weekly, or monthly, leave this parameter blank.

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Parameter	Description
reportingPeriod	The start date and end date you want to generate the custom usage report for. The date format is YYYYMMDD. For example: "frequency": "One Time", "billingCycle": "Date Range", "reportPeriod":{ "startDate":"2025-08-23", "endDate":"2025-08-24" }, "generationDate": "2025-06-30" If Control Center generates the report daily, weekly, or monthly, leave this parameter blank.
generationDate	The generation date that you want to retrieve usage details for. The date format is MMYYYY. For example: "frequency": "One Time", "billingCycle": "Jul 2025", "generationDate": "2025-06-30" If Control Center generates the report daily, weekly, or monthly, leave this parameter blank.

Settings Parameters

Use these parameters to specify the report file format. In addition, you can specify whether Control Center sends an email notification when the report is ready and/or delivers the report through sFTP.

Parameter	Description
reportFormat	The file format for the report. Possible values include: • *CSV* • *DAT*
notificationEmail	
enabled	To receive notifications about generated reports via email, set this value to *true.* When the report is available, Control Center sends an email containing a download link for the report.
emailIds	The email address that Control Center sends the notification email to. Separate multiple entries with a comma.

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Parameter	Description
sftp	
enabled	To receive the report through sFTP, set this value to *true.* The service provider must have sFTP enabled to use this feature.

Field Parameters

Use this parameter to specify the fields for the report.

Parameter	Description
fields	Specify the fields to include on the report. Separate entries with a comma. Available fields vary based on the chosen category and metrics. The following are required fields that you must include on the report: • *opAcctId* • *acctId* • *acctName* • *operatorId* • *dataUsage* • *smsUsage* • *voiceUsage* • *IMSI* The following are optional fields that you can choose to include or exclude from the report: • *cycleId* • *ratePlanName* • *country* • *ratingZoneName* • *deviceCount* • *isoffpeak* • *ratType* • *totalrawDataUsage* • *bytesuplinkUsage* • *bytesdownlinkUsage* • *apn* • *carrierName* • *sgsnaddr* • *ratePlanType* • *servedpdpaddr* • *servedpdpaddripv6* • *msisdn*

Filter Parameters

Use these parameters to filter the report by specific fields on the report (only some fields are available). For example, if you want to filter the

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report by an ICCID range, enter *ICCID* for the field parameter, enter *Range* for the operator parameter, and enter the lower and upper values of the ICCID (separated by a comma) for the values parameter. You can add multiple filters to the report. For each filter, you must specify the field, operator, and values. See the *Request Example on page 197*.

Parameter	Description
operator	Always set this field to *AND*.
items	
field	The fields to apply the filters to. You can filter by any one of these fields as long as you include it on the report: • *opAcctId* • *acctId* • *acctName* • *operatorId* • *dataUsage* • *smsUsage* • *voiceUsage* • *IMSI* • *cycleId* • *ratePlanName* • *country* • *ratingZoneName* • *deviceCount* • *isoffpeak* • *ratType* • *totalrawDataUsage* • *bytesuplinkUsage* • *bytesdownlinkUsage* • *apn* • *carrierName* • *sgsnaddr* • *ratePlanType* • *servedpdpaddr* • *servedpdpaddripv6* • *msisdn*
operator	The filter options include operators. Possible values are: • *Range - Specify lower and upper values.* • *=* • *>* • *<*

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Parameter	Description
values	The value to filter the report by.
unit	Enter a required unit value for the usage filter. Possible unit values can be specified as: • *KB* • *MB* • *GB* • *TB*

Response Parameters

Parameter	Description
requestId	A unique ID that Control Center generates when it accepts the report request.
reportName	The name of the report.
dateCreated	Date the report was created.

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Request Example

```
curl -X POST \
  --header "Content-Type: application/json" \
  --header "Accept: application/json" \
  --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>" \
  -d '{
       "reportName": "Async_Usage_report",
       "reportType": "Usage",
       "category": "Account",
       "metrics": ["data"],
       "schedule": {
       "frequency": "One Time",
       "billingCycle": "Jul 2025",
       "generationDate": "2025-10-29"
       },
       "settings": {
       "reportFormat": "csv",
       "notificationEmail": {
       "enabled": "true",
       "emailIds": [
         "someone@somedomain.com",
         "someone2@somedomain.com",
       ]
       },
       "sftp": {
       "enabled": "false"
       }
       },
       "fields": ["acctId","acctName","operatorId","dataUsage"],
       "filters": {
       "operator": "AND",
       "items": [
       {
         "field": "dataUsage",
         "operator": "Range",
         "values": ["141","1412"],
         "unit":"MB"
       }
       ]
   }
  }'
```

Response Example

```
{
  "requestId": 473720,
  "reportName": "Async_usage_report",
  "dateCreated": "2025-08-29 08:03:36:503+0530"
}
```

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Errors

Error Code	HTTP Code	Error Message
15000020	400	Requested report type is not valid.
15000022	400	Requested field name is not valid.
15000026	400	OperatorId and acctId are mandatory fields.
15000027	400	Requested accountId(s) in the filters section do not belong to the user.
15000028	400	Both operatorId and accountId can not be null or empty.
15000029	400	Incorrect filter value(s) provided for {fieldNames}.
15000030	400	Requested category is not valid.
15000031	400	Category is empty.
15000032	400	Requested metrics is not valid.
15000033	400	Internal system error.
15000034	400	Requested filter is not present in the fields.
15000035	400	Metric is empty.
15000036	400	GenerationEndDate cannot be before GenerationStartDate.
15000037	400	Mandatory Fields are missing for the report.
15000038	400	Past dates are not allowed. Please choose a valid date.
15000039	400	Invalid date format.
15000040	400	Invalid report format. Only CSV and DAT format are supported.
15000041	400	Invalid Frequency.
15000042	400	Please contact Cisco Support to enable SFTP.
15000043	400	Email Id(s) must be provided.
15000044	400	Invalid email provided for email notification.

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Error Code	HTTP Code	Error Message
15000047	400	Generation start date and end date are required for daily, weekly, and monthly requests.
15000048	400	Filtering data from other accounts is not permitted.
15000049	400	For the filter section allowed options for the operator are Range, <, >, =.
15000050	400	Only one metric can be selected at a time.
15000052	400	Sub-Category is only applicable for Billing account level report.
15000053	400	'Subscription Charges' and 'Rating Zone ID' cannot be provided together.
15000054	400	'Rating Zone ID' cannot be provided with any Prepaid Term fields.
15000055	400	Requested billing cycle is not valid.
15000057	400	Filter section value can not be empty.
15000058	400	Only one value is allowed when using the <, > or = operator.
15000059	400	The Range operator requires exactly two values.
15000060	400	Billing cycle is not allowed for this report type.
15000061	400	Billing cycle is empty.
15000062	400	Unit is not valid.
15000063	400	Unit is mandatory for the usage filter.
15000064	400	Start date and end date cannot be empty for a date range.
15000065	400	End Date can not be before the Start Date.
15000066	400	Report period is not allowed.

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Error Code	HTTP Code	Error Message
15000067	400	Only one-time frequency supports a date range and report period for the usage report.
15000068	400	DaysOfMonth and DaysOfWeek are only allowed for weekly or monthly reports.
15000069	400	Requested category, subcategory, or metrics configuration is invalid for Device report type.
15000070	400	GenerationDate can not be before EndDate.
15000071	400	Requested start and end dates must be within the last 6 months.

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Get Report Status

Description

This API allows you to check the status of a report that was created using the Create Device Report API. Along with the current status of the report, this API returns additional information such as the report name, creation date, and a download link for the report.

When the report is available (it has a status of *Completed*), you can access it in several ways:

- Use the link from this API to download the report.
- Open the Dynamic Reporting page in Control Center to download the report. See *Downloading a Dynamic Report* in the *Control Center User Guide*.
- Click the link in the body of the email notification. You must enable email delivery when running the Create Device Report API to use this option.
- Access the report through an sFTP server. The service provider must have sFTP enabled to use this feature.

You must have a Service Provider Admin or Account Admin role to execute this function.

Resource URL

GET BaseURL/v{apiVersion}/dynareport/requests/{requestId}

Request Parameters

Parameter	Description
apiVersion (Path)	The version number for this API. The current version for all functions is 1.
requestId (Query)	The Request ID for the report you want to view the status for. The API that created the report includes this ID in the response parameters.

Response Parameters

Return Value	Description
requestId	The request ID for the device report.
reportName	The name of the report.

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Return Value	Description
status	The report status. Possible values include: • *Completed* • *Processing* • *Cancelled*
errorMessage	(Optional) Error message for a failed request.
dateCreated	Date the report was created.
reportHistoryId	The ID for this instance of the report.
fileName	The generated report file name.
downloadUrl	The download URL for the report, which is valid for 30 days.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/dynareport/requests/473720"
```

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Response Example

When the report is generated:

```
{
   "requestId": 526720,
   "reportName": "device_monthly_18June2025",
   "status": "Completed",
   "dateCreated": "2025-07-02 05:19:01:481-0700",
   "reports": [
     {
        "reportHistoryId": 448740,
        "fileNames": [
            {
              "fileName": "generic_device_ONETimeacctidIN2_30131703_2025-07-02_
1751459514.csv",
              "downloadUrl":
"https://<DOMAIN>/provision/api/v1/dynaReport/downloadFile/555820/generic_
device_ONETimeacctidIN2_30131703_2025-07-02_1751459514.csv"
            },
            {
              "fileName": "generic_device_ONETimeacctidIN2_30131703_2025-07-02_
1751459514.ctl",
              "downloadUrl":
"https://<DOMAIN>/provision/api/v1/dynaReport/downloadFile/555820/generic_
device_ONETimeacctidIN2_30131703_2025-07-02_1751459514.ctl"
            }
         ]
      }
   ]
}
```

When the report is not generated:

```
{
   "requestId": 488420,
   "reportName": "TestReport",
   "status": "Processing",
   "dateCreated": "2025-04-25 01:45:01:021-0700"
}
```

Errors

Error Code	HTTP Code	Error Message
15000025	400	Request does not belong to the user.
15000028	400	Both operatorId and accountId can not be null or empty.
15000033	400	Internal system error.

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Devices

Search Devices

Description

For a given enterprise account, returns a list of devices that have changed within a specific time period. The time period must be one year or less. To get more than a year of data, you can call the API multiple times, changing the time period for each call. You can filter the list by SIM status, as well as custom device fields.

The following table shows several use cases where the current date is July 5, 2024. The modifiedSince and modifiedTill parameters define the start and end of the time window.

Start Date (modifiedSince)	End Date (modifiedTill)	API Behavior
January 6, 2024	Not provided	Returns a list of device changes from January 6, 2024 to July 5, 2024 (the current date).
April 6, 2023	January 1, 2024	Returns a list of device changes from April 6, 2023 to January 1, 2024.
April 6, 2023	Not provided	Returns a list of device changes from April 6, 2023 to April 5, 2024 (one year later).

Resource URL

GET BaseURL/v{apiVersion}/devices

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	(Optional) Unique identifier for the account that owns the devices you're searching for. If you omit this parameter, Control Center uses the account associated with the user name.

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Parameter	Description
modifiedSince	The start date for the time period. You can specify an end date for the time period using modifiedTill. If you do not specify an end date, Control Center defaults to a one year time period. The function returns a list of devices that have been modified during the time period. Use the ISO 8601 format for both the start and end dates.
modifiedTill	(Optional) The end date for the time period. You must also specify the start date for the time period using modifiedSince. The modifiedTill date/time must be greater than the modifiedSince date and no more than one year after the modifiedSince date/time. The default modifiedTill date is one year from the modifiedSince date. The function returns a list of devices that have been modified during the time period. Use the ISO 8601 format for both the start and end dates.
status	(Optional) A SIM status value. The function returns only devices with this status. For a list of valid values, see *SIM Status Values on page 50*.
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.
pageNumber	(Optional) Specifies the number of the response page to return. The value defaults to 1. See *Pagination on page 48*.
accountCustom1-10	(Optional) Custom device fields the enterprise has created in Control Center. These fields use the VARCHAR data type and can be up to 2000 characters in length. In the API, the parameters use the default label (accountCustom1-10), not the custom label. Both service provider and account users can use these parameters. If the custom field is a combo box in the user interface, then the value in the custom field must be either a blank value or one of the pre-defined values. Wildcards (*) are supported for text fields.

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Parameter	Description
operatorCustom1-5	(Optional) Custom device fields the service provider has created in Control Center. These fields use the VARCHAR data type and can be up to 2000 characters in length. In the API, the parameters use the default label (operatorCustom1-5), not the custom label. Only service provider users can use these parameters. If the custom field is a combo box in the user interface, then the value in the custom field must be either a blank value or one of the pre-defined values. Wildcards (*) are supported for text fields.
customerCustom1-5	(Optional) Custom device fields the customer has created in Control Center. These fields use the VARCHAR data type and can be up to 2000 characters in length. In the API, the parameters use the default label (customerCustom1-5), not the custom label. Both service provider and customer users can use these parameters. If the custom field is a combo box in the user interface, then the value in the custom field must be either a blank value or one of the pre-defined values. Wildcards (*) are supported for text fields.

Response Parameters

The function returns an array of devices with the information below. Records are sorted by modification date in ascending order with the oldest device change listed first.

Return Value	Description
devices	The array containing the device list.
iccid	The ICCID of the device.
status	The device's SIM status. For a list of valid values, see *SIM Status Values on page 50*.
ratePlan	The name of the rate plan associated with the device.
communicationPlan	The name of the communication plan associated with the device.
pageNumber	An integer specifying the number of the current response page.
lastPage	A true or false value indicating whether the current response page is the last in the series.
totalCount	Total records fetched.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/v1/devices?
accountId=100020620&accountCustom1=customValue&modifiedSince=2023-04-
18T17%3A31%3A34%2B00%3A00&pageSize=50&pageNumber=1"
```

Response Example

```
{
    "pageNumber": 1,
    "devices": [
        {
            "iccid": "8988216716970004971",
            "status": "ACTIVATION_READY",
            "ratePlan": "hphlr rp1",
            "communicationPlan": "CP_Basic_ON",
        },
        {
            "iccid": "8988216716970004975",
            "status": "ACTIVATED",
            "ratePlan": "hphlr rp1",
            "communicationPlan": "CP_Basic_ON",
        }
    ],
    "totalCount": 2,
    "lastPage": true
}
```

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Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/?modifiedSince=2023-04-
18T17%3A31%3A34%2B00%3A00'
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-URL>/rws/api/v1/devices/?modifiedSince=2023-
04-18T17%3A31%3A34%2B00%3A00').auth('username', 'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/devices/?modifiedSince=2023-04-
18T17%3A31%3A34%2B00%3A00'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000002	400	AccountId is required.
10000003	400	ModifiedSince is required.
10000004	400	Invalid accountId.
10000005	400	Invalid status.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000008	400	Your role does not have access to operator and customer custom fields <field name>
10000009	400	Your role does not have access to account custom fields <field name>
10000010	400	Your role does not have access to customer custom field <field name>
10000012	400	Invalid date format.
10000024	400	Invalid apiVersion.

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Error Code	HTTP Code	Error Message
10000045	400	Invalid date. This date/time cannot be more than one year in the past.
10002117	400	modifiedTill date is earlier than modifiedSince date.
10002118	400	modifiedTill date is after the max allowed modifiedTill date.
10002119	400	modifiedSince is a future date.
30000001	500	Unknown server error.

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Get Device Details

Description

Returns device details for a specific ICCID.

Resource URL

GET BaseURL/v{apiVersion}/devices/{iccid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want information about.

Response Parameters

Return Value	Description
iccid	The ICCID of the device.
imsi	The device IMSI.
msisdn	The device MSISDN or phone number.
imei	The device IMEI.
status	The device SIM status. For a list of valid values, see *SIM Status Values on page 50*.
ratePlan	The name of the rate plan associated with the device.
communicationPlan	The name of the communication plan associated with the device.
customer	The name of the customer (generally an enterprise or business unit), if any, associated with this device.
endConsumerId	The ID of the person, if any, associated with this device.
dateActivated	The date when the device was first activated. See Date Formats for more details.
dateUpdated	The date when the last device information update occurred. See Date Formats for more details.
dateShipped	The date when the device SIM was transferred from the service provider inventory into the enterprise account. See Date Formats for more details.

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Return Value	Description
accountId	The ID of the enterprise account associated with the device.
fixedIPAddress	The IPv4 address assigned to the device. This field may be null if the device communication plan uses dynamic IP addresses or if the device uses an IPv6 address instead.
fixedIpv6Address	The IPv6 address assigned to the device. This field may be null if the device communication plan uses dynamic IP addresses or if the device uses an IPv4 address instead.
operatorCustom1-5	Values for any custom device fields the service provider has created in Control Center. This information is applicable to users with service provider roles only.
accountCustom1-10	Values for any custom device fields the enterprise has created in Control Center. This information is applicable to users with account roles only.
customerCustom1-5	Values for any custom device fields the customer has created in Control Center. This information is applicable to users with customer roles only.
simNotes	Information the service provider has added about the device.
euiccid	The unique identifier for the eUICC chip that contains the bootstrap profile.
deviceID	Optional identifier that an account or customer can give to a device.
modemID	Identifies the modem used by the device to transmit data.
globalSIMType	For enterprises taking advantage of Control Center's Global SIM feature, this value indicates whether the device is using a primary SIM from the lead service provider or a virtual subscription from a partner service provider.
eid	eSIM identifier that associates a profile in Control Center with an eSIM. A profile consists of an ICCID, IMSI, and MSISDN and, in Control Center, is equivalent to a device.

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Return Value	Description
simProfileId	An identifier that defines the type of SIM provided by the manufacturer. The values are specific to the service provider and the manufacturer.
mec	Indicates whether MEC (Mobile Edge Computing) is enabled in the communication profile for the device. Possible values are *Yes* or *No*. This field may be null if MEC does not apply to the device.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8988216716970004975"
```

Response Example


```
{
   "iccid": "8988216716970004975",
   "imsi": "901161697004975",
   "msisdn": "882351697004975",
   "imei": "",
   "status": "ACTIVATED",
   "ratePlan": "hphlr rp1",
   "communicationPlan": "CP_Basic_ON",
   "customer": null,
   "endConsumerId": null,
   "dateActivated": "2016-06-29 00:21:33.339+0000",
   "dateAdded": "2014-07-22 23:31:46.571+0000",
   "dateUpdated": "2016-07-06 22:04:04.380+0000",
   "dateShipped": "2016-06-27 07:00:00.000+0000",
   "accountId": "100020620",
   "fixedIPAddress": null,
   "fixedIpv6Address": null,
   "accountCustom1": "78",
   "accountCustom2": "",
   "accountCustom3": "",
   "accountCustom4": "",
   "accountCustom5": "",
   "accountCustom6": "",
   "accountCustom7": "",
   "accountCustom8": "",
   "accountCustom9": "",
   "accountCustom10": "",
   "simNotes": null,
   "euiccid": null,
   "deviceID": null,
   "modemID": "2221",
   "globalSimType": "",
   "eid": null,
   "simProfileId": ""
```


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```
   "mec": null,
}
```

Code Samples

Make sure to use your own URL and user credentials.

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-
URL>/rws/api/v1/devices/8988216716970004975').auth('username', 'password',
false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/'+input("iccid: ")
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
       # Loading the response data into a dict variable
       # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
       myResponse.raise_for_status()
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/devices/89011704252318147060'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Get Device Details by EID

Description

Returns device details for a specific EID.

Resource URL

GET BaseURL/v{apiVersion}/devices/eid/{eid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
eid	The EID of the device you want information about. The EID is the eSIM identifier that associates a profile in Control Center with an eSIM.

Response Parameters

Return Value	Description
iccid	The ICCID of the device.
imsi	The device IMSI.
msisdn	The device MSISDN or phone number.
imei	The device IMEI.
status	The device SIM status. For a list of valid values, see *SIM Status Values on page 50*.
ratePlan	The name of the rate plan associated with the device.
communicationPlan	The name of the communication plan associated with the device.
customer	The name of the customer (generally an enterprise or business unit), if any, associated with this device.
endConsumerId	The ID of the person, if any, associated with this device.
dateActivated	The date when the device was first activated. See Date Formats for more details.
dateUpdated	The date when the last device information update occurred. See Date Formats for more details.
dateShipped	The date when the device SIM was transferred from the service provider inventory into the enterprise account. See Date Formats for more details.

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Return Value	Description
accountId	The ID of the enterprise account associated with the device.
fixedIPAddress	The IPv4 address assigned to the device. This field may be null if the device communication plan uses dynamic IP addresses or if the device uses an IPv6 address instead.
fixedIpv6Address	The IPv6 address assigned to the device. This field may be null if the device communication plan uses dynamic IP addresses or if the device uses an IPv4 address instead.
operatorCustom1-5	Values for any custom device fields the service provider has created in Control Center. This information is applicable to users with service provider roles only.
accountCustom1-10	Values for any custom device fields the enterprise has created in Control Center. This information is applicable to users with account roles only.
customerCustom1-5	Values for any custom device fields the customer has created in Control Center. This information is applicable to users with customer roles only.
simNotes	Information the service provider has added about the device.
euiccid	The unique identifier for the eUICC chip that contains the bootstrap profile.
deviceID	Optional identifier that an account or customer can give to a device.
modemID	Identifies the modem used by the device to transmit data.
globalSIMType	For enterprises taking advantage of Control Center's Global SIM feature, this value indicates whether the device is using a primary SIM from the lead service provider or a virtual subscription from a partner service provider.
eid	eSIM identifier that associates a profile in Control Center with an eSIM. A profile consists of an ICCID, IMSI, and MSISDN and, in Control Center, is equivalent to a device.
simProfileId	An identifier that defines the type of SIM provided by the manufacturer. The values are specific to the service provider and the manufacturer.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/eid/89012345678901234567890123456789"
```

Response Example


```
{
   "resultCount": 2,
   "devices": [
  {
   "iccid": "8988216716970004975",
   "imsi": "901161697004975",
   "msisdn": "882351697004975",
   "imei": "",
   "status": "ACTIVATED",
   "ratePlan": "hphlr rp1",
   "communicationPlan": "CP_Basic_ON",
   "customer": null,
   "endConsumerId": null,
   "dateActivated": "2016-06-29 00:21:33.339+0000",
   "dateAdded": "2014-07-22 23:31:46.571+0000",
   "dateUpdated": "2016-07-06 22:04:04.380+0000",
   "dateShipped": "2016-06-27 07:00:00.000+0000",
   "accountId": "100020620",
   "fixedIPAddress": null,
   "fixedIpv6Address": null,
   "accountCustom1": "78",
   "accountCustom2": "",
   "accountCustom3": "",
   "accountCustom4": "",
   "accountCustom5": "",
   "accountCustom6": "",
   "accountCustom7": "",
   "accountCustom8": "",
   "accountCustom9": "",
   "accountCustom10": "",
   "simNotes": null,
   "euiccid": null,
   "deviceID": null,
   "modemID": "2221",
   "globalSimType": "",
   "eid": "89012345678901234567890123456789",
   "simProfileId": null
  },
  {
   "iccid": "8988216716970004971",
   "imsi": "901161697004971",
   "msisdn": "882351697004971",
   "imei": "",
   "status": "ACTIVATED",
   "ratePlan": "hphlr rp1",
   "communicationPlan": "CP_Basic_ON",
   "customer": null,
```


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```
   "endConsumerId": null,
   "dateActivated": "2016-06-29 00:21:33.339+0000",
   "dateAdded": "2014-07-22 23:31:46.571+0000",
   "dateUpdated": "2016-07-06 22:04:04.380+0000",
   "dateShipped": "2016-06-27 07:00:00.000+0000",
   "accountId": "100020620",
   "fixedIPAddress": null,
   "fixedIpv6Address": null,
   "operatorCustom1": "",
   "operatorCustom2": "",
   "operatorCustom3": "",
   "operatorCustom4": "",
   "operatorCustom5": "",
   "accountCustom1": "",
   "accountCustom2": "",
   "accountCustom3": "",
   "accountCustom4": "",
   "accountCustom5": "",
   "accountCustom6": "",
   "accountCustom7": "",
   "accountCustom8": "",
   "accountCustom9": "",
   "accountCustom10": "",
   "customerCustom1": "",
   "customerCustom2": "",
   "customerCustom3": "",
   "customerCustom4": "",
   "customerCustom5": "",
   "simNotes": null,
   "deviceID": null,
   "modemID": "2221",
   "globalSimType": "",
   "eid": "89012345678901234567890123456789",
   "simProfileId": ""
   }]
}
```


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Code Samples

Make sure to use your own URL and user credentials.

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-
URL>/rws/api/v1/devices/eid/89012345678901234567890123456789').auth
('username', 'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/'+input("eid: ")
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
       # Loading the response data into a dict variable
       # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
       myResponse.raise_for_status()
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-
URL>/rws/api/v1/devices/eid/89012345678901234567890123456789'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
30000001	500	Unknown server error.
101000013	400	Invalid eSIM ID (EID)
101000017	404	eSIM ID (eid) not found.

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Edit Device Details

Description

Modifies device attributes such as SIM status, rate plan, communication plan, and custom fields for a specified device.

The API supports several different parameters for changing a device rate plan, depending on the current rate plan type and the target rate plan type. Both current and target rate plans must be base rate plans.

Current Rate Plan Type	Target Rate Plan Type	Parameters
Monthly	Monthly	ratePlan
Monthly	Prepaid	ratePlan
Prepaid	Prepaid	termEndDate, renewalMode, renewalRatePlan
Prepaid	Monthly	termEndDate, renewalMode, renewalRatePlan

See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API. Be aware that not all roles can edit all fields. To verify field access for a particular role, log into the web interface as a user with the same role (who also has web interface access) and open the Edit Device pop-up (Devices > Device List > *ICCID* link > Edit button). Typically, the user can edit only the fields that appear in the pop-up. Exceptions are noted in the parameter descriptions.

Resource URL

PUT BaseURL/v{apiVersion}/devices/{iccid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want to edit.
effectiveDate	(Optional) The date when the change takes effect (see Date Formats). If you omit this parameter, the change occurs immediately. If you choose a future date, it cannot be more than 365 days in the future.
At least one of the following parameters is required	

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Parameter	Description
status	The device SIM status. For a list of valid values, see *SIM Status Values on page 50*. If this value is the same as the current value, the API does not change the device attribute or create a change record in the device audit trail. However, the API does return an HTTP 200 Success message.
ratePlan	The name of a base rate plan. You cannot assign a supplemental rate plan to a device using this API. You can use this field to change a device from a monthly rate plan to either a monthly or prepaid rate plan. Both rate plans must be base rate plans. To move a device from a prepaid rate plan to either a monthly or a prepaid rate plan, use the termEndDate, renewalMode, and renewalRatePlan parameters.
communicationPlan	The name of a communication plan. If this value is the same as the current value, the API does not change the device attribute or create a change record in the device audit trail. However, the API does return an HTTP 200 Success message.
customer	The name of a customer. Customer users may not change this value.
deviceID	Optional identifier that an account or customer can give to a device.
modemID	A modem ID number.
overageLimitOverride	Determines the device behavior when it reaches the data limit defined in the rate plan. Valid values are: • DEFAULT. The device cannot exceed the data limit. • TEMPORARY_OVERRIDE. The device can use any amount of data until the end of the current billing cycle, at which point Control Center begins enforcing the data limit set in the rate plan. • PERMANENT_OVERRIDE. The device can use any amount of data, regardless of the data limit defined in the rate plan. Not used for Essential or Lite accounts.

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Parameter	Description
ipv4Address	The IPv4 address assigned to the device. Do not assign an IPv4 address unless the service provider also supports fixed IPv6 addresses and the device communication plan supports fixed IP addresses. A device can have both an IPv4 and an IPv6 address depending on the APNs in the communication plan. If you assign an IPv4 address value, you must also provide values for the apn and pdpId parameters. Users can edit this parameter in the API, but not in the web interface.
ipv6Address	The IPv6 address assigned to the device. Do not assign an IPv4 address unless the service provider also supports fixed IPv6 addresses and the device communication plan supports fixed IP addresses. A device can have both an IPv4 and an IPv6 address depending on the APNs in the communication plan. If you assign an IPv6 address value, you must also provide values for the apn and pdpId parameters. Users can edit this parameter in the API, but not in the web interface.
apn	If you change the IPv4 or IPv6 address fields, you must supply an APN value to identify the IP address. This value does not change the APN used by the device. Users can edit this parameter in the API, but not in the web interface.
pdpId	If you change the IPv4 or IPv6 address fields, you must supply a PDP ID value to identify the IP address. This value does not change the PDP ID used by the device. Users can edit this parameter in the API, but not in the web interface.
operatorCustom1-5	Custom device fields the service provider has created in Control Center. These fields use the VARCHAR data type and can be up to 2000 characters in length. Only service provider users can modify these attributes. If the custom field is a combo box in the user interface, then the value in the custom field must be either a blank value or one of the pre-defined values.

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Parameter	Description
accountCustom1-10	Custom device fields the enterprise has created in Control Center. These fields use the VARCHAR data type and can be up to 2000 characters in length. Both service provider and account users can modify these attributes. Be aware that service provider users do not have edit privileges in the web interface. If the custom field is a combo box in the user interface, then the value in the custom field must be either a blank value or one of the pre-defined values.
customerCustom1-5	Custom device fields the customer has created in Control Center. These fields use the VARCHAR data type and can be up to 2000 characters in length. Both service provider and customer users can modify these attributes. Be aware that service provider users do not have edit privileges in the web interface. If the custom field is a combo box in the user interface, then the value in the custom field must be either a blank value or one of the pre-defined values.
endConsumerName	The name of the person in possession of the device. Enterprises typically specify an end consumer for devices such as phones and tablets. Not all devices will have an end consumer. This string can be up to 60 characters in length and cannot contain commas (,) or asterisks (*). Not supported for Lite accounts.
endConsumerId	A unique identifier for the end consumer. This string can be up to 50 characters in length and cannot contain commas (,) or asterisks (*). Not supported for Lite accounts.
primaryContact	The name of the person who is responsible for the device, typically an operations person. This person may or may not have physical possession of the device. This string can be up to 30 characters in length and cannot contain commas (,) or asterisks (*). Not supported for Lite accounts.
contactPhone	Phone number for the primary contact. This string can be up to 25 characters in length and cannot contain commas (,) or asterisks (*). Not supported for Lite accounts.

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Parameter	Description
contactFax	Fax number for the primary contact. This string can be up to 320 characters in length and cannot contain commas (,) or asterisks (*). Not supported for Lite accounts.
contactEmail	Email address for the primary contact. This string can be up to 25 characters in length and cannot contain commas (,) or asterisks (*). Not supported for Lite accounts.
eid	eSIM identifier that associates a profile in Control Center with an eSIM. A profile consists of an ICCID, IMSI, and MSISDN and, in Control Center, is equivalent to a device. This string must be 32 numeric characters in length. Entering fewer or more numeric characters results in an error. Service provider users cannot set the EID for a device in an inventory account. When a service provider user transfers a device from an enterprise account back into inventory, Control Center removes any EID value.
termEndDate	To move a device from a prepaid rate plan to another prepaid rate plan or to a monthly rate plan, you must use three parameters: termEndDate, renewalMode, and renewalRatePlan parameters. All prepaid rate plans must be base rate plans. The term end date specifies the date and time when you want the term of the current prepaid rate plan to end so the device can move to another rate plan. This date must be after the term start date and the same as or before the original term end date. If you leave the field blank, Control Center uses the original term end date. Be aware that the account must pay the full subscription fee for the original prepaid rate plan. Control Center does not offer any refunds or fee proration. To prevent unintended behavior, do not use effectiveDate in combination with termEndDate. Date format: yyyy-MM-dd'T'HH:mm:ssZ.

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Parameter	Description
renewalMode	To move a device from a prepaid rate plan to another prepaid rate plan or to a monthly rate plan, you must use three parameters: termEndDate, renewalMode, and renewalRatePlan parameters. The renewal mode determines the rate plan renewal behavior when the term of the currently assigned prepaid rate plan expires. This API supports only one option: *N* for NamedPlan, which allows you to assign a specific rate plan to the device when the current rate plan term expires. If the device has a rate plan queue, Control Center overrides the queue and uses the named plan (see renewalRatePlan) instead.
renewalRatePlan	To move a device from a prepaid rate plan to another prepaid rate plan or to a monthly rate plan, you must use three parameters: termEndDate, renewalMode, and renewalRatePlan parameters. The renewal rate plan is the name of the base rate plan you want to assign to the device when the current rate plan term expires. Control Center uses the most recent published version of the named rate plan. This parameter is required if renewal mode is N (Named Plan). If not, the API ignores the parameter.

Response Parameters

Return Value	Description
iccid	A unique identifier for the device Control Center updated.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
{curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>"
-d "{
   \"status\": \"DEACTIVATED\"
}" "https://<YOUR-BASE-URL>/rws/api/v1/devices/9088217871987000001"
```

```
{curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>"
-d "{
   \"ipv6Address\": \"2003:2002:1001:114a::\",
   \"apn\": \"awt-v6-s1\",
   \"pdpId\": \"2098\"
}" "https://<YOUR-BASE-URL>/rws/api/v1/devices/9088217871987000001"
```

Response Example

For changes that are effective immediately, this function returns an HTTP status of 200. For changes that are scheduled in the future, the HTTP status code is 202.

```
{
   "iccid": "9088217871987000001"
}
```

Code Samples

Make sure to use your own URL and user credentials.

Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/devices/89011704252318147060'
response = RestClient::Request.execute(
 method: :put,
 url: url,
 user: 'dpSKit20',
 password: 'b09c4266-83c6-411a-a475-ca4925b3bb4a',
 :payload => '{"accountCustom1":"test acctCustom1: sent by ruby"}',
 :headers => {:accept => :json,
 :content_type => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

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Node.js

```
var request = require('request');
var body = [];
var auth = "Basic " + new Buffer("username:password").toString("base64");
request({
 method: 'PUT',
 url: 'https://<your-base-URL>/rws/api/v1/devices/8988216716970004975',
 headers : {"Authorization" : auth},
 body: {"customerCustom1":"CustCustom11111"},
 json: true
},
 function (error, response, body) {
  if(error) {
   console.log('Error:', error);
   return;
  } else {
   // return statusCode
   console.log(response.statusCode);
   // return contentType
   console.log(response.headers['content-type']);
   console.log(body);
  }
 }
)
```

Python

```
#!/usr/local/bin/python3
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/'+input("iccid: ")
data = {'customerCustom1':'CustCustom11111'}
myResponse = requests.put(url,auth=(input("username: "),input("api_key:
")),json=data)
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
  # Loading the response data into a dict variable
  # json.loads takes in only binary or string variables so using content to
fetch binary content
  # Loads (Load String) takes a Json file and converts into Python data
structure (dict or list, depending on JSON)
  jData = json.loads(myResponse.content)
  pp=pprint.PrettyPrinter(indent=4)
  pp.pprint(jData)
else:
  # If response code is not ok (200), print the resulting http error code with
description
  print("Failure")
  myResponse.raise_for_status()
```

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Errors

Error Code	HTTP Code	Error Message
1100200	400	Term end date must be greater than term start date.
1100400	400	Renewal Rate Plan is not valid for the device. Description: The target rate plan is not available to the device. The rate plan may be associated with a different account. Or, in the case of a shared rate plan, you may need to assign the rate plan to the account.
1100500	400	Renewal mode is not valid.
1100702	400	Renewal rate plan mandatory for NamedPlan renewal mode.
1100704	400	Term not found. Description: The event rate plan does not have an active term.
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000005	400	Invalid status.
10000008	400	Your role does not have access to operator and customer custom fields <field name>
10000009	400	Your role does not have access to account custom fields <field name>
10000010	400	Your role does not have access to customer custom field <field name>
10000011	400	One or more required fields are missing.
10000012	400	Invalid date format.
10000013	404	Invalid ratePlan.
10000014	400	Invalid communicationPlan.

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Error Code	HTTP Code	Error Message
10000015	400	Invalid customer.
10000016	400	Invalid overageLimitOverride.
10000023	400	The JSON in the request is not well formed. Please ensure that commas, colons, braces etc. are formatted properly.
10000024	400	Invalid apiVersion.
10000025	400	Invalid deviceID.
10000026	400	Invalid modemID.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.
10000029	400	This SIM may not be moved back to a Pre Activation status.
10000030	400	Your role does not have access to this API function.
10000031	400	Invalid Zone.
10000124	400	Invalid end consumer name. Number of characters allowed 0-60 and cannot contain commas (,) or asterisks (*).
10000125	400	Invalid end consumer identity. Number of characters allowed 0-50 and cannot contain commas (,) or asterisks (*).
10000126	400	Invalid primary contact. Number of characters allowed 0-30 and cannot contain commas (,) or asterisks (*).
10000127	400	Invalid contact phone. Number of characters allowed 0-25 and cannot contain commas (,) or asterisks (*).
10000128	400	Invalid contact email. Number of characters allowed 0-320 and cannot contain commas (,) or asterisks (*).

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Error Code	HTTP Code	Error Message
10000129	400	Invalid contact fax. Number of characters allowed 0-25 and cannot contain commas (,) or asterisks (*).
10000517	400	The ipv4Address/ipv6Address is already in use, please assign a different one.
10000700	400	Effective Date exceeds maximum - Effective Date must be within 365 days from today
10030000	400	Cannot activate a SIM which is already activated.
15000001	400	Start date should be paired with end date.
15000002	400	Either provide start and end date or billing cycle.
15000003	400	End date should be greater than start date.
15000004	400	Maximum duration between start date and end date is allowed 30 days.
15000005	400	Records count exceeded the maximum permissible limit in response(15k).
15000006	400	Provided login session is not valid.
15000007	400	Not a valid group by element.
15000008	400	Not a valid metric element.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.
101000013	400	Invalid eSIM ID (EID)
102000002	400	Cannot update the EID for a device in the service provider inventory account.
102000001	400	If you specify the EID parameter, you must supply a value.
102000003	400	You do not have access to the EID attribute.

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Get Device Audit History

Description

For a specified device, returns information about changes that occurred within a given time period as measured from today's date. You can request up to a year's worth of data (365 days).

Resource URL

GET BaseURL/v{apiVersion}/devices/{iccid}/auditTrails

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want information about.
daysOfHistory	(Optional) Specifies the audit history time period as measured from today. You can request up to a year's worth of data (no more than 365 days). If you do not specify the number of days, Control Center returns data from the past 30 days by default.
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48* for details.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.

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Response Parameters

The function returns an array of audit history records, with the most recent listed first.

Return Value	Description
iccid	The ICCID of the device.
timestamp	The time when the audit information was returned. The date format is yyyy-MM-ddTHH:mm:ss.SSSZ. See Date Formats for more details.
pageNumber	An integer specifying the number of the current page. See *Pagination on page 48* for details.
lastPage	A true or false value indicating whether the current page is the last in the series. See *Pagination on page 48* for details.
deviceAuditTrails	An array of change records for the specified device.
Fields within each change record	
field	The name of the field whose value was changed.
priorValue	The previous value of the field. No value indicates that this is the first time the field was set.
value	The current value of the field.
effectiveDate	The date and time the change occurred. The date format is yyyy-MM-ddT HH:mm:ss.SSSZ. See Date Formats for more details.
status	The status of the change. Valid values include: *Pending*, *Running*, *Executed*, *Error*, *Deleted*, *Cancelled*, and *WaitingRetry*.
userName	The name of the user who performed the change.
delegatedUser	The name of the user, if any, who acted on behalf of the person who performed the change. This information will alert you if one user was mirroring another when the change occurred.
ipAddress	The IP address of the machine used by the person who made the change.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/
v1/devices/9901310650450118011/auditTrails?daysOfHistory=30&pageSize=50&pageNu
mber=1"
```

Response Example


```
{
  "iccid":"9901310650450118011",
  "timeStamp":"2016-12-06T15:58:06.466Z",
  "pageNumber": 1,
  "lastPage": true,
  "deviceAuditTrails":[
       {
           "field":"Usage Limit Reached",
               "priorValue":"false",
               "value":"false",
               "effectiveDate":"2016-12-06T01:34:16.613Z",
               "status":"Executed",
               "userName":"simUsageManagementUser",
               "delegatedUser":"",
               "ipAddress":"10.106.232.184"
       },
       {
               "field":"Rate Plan",
               "priorValue":"Integration Test -- SP1 Default RP",
               "value":"Integration Test -- SP1 PrepaidTerm",
               "effectiveDate":"2016-11-11T03:39:06.570Z",
               "status":"Executed",
               "userName":"businessRuleUser",
               "delegatedUser":"",
               "ipAddress":"127.0.0.1"
       }
  ]
}
```


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Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/'+input("iccid:
")+'/auditTrails'
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-
URL>/rws/api/v1/devices/8988216716970004975/auditTrails').auth('username',
'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-
URL>/rws/api/v1/devices/89011704252318147060/auditTrails'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000024	400	Invalid apiVersion.
10000049	400	The daysOfHistory must be less than or equal to 365.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Get Device Location

Description

Returns location information for a given device during a specified time period. Control Center collects location details from data session information via RADIUS accounting records. Control Center does not track device location information for SMS or voice usage.

The Location Based Services feature must be enabled for the account and cell ID tracking must be turned on for the device. Only users with the AccountLBSUser role can perform this function.

Resource URL

GET BaseURL/v{apiVersion}/devices/{iccid}/locationHistory

Request Parameters

Use the fromDate and toDate parameters to define a reporting time period. The standard pagination parameters (pageSize and pageNumber) allow you to control the amount of data that is returned.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID for the device whose location history you want to retrieve.
fromDate	(Optional) A date and time indicating the beginning of the time period you want to report on. If there is no fromDate, the function returns an array with two records: the most recent device location and the location immediately prior to that. The date format is yyyy-MM-ddTHH:mm:ssZ. For example, 2018-07-18T17:31:34+00:00.
toDate	(Optional) A date and time indicating the end of the time period you want to report on. If there is no toDate, the function returns the location history from the date specified in fromDate to the current date and time. If there is no fromDate, the function returns an array with two records: the most recent device location and the location immediately prior to that. The date format is yyyy-MM-ddTHH:mm:ssZ. For example, 2018-07-30T17:31:34+00:00.

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Parameter	Description
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48* for more information.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.

Response Parameters

The function returns an array of locations from the specified time period. All return values are strings unless otherwise specified.

Return Value	Description
pageNumber	An integer specifying the number of the current page.
simlocations	An array of location records. See below for details. If cell ID tracking is enabled for the device, but no information is available, this array will be empty. If cell ID tracking is disabled for the device, Control Center returns one record with "-1" values for several of the fields. See the Response Examples below.
lastPage	A true or false value indicating whether the current page is the last in the series.
Parameters in the simLocations array	
iccid	The ICCID of the device.
dateReceived	The date associated with the cell ID. The date format is yyyy-MM-ddTHH:mm:ssZ. For example, 2018-07-12T17:31:34+00:00.
cellId	The cell ID associated with a data session (from the Radius accounting record). If cell ID tracking is disabled for the device, Control Center returns "-1" in this field. Data type: number.
cellLac	The location area code (LAC) associated with the cell ID. If cell ID tracking is disabled for the device, Control Center returns "-1" in this field. Data type: number.
servingMcc	The mobile country code (MCC) associated with the cell ID. If cell ID tracking is disabled for the device, Control Center returns "-1" in this field. Data type: number.
servingMnc	The mobile network code (MNC) associated with the cell ID. If cell ID tracking is disabled for the device, Control Center returns "-1" in this field. Data type: number.

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Return Value	Description
latitude	The latitude associated with the cell ID. This field is null if the service provider doesn't upload a cell ID mapping file. Data type: number (floating point)
longitude	The longitude associated with the cell ID. This field is null if the service provider doesn't upload a cell ID mapping file. Data type: number (floating point)
accuracy	Indicates the accuracy of the latitude/longitude values. This number comes from the cell ID mapping file which determines the unit of measure. Data type: number.
city	The city associated with the cell ID.
state	The state associated with the cell ID.
country	The country associated with the cell ID.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8988216716970004975/locationHistory?fromDate%3D2022-
04-01T17%3A31%3A34%2B00%3A00%26toDate%3D2022-04-
18T17%3A31%3A34%2B00%3A00%26pageSize%3D100"
```

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Response Example

For a device with cell ID tracking enabled and some available location information.

```
{
  "pageNumber": 1,
  "simLocations": [{
    "iccid": "89014103255100000055",
    "dateReceived": "2018-07-25T01:13:34+0000",
    "cellId": 43761,
    "cellLac": 10000,
    "servingMcc": 302,
    "servingMnc": 720,
    "latitude": 37.7749,
    "longitude": -122.4194,
    "accuracy": 20,
    "city": "Boston",
    "state": "MA",
    "country": "USA"
  }, {
    "iccid": "89014103255100000055",
    "dateReceived": "2018-07-25T01:13:34+0000",
    "cellId": 28312832,
    "cellLac": 65534,
    "servingMcc": 302,
    "servingMnc": 220,
    "latitude": 42.3601,
    "longitude": -71.0589,
    "accuracy": 50,
    "city": "SF",
    "state": "CA",
    "country": "USA"
  }],
  "lastPage": true
}
```

For a device with cell ID tracking enabled, but no available location information.

```
{
  "pageNumber": 1,
  "simLocations": [],
  "lastPage": true
}
```

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For a device with cell ID tracking disabled.

```
{
  "simLocations": [{
    "iccid": "89014103255100000055",
    "dateReceived": "2018-07-25T01:13:34+0000",
    "cellId": -1,
    "cellLac": -1,
    "servingMcc": -1,
    "servingMnc": -1,
  }],
  "pageNumber": 1,
  "lastPage": true
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000021	400	Invalid iccid.
10000022	400	FromDate is required.
10000024	400	Invalid apiVersion.
10000027	400	ToDate must be after fromDate.
10000030	400	Your role does not have access to this API function.
10000062	400	Access Denied.
10000079	400	Account does not have access to this API.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Get Device Usage

Description

Returns cycle-to-date usage information for a specified device.

Resource URL

GET BaseURL/v{apiVersion}/devices/{iccid}/ctdUsages

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want information about.

Response Parameters

Return Value	Description
iccid	The ICCID of the device.
imsi	The device IMSI.
msisdn	The device MSISDN or phone number.
imei	The device IMEI.
status	The device SIM status. For a list of valid values, see *SIM Status Values on page 50*.
ratePlan	The name of the rate plan associated with the device.
communicationPlan	The name of the communication plan associated with the device.
ctdDataUsage	The amount of data used (in bytes) since the beginning of the billing cycle.
ctdSMSUsage	A count of the mobile-originated and mobile-terminated messages since the beginning of the billing cycle.
ctdVoiceUsage	The number of voice seconds used since the beginning of the billing cycle.
ctdSessionCount	The number of data sessions since the beginning of the billing cycle.

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Return Value	Description
overageLimitReached	A true/false value indicating whether the device has reached the data limit set in the rate plan.
overageLimitOverride	Indicates whether the device can exceed the data limit specified in the rate plan. The possible values are: • DEFAULT. The device cannot exceed the data limit. • TEMPORARY_OVERRIDE. The device can use any amount of data until the end of the current billing cycle, at which point Control Center will begin enforcing the data limit set in the rate plan. • PERMANENT_OVERRIDE. The device can use any amount of data, regardless of the data limit defined in the rate plan.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8988216716970004975/ctdUsages"
```

Response Example

```
{
       "iccid": "8988216716970004975",
       "imsi": "901161697004975",
       "msisdn": "882351697004975",
       "imei": "",
       "status": "ACTIVATED",
       "ratePlan": "hphlr rp1",
       "communicationPlan": "CP_Basic_ON",
       "ctdDataUsage": 0,
       "ctdSMSUsage": 0,
       "ctdVoiceUsage": 0,
       "ctdSessionCount": null,
       "overageLimitReached": false,
       "overageLimitOverride": "DEFAULT"
}
```

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Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/'+input("iccid:
")+'/ctdUsages'
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
       # Loading the response data into a dict variable
       # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
       pp=pprint.PrettyPrinter(indent=4)
       pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
       myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-
URL>/rws/api/v1/devices/8988216716970004975/ctdUsages').auth('username',
'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-
URL>/rws/api/v1/devices/89011704252318147060/ctdUsages'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
10000031	400	Invalid Zone.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Get Device Usage by Zone

Description

For a specified device, returns usage information for a single billing cycle, broken down by rate plan and zone. You can filter the usage information by zone and/or rate plan, and specify a particular cycle start date. If no billing cycle is specified, the API returns usage for the current billing cycle by default. All usage is pre-invoiced and applies to wholesale rate plans. To retrieve retail rate plan usage, you must examine the corresponding wholesale rate plan usage.

Resource URL

GET BaseURL/v{apiVersion}/devices/{iccid}/usageInZone

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want information about.
cycleStartDate	(Optional) The first day of a billing cycle. This date must correspond to one of the last three billing cycles (the current billing cycle plus the two preceding cycles). If there is no date, the function returns usage information for the current billing cycle only. The date format is yyyy-MM-ddZ. See Date Formats for details.
ratePlan	(Optional) The name of a rate plan. Use this parameter to retrieve usage information that's linked to a specific rate plan. If there's no rate plan listed, the function returns usage information for any rate plans used by the device during the specified time period. You can specify only one rate plan per function call.

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Parameter	Description
zone	(Optional) The name of a billing zone. Use this parameter to track usage for a specific zone. To get usage information for all zones specified in the rate plans used by the device, omit this parameter. You can specify only one zone per function call.
isActiveRatePlan	(Optional) Indicates whether the reported rate plans have active terms during the specified billing cycle. This parameter allows you to exclude inactive rate plans and their corresponding zones. • *true*. Includes rate plans with active terms and excludes rate plans with inactive terms. • *false*. Includes all rate plans associated with the device, regardless of whether the terms are active or inactive. If you do not specify a value for this parameter, the API returns all rate plans.

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Response Parameters

The function returns an array of usage records sorted by rate plan and zone.

Return Value	Description
iccid	The ICCID of the device.
timestamp	The point in time when the usage information was returned. The date format is yyyy-MM-ddTHH:mm:ss.SSSZ. See Date Formats for more details.
cycleStartDate	The beginning of the billing cycle associated with the usage. The date format is yyyy-MM-ddTHH:mm:ss.SSSZ. See Date Formats for more details.
cycleEndDate	The end of the billing cycle associated with the usage. The date format is yyyy-MM-ddTHH:mm:ss.SSSZ. See Date Formats for more details.
deviceCycleUsageInZones	An object containing a list of rate plans used by the device during the billing cycle.
deviceCycleUsageInZones	
Rate Plan Name, Version Number	Each rate plan contains an array of zones where usage occurred. If the device did not use service in a particular zone, that zone does not appear in the array. The function returns the rate plan version number, because in the case of prepaid rate plans it's possible for a device to use two versions of the same rate plan during the billing cycle.
ratePlan	The name of the rate plan associated with the device usage.
ratePlanVersion	The version number of the rate plan.
zone	The name of the zone associated with the device usage.
dataUsage	The amount of data used during the specified time period.
dataUsageUnit	The unit of measure for the data usage. Valid values include: bytes.
voiceMTUsage	The number of mobile-terminated voice seconds used since the specified start date.

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Return Value	Description
voiceMTUsageUnit	The unit of measure for the voice MT usage. Valid values include: seconds.
voiceMOUsage	The number of mobile-originated voice seconds used since the specified start date.
voiceMOUsageUnit	The unit of measure for the voice MO usage. Valid values include: seconds.
smsmtusage	The number of mobile-terminated messages sent since the specified start date.
smsmousage	The number of mobile-originated messages sent during the specified time period.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

The first example requests all usage records for a device during the current billing cycle. See the Response Example for the response.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8988217755520000031/usageInZone"
```

To get usage for a specific cycle start date:

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8988217755520000031/usageInZone?cycleStartDate=2016-
11-10Z"
```

To get usage for a specific cycle start date, rate plan, and zone:

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8988217755520000031/usageInZone?cycleStartDate=2016-
11-10Z&ratePlan=Rate%20Plan%20A&zone=Zone%20B
```

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Response Example

The response contains an array of rate plans. Each rate plan contains an array of usage records. This response matches the first request example.


```
{
  "iccid":"8988217755520000031",
  "timeStamp":"2016-11-10T05:31:17.557Z",
  "cycleStartDate":"2016-10-31T10:30:00.000Z",
  "cycleEndDate":"2016-11-30T10:29:59.000Z",
  "deviceCycleUsageInZones":{
       "Rate Plan A, 1":[
               {
               "ratePlan":"Rate Plan A",
               "ratePlanVersion":"1",
               "zone":"default zone",
               "dataUsage":null,
               "dataUsageUnit":null,
               "voiceMTUsage":null,
               "voiceMTUsageUnit":null,
               "voiceMOUsage":600,
               "voiceMOUsageUnit":"seconds",
               "smsmtusage":null,
               "smsmousage":null
               }
           ],
       "Rate Plan B, 2":[
               {
               "ratePlan":"Rate Plan B",
               "ratePlanVersion":"2",
               "zone":"default zone",
               "dataUsage":943718,
               "dataUsageUnit":"bytes",
               "voiceMTUsage":null,
               "voiceMTUsageUnit":null,
               "voiceMOUsage":943718,
               "voiceMOUsageUnit":"seconds",
               "smsmtusage":null,
               "smsmousage":null
               }
           ]
       }
}
```


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Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/'+input("iccid:
")+'/usageInZone'
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-
URL>/rws/api/v1/devices/8988216716970004975/usageInZone').auth('username',
'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-
URL>/rws/api/v1/devices/89011704252318147060/usageInZone'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000012	400	Invalid date format.
10000013	404	Invalid ratePlan.
10000024	400	Invalid apiVersion.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.
10000031	400	Invalid Zone.
10000032	400	Cycle Start Date should be greater than or equal to 20xx-xx-xx. Description: The cycleStartDate must specify one of the last three billing cycles (including the current cycle).
10000256	400	No active rate plan found for the device.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Get Session Details

Description

Returns information about the current or most recent data session for a given device.

If a device supports multiple simultaneous sessions, and the most recent session is terminated, this API will return information for the next most recent active session.

Resource URL

GET BaseURL/v{apiVersion}/devices/{iccid}/sessionInfo

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want information about.

Response Parameters

Return Value	Description
iccid	The ICCID of the device.
dateSessionStarted	The time when the current or most recent data session began. For a device that has never had a session, this date is null.
dateSessionEnded	The time when the most recent data session ended. For a device currently in a session, this field is null. For a device that has never had a session, this date is null.
ipAddress	The IPv4 address used by the device during the current or most recent session. A session has at least one IP Address value (ipAddress or ipv6Address) and, in some cases, may have values for both fields. This field is always returned, even if the device does not have an IPv4 address. For a device that has never been in session, this field is null.

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Return Value	Description
ipv6Address	The IPv6 address used by the device during the current or most recent session. A session has at least one IP Address value (ipAddress or ipv6Address) and, in some cases, may have values for both fields. This field is returned only when the device has an IPv6 address. The field never has a null value.
apn	The APN used by the device during the current or most recent session.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8668216191110811771/sessionInfo"
```

Response Example

```
{
       "iccid": "8668216191110811771",
       "ipAddress": "192.168.5.66",
       "dateSessionStarted": "2025-03-05 21:41:26.312+0000",
       "dateSessionEnded": null,
       "apn": "ndo1",
       "ipv6Address": "2001:db8:85a3::/64"
}
```

Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+cobrandURL+'/rws/api/v1/devices/'+input("iccid:
")+'/sessionInfo'
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

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Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-
URL>/rws/api/v1/devices/8988216716970004975/sessionInfo').auth('username',
'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-
URL>/rws/api/v1/devices/8901170425231814706 0/sessionInfo'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
10000031	400	Invalid Zone.

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Error Code	HTTP Code	Error Message
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Transfer Device to Account

Description

Transfers up to 5,000 devices from one enterprise account to another.

Supported transfer types include:

- Inventory to Advantage/Essential/Lite
- Advantage/Essential/Lite to Inventory
- Advantage to Advantage
- Essential to Essential
- Lite to Lite

You can supply a target rate plan and communication plan for all devices in the transfer. If you do not specify this information, Control Center uses the default rate plan and/or communication plan associated with the target account.

For transfers between accounts and for transfers from inventory to account, you can specify a target SIM state to apply to all devices in the transfer. If you do not specify a SIM state, Control Center uses the default SIM state defined at the account level. For return to inventory transfers, Control Center changes the SIM status to *Start*.

In addition, Control Center adjusts other device attributes as follows:

- **Date Added.** The Date Added changes to the current date.
- **Current Usage.** Cycle-to-date usage values drop to zero in the target account.
- **Fixed IP Addresses**. For devices using fixed IP addresses, Control Center automatically releases the fixed IP address during the transfer. If needed, you can reassign the IP address to the device using the Edit Device Details API when the transfer is complete.
- **Custom Field Values**. For transfers between enterprise accounts, Control Center deletes the accounts custom field values, but maintains the service provider custom field values. For transfers from an enterprise account to inventory, Control Center deletes all custom field values.
- **Audit Trail**. Device audit trail is deleted.
- **Other Attribute Values**. With the exception of the rate plan and communication plan, all other device attribute values remain the same.

After you submit the API request, Control Center runs multiple validations on the supplied information, schedules the transfer, and then returns status information for each device. If the user includes a callback URL, Control Center sends a callback notification with status details after the

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job is complete. For more information about callbacks, see *Asynchronous Responses on page 45*.

This API is available to service providers and Advantage accounts only. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Before an enterprise can use this API, the service provider must map the accounts that are transferring devices. During the mapping process, you:

- Select the accounts that an enterprise can transfer devices to.
- Specify whether the account can use an API to transfer devices. If the Allow API Transfer field on the SIM Transfer Mapping page is set to *No*, the enterprise will receive an error when running this API.
- Determine whether the initiating account can change the SIM state during the device transfer.

Resource URL

POST BaseURL/v{apiVersion}/devices/transferDeviceToAccount

Request Parameters

Parameter	Description
accountId	A unique identifier for the target account.
iccids	An array of ICCIDs for the devices you want to transfer.
ratePlanName	(Optional) The name of the rate plan the devices receive in the target account. If this parameter is missing, devices receive the default account rate plan.
commPlanName	(Optional) The name of the communication plan the devices receive in the target account. If this parameter is missing, devices receive the default account communication plan.
callbackURL	(Optional) The destination URL Control Center uses when sending the callback notification. For general information about callback notifications, see *Asynchronous Responses on page 45*.
simState	(Optional) The SIM state the device receives when transferred to a new account. If you do not specify a SIM state, Control Center uses the SIM state defined at the account level.

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Response Parameters

Synchronous Response

This function returns an HTTP 202 Accepted code to indicate that the request is scheduled for processing. The job generally finishes within several seconds. The response contains information about each device in the list. If there is an authentication or formatting error, Control Center returns an error message. See *Error Handling on page 50*.

Return Value	Description
iccid	The device ICCID.
transferStatus	Indicates whether the transfer should be successful. Valid values are *success* and *failure*.
errorMessage	If Control Center detects authentication or formatting errors, it cancels the transfer for the given device and returns an error message immediately. If Control Center anticipates a successful transfer, there is no error message.

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Asynchronous Response

If you supply a callback URL in the request, the API sends an JSON callback notification as soon as the processing finishes. The notification contains information about each device in the list.

Return Value	Description
correlationId	The target account ID. You can use this information to determine which request is associated with the asynchronous response.
apiCallbackNotificationId	An internal sequence number that uniquely identifies this asynchronous response.
sourceApi	The name of the API associated with the callback message. In this case, the value is: TransferDevicesToAcctResource.
requestDate	The date when the API request was sent. The date format is YYYY-MM-DDTHH:MM:SS.SSSZ, for example: 2022-04-27T06:41:20.291Z,
transferDeviceToAccountStatusList	An array of ICCID/transfer status information.
transferDeviceToAccountStatus	Information about each device you attempt to transfer.
iccid	The ICCID for a device.
transferStatus	Indicates whether the transfer was successful. Valid values are *success* and *failure*.
errorMessage	An error message describing the problem if the transfer fails.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
{curl -X POST --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>"
-d "{
   \"accountId\": 130222403,
   \"iccids\": [\"99999999999999990648\",\"99999999999999990649\"]
   \"ratePlanName\": \"Monthly Individual 3MB\",
   \"commPlanName\": \"Data Only\",
   \"callbackUrl\": \"https://1dd78df098f4f05fa451cce590ddc2f7.callback.net\"
}" "https://<YOUR-BASE-URL>/rws/api/v1/devices/transferDeviceToAccount"
```

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Response Example

Synchronous Response Example

```
[
  {
     "iccid": "99999999999999990648",
     "transferStatus": "success",
     "errorMessage": ""
  },
  {
     "iccid": "99999999999999990649",
     "transferStatus": "failure",
     "errorMessage": "100101 Invalid ICCID"
  }
]
```

Asynchronous Response Example

Here is the data portion of the callback notification. For information about the other parameters in the notification, see *Callback Notification Format on page 46*.

```
{
   ...
   "data": {
      "correlationId": "1",
      "apiCallbackNotificationId": "4205",
      "sourceApi": "TransferDevicesToAcctResource",
      "requestDate": "148413074253",
      "transferDeviceToAccountStatusList": {
         "transferDeviceToAccountStatus": [
         {
           "iccid": "99999999999999990648",
           "transferStatus": "success",
           "errorMessage": ""
         },
         {
           "iccid": "99999999999999990649",
           "transferStatus": "failure",
           "errorMessage": "1000512 SIM is not in Purged State"
         }
         ]
      }
   },
   ...
}
```

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Errors

Error Code	HTTP Code	Error Message
100100	206	Terminal not found.
100101	202	Invalid ICCID. ICCID-<ICCID>. Description: The ICCID doesn't exist in Control Center.
400201	202	Access violation. Description: The user does not have access to the requested resource.
400400	400	Too many entries. Requested iccids more than 5000. Description: The request contains more than 5000 devices.
1000100	400	Account not found. accountid=<AcctId>. Description: The target account doesn't exist in Control Center.
1000100	202	Account not found. accountid=<AcctId>. Description: The source account doesn't exist in Control Center. Be aware that there may be multiple source accounts, depending on the devices you transfer.
1000508	202	Cannot Transfer SIM to this Account.
1000509	202	Cannot Transfer SIM to same account. Description: The source and target accounts must be different.
1000510	202	Cannot Transfer SIM.
1000512	202	SIM is not in Purged State. When you return a device to inventory, it must have the *Purged* SIM state.

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Error Code	HTTP Code	Error Message
1000513	202	Cannot Transfer SIM when source account is inventory account. Description:You cannot transfer a device to inventory when it's already in the inventory account.
1000514	202	Cannot Transfer SIM when source and destination account doesn't belong to same service provider. Description: The source and target accounts must be located in the same service provider.
1000523	202	Rate Plan is Not Eligible. RatePlanType and acctType mismatch. Description: The rate plan is not available for the target account.
1000526	202	Communication Plans and account type mismatch. CommPlanType and acctType mismatch. Description: The communication plan is not available for the target account.
1000550	202	Transfer of devices is allowed across Essential accounts only. Description: You cannot transfer devices from an Essential account to an Advantage account.
1000551	202	Transfer of devices is allowed across Advantage accounts only. Description: You cannot transfer devices from an Advantage account to an Essential account.
1000552	202	Transfer of devices is not supported for Essential accounts. Description: You cannot transfer devices to or from an Essential account.

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Error Code	HTTP Code	Error Message
1000559	202	Target SIM State not allowed for Return to Inventory Transfer.
1000600	202	CommPlan and Account Voice WL flag mismatch. Description: The communication plan supports Voice Permit List, but the target account does not.
1000700	202	CommPlan and Account VoLTE RUG flag mismatch. Description: The communication plan supports VoLTE RUG, but the target account does not.
1000705	400	Device transfer for CIP accounts not supported.
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000011	400	One or more required fields are missing. Description: An account ID is required. There must be at least one ICCID.
10000024	400	Invalid apiVersion.
10000062	400	Access Denied. Description: The user doesn't have access to the target account.
10000080	403	The API access is forbidden. Description: This error can occur if your role doesn't have access to the function.
10000131	400	Invalid SIM State.
10000800	202	Validation failed due to invalid target account or invalid target sim state.
30000001	500	Unknown server error.

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Echo

Echo

Description

Returns any value passed to the function. This information can be useful for debugging.

Resource URL

GET BaseURL/v{apiVersion}/echo/{param}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
param	The string you want to echo.

Response Parameters

Return Value	Description
context	The string that you passed into the function.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/v1/echo/hello-
world"
```

Response Example

```
{
   "context": "hello-world"
}
```

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Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64

cobrandURL=input("Cobrand URL: ")
params = input("params: ")
url = 'https://'+cobrandURL+'/rws/api/v1/echo/'+params
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
       # Loading the response data into a dict variable
       # json.loads takes in only binary or string variables so using content
to fetch binary content
       # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
       print(jData)
       print("The response contains {0} properties".format(jData))
       print("\n")
       for key in jData:
        print (key + " : " + jData[key])
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
       myResponse.raise_for_status()
```

Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/echo/hello%20World'
response = RestClient.get(url)
puts JSON.parse(response)
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-URL>/rws/api/v1/echo/hello%20world!').auth
('username', 'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
```

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```
.on('end',function(){
    body = Buffer.concat(body).toString();
    console.log(body);
});
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000011	400	One or more required fields are missing.
10000024	400	Invalid apiVersion.
30000001	500	Unknown server error.

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Multi-Party Billing

Get Partner Accounts for Device

Description

Used in Multi-Party Billing , this function returns information about the partner accounts currently associated with a particular device. To get information about past associations, you must use the web interface. See Viewing Partner Accounts for a Device in the Multi-Party Billing section of the *Control Center User Guide*.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

GET BaseURL/v{apiVersion}/devices/{iccid}/partnerAccounts

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want information about.

Response Parameters

Return Value	Description
iccid	The ICCID of the device you want information about.
partnerAccounts	An array containing information about each partner account associated with the device. If the device doesn't have any partner accounts, the function returns an empty array and a successful HTTP code (200).
Parameters in the partnerAccounts array	
partnerAcctId	A Control Center-generated identifier for the account.
partnerAcctName	The account name appears throughout Control Center as an identifier for the account.
associationDate	The date and time when the device was assigned to the partner account. If the device and the account have been associated more than once, this timestamp reflects the most recent association.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/9846897774444000099/partnerAccounts"
```

Response Example

```
{

  "iccid": "9846897774444000099",
  "partnerAccounts": [
  {
      "partnerAcctId": 130222403,
      "partnerAcctName": "QaAwtBillOprMPBAC001",
      "associationDate": "2019-01-31 17:44:16+0000"
   },
   {
      "partnerAcctId": 130222503,
      "partnerAcctName": "QaAwtBillOprMPB_AC002",
      "associationDate": "2019-02-01 03:05:57+0000"
   }
   ]
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000062	400	Access Denied.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Assign Device to Partner Account

Description

Used in Multi-Party Billing , this function assigns a device to a partner account. As part of the function call, you can also specify a partner rate plan. The device's enterprise account must have a trust relationship with the partner account. At the end of the current billing cycle, Control Center determines whether the device is billable and calculates the subscription fee and any overage charges.

This function assigns one device to one partner account. To assign multiple devices to a partner account or a single device to multiple accounts, you must use several function calls.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

POST BaseURL/v{apiVersion}/devices/{iccid}/partnerAccounts

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want to assign to a partner account.
partnerAcctId	An identifier for the partner account.
partnerRatePlanName	(Optional) The name of the rate plan Control Center will use to rate partner services usage. This partner rate plan must be associated with the partner account. If you omit this parameter, Control Center uses the default rate plan for the partner account.
effectiveDate	(Optional) This field is not currently supported. If you do supply a value, the date format is yyyy-MM-ddZ.

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Response Parameters

This function returns an HTTP 202 Accepted code to indicate that the request is scheduled for processing and the task should be complete within several seconds.

Return Value	Description
iccid	The device ICCID.
partnerAcctId	The partner account ID.
partnerRatePlanName	The name of the partner rate plan.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X POST --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"partnerAcctId\": 130222403,
  \"partnerRatePlanName\": \"QaAwtBillOprMPB_RP003\"
}" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/304689444405000/partnerAccounts"
```

Response Example

```
{
  "iccid": "9846897774444000095",
  "partnerAcctId": 130222403,
  "partnerRatePlanName": "QaAwtBillOprMPB_RP003"
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000002	400	AccountId is required.
10000012	400	Invalid date format.
10000013	404	Invalid ratePlan.

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Error Code	HTTP Code	Error Message
10000023	400	The JSON in the request is not well formed. Please ensure that commas, colons, braces etc. are formatted properly.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000045	400	Invalid date.
10000062	400	Access Denied.
10000074	400	Account is not of Partner Type.
10000075	400	Trust Relationship is not established with Partner Account.
10000076	400	Trust Relationship is not active.
10000077	400	Device is already assigned to Partner Account.
20000001	404	Resource not found - Invalid ICCID.
20000003	404	Resource not found - Invalid accountId.
30000001	500	Unknown server error.

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Change Partner Rate Plan for Device

Description

Used in Multi-Party Billing , this function changes the partner rate plan for a device associated with a particular partner account. You must supply the name of a different partner rate plan linked to the partner account.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

PUT BaseURL/v{apiVersion}/devices/{iccid}/partnerAccounts/{partnerAcctId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device whose partner rate plan you want to change.
partnerAcctId	An identifier for the partner account.
partnerRatePlanName	The name of the partner rate plan you want the device to start using. This rate plan must be associated with the specified partner account and the rate plan must be different from the current partner rate plan.

Response Parameters

Return Value	Description
iccid	The ICCID of the device with the changed partner rate plan.
partnerAcctId	The identifier for the partner account associated with the partner rate plan.
partnerRatePlanName	The name of the new partner rate plan associated with the device.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{ \"partnerRatePlanName\": \"QaAwtBillOprMPB_RP005\" }"
"https://<YOUR-BASE-
URL>/rws/api/v1/devices/9856897774444002318/partnerAccounts/130307603"
```

Response Example

```
{
   "iccid": "9856897774444002318",
   "partnerAcctId": 130307603,
   "partnerRatePlanName": "QaAwtBillOprMPB_RP005"
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000004	400	Invalid accountId.
10000013	404	Invalid ratePlan.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000045	400	Invalid date.
10000062	400	Access Denied.
10000074	400	Account is not of Partner Type.
10000078	400	Device is not assigned to Partner Account.
20000001	404	Resource not found - Invalid ICCID.
20000003	404	Resource not found - Invalid accountId.
30000001	500	Unknown server error.

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Remove Device from Partner Account

Description

Used in Multi-Party Billing , this function removes a device from a given partner account. Control Center charges the full subscription fee at the end of the billing cycle. There is no proration for subscription fees in the last billing cycle.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

DELETE BaseURL/v{apiVersion}/devices/{iccid}/partnerAccounts/{partnerAcctId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want to remove.
partnerAcctId	The identifier for the partner account.
effectiveDate	(Optional) This field is not currently supported. The date and time when the removal takes effect. You can select a time in the future, but not in the past. The date format is yyyy-MM-ddZ.

Response Parameters

This function returns an HTTP 202 Accepted code to indicate that the request is scheduled for processing and the task should be complete within several seconds.

Return Value	Description
iccid	The ICCID of the target device.
partnerAcctId	The identifier for the target partner account.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X DELETE --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{ \"effectiveDate\": \"2019-06-17\" }" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/8988216716970004975"
```

Response Example

```
{
   "iccid": "9856897774444002318",
   "partnerAcctId": 130307603
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000004	400	Invalid accountId.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000062	400	Access Denied.
10000078	400	Device is not assigned to Partner Account.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Assign Partner Event Rate Plan to Device

Description

Used in Multi-Party Billing , this function assigns a prepaid event plan to a partner device. The device must have an *Activated* SIM status.

A prepaid event plans allow a service provider to charge a daily subscription fee to a partner. You can activate up to 30 sequential instances of an event rate plan for a device.

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

PUT BaseURL/v{apiVersion}/partnereventplan/{iccid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you are activating the partner event rate plan for.
partnerEventName	The name of a partner event rate plan you are activating on the device.
effectiveDate	Effective date of the event rate plan activation. If there is no date, the API will activate the rate plan immediately. Date format: yyyy-MM-ddTHH:mm:ssZ.
Instances	The number of instances you want to assign to the device. Valid values are integers from 1-30. If you omit this parameter, the API assigns only one instance.

Response Parameters

Return Value	Description
iccid	The ICCID of the device receiving the event rate plan.
eventName	The name of an event rate plan.
Status	Indicates whether the Control Center successfully activated the partner event plan for the device. Possible values include *Success* and *Failure*.
Details	The activated prepaid event rate plan details.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type:application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{ \"eventName\": \"JasperRP\", \"effectiveDate\": \"2023-
10-10T10:34:06Z\", \"instances\":10}" https://<YOUR-BASE-
URL>/rws/api/v1/partnereventplan/9088217871987000001"
```

Response Example

```
{
       "ICCID":    "1001"
       "EventName": "EV1"
       "Status": "Success"
       "Details": ""
}
```

Errors

Error Code	HTTP Code	Error Message
100100	206	Terminal not found.
1200100	400	Invalid Event Name.
1200101	400	Invalid effectiveDate.
1200102	400	SIM State is not Activated.
1200104	400	This event instance has been created already.
1200400	400	SIM is not part of a Partner Account or Rate Plan.
10000012	400	Invalid date format.
10000024	400	Invalid apiVersion.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Get Partner Event Rate Plans

Description

Used in Multi-Party Billing , this function returns partner event rate plans assigned to a device. The device must have an Activated SIM status.

- For each instance of an event rate plan, the API returns the effective date and time.
- If the event rate plan has multiple instances, the API also returns the total number of instances, the number of pending instances, and the Instance ID for the current instance (if the instance is active).

You must have a ServiceProviderAdmin role to execute this function.

Resource URL

GET BaseURL/v{apiVersion}/partnereventplan/{iccid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device with the event rate plans.
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.
pageNumber	(Optional) Specifies the number of the response page to return. The value defaults to 1. See *Pagination on page 48*.

Response Parameters

Return Value	Description
lastpage	Indicates whether the response is the last page of data. This value will always be *true*.
iccid	The ICCID of the device with the event rate plans.
eventName	The name of an event rate plan assigned to a device.
eventID	A unique identifier for this particular combination of an event rate plan and a device. If you assign the same event rate plan to the same device at a different time, this ID will change.

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Return Value	Description
instanceEventID	For event rate plans with multiple instances, a unique identifier for a particular instance of this association between the event rate plan and the device. All instances have the same event ID. This parameter is not returned if there is only one instance of the event rate plan.
TotalInstances	The total number of event rate plan instances assigned to the device, including past, current, and future instances. For event rate plans with only one instance, this parameter is not returned.
PendingInstances	The number of future event rate plan instances assigned to the device. For event rate plans with only one instance, this parameter is not returned.
effectiveDate	The date when the event rate plan term began. For event rate plans with multiple instances, this date is for the current instance. If the event rate plan is not active, this date indicates when the term will begin. Date format: yyyy-MM-ddTHH:mm:ssZ.
Active	A flag indicating whether the current instance of the event rate plan has an active term. An event rate plan with a future-dated term is considered inactive.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET "Content-Type:application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/partnereventplan/9088217871987000001"
```

Response Example

```
{
  "ICCID": "1001"
  "Events": [
       {
          "EventName": "EV1",
          "Instances": "10",
          "EventID": 2001,
          "InstanceEventID": 3001,
          "TotalInstances": 10,
          "PendingInstances": 8
       },
       {
          "EventName": "EV2",
```

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```
            "EventID": 4001,
        }
    ]
}
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000062	400	Access Denied.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.

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Rate Plans

Control Center supports several APIs for managing rate plans. Using the event rate plan APIs, enterprise accounts can create subscription services with prepaid offers, auto-renewals, and cancellations.

Rate Plan REST API	Description
Assign Event Rate Plan to Device	Assigns an event rate plan to a device. You can specify one or more instances.
Get Event Rate Plans Assigned to Device	Returns information about the event rate plans assigned to a device, including usage in the current term and the number of pending instances.
Edit Event Rate Plan/Device Assignment	Allows you to modify the assignment of an event rate plan to a device. For example, you can cancel future instances of the event rate plan.
Get Rate Plans	Retrieves information about published rate plans. You can filter the rate plans by account or request a list of all shared rate plans.

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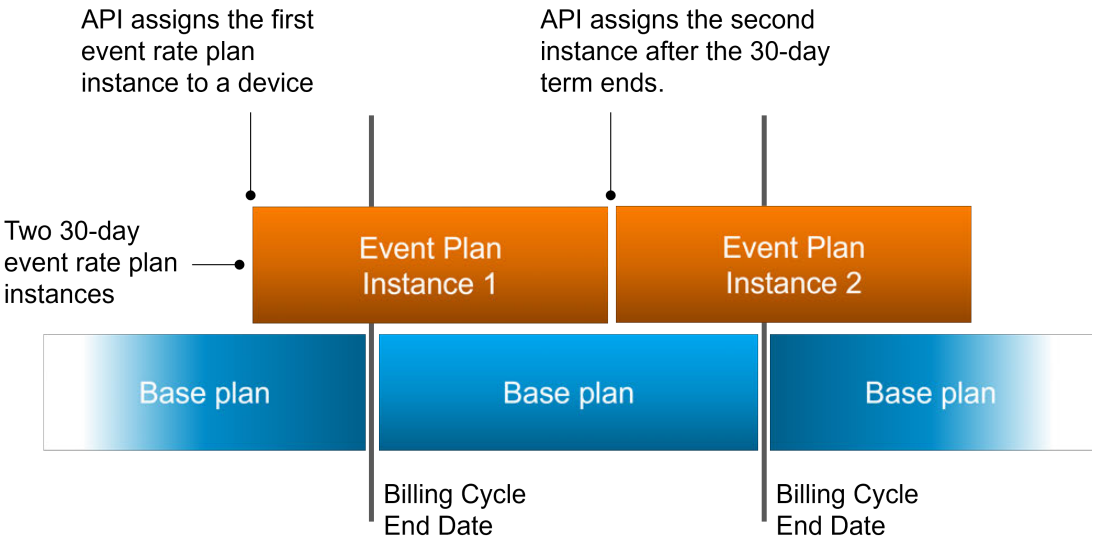
Assign Event Rate Plan to Device

Description

Assigns an event rate plan to a device with an *Activated* SIM status. This API allows users to create subscription services with prepaid offers and automatic renewals. To implement subscription service cancellations, see *Edit Event Rate Plan/Device Assignment on page 289*.

Users can assign up to 36 sequential instances of an event rate plan to a device. Control Center applies each instance to the device at the end of the previous rate plan term as shown in the diagram.



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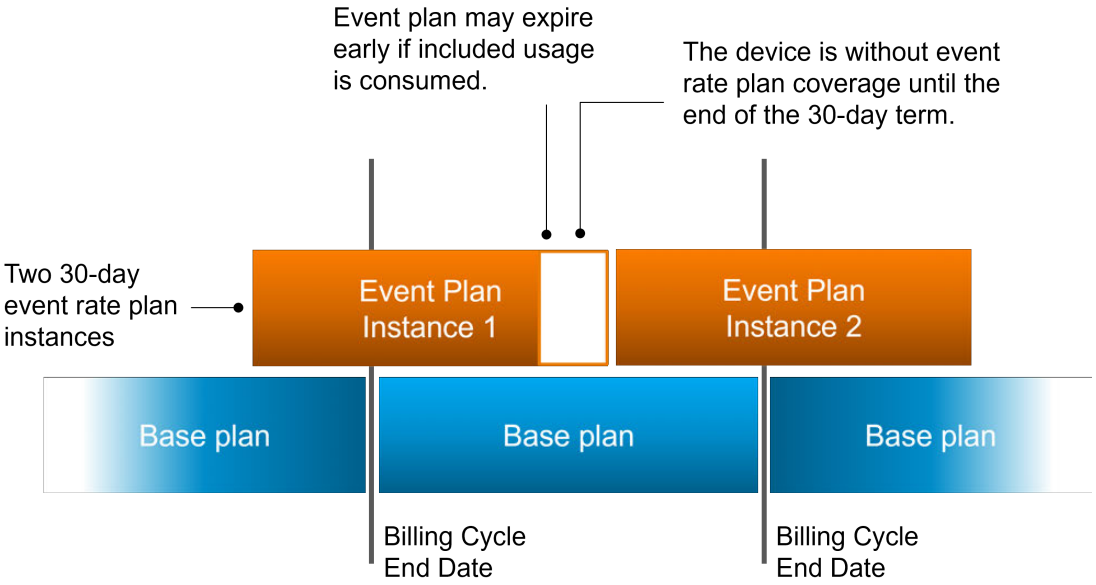
If the event rate plan expires early when the included usage is consumed, Control Center waits to assign the next instance until after the specified term ends. In this case, the device will be without event rate plan coverage until the predefined term ends.



Automated Notifications. To receive automatic notifications whenever an event rate plan term begins or ends, you can define a SIM Event Plan automation rule.

You cannot use this API to assign stacked event rate plans to a device. Only event rate plans work with this API.

This API is available to both service provider and account users. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

PUT BaseURL/v{apiVersion}/eventplan/{iccid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device receiving the event rate plan.
eventName	The name of an event rate plan.

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Parameter	Description
effectiveDate	(Optional) Effective date of event rate plan assignment. If there is no date, the API will assign the rate plan immediately. Date format: yyyy-MM-ddTHH:mm:ssZ.
instances	(Optional) The number of instances you want to assign to the device. Valid values are integers from 1-36. If you omit this parameter, the API assigns only one instance.

Response Parameters

Return Value	Description
iccid	The ICCID of the device receiving the event rate plan.
EventName	The name of an event rate plan.
EventId	A unique identifier for this particular combination of an event rate plan and a device.
InstanceEventId	A unique identifier for the current instance of this event rate plan. The API does not return this parameter if there is only one instance of the event rate plan.
EffectiveDate	The date and time when the event rate plan instance term begins. Date format: yyyy-MM-ddTHH:mm:ssZ.

Request Example

Make sure to use your own URL and user credentials.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"eventName\": \" Prepaid Event Plan 1\",
  \"effectiveDate\": \"2023-02-20T10:34:06Z\"
  \"instances\": \"10\"
}" "https://<YOUR-BASE-URL>/rws/api/v1/eventplan/9088217871987000001"
```

Response Example

```
{
   "iccid": "9088217871987000001",
   "EventName": "EV1",
   "EventId”: 2010,
   "InstanceEventId": 2010,
   "EffectiveDate": "2023-02-20T10:34:06Z"
}
```

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Errors

Error Code	HTTP Code	Error Message
1100704	400	Term not found. Description: The event rate plan does not have an active term.
1200100	400	Invalid Event Name.
1200101	400	Invalid effectiveDate.
1200102	400	SIM State is not Activated.
1200104	400	This event instance has been created already.
2000114	400	Recurring event number of instances exceeds max count.
2000116	400	Invalid Request. Instance count should be greater than 0.
2000117	400	Invalid Event Type for recurring instances. Ensure the rate plan type is Event, not Stacked Event, or Add-on.
10000012	400	Invalid date format.
10000024	400	Invalid apiVersion.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Edit Event Rate Plan/Device Assignment

Description

Edits the association between an event rate plan and a device. Currently, you can use this API to cancel current and future instances of an event rate plan assigned to a device.

Typically, you would run the Get Event Rate Plans Assigned to a Device API to get the event ID for the event rate plan you want to cancel. See *Get Event Rate Plans Assigned to Device on page 292* for details.

To receive an automated notification whenever you cancel an event rate plan with an active term, you can define a SIM Event Plan automation rule.

This API is available to both service provider and account users. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

PUT BaseURL/v{apiVersion}/editeventplan/{iccid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device.
action	Specify the edit action. Currently, there are two supported actions: • *CancelPendingEvents*. For an event rate plan with multiple instances, cancel all pending instances. For an event rate plan with a pending single instance, cancel the pending instance. When using this option, you must supply the ICCID and either an event ID or an event name. • *CancelEvent*. For an event rate plan with multiple instances, cancel the current instance as well as any pending instances. For an event rate plan with a single instance, cancel the instance, regardless of whether it is current or pending. When using this option, you must supply the ICCID, but you do not need to supply an event ID or an event name.

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Parameter	Description
eventID	(Optional) A unique identifier for this particular combination of an event rate plan and a device. This parameter is optional, but you must supply either an event ID or an event name in order to cancel pending instances. To cancel an event, only the ICCID is required.
eventName	(Optional) The name of an event rate plan assigned to the device. This parameter is optional, but you must supply either an event ID or an event name in order to cancel pending instances. To cancel an event, only the ICCID is required.

Response Parameters

Return Value	Description
iccid	The ICCID of the device you want information about.
eventName	The name of an event rate plan assigned to the device.
eventID	A unique identifier for this particular combination of an event rate plan and a device.
TotalInstances	The total number of event rate plan instances originally assigned to the device. For event rate plans with only one instance, this parameter is not returned.
PendingInstances	The number of future event rate plan instances assigned to the device after the cancellation process. For event rate plans with only one instance, this parameter is not returned.

Request Example

Make sure to use your own URL and user credentials.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"eventName\": \" Prepaid Event Plan 1\",
  \"action\": “CancelPendingEvents\”
}" "https://<YOUR-BASE-URL>/rws/api/v1/editeventplan/9088217871987000001"
```

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Response Example

```
{
       "iccid": "9088217871987000001",
       "eventName": "Prepaid Event Plan 1",
       "eventID": "2010",
       "TotalInstances": "3"
       "PendingInstances": "0"
}
```

Errors

Error Code	HTTP Code	Error Message
1100704	400	Term not found. Description: The event rate plan does not have an active term.
1200100	400	Invalid Event Name.
1200200	400	Event Instance not found.
2000115	400	No pending instances found for the event.
10000024	400	Invalid apiVersion.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Get Event Rate Plans Assigned to Device

Description

Gets a list of event rate plans assigned to a device. For each event rate plan, the response provides additional details about each instance:

- For each instance of an event rate plan, the API returns the effective date and time, total usage for the instance (in bytes), and term-to-date usage (in bytes). If the instance is inactive, the term-to-date usage is always zero.
- If the event rate plan has multiple instances, the API also returns the total number of instances, the number of pending instances, and the Instance ID for the current instance (if the instance is active).

This API is available to both service provider and account users. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

GET BaseURL/v{apiVersion}/eventplan/{iccid}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device with the event rate plans.

Response Parameters

Return Value	Description
iccid	The ICCID of the device with the event rate plans.
lastPage	Indicates whether the response is the last page of data. This value will always be *true*.
eventName	The name of an event rate plan assigned to the device.
EventID	A unique identifier for this particular combination of an event rate plan and a device. If you assign the same event rate plan to the same device at a different time, this ID will change.

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Return Value	Description
InstanceEventID	For event rate plans with multiple instances, a unique identifier for a particular instance of this association between the event rate plan and the device. All instances have the same event ID. This parameter is not returned if there is only one instance of the event rate plan.
EffectiveDate	The date when the event rate plan term began. For event rate plans with multiple instances, this date is for the current instance. If the event rate plan is not active, this date indicates when the term will begin. Date format: yyyy-MM-ddTHH:mm:ssZ.
Active	A flag indicating whether the current instance of the event rate plan has an active term. An event rate plan with a future-dated term is considered inactive. Valid values are: *Y* and *N*.
TotalUsage	The total amount of usage (in bytes) defined for the event rate plan instance.
CurrentUsage	The term-to-date usage (in bytes) consumed by the device for the current event rate plan instance.
TotalInstances	The total number of event rate plan instances assigned to the device, including past, current, and future instances. For event rate plans with only one instance, this parameter is not returned.
PendingInstances	The number of future event rate plan instances assigned to the device. For event rate plans with only one instance, this parameter is not returned.
PageNumber	The current page number. This value is always one.

Request Example

Make sure to use your own URL and user credentials.

```
curl -X GET --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>"
"https://<YOUR-BASE-URL>/rws/api/v1/eventplan/9088217871987000001"
```

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Response Example

This response shows two different event rate plans. The first one has an active instance with two pending instances. The second one is inactive.


```
{
   “lastPage”:true,
   "iccid": "9088217871987000001",
   "Events": [
       {
         “EventName":”EV1”,
         “EventID”:2010,
         “InstanceEventID”:2010,
         “TotalInstances”:3,
         “PendingInstances”:2,
         “EffectiveDate”:”2023-01-10T10:34:06Z”,
         “Active”:”Y”,
         “TotalUsage”:2147483648,
         “CurrentUsage”:20971520
       },
       {
         “EventName":”EV2”,
         “EventID”:4001,
         “EffectiveDate”:”2023-02-20T10:34:06Z”,
         “Active”:”N”,
         “TotalUsage”:20971520,
         “CurrentUsage”:0
       }
   ],
   “PageNumber”:1
}
```


Errors

Error Code	HTTP Code	Error Message
1100704	400	Term not found. Description: The event rate plan does not have an active term.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000024	400	Invalid apiVersion.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Get Rate Plans

Description

Returns detailed information about a group of rate plans. The user can specify the rate plans of interest:

- All rate plans (shared and account-specific)
- All shared rate plans
- All rate plans available to a particular account (account-specific rate plans and shared rate plans assigned to the account).

Control Center returns only published rate plans. Retail rate plans are excluded from the list.

You must have the ServiceProviderAdmin or AccountAdmin role to perform this function. Account users can access rate plan information for their own account, but not for any other account.

Resource URL

GET BaseURL/v{apiVersion}/rateplans

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	(Optional) The ID of the account whose rate plans you want to retrieve. The response includes account-specific and shared rate plans associated with the account. If you do not specify an account, the response includes all rate plans for the service provider. If an account user provides this parameter, the account ID must match their own account ID. If the account user omits this parameter, the API automatically supplies their account ID.
shared	(Optional) If set to *true*, the function returns all shared rate plans. Do not use this parameter together with accountId; an error will result. You can use accountId alone to get a list of shared rate plans for that account. Do not set this parameter to *false*; an error will occur. To get a list of account-specific rate plans, you must specify a particular account ID.

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Parameter	Description
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.

Response Parameters

Different rate plan types use different parameters. The function returns the parameters that apply to each rate plan. Be aware that some parameters appear in different arrays, depending on the rate plan type.

For more information about each rate plan type, refer to the Rate Plan Types section in the Knowledge Base or *Control Center User Guide*. The Rate Plan Comparison matrix provides a high-level overview of each rate plan type and the various rate plan components.

Return Value	Description
operatorName	The name of the service provider associated with the logged in user.
operatorId	The identification number of the service provider.
timeStamp	The point in time when the rate plan information was returned. The date format is yyyy-MM-ddTHH:mm:ss.SSSZ. See Date Formats for more details.
pageNumber	An integer specifying the number of the current response page. See *Pagination on page 48*.
Rate Plans	
ratePlans	An array of rate plans.
name	Descriptive rate plan name. Names are unique within a service provider.
id	Identifies a particular rate plan. Because a rate plan can have multiple versions, this value might not be unique. Version ID is the only unique rate plan identifier.
accountName	The name of the account using the rate plan. If the rate plan is available for multiple accounts to use, this value is *Shared*.

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Return Value	Description
versionId	Uniquely identifies a particular version of a rate plan in relation to every other rate plan version in the system, including different versions of the same rate plan and different versions of other rate plans. Because a rate plan may have multiple active versions at any one time, the versionId, not the id, is the only truly unique rate plan identifier.
version	Specifies the version of the current rate plan. Two different versions of the same rate plan may coexist if there are billable devices assigned to each version (for prepaid plans only). All rate plans use the same version number sequence (1, 2, 3, and so on).
status	The function returns only published rate plans with the *Published* status value. Published rate plans are approved and available for use by devices.
type	The type refers to the payment method (monthly vs. prepaid) and included usage type (individual vs. pooled vs. event).
subscriptionCharge	For monthly rate plans, this charge is the monthly subscription fee for each of the subscribers in the tier. For prepaid rate plans, this is a one-time charge per subscriber that covers the entire term of the plan.
numberOfTiers	The number of tiers specifies how many subscription tiers the rate plan has. Each tier can have different subscription fees, account charges (for certain plan types), usage allotments, and overage charges.
tierTreatment	If this value is *Retroactive*, then all subscribers are charged the rate defined in the highest tier used. If the value is *Incremental*, each subscriber is charged the fee for its own tier. Fixed pool rate plans use special tier treatment. See Subscription Tiers and Fixed Pools in the Knowledge Base or *Control Center User Guide*.
tiers	See *Tiers on page 299*.
expireTermBasedOnUsage	For prepaid rate plans, if set to *true* the rate plan is capped. When the device uses up the included usage allotment in the first zone, Control Center blocks network access and ends the subscription term.

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Return Value	Description
lengthOfTerm	For prepaid rate plans, length of term is the term length measured in days or minutes. For monthly plans or add-on plans, which are limited to the current billing cycle only, this parameter has a zero value.
lengthOfTermUnit	The unit of measure for the term length. This parameter appears only if lengthOfTerm is present and has a non-zero value.
frequencyOfPer SubscriberCharge	For prepaid fixed pool rate plans, this value specifies how often the per-subscriber fee is charged: once during the term or at the end of each month. The subscriber must be active at the beginning of the month in order for the per subscriber charge to be charged at month-end. This parameter appears for prepaid fixed pool rate plans only.
subscriptionChargeUnit	The currency associated with the subscription. For more information about valid currency values, see *Currency Values on page 51*.
zones	See *Zones on the next page*.
ratePlanCustomField	See *Get Rate Plans on page 295*.
dataUsage	See *Data Usage on the next page*.
smsUsage	See *SMS Usage on page 302*.
voiceUsage	See *Voice Usage on page 306*.
lastPage	(Optional) A true or false value indicating whether the current response page is the last in the series. See *Pagination on page 48*.

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Tiers

Parameter	Description
tiers	An array of tiers that specify subscription charges.
tierLevel	This number identifies the tier level (1 - 5). If no tiers are used, the value is 0.
subscriberThreshold	This number identifies the first subscriber in the tier.
subscriberCapacity	This number identifies the last subscriber in the tier.
accountCharge	Fixed pool (both monthly and prepaid) and add-on rate plans may have an account charge. This parameter appears only when there is a non-zero value.
perSubscriberCharge	This charge is the monthly subscription fee for each of the subscribers in the tier. For prepaid rate plans, this is a one-time charge that covers the entire term of the plan.

Zones

Parameter	Description
zones	Indicates an array of zones will follow.
reportOverageAsRoaming	An array of zones. This array contains the default zone (called "*default zone*") as well as any others defined for the rate plan.
<zone name>	Each zone has a value of *true* or *false* that indicates whether or not Control Center categorizes the usage in the zone as roaming on the invoice.

Data Usage

Parameter	Description
dataUsage	The rates and settings for data usage.
useDefaultRating	If *true*, Control Center uses the rates and settings specified in the service provider default pricing rate plan. Be aware there are no subscriber tiers or zones in the default pricing rate plan. If *false*, Control Center uses the rates and settings specified in the rate plan. See *Data Default Pricing on page 302*.

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Parameter	Description
usageLimit	This parameter restricts the total amount of data a device can send or receive during a billing cycle. When a device reaches the usage limit, Control Center shuts down data access to the system for the rest of the billing cycle. If this parameter is missing, there is no usage limit.
usageLimitUnit	The data usage limit unit is MB per billing cycle (*MB per cycle*).
Zones	
zones	An array of zones associated with the rate plan.
<zone name>	The name of a particular zone.
includedData	The included data value specifies the amount of data a device receives as part of the subscription fee. For fixed-pool rate plans, the included data appears in the sharedData parameter in each tier. The includedData parameter does not appear at all.
includedDataUnit	The included data unit is *MB* for monthly rate plans and *MB per term* for prepaid rate plans. This parameter is not included for fixed-pool plans.
zoneUsageLimitUnit	The zone usage limit unit is MB per subscriber (*MB per subscriber*).
bulkOverageEnabled	The bulk overage feature is available for monthly plans only. If set to *true*, the rate plan charges for data overage in discrete chunks rather than charging for overage on a per kilobyte basis. For example, the rate plan might charge a set fee for every gigabyte of overage. The account would pay for 1 gigabyte of overage regardless of whether the device on the rate plan had 4MB of overage or 800MB of overage. Similarly, the account would pay for two gigabytes of data if the device used between 1 and 2 gigabytes of overage. The rate plan can define data chunks of virtually any size (KB, MB, GB, and TB) for each zone in the rate plan.
bulkOverage	A number representing the size of the overage chunk. This parameter appears only when bulkOverageEnabled is set to *true*.

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Parameter	Description
bulkOverageUnit	The unit of measure for the overage chunk. This parameter appears only when bulkOverageEnabled is set to *true*.
useTheseDataRounding SettingsForAllZones	If set to *true*, the rounding rules apply to all zones. If set to *false*, each zone has its own rounding rules.
dataRoundingUnit	The data rounding unit determines whether Control Center rounds the amount of data up to the nearest 1KB, 10KB, 50KB or 1 MB before calculating the overage charges in a particular zone. This return value can also be *None*.
dataRoundingFrequency	The data rounding frequency determines how often Control Center rounds the data values — every day, for each data session, or for each call detail record. The data rounding frequency values are *Per CDR*, *Per Session* or *Daily*.
Tiers	
tiers	An array of subscription tiers. Each tier level can have different data overage charges.
tierLevel	This number identifies the tier level (1 - 5). If no tiers are used, the value is 0.
subscribersMoreThan	This number identifies the first subscriber in the tier.
subscribersUpTo	This number identifies the last subscriber in the tier.
sharedData	This parameter appears for fixed pool plans only. The number specifies the size of the pool that the devices on the plan can use.
sharedDataUnit	This parameter appears for fixed pool plans only. The shared data unit is always *MB*.
dataOverage	This number specifies the cost of each kilobyte of data used by the subscriber over the included amount specified in the includedData parameter. The unit of measure for this value is the default account currency. If the plan is pooled and there is surplus usage available, a single subscriber may exceed the individual usage allotment without being charged for overage. These numbers are per KB, not per MB. If useDefaultRating is *true*, Control Center supplies the data overage charge from the default pricing rate plan.
dataOverageUnit	The data overage unit is always KB (*per KB*).

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Data Default Pricing

For rate plans that use default pricing for data service, the function returns fewer parameters.

Parameter	Description
useDefaultRating	In this scenario, the parameter is always *true*.
dataOverage	This number specifies the cost of each kilobyte of data used by the subscriber. The unit of measure for this value is the default account currency.
dataOverageUnit	The data overage unit is a KB (*per KB*).

SMS Usage

Parameter	Description
smsUsage	The rates and settings for SMS usage.
useDefaultRating	If *true*, Control Center uses the SMS rates and settings specified in the default pricing rate plan. Be aware there are no subscriber tiers or zones in the default pricing rate plan. If *false*, Control Center uses the SMS rates and settings specified in the rate plan. See *SMS Default Pricing on page 306*.
type	The type of rating for SMS usage: *Subscription Tiers* or *Destination Based Rating*. Subscription Tiers specifies different overage rates based on the number of subscribers using the rate plan. Destination Based Rating specifies different overage rates and included usage based on the SMS destination (for SMS MO). Each zone may contain one or more regions, each with different included usage and overage charges.
moAndMtRating	Available for rate plans with subscription tier-based SMS rating only. This parameter specifies whether SMS MO and SMS MT messages are rated together (*Rate Both Together*) or separately (*Rate Both Separately*).
poolSMSUsage	Available for rate plans with destination-based SMS rating only. If *true*, the rate plan pools SMS usage for all the subscribers using this rate plan.

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Parameter	Description
poolSMSMTUsage	Available for rate plans with destination-based SMS rating only. If *true*, the rate plan pools SMS MT usage for all the subscribers using this rate plan.
poolSMSMOUsage	Available for rate plans with destination-based SMS rating only. If *true*, the rate plan pools SMS MO usage for all the subscribers using this rate plan.
Zones	
zones	An array of zones associated with the rate plan.
<zone name>	The name of a particular zone.
includedSMS	The included SMS value specifies the number of text messages a device can send or receive as part of the subscription fee. If mobile-terminated (SMS MT) and mobile-originated (SMS MO) messages are rated separately, this parameter is replaced by separate parameters for includedSMSMO and includedSMSMT. For fixed-pool rate plans, the included SMS appears in the sharedSMS parameter in each tier. The includedSMS parameter does not appear at all. If the rate plan type is *Destination Based Rating*, the included usage is for mobile-terminated messages only (includedSMSMT). The included usage for mobile-originated messages appears in a group array within each zone.
includedSMSUnit	The included SMS unit (SMS MT or SMS MO) is a message (*msgs*). If mobile-terminated (SMS MT) and mobile-originated (SMS MO) messages are rated separately, this parameter is replaced by separate parameters for includedSMSMOUnit and includedSMSMTUnit. This parameter is not included for fixed-pool plans. If the rate plan type is *Destination Based Rating*, the included SMS unit is for mobile-terminated messages only (includedSMSMTUnit). The included SMS unit for mobile-originated messages appears in a group array within each zone.

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Parameter	Description
smsOverage	This number specifies the cost of each mobile-terminated message used by the subscriber in excess of the number specified in the includedSMSMT field. The unit of measure for this value is the default account currency. The parameter appears in this location (the zones array) only when the rate plan type is *Destination Based Rating*. In this scenario, the overage cost for mobile-originate messages appears in the groups array.
smsOverageUnit	The SMS MT overage unit is a message (*per MT msg*). The parameter appears in this location (the zones array) only when the rate plan type is *Destination Based Rating*. In this scenario, the overage unit for mobile-originated messages appears in the groups array.
Tiers	
tiers	An array of subscription tiers. Each tier level can have different SMS overage charges. Tiers are used only when the rate plan type is *Subscription Tiers*.
tierLevel	This number identifies the tier level (1 - 5). If no tiers are used, the value is 0.
subscribersMoreThan	This number identifies the first subscriber in the tier.
subscribersUpTo	This number identifies the last subscriber in the tier.
sharedSMS	For fixed pool rate plans only. This number specifies the size of the pool that the devices on the plan can use. If mobile-terminated (SMS MT) and mobile-originated (SMS MO) messages are rated separately, this parameter is replaced by separate parameters for sharedSMSMO and sharedSMSMT.
sharedSMSUnit	For fixed pool rate plans only. The shared SMS unit is a message (*msgs*). If mobile-terminated (SMS MT) and mobile-originated (SMS MO) messages are rated separately, this parameter is replaced by separate parameters for sharedSMSMOUnit and sharedSMSMTUnit.

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Parameter	Description
smsOverage	This number specifies the cost of each message used by the subscriber in excess of the number specified in the includedSMS field. The unit of measure for this value is the default account currency. If mobile-terminated (SMS MT) and mobile-originated (SMS MO) messages are rated separately, this parameter is replaced by separate parameters for smsOverageMO and smsOverageMT.
smsOverageUnit	The SMS overage unit is a message (*per msg*). If mobile-terminated (SMS MT) and mobile-originated (SMS MO) messages are rated separately, this parameter is replaced by separate parameters for smsOverageMOUnit and smsOverageMTUnit with the values *per MO msg* and *per MT msg*.
Groups	
groups	An array of billing groups. This array appears only when the rate plan type is *Destination Based Rating*.
name	The name of a billing group.
includedSMSMO	The included SMS value specifies the number of text messages a device can send as part of the subscription fee. For fixed-pool rate plans, the included SMS MO appears in the sharedSMSMO parameter. The includedSMSMO parameter does not appear at all.
includedSMSMOUnit	The included SMS MO unit is a message (*msgs*). This parameter is not included for fixed-pool plans.
sharedSMSMO	For fixed pool rate plans only. This number specifies the size of the SMS MO pool that the devices on the plan can use.
sharedSMSMOUnit	For fixed pool rate plans only. The shared SMS unit is a message (*msgs*).
smsMOOverage	This number specifies the cost of each message used by the subscriber in excess of the number specified in the includedSMSMO field. The unit of measure for this value is the default account currency.
smsMOOverageUnit	The SMS MO overage unit is a message (*msgs*).

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SMS Default Pricing

For rate plans that use default pricing for SMS service, the function returns fewer parameters.

Parameter	Description
useDefaultRating	In this scenario, the parameter is always *true*.
poolSMSUsage	These three pooled usage parameters are always included and always set to *false*.
poolSMSMOUsage	
poolSMSMTUsage	
smsMOOverage	This number specifies the cost of each message sent by the subscriber. The unit of measure for this value is the default account currency.
smsMOOverageUnit	The SMS MO overage unit is a message (*per MO msg*).
smsMTOverage	This number specifies the cost of each message received by the subscriber. The unit of measure for this value is the default account currency.
smsMTOverageUnit	The SMS MT overage unit is a message (*per MT msg*).

Voice Usage

Parameter	Description
voiceUsage	The rates and settings for voice usage.
useDefaultRating	If *true*, Control Center uses the voice rates and settings specified in the default pricing rate plan. Be aware there are no subscriber tiers or zones in the default pricing rate plan. If *false*, Control Center uses the voice rates and settings specified in the rate plan. See *Voice Usage above*.
type	The type of rating for voice usage: *Subscription Tiers* or *Destination Based Rating*. Subscription Tiers specifies different overage rates based on the number of subscribers using the rate plan. Destination Based Rating specifies different overage rates and included usage based on the voice destination (for voice MO). Each zone may contain one or more regions, each with different included usage and overage charges.

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Parameter	Description
moAndMtRating	Available for rate plans with tier-based voice rating only. This parameter specifies whether voice MO and voice MT minutes are rated together (*Rate Both Together*) or separately (*Rate Both Separately*).
poolVoiceUsage	Available for rate plans with destination-based voice rating only. If *true*, the rate plan pools voice usage for all the subscribers using this rate plan.
poolVoiceMTUsage	Available for rate plans with destination-based voice rating only. If *true*, the rate plan pools voice MT usage for all the subscribers using this rate plan.
poolVoiceMOUsage	Available for rate plans with destination-based voice rating only. If *true*, the rate plan pools voice MO usage for all the subscribers using this rate plan.
Zones	
zones	An array of zones associated with the rate plan.
<zone name>	The name of a particular zone.
includedVoice	The included voice value specifies the number of voice minutes a device receives as part of the subscription fee. If mobile-terminated (voice MT) and mobile-originated (voice MO) calls are rated separately, this parameter is replaced by separate parameters for includedVoiceMO and includedVoiceMT. For fixed-pool rate plans, the included voice appears in the sharedVoice parameter in each tier. The includedVoice parameter does not appear at all. If the rate plan type is Destination Based Rating, the included usage is for mobile-terminated calls only (includedVoiceMT). The included usage for mobile-originated calls appears in a group array within each zone.

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Parameter	Description
includedVoiceUnit	The included voice unit (voice MT or voice MO) is minutes:seconds (*m:ss)*. If mobile-terminated (voice MT) and mobile-originated (voice MO) messages are rated separately, this parameter is replaced by separate parameters for includedVoiceMOUnit and includedVoiceMTUnit. This parameter is not included for fixed-pool plans. If the rate plan type is Destination Based Rating, the included voice unit is for mobile-terminated calls only (includedVoiceMTUnit). The included voice unit for mobile-originated calls appears in a group array within each zone.
voiceOverage	This number specifies the cost of each mobile-terminated minute used by the subscriber in excess of the number specified in the includedVoiceMT field. The unit of measure for this value is the default account currency. The parameter appears in this location (the zones array) only when the rate plan type is Destination Based Rating. In this scenario, the overage cost for mobile-originated minutes appears in the groups array.
voiceOverageUnit	The voice MT overage unit is a minute (*per MT min*). The parameter appears in this location (the zones array) only when the rate plan type is Destination Based Rating. In this scenario, the overage unit for mobile-originated messages appears in the groups array.
useTheseDataRoundingSettingsForAllZones	If set to *true*, the voice rounding rules apply to all zones. If set to *false*, each zone has its own rounding rules.
voiceRoundingUnit	Control Center rounds the voice usage up to the nearest minute or second before calculating the overage charges in a particular zone. Valid values are minutes (*minutes*) and seconds (*seconds*).
Tiers	
tiers	An array of subscription tiers. Each tier level can have different voice overage charges. This array appears only when the rate plan type is Subscription Tiers.
tierLevel	This number identifies the tier level (1 - 5). If no tiers are used, the value is 0.

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Parameter	Description
subscribersMoreThan	This number identifies the first subscriber in the tier.
subscribersUpTo	This number identifies the last subscriber in the tier.
sharedVoice	This parameter appears for fixed pool rate plans only. The number specifies the size of the pool that the devices on the plan can use. If mobile-terminated (voice MT) and mobile-originated (voice MO) calls are rated separately, this parameter is replaced by separate parameters for sharedVoiceMO and sharedVoiceMT.
sharedVoiceUnit	This parameter appears for fixed pool rate plans only. The shared voice unit is minutes:seconds (*m:ss*). If mobile-terminated (voice MT) and mobile-originated (voice MO) calls are rated separately, this parameter is replaced by separate parameters for sharedVoiceMOUnit and sharedVoiceMTUnit.
voiceOverage	This parameter specifies the cost of each minute used by the subscriber in excess of the number specified in the includedVoice parameter. The unit of measure for this value is the default account currency. If mobile-terminated (voice MT) and mobile-originated (voice MO) calls are rated separately, this parameter is replaced by separate parameters for voiceOverageMO and MTvoiceOverage. For rate plans with destination-based rating, voiceOverage applies to voice MT overage only.
voiceOverageUnit	The data overage unit is a minute (*per min*). If mobile-terminated (voice MT) and mobile-originated (voice MO) calls are rated separately, this parameter is replaced by separate parameters for voiceOverageMOUnit and voiceOverageMTUnit with the values *per MO min* and *per MT min*. For rate plans with destination-based rating, voiceOverageUnit applies to voice MT overage only with a value of *per MT min*.
Groups	
groups	An array of billing groups. This array appears only when the rate plan type is *Destination Based Rating*.
name	The name of a billing group.

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Parameter	Description
includedVoiceMO	The included voice value specifies the number of call minutes a device can use as part of the subscription fee. For fixed-pool rate plans, the included voice MO appears in the sharedVoiceMO parameter. The includedVoiceMO parameter does not appear at all.
includedVoiceMOUnit	The included voice MO unit is minutes:seconds (*m:ss*). This parameter is not included for fixed-pool plans.
sharedVoiceMO	For fixed pool rate plans only. This number specifies the size of the voice MO pool that the devices on the plan can use.
sharedVoiceMOUnit	For fixed pool rate plans only. The shared voice unit is minutes:seconds (*m:ss*).
voiceMOOverage	This number specifies the cost of each minute used by the subscriber in excess of the number specified in the includedVoiceMO field. The unit of measure for this value is the default account currency.
voiceMOOverageUnit	The voice MO overage unit is a minute (*Per Min*).

Voice Default Pricing

For rate plans that use default pricing for voice service, the function returns fewer parameters.

Parameter	Description
useDefaultRating	In this scenario, the parameter is always *true*.
poolVoiceUsage	These three pooled usage parameters are always included and always set to *false*.
poolVoiceMOUsage	
poolVoiceMTUsage	
voiceMOOverage	This number specifies the cost of each minute used by the subscriber during an outgoing call. The unit of measure for this value is the default account currency.
voiceMOOverageUnit	The voice MO overage unit is a minute (*per MO min*).
voiceMTOverage	This number specifies the cost of each minute used by the subscriber during an incoming call. The unit of measure for this value is the default account currency.
voiceMTOverageUnit	The voice MT overage unit is a minute (*per MT min*).

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Request Example

Make sure to use your own URL and user credentials.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/rateplans?pageSize=50&pageNumber=1"
```

Response Example


```
{
   "operatorName": "Jasper Systems",
   "operatorId": 8410,
   "timeStamp": "2019-03-15T18:02:25.013Z",
   "pageNumber": 1,
   "ratePlans": [
   {
      "name": "QaAwtRP-180421030819",
      "id": 30884203,
      "accountName": "Shared",
      "versionId": 30885703,
      "version": 1,
      "status": "Published",
      "type": "Monthly - Individual Subscriber",
      "subscriptionCharge": 0,
      "numberOfTiers": 1,
      "tierTreatment": "Incremental",
      "tiers": [
      {
         "tierLevel": 0,
         "subscriberThreshold": 0,
         "subscriberCapacity": "And Up",
         "perSubscriberCharge": 0
      }
      ],
      "expireTermBasedOnUsage": false,
      "lengthOfTerm": 0,
      "subscriptionChargeUnit": "USD",
      "zones":
      {
         "reportOverageAsRoaming":
         {
            "default zone": "false"
         }
      },
      "dataUsage":
      {
         "useDefaultRating": false,
         "usageLimitUnit": "MB per cycle",
         "zones":
         {
            "default zone":
               {
                  "includedData": 0,
                  "includedDataUnit": "MB",
                  "zoneUsageLimitUnit": "MB per subscriber",
                  "bulkOverageEnabled": false,
                  "useTheseDataRoundingSettingsForAllZones": true,
                  "dataRoundingUnit": "None",
```


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```
                "dataRoundingFrequency": "Per CDR",
                "tiers": [
                {
                   "tierLevel": 0,
                   "subscribersMoreThan": 0,
                   "subscribersUpTo": "And Up",
                   "sharedData":0,
                   "sharedDataUnit":"MB",
                   "dataOverage": 0,
                   "dataOverageUnit": "per KB"
                }
                ]
             }
          }
       },
       "smsUsage":
       {
          "useDefaultRating": false,
          "type": "Subscription Tiers",
          "moAndMtRating": "Rate Both Together",
          "poolSMSUsage": false,
          "poolSMSMOUsage": false,
          "poolSMSMTUsage": false,
          "zones":
          {
             "default zone":
             {
                "includedSMS": 0,
                "includedSMSUnit": "msgs",
                "tiers": [
                   {
                      "tierLevel": 0,
                      "subscribersMoreThan": 0,
                      "subscribersUpTo": "And Up",
                      "sharedSMS":0
                      "sharedSMSUnit":"msgs",
                      "smsOverage": 0,
                      "smsOverageUnit": "per msg"
                   }
                   ]
                }
             }
          },
          "voiceUsage": {
             "useDefaultRating": false,
             "type": "Subscription Tiers",
             "moAndMtRating": "Rate Both Together",
             "poolVoiceUsage": false,
             "poolVoiceMTUsage": false,
             "poolVoiceMOUsage": false,
             "zones": {
                "default zone": {
                   "includedVoice": "0:00",
                   "includedVoiceUnit": "m:ss",
                   "useTheseDataRoundingSettingsForAllZones": true,
                   "voiceRoundingUnit": "minutes",
                   "tiers": [
                      {
                         "tierLevel": 0,
```

```
                            "subscribersMoreThan": 0,
                            "subscribersUpTo": "And Up",
                            "voiceOverage": 0.03,
                            "voiceOverageUnit": "per min"
                        }
                    ]
                }
            }
        },
        "ratePlanCustomField":{
                                "custom1": "TestCustome000001",
                                "custom2": "TestCustom99999...99",
                                "custom3": "TestCustome000002.9999999",
                                "custom4": "Test Custom 1",
                                "custom5": "TestCustome000001$$%$%$%"
                        }
        }
    ],
    "lastPage": true
}
```


Code Samples

Make sure to use your own URL and user credentials.

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-URL>/rws/api/v1/rateplans?accountId=acct_
id&pageSize=page_size&pageNumber=page_number').auth('user_name', 'password',
false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Python

```
import requests
import json
import base64
import pprint

apiUrl= 'https://<your-base-URL>/rws/api/v1/rateplans'

parameters= {'accountId':"acct_id", 'pageSize': "page_size", 'pageNumber':
"page_number"}

myResponse = requests.get(url=apiUrl, auth=(raw_input("username: "),raw_input
("api_key: ")), params=parameters)
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
       # Loading the response data into a dict variable
       # json.loads takes in only binary or string variables so using content to
fetch binary content
       # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
       pp = pprint.PrettyPrinter(indent=4)
       jData = json.loads(myResponse.content)
       pp.pprint(jData)
else:
       # If response code is not ok (200), print the resulting http error code with
description
       myResponse.raise_for_status()
```

Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/rateplans?accountId=acct_
id&pageSize=page_size&pageNumber=page_number'
response = RestClient::Request.execute(
       method: :get,
       url: url,
       user: 'user_name',
       password: 'password',
       :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000004	400	Invalid accountId.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000024	400	Invalid apiVersion.

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Error Code	HTTP Code	Error Message
10000436	400	Shared can only be true.
10000437	400	Shared can not be used along with accountId.
20000000	404	Requested resource not found.
30000001	500	Unknown server error.

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SMS Messages

Search SMS

Description

Returns a list of SMS message IDs associated with a particular account during a specified time period. Optionally, you can restrict the search to a specific device.

The function returns messages between Control Center and a device. You cannot get information about messages exchanged between a device and a non-Control Center SMS client.

Resource URL

GET BaseURL/v{apiVersion}/smsMessages

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
accountId	(Optional) Unique identifier for the account associated with the messages you want to retrieve.
iccid	(Optional) Restricts the list of returned messages to just those messages sent or received by a particular device. This device must be owned by the specified account.
fromDate	A date and time using the ISO 8601 format (see Date Formats). The function returns any messages that were sent or received on or after this date.
toDate	(Optional) A date and time using the ISO 8601 format (see Date Formats). Use this parameter to specify a particular period of time for the messages. If you do not specify a toDate, Control Center returns all messages up to the current date and time.
msgType	(Optional) Indicates whether the message was sent by the device (*MO*, mobile-originated) or received by the device (*MT*, mobile-terminated).
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.

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Response Parameters

The function returns an array of SMS message IDs. Records are sorted by modification date in ascending order with the oldest message listed first.

Return Value	Description
smsMsgIds	An array of message identifiers. You must use the Get SMS Details API to retrieve the content of each message.
pageNumber	An integer specifying the number of the current page.
lastPage	A true or false value indicating whether the current page is the last in the series.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-URL>/rws/api/v1/smsMessages"
```

Response Example

```
{
       "smsMsgIds": [
               106184,
               105025
       ],
       "pageNumber": 1,
       "lastPage": true
}
```

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Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+input("Cobrand URL: ")+'/rws/api/v1/smsMessages?accountId='
+input("Account ID: ")+'&fromDate=2016-04-
18T17%3A31%3A34%2B00%3A00&pageSize=50&pageNumber=1'
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-
URL>/rws/api/v1/devices/smsMessages?accountId=147060908&fromDate=2016-04-
18T17%3A31%3A34%2B00%3A00&pageSize=50&pageNumber=1').auth('username',
'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-
URL>/rws/api/v1/smsMessages?accountId=147060908&fromDate=2016-04-
18T17%3A31%3A34%2B00%3A00&pageSize=50&pageNumber=1'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000002	400	AccountId is required.
10000004	400	Invalid accountId.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000012	400	Invalid date format.
10000021	400	Invalid iccid.
10000022	400	FromDate is required.
10000024	400	Invalid apiVersion.
10000027	400	ToDate must be after fromDate.
30000001	500	Unknown server error.
101000022	400	Invalid message type.

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Get SMS Details

Description

Returns detailed information about a particular SMS message sent by Control Center to a device or sent by a device to Control Center. You cannot get information about messages exchanged between a device and a non-Control Center SMS client.

Resource URL

GET BaseURL/v{apiVersion}/smsMessages/{smsMsgId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
smsMsgId	A unique identifier for the SMS message you want to retrieve.
messageEncoding	(Optional) The type of message encoding used. Valid values are: LITERAL (default) or BASE64.

Response Parameters

Return Value	Description
smsMsgId	A unique identifier for the SMS message.
status	The message delivery status. Valid values depend on the message type. • Mobile-Originated (MO): *Received* • Mobile-Terminated (MT): *Cancelled*, *CancelFailed*, *CancelPending*, *Delivered*, *Pending*, *Failed*, *Unknown*.
messageText	The content of the SMS message.
senderLogin	Identifies the message sender. For mobile-originated messages, this value is *Mobile Device*. For mobile-terminated messages, the value is the Control Center user name.
sentTo	Identifies the recipient of the message. If the recipient is Control Center, the value is *Server*. Otherwise, the recipient device's MSISDN, the equivalent of a phone number, appears.

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Return Value	Description
sentFrom	Identifies the device or computer that sent the message. If the sender is Control Center, the value is *Server*. Otherwise, the sending device's MSISDN, the equivalent of a phone number, appears.
msgType	Message Type indicates whether the message was sent by the device (*MO*, mobile-originated) or received by the device (*MT*, mobile-terminated).
dateSent	The date and time (including the time zone) when the message was sent. See Date Formats.
dateReceived	The date and time (including the time zone) when Control Center received the message. See Date Formats.
dateModified	The date and time (including the time zone) when the delivery status changed. See Date Formats.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/smsMessages/106184"
```

Response Example

```
{
       "smsMsgId": 106184,
       "status": "Pending",
       "messageText": "Hello world",
       "senderLogin": "dpTrialUser2",
       "iccid": "8988216716970004975",
       "sentTo": "882351697004975",
       "sentFrom": "Server",
       "msgType": "MT",
       "dateSent": "2016-07-06 16:05:16.280-0700",
       "dateModified": "2016-07-06 16:05:16.522-0700"
}
```

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Code Samples

Make sure to use your own URL and user credentials.

Python

```
import requests
import json
import base64
import pprint

cobrandURL=input("Cobrand URL: ")
url = 'https://'+input("Cobrand URL: ")+'/rws/api/v1/smsMessages/'+input("SMS
Message ID: ")
myResponse = requests.get(url,auth=(input("username: "),input("api_key: ")))
# For successful API call, response code will be 200 (OK)
if(myResponse.ok):
    # Loading the response data into a dict variable
    # json.loads takes in only binary or string variables so using content
to fetch binary content
    # Loads (Load String) takes a Json file and converts into python data
structure (dict or list, depending on JSON)
    jData = json.loads(myResponse.content)
    pp=pprint.PrettyPrinter(indent=4)
    pp.pprint(jData)
else:
    # If response code is not ok (200), print the resulting http error
code with description
    print("Failure")
    myResponse.raise_for_status()
```

Node.js

```
var request = require('request');
var body = [];
request.get('https://<your-base-URL>/rws/api/v1/smsMessages/23178931802').auth
('username', 'password', false)
    .on('error', function(error){
        console.log('Error:', error);
    })
    .on('response', function(response) {
        console.log(response.statusCode); // return statusCode
        console.log(response.headers['content-type']); // return contentType
    })
    .on('data',function(chunk){
        body.push(chunk);
    })
    .on('end',function(){
        body = Buffer.concat(body).toString();
        console.log(body);
    });
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-URL>/rws/api/v1/smsMessages/25830319302'
response = RestClient::Request.execute(
 method: :get,
 url: url,
 user: 'username',
 password: 'password',
 :headers => {:accept => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000017	400	Invalid messageEncoding.
10000024	400	Invalid apiVersion.
20000002	404	Resource not found - Invalid smsMsgId.
30000001	500	Unknown server error.

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Send SMS

Description

Sends an SMS message to a given device or short code.

This function is available to users with the following roles: ServiceProviderAdmin, ServiceProviderDemo, AccountAdmin, AccountAPIOnly, TrialUser, AccountDemo, and CustomerAdmin.

Resource URL

POST BaseURL/v{apiVersion}/devices/{iccid}/smsMessages

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device you want information about.
messageText	The SMS message text. The maximum length of the message depends on the data coding type (see dataCoding). • 0 - 160 characters • 1 - 160 characters • 3 - 160 characters • 4 - 140 characters • 8 - 70 characters If the API does not specify the data coding type, the default maximum message length is 320 characters. Messages that exceed 160 characters appear on two lines.
messageEncoding	(Optional) The type of message encoding used. Valid values are: LITERAL (default) or BASE64.
dataCoding	(Optional) The type of data encoding used. • 0 - SMSC default alphabet; often GSM encoding • 1 - IA5/ASCII, but sometimes GSM encoding, depending on the SMSC implementation • 3 - Latin 1 (ISO-8859-1) • 4 - Binary SMS • 8 - Unicode UCS2
tpvp	(Optional) The length of time the message is available before expiring. For more information about setting this value, see SMS Validity Period.

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Response Parameters

Return Value	Description
smsMessageId	A unique identifier for the SMS message. You can use this ID to get details about a particular message with the Get SMS Details API.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X POST --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-CREDENTIALS>"
-d "{
   \"messageText\": \"Hello world\"
}" "https://<YOUR-BASE-
URL>/rws/api/v1/devices/89302720396916018117/smsMessages"
```

Response Example

```
{
       "smsMessageId": 106184
}
```

Code Samples

Make sure to use your own URL and user credentials.

Node.js

```
var request = require('request');
var body = [];
var auth = "Basic " + new Buffer("username:password").toString("base64");
request.post({
 url: 'https://<your-base-
URL>/rws/api/v1/devices/8988216716970004975/smsMessages',
 headers : {"Authorization" : auth},
 body: {"messageText": "Test"},
 json: true
},
 function(error, response, body) {
  if(error) {
   console.log('Error:', error);
   return;
  } else {
   // return statusCode
   console.log(response.statusCode);
   // return contentType
   console.log(response.headers['content-type']);
   console.log(body);
  }
 }
)
```

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Ruby

```
#!/usr/bin/ruby -w
require 'rest-client'
require 'json'

url = 'https://<your-base-
URL>/rws/api/v1/devices/89011704252318147060/smsMessages'
response = RestClient::Request.execute(
 method: :post,
 url: url,
 user: 'dpSKit20',
 password: 'b09c4266-83c6-411a-a475-ca4925b3bb4a',
 :payload => '{"messageText":"testing from minde: sent by ruby"}',
 :headers => {:accept => :json,
  :content_type => :json}
)
puts JSON.pretty_generate(JSON.parse(response))
```

Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000017	400	Invalid messageEncoding.
10000018	400	Invalid dataCoding.
10000019	400	Invalid tpvp. Description: The validityPeriod is invalid.
10000020	400	Message length exceeds the maximum permissible length.
10000023	400	The JSON in the request is not well formed. Please ensure that commas, colons, braces etc. are formatted properly.
10000024	400	Invalid apiVersion.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.

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Error Code	HTTP Code	Error Message
10000030	400	Your role does not have access to this API function.
10000031	400	Invalid Zone.
20000001	404	Resource not found - Invalid ICCID.
30000001	500	Unknown server error.
30000002	500	Control Center failed to submit the message to the SMSC.

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Usages

Get Usage by Rate Plan

Description

Returns usage information associated with a single rate plan during a specified billing cycle. By default, the function returns cycle-to-date data, SMS, and voice usage per zone for each device using the current version of the rate plan. However, you can request a particular usage type, a past billing cycle, or a different rate plan version.

The response includes only devices with usage during the specified period. You cannot use this API to get a list of all devices associated with a particular rate plan.

No Support for Shared or Unpublished Rate Plans. Be aware that you cannot request usage information for a shared rate plan. In addition, rate plans must be published.

This function is available for service providers, accounts, and customers. A user can request usage information about rate plans associated with accounts they have access to. Customer users can view usage for their own devices, but not for devices in the same account, using the same rate plan, that are owned by a different customer.

See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

GET BaseURL/v{apiVersion}/usages

Request Parameters

The rate plan name and API version number are required. All other parameters are optional.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
ratePlanName	The name of the desired rate plan.

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Parameter	Description
cycleStartDate	(Optional) The start date of a desired billing cycle. The format is yyyy-MM-dd. The function returns cycle-to-date usage information if this parameter is not used. You can get the cycle start day for an account using the Get Account Billing Settings API (currentBillingCycleStart response parameter). Or, in the web interface, see the Billing Settings (Admin > Accounts > Account ID link > Billing Settings subtab > Current Billing Cycle Start).
usageType	(Optional) The type of usage information the function should return. Options are: DATA, SMS, and VOICE. You can specify only one type. However, the function returns all types of usage information if this parameter is not present.
ratePlanVersion	(Optional) The desired rate plan version. It's possible for two versions of the same rate plan to be active during the same billing cycle. This parameter allows you to specify which version you want to use. The function returns information for the most recent version of the rate plan if this parameter is not present.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.

Response Parameters

The function returns an array of zones, where each zone contains an array of devices. For each device, there are total usage numbers for the requested usage types. A device may have usage in more than one zone and, as a result, may appear multiple times in the response. Be aware that devices with no usage in a zone do not appear in the response for that zone.

Return Value	Description
accountId	The ID of the account using the rate plan.
accountName	The name of the account using the rate plan.
ratePlanName	The name of the rate plan.
ratePlanVersion	The version of the rate plan.

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Return Value	Description
pageNumber	An integer specifying the number of the current response page.
lastPage	A true or false value indicating whether the current response page is the last in the series.
Zones	
zones	An array of zones used in the rate plan.
zone	The name of a zone.
Devices	
devices	An array of devices that used network services in the zone. The following information is returned for each device.
deviceId	The ICCID for the device. This field does not contain the Device ID found on the Device Details page. The ICCID is a unique identifier for a device in Control Center. Device ID is an optional identifier that the account or customer can give to each device.
usage	A label that indicates the usage fields for the current device. The function returns all usage information unless you specify a particular usage type. If you request one type of usage, the function returns all usage fields, but inserts zero values in the non-requested usage fields, regardless of the actual values.
dataUsage	The volume of data (in bytes) used by the device during the time period (cycle to date or a particular billing cycle). If you request a different usage type, the value in this field will be zero, regardless of the actual value.
smsMOUsage	The number of messages sent by the device during the time period (cycle to date or a particular billing cycle). If you request a different usage type, the value in this field will be zero, regardless of the actual value.
smsMTUsage	The number of messages received by the device during the time period (cycle to date or a particular billing cycle). If you request a different usage type, the value in this field will be zero, regardless of the actual value.

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Return Value	Description
voiceMOUsage	The number of seconds a device used for outbound calls during the time period (cycle to date or a particular billing cycle). If you request a different usage type, the value in this field will be zero, regardless of the actual value.
voiceMTUsage	The number of seconds a device used for inbound calls during the time period (cycle to date or a particular billing cycle). If you request a different usage type, the value in this field will be zero, regardless of the actual value.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization: Basic
<YOUR-ENCRYPTED-CREDENTIALS>"https://<YOUR-BASE-
URL>/rws/api/v1/usages?ratePlanName=Monthly-Flexible-
Pool&pageNumber=1&pageSize=50"
```

Response Example


```
{
   "accountId": 100477039,
   "accountName": "Acme Corp",
   "ratePlanName": "Monthly-Flexible-Pool",
   "ratePlanVersion": 1,
   "zones": [
   {
      "zone": "Zone A",
         "devices": [
         {
            "deviceId": "8988209000000000723",
            "usage": {
               "dataUsage": 54765,
               "smsMOUsage": 37,
               "smsMTUsage": 42,
               "voiceMOUsage": 0,
               "voiceMTUsage": 0
               }
             },
         {
            "deviceId": "8988209000000000747",
            "usage": {
               "dataUsage": 23456,
               "smsMOUsage": 15,
               "smsMTUsage": 11,
               "voiceMOUsage": 0,
               "voiceMTUsage": 0
               }
            }
```


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```
        ]
      "zone": "Zone B",
         "devices": [
         {
            "deviceId": "8988209000000000859",
            "usage": {
               "dataUsage": 94480,
               "smsMOUsage": 0,
               "smsMTUsage": 0,
               "voiceMOUsage": 900,
               "voiceMTUsage": 0
               }
            }
         ]
      }
   ]
   "pageNumber": 2,
   "lastPage": true
}
```


Errors

Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000012	400	Invalid date format.
10000013	404	Invalid ratePlan.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000062	400	Access Denied.
10000441	400	Invalid RatePlan Version.
10000442	400	More than one RatePlan having the same name. Description: The rate plan name is not unique. Try using the rate plan version number along with the name to indicate a unique rate plan.

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Error Code	HTTP Code	Error Message
10000443	400	Invalid BillingCycleStart. Description: The billing cycle start date doesn't correspond to a valid billing cycle.
20000058	400	Requested resource not available for Shared Rate plan. Description: You cannot use shared rate plans in this function.
30000001	500	Unknown server error.
30000003	400	Invalid Usage Type.

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Users

Get User Details

Description

Returns the user details for a given user ID. All users can request their own user details. Account administrators can view details for all account and customer users in their account. Service provider administrators can view details for all service provider, account, and customer users. Other roles may have access to user information as well. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

If you have a user name, but not a user ID, you can use the Get All User Details API to return information for a single user. See *Get All User Details on page 338*.

Resource URL

GET BaseURL/v{apiVersion}/users/{userId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
userId	The ID of the user for which you want information. The user ID is not available in the Control Center web interface, but you can use the Get All User Details API to find a user ID. Use a partial response to return the username and userId only. For more information, see Partial Response Guidelines.

Response Parameters

Return Value	Description
username	A unique identifier for the person within Control Center. A user must supply this user name and a password to log into Control Center. The user name is unique across all accounts in Control Center.
userId	A unique identifier for the user.
accountName	The name of the account to which the user is assigned.
accountId	A Control Center-generated identifier for the account.
operatorName	The name of the service provider.

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Return Value	Description
operatorId	A unique identifier for the service provider.
roleName	The role name defines the application privileges for the user. See *User Role Values on page 52* for a list of possible values.
status	The user status valid values include: • *Active*. The user is active in the system. • *Deactivated*. The user hasn't reset their password within the given time frame. The user can be activated. • *Deleted*. The user is no longer active in the system. • *PasswordReset*. This value indicates that a password reset is in process. See *Reset User Password on page 350* for details. • *LockedOut*. The user cannot log in because they have had too many failed login attempts. To provide user access before the lock out period ends, you can initiate a password reset using the Reset User Password API or set the userLocked parameter to *true* using the Modify User Details API. • *Unknown*. The user has no status value. Control Center automatically deactivates users that haven't reset their password within a given time frame and have exceeded the user deactivation grace period.
userLocked	Indicates whether the user is locked out of their account because of too many incorrect login attempts. If this value is *true*, you can change it to *false* to restore user access before the lockout period ends.
accessType	Indicates the method(s) a user can use to access Control Center: *UI*, *API*, or *API_UI* (both).
firstName	The user first name.
lastName	The user last name.
email	The user email address.
phone	The user phone number.
language	The language used to display all the text in the web interface.
timeZone	Control Center uses this value to determine the date and time to display throughout the web interface. Valid values are Java 7-supported time zones.

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Return Value	Description
customerName	The customer with which this user is associated, if applicable. Applies only to accounts that use customers, and only if this user is a customer user. Otherwise, the parameter is null.
customerId	A unique identifier for the customer with which this user is associated, if applicable. Applies only to accounts who use customers, and only if this user is a customer user.
customerGroup	The customer group to which the user is assigned, if any. A customer group contains one or more customers belonging to a common enterprise account. For account users, a customer group restricts their view of device information to just the customers in the group. For customer users, a customer group expands their view of device information to all the customers in the group. The field is always null for service provider users.
accountGroup	For service provider users only, indicates the name of the service provider account group, if any, associated with the user. Account groups filter the information a user can view. This field is always null for account and customer users.
lastLogin	The date and time the user last accessed the system using the web interface.
lastPasswordResetDate	The date and time when the user last reset the password. Timestamp format: yyyy-MM-dd HH:mm:ss:SSSZ.
passwordExpirationInDays	The number of days before the user's password expires. Seven days before the expiration, Control Center begins sending email notifications to the user asking them to reset their password.
liveUpdateEnabled	Indicates whether Control Center automatically pushes device changes to the web interface without requiring a manual page refresh (*true* or *false*).
dateAdded	The date and time when the user was created. Timestamp format: yyyy-MM-dd HH:mm:ss:SSSZ.
dateModified	The date and time of the last change to the user details. Timestamp format: yyyy-MM-dd HH:mm:ss:SSSZ.

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Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/users/84176"
```

Response Example

```
{
    "username": "JohnDoe",
    "userId": 84176,
    "accountName": "Acme Inc",
    "accountId": 100210811,
    "operatorName": null,
    "operatorId": null,
    "roleName": "ACCOUNTADMIN",
    "status": "Active",
    "userLocked": false,
    "accessType": "API_UI",
    "firstName": "John",
    "lastName": "Doe",
    "email": "john.doe@jasper.com",
    "phone": null,
    "language": "English",
    "timeZone": "India Standard Time",
    "customerName": null,
    "customerId": null,
    "customerGroup": null,
    "accountGroup": null,
    "lastLogin": "2017-09-18 12:28:58:828+0000",
    "lastPasswordResetDate": "2017-09-18 12:26:02:000+0000",
    "passwordExpirationInDays": 180,
    "liveUpdateEnabled": true,
    "userPrivacyEnabled": true,
    "dateAdded": "2017-09-18 12:26:01:789+0000",
    "dateModified": "2018-01-10 08:40:43:000+0000"
}
```

Errors

Error Code	HTTP Code	Error Message
1400109	404	User not found.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Get All User Details

Description

Returns the user details for all users. Optionally, you can supply a particular user name if you want to get details for only one user.

Account administrators can view details for all account and customer users in their account. Service provider administrators can view details for all service provider, account, and customer users. Other roles may have access to user information as well. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

No Partial Response Support. This GET API does not support partial responses. See *Partial Responses on page 36*.

Resource URL

GET BaseURL/v{apiVersion}/users

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
username	The name of a particular user. If you provide this value, the API returns details for only the user with this name. No other user details are returned. If Control Center cannot find a user with this name, the response is empty.
pageSize	(Optional) Specifies the number of records returned in each response page. The maximum value is 50. The value defaults to 50. See *Pagination on page 48*.
pageNumber	(Optional) Specifies the number of the response page to return. This value defaults to 1. See *Pagination on page 48*.

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Response Parameters

Returns an array of users along with the pagination parameters. For more information about the user parameters, see *Get User Details on page 334*.

Return Value	Description
pageNumber	An integer specifying the number of the current response page.
users	An array of users with details about each. See *Get User Details on page 334* for information about the response parameters.
lastPage	A true or false value indicating whether the current response page is the last in the series.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X GET --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/users?pageSize=50&pageNumber=1"
```

Response Example


```
{
  "pageNumber": 1,
  "users": [{
    "username": "JohnDoe",
    "userId": 84176,
    "accountName": "Acme Inc",
    "accountId": 100210811,
    "operatorName": null,
    "operatorId": null,
    "roleName": "ACCOUNTADMIN",
    "status": "Active",
    "userLocked": false,
    "accessType": "API_UI",
    "firstName": "John",
    "lastName": "Doe",
    "email": "john.doe@jasper.com",
    "phone": null,
    "language": "English",
    "timeZone": "India Standard Time",
    "customerName": null,
    "customerId": null,
    "customerGroup": null,
    "accountGroup": null,
    "lastLogin": "2017-09-18 12:28:58:828+0000",
    "lastPasswordResetDate": "2017-09-18 12:26:02:000+0000",
    "passwordExpirationInDays": 180,
    "liveUpdateEnabled": true,
    "userPrivacyEnabled": true,
```


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```
    "dateAdded": "2017-09-18 12:26:01:789+0000",
    "dateModified": "2018-01-10 08:40:43:000+0000"
  }, {
    "username": "JaneAdmin",
    "userId": 75677,
    "accountName": "Network J",
    "accountId": 100530743,
    "operatorName": null,
    "operatorId": null,
    "roleName": "ACCOUNTADMIN",
    "status": "Active",
    "userLocked": false,
    "accessType": null,
    "firstName": "Jane",
    "lastName": "Doe",
    "email": "jane.doe@cisco.com",
    "phone": null,
    "language": "English",
    "timeZone": "Pacific Standard Time",
    "customerName": null,
    "customerId": null,
    "customerGroup": null,
    "accountGroup": null,
    "lastLogin": "2018-04-02 08:39:53:318+0000",
    "lastPasswordResetDate": "2017-07-25 12:04:27:000+0000",
    "passwordExpirationInDays": 180,
    "liveUpdateEnabled": true,
    "userPrivacyEnabled": false,
    "dateAdded": "2016-07-21 08:54:09:078+0000",
    "dateModified": "2018-04-02 08:40:17:103+0000"
  }],
    "lastPage": false
}
```


Errors

Error Code	HTTP Code	Error Message
1400109	404	User not found.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Create User

Description

Creates a new user. Control Center sends a welcome email to the new user with a link to the password set-up page in the product. If you create a passwordless API user, Control Center does not send a welcome email. You must send the user name and API key to the new user.

See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

POST BaseURL/v{apiVersion}/users

Request Parameters

All parameters are required unless they are marked as optional.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
username	The user name is a unique identifier for the person within Control Center. A user must supply this user name and a password to log in to Control Center. The name must be 6-25 alphanumeric characters, with no special characters or punctuation. No spaces are allowed at the beginning, middle, or end of the string. The user name must be unique across all accounts in Control Center.
roleName	The role name defines the application privileges for the user. For a list of valid values, see *User Role Values on page 52*. Be aware that not all users can create all types of users. The types of users you can create depend on your user role.

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Parameter	Description
accessType	Users can access Control Center information through a web interface or an API. This value determines which method is available to the user: *UI*, *API*, or *API_UI* (both). If you create an *API Only* user, you must provide them with API documentation because they will not be able to access it online. For users who log into Control Center through an external system that provides single sign-on access, this value is irrelevant. Control Center treats the user as if the accessType were *API_UI*, regardless of the actual value. To avoid API and web interface abuse, we recommend that you select one or the other option rather than giving the user access to both. The combination option is intended for developers who are coding with APIs and want to view test results in the web interface. If necessary, the service provider can remove the *Both API and UI* option from the drop down.
accountName	The name of the account to which the user is assigned.
customerName	(Optional) The customer with which this user is associated, if applicable. Applies only to accounts that use customers, and only if this user is a customer user. Otherwise, the parameter is null.
firstName	The user first name.
lastName	The user last name.
email	Control Center uses this email address to send the new user a welcome email with a link to the password set-up page. The email domain must be an allowed domain if the Email Domain Permit List feature is enabled.
phone	(Optional) You can provide a phone number for the new user.
language	Control Center uses the supplied language to display all the text in the user interface. The new user can change this setting in their user profile. See *Language Values on page 51* for a list of valid values.
timeZone	Control Center uses this value to determine the date and time to display throughout the web interface. The new user can change this setting in their user profile. Valid values are Java 7-supported time zones.

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Parameter	Description
liveUpdateEnabled	(Optional) Indicates whether Control Center automatically pushes device changes to the web interface without requiring a manual page refresh (*true* or *false*).
passwordlessApiUser	(Optional) Indicates that Control Center creates the API-only user without a password (*true* or *false*).

Response Parameters

Return Value	Description
userId	A unique identifier for the new user.
timestamp	The date and time when the user was created. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
{ curl -X POST --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"username\": \"AcmeUser\",
  \"roleName\": \"ACCOUNTUSER\",
  \"accessType\": \"UI\",
  \"accountName\": \"Acme Inc\",
  \"firstName\": \"Acme\",
  \"lastName\": \"User\",
  \"email\": \"acme_user@acme.com\",
  \"language\": \"English\",
  \"timeZone\": \"Pacific Standard Time\"
}" "https://<YOUR-BASE-URL>/rws/api/v1/users"
```

Response Example

```
{
  "userId": 30466103,
  "timestamp": 1527590173
}
```

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Errors

Error Code	HTTP Code	Error Message
1400103	400	Could not create user.
1400112	400	Email is blank.
1400113	400	First Name is blank.
1400114	400	Last Name is blank.
1400118	400	Timezone is missing.
1400119	400	Language is missing.
1400121	400	First Name invalid.
1400122	400	Last Name invalid.
1400123	400	Phone invalid.
1400128	400	Invalid entry. User name must be 6 to 25 alphanumeric characters.
1400129	400	Invalid user access type specified. Permissible access types are API, UI, API_UI.
1400206	400	Email domain not allowed. Please provide an email address from a permitted domain.
1700281	400	Invalid Access type for Passwordless API user.
1700282	400	Passwordless API user cannot be created as feature is not enabled for requested service provider.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
10000567	404	Invalid Account Name.
10000568	400	Username already used. Please try a different one.
10000569	400	Invalid Authority.
10000570	400	Authority is required.
10000571	400	Account Name is required.
10000572	400	Service Provider Name is required.

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Error Code	HTTP Code	Error Message
10000574	400	AccessType is required.
10000575	400	Invalid Language value provided. For permissible values, please check the API documentation.
20000005	404	Resource not found - Invalid Role.
30000001	500	Unknown server error.

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Edit User Details

Description

Changes one or more attributes for a given user. See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

PUT BaseURL/v{apiVersion}/users/{userId}

Request Parameters

You can change any of the user attributes shown below. One function call can update multiple attributes.

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
userId	An identifier for the user you want to modify. You cannot change this attribute. The user ID is not available in the Control Center web interface, but you can use the Get All User Details API to find a user ID. Use a partial response to return the username and userId only. For more information, see Partial Response Guidelines.
roleName	The role name defines the application privileges for the user. For a list of valid values, see *User Role Values on page 52*. Be aware that not all users can assign all user roles. The types of roles you can assign depend on your user role.

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Parameter	Description
accessType	Users can access Control Center information through a web interface or an API. This value determines which method is available to the user: *UI*, *API*, or *API_UI* (both). If you create an *API Only* user, you must provide them with API documentation because they will not be able to access it online. For users who log into Control Center through an external system that provides single sign-on access, this value is irrelevant. Control Center treats the user as if the accessType were *API_UI*, regardless of the actual value. To avoid API and web interface abuse, we recommend that you select one or the other option rather than giving the user access to both. The combination option is intended for developers who are coding with APIs and want to view test results in the web interface. If necessary, the service provider can remove the *Both API and UI* option from the drop down.
userLocked	Indicates whether the user is locked out of their account because of too many incorrect login attempts. If this value is *true*, you can change it to *false* to restore user access before the lockout period ends.
liveUpdateEnabled	Indicates whether Control Center automatically pushes device changes to the web interface without requiring a manual page refresh (*true* or *false*).
firstName	The user first name.
lastName	The user last name.
email	The user email address. The email domain must be an allowed domain if the Email Domain Permit List feature is enabled. When the email address for a Control Center account changes, an email is sent to the user's prior email address notifying them of the change. The email advises users to contact their administrator and secure the account if they do not recognize the activity.
phone	The user phone number.

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Parameter	Description
language	Control Center uses the supplied language to display all the text in the user interface. The new user can change this setting in their user profile. See *Language Values on page 51* for a list of valid values.
timeZone	Control Center uses this value to determine the date and time to display throughout the web interface. The new user can change this setting in their user profile. Valid values are Java 7-supported time zones.
passwordlessApiUser	Indicates that Control Center creates the API-only user without a password (*true* or *false*).

Response Parameters

Return Value	Description
userId	The ID of the user whose information was updated.
timestamp	The date and time when the user details were modified. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  \"phone\": \"925-457-0399\",
  \"liveUpdateEnabled\": true
}" "https://<YOUR-BASE-URL>/rws/api/v1/users/30466103"
```

Response Example

```
{
       "userId": 30466103,
       "timestamp": 1527590173
}
```

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Errors

Error Code	HTTP Code	Error Message
1400109	404	User not found.
1400116	400	Email is not valid.
1400117	400	Could not update user.
1400206	400	Email domain not allowed. Please provide an email address from a permitted domain.
1700281	400	Invalid Access type for Passwordless API user.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Reset User Password

Description

Resets the web interface password for a given user. Control Center sends an email to the user with instructions for resetting their password. The user status automatically changes to *PasswordReset* until the user supplies a new password.

See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

PUT BaseURL/v{apiVersion}/users/{userId}/resetPassword

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
userId	The ID of the user whose password you want to reset. The user ID is not available in the Control Center web interface, but you can use the Get All User Details API to find a user ID. Use a partial response to return the username and userId only. For more information, see Partial Response Guidelines.

Response Parameters

Return Value	Description
userId	The ID of the user affected by the password reset.
timestamp	The date and time when the password reset process was initiated. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d
"https://<YOUR-BASE-URL>/rws/api/v1/users/30466103/resetPassword"
```

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Response Example

```
{
        "userId": 30466103,
        "timestamp": 1527590173
}
```

Errors

Error Code	HTTP Code	Error Message
1400109	404	User not found.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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Delete User

Description

Deletes a given user so the user details are no longer available through the web interface or the APIs. Control Center retains the data throughout the service provider's default data retention period and then begins the data erasure process.

See the API/Role matrix in the Knowledge Base for a complete list of roles that can use this API.

Resource URL

DELETE BaseURL/v{apiVersion}/users/{userId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
userId	An identifier for the user you want to delete. The user ID is not available in the Control Center web interface, but you can use the Get All User Details API to find a user ID. Use a partial response to return the username and userId only. For more information, see Partial Response Guidelines.

Response Parameters

Return Value	Description
userId	The ID of the deleted user.
timestamp	The date and time when the user was deleted. The timestamp displays in UNIX Epoch clock format. See *Date/Time Formats on page 49* for more information.

Request Example

Make sure to use your own URL and user credentials. See *Authentication on page 33* for information about creating an authorization header.

```
curl -X DELETE --header "Accept: application/json" --header "Authorization:
Basic <YOUR-ENCRYPTED-CREDENTIALS>" "https://<YOUR-BASE-
URL>/rws/api/v1/users/30466103"
```

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Response Example

```
{
        "userId": 30466103,
        "timestamp": 1527590173
}
```

Errors

Error Code	HTTP Code	Error Message
1400109	404	User not found.
10000024	400	Invalid apiVersion.
10000030	400	Your role does not have access to this API function.
30000001	500	Unknown server error.

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REST Error Messages

Error Code	HTTP Code	Error Message
400	400	ChangeImeiRequest must have Imei field set.
400	400	ChangeImeiRequest must have Imsi or Msisdn field set.
400	400	ChangeImeiRequest parameter Msisdn doesn't match the value for Msisdn configured in system.
400	400	Mandatory headers are missing in request.
400	400	Required attributes do not match CSR.
400	400	Invalid CSR.
400	400	Invalid request.
400	400	Unsupported grant type.
400	400	AccountId is mandatory.
400	400	Access Denied.
401	401	Invalid credentials.
401	401	Invalid client.
404	404	SIM with Msisdn not found
404	404	SIM with Imsi not found.
404	404	Connection Cert not found for the device.
404	404	Device could not be found.
404	404	Cloud Connector policy not found.
404	404	Certificate not found.
409	409	Listener already registered. Please delete existing one to register new.
500	500	Error while updating Imei.
500	500	Unable to process request. Retry request after sometime.
500	500	Unknown server error.

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Error Code	HTTP Code	Error Message
100100	206	Terminal not found.
100101	202	Invalid ICCID. ICCID-<ICCID>. Description: The ICCID doesn't exist in Control Center.
400201	202	Access violation. Description: The user does not have access to the requested resource.
400300	500	System internal error.
400400	400	Too many entries. Requested iccids more than 5000. Description: The request contains more than 5000 devices.
1000100	400	Account not found. accountid=<AcctId>. Description: The target account doesn't exist in Control Center.
1000100	202	Account not found. accountid=<AcctId>. Description: The source account doesn't exist in Control Center. Be aware that there may be multiple source accounts, depending on the devices you transfer.
1000508	202	Cannot Transfer SIM to this Account.
1000509	202	Cannot Transfer SIM to same account. Description: The source and target accounts must be different.
1000510	202	Cannot Transfer SIM.
1000512	202	SIM is not in Purged State. When you return a device to inventory, it must have the *Purged* SIM state.

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Error Code	HTTP Code	Error Message
1000513	202	Cannot Transfer SIM when source account is inventory account. Description:You cannot transfer a device to inventory when it's already in the inventory account.
1000514	202	Cannot Transfer SIM when source and destination account doesn't belong to same service provider. Description: The source and target accounts must be located in the same service provider.
1000523	202	Rate Plan is Not Eligible. RatePlanType and acctType mismatch. Description: The rate plan is not available for the target account.
1000526	202	Communication Plans and account type mismatch. CommPlanType and acctType mismatch. Description: The communication plan is not available for the target account.
1000550	202	Transfer of devices is allowed across Essential accounts only. Description: You cannot transfer devices from an Essential account to an Advantage account.
1000551	202	Transfer of devices is allowed across Advantage accounts only. Description: You cannot transfer devices from an Advantage account to an Essential account.
1000552	202	Transfer of devices is not supported for Essential accounts. Description: You cannot transfer devices to or from an Essential account.

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Error Code	HTTP Code	Error Message
1000559	202	Target SIM State not allowed for Return to Inventory Transfer.
1000591	400	Setting P5G attributes are not allowed since P5GaaS is not enabled for the operator Description: You cannot set the p5gCommercialStatus field because the service provider does not support the Cisco IoT Private 5G (P5G) service.
1000593	400	Setting p5gCommercialStatus access is not allowed for the user
1000594	400	Updating from commercial status is not allowed
1000600	202	CommPlan and Account Voice WL flag mismatch. Description: The communication plan supports Voice Permit List, but the target account does not.
1000621	400	Invalid Industry Vertical.
1000700	202	CommPlan and Account VoLTE RUG flag mismatch. Description: The communication plan supports VoLTE RUG, but the target account does not.
1000705	400	Device transfer for CIP accounts not supported.
1100200	400	Term end date must be greater than term start date.
1100400	400	Renewal Rate Plan is not valid for the device. Description: The target rate plan is not available to the device. The rate plan may be associated with a different account. Or, in the case of a shared rate plan, you may need to assign the rate plan to the account.
1100500	400	Renewal mode is not valid.

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Error Code	HTTP Code	Error Message
1100702	400	Renewal rate plan mandatory for NamedPlan renewal mode.
1100704	400	Term not found. Description: The event rate plan does not have an active term.
1200100	400	Invalid Event Name.
1200101	400	Invalid effectiveDate.
1200102	400	SIM State is not Activated.
1200104	400	This event instance has been created already.
1200200	400	Event Instance not found.
1200400	400	SIM is not part of a Partner Account or Rate Plan.
1400103	400	Could not create user.
1400109	404	User not found.
1400112	400	Email is blank.
1400113	400	First Name is blank.
1400114	400	Last Name is blank.
1400116	400	Email is not valid.
1400117	400	Could not update user.
1400118	400	Timezone is missing.
1400119	400	Language is missing.
1400121	400	First Name invalid.
1400122	400	Last Name invalid.
1400123	400	Phone invalid.
1400128	400	Invalid entry. User name must be 6 to 25 alphanumeric characters.
1400129	400	Invalid user access type specified. Permissible access types are API, UI, API_UI.

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Error Code	HTTP Code	Error Message
1400206	400	Email domain not allowed. Please provide an email address from a permitted domain.
1700102	400	Invalid IP address.
1700281	400	Invalid Access type for Passwordless API user.
1700282	400	Passwordless API user cannot be created as feature is not enabled for requested service provider.
1800218	404	Invalid UMTS or LTE apn.
2000114	400	Recurring event number of instances exceeds max count.
2000115	400	No pending instances found for the event.
2000116	400	Invalid Request. Instance count should be greater than 0.
2000117	400	Invalid Event Type for recurring instances. Ensure the rate plan type is Event, not Stacked Event, or Add-on.
2310000	400	Invalid jobId - Value must be a number.
2310001	400	Invalid duration - Value must be between 15 minutes and 4 hours (in seconds).
2310002	500	Error while fetching entitlement.
2310003	409	A job is in progress.
2310004	404	Job does not exist.
2310005	400	Invalid update operation.
2310006	409	Job not qualified for revocation.
2310007	409	SIM not active.

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Error Code	HTTP Code	Error Message
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000002	400	AccountId is required.
10000003	400	ModifiedSince is required.
10000004	400	Invalid accountId.
10000005	400	Invalid status.
10000006	400	Invalid pageSize.
10000007	400	Invalid pageNumber.
10000008	400	Your role does not have access to operator and customer custom fields <field name>
10000009	400	Your role does not have access to account custom fields <field name>
10000010	400	Your role does not have access to customer custom field <field name>
10000011	400	One or more required fields are missing. Missing query fields. Description: An account ID is required. There must be at least one ICCID.
10000012	400	Invalid date format.
10000013	404	Invalid ratePlan. Description: Two scenarios can cause this error: • The rate plan name doesn't exist. • The specified rate plan is not allowed with the device's communication plan.
10000014	400	Invalid communicationPlan.
10000015	400	Invalid customer.

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Error Code	HTTP Code	Error Message
10000016	400	Invalid overageLimitOverride.
10000017	400	Invalid messageEncoding.
10000018	400	Invalid dataCoding.
10000019	400	Invalid tpvp. Description: The validityPeriod is invalid.
10000020	400	Message length exceeds the maximum permissible length.
10000021	400	Invalid iccid.
10000022	400	FromDate is required.
10000023	400	The JSON in the request is not well formed. Please ensure that commas, colons, braces etc. are formatted properly.
10000024	400	Invalid apiVersion.
10000025	400	Invalid deviceID.
10000026	400	Invalid modemID.
10000027	400	ToDate must be after fromDate.
10000028	400	Invalid request. The request contained one or more unrecognized parameters.
10000029	400	This SIM may not be moved back to a Pre Activation status.
10000030	400	Your role does not have access to this API function.
10000031	400	Invalid Zone.
10000032	400	Cycle Start Date should be greater than or equal to 20xx-xx-xx. Description: The cycleStartDate must specify one of the last three billing cycles (including the current cycle).
10000033	400	Invalid contactName.

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Error Code	HTTP Code	Error Message
10000045	400	Invalid date. This date/time cannot be more than one year in the past.
10000049	400	The daysOfHistory must be less than or equal to 365.
10000062	400	Access Denied. Description: The user doesn't have access to the target account.
10000074	400	Account is not of Partner Type.
10000075	400	Trust Relationship is not established with Partner Account.
10000076	400	Trust Relationship is not active.
10000077	400	Device is already assigned to Partner Account.
10000078	400	Device is not assigned to Partner Account.
10000079	400	Account does not have access to this API.
10000080	403	The API access is forbidden. Description: This error can occur if your role doesn't have access to the function. Description: This error can occur if your role doesn't have access to the function or if the service provider doesn't have access to the Custom Charges REST APIs.
10000124	400	Invalid end consumer name. Number of characters allowed 0-60 and cannot contain commas (,) or asterisks (*).
10000125	400	Invalid end consumer identity. Number of characters allowed 0-50 and cannot contain commas (,) or asterisks (*).

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Error Code	HTTP Code	Error Message
10000126	400	Invalid primary contact. Number of characters allowed 0-30 and cannot contain commas (,) or asterisks (*).
10000127	400	Invalid contact phone. Number of characters allowed 0-25 and cannot contain commas (,) or asterisks (*).
10000128	400	Invalid contact email. Number of characters allowed 0-320 and cannot contain commas (,) or asterisks (*).
10000129	400	Invalid contact fax. Number of characters allowed 0-25 and cannot contain commas (,) or asterisks (*).
10000131	400	Invalid SIM State.
10000256	400	No active rate plan found for the device.
10000301	400	Test Ready Data Limit length must be between 0 and 10.
10000302	400	Test Ready Sms Limit length must be between 0 and 10.
10000303	400	Test Ready Voice Limit length must be between 0 and 10.
10000304	400	Test Ready Csd Limit length must be between 0 and 10.
10000305	400	Test Ready Data Limit must be a positive number.
10000306	400	Test Ready Sms Limit must be a positive number.
10000307	400	Test Ready Voice Limit must be a positive number.
10000308	400	Test Ready Csd Limit must be a positive number.
10000309	400	Invalid Default Sim State.
10000310	400	Invalid Test ready data state.
10000311	400	Invalid File export method

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Error Code	HTTP Code	Error Message
10000312	400	No support information to show. Either Account Support Email and/or Account Support Phone must be specified.
10000313	400	Account Support Email must be a correctly formatted email address and cannot contain commas.
10000314	400	Account Support Phone cannot be more than 25 characters and cannot contain commas (,) or asterisks (*).
10000315	400	Invalid SMS Enabled value Only true/false allowed.
10000316	400	Invalid Voice Enabled value Only true/false allowed.
10000321	400	Application feature cannot be null.
10000322	400	Sim feature cannot be null.
10000323	400	Site configuration cannot be null.
10000324	400	Support cannot be null.
10000325	400	AccessCommPlan cannot be null.
10000326	400	CsdAccess cannot be null.
10000327	400	BusinessRulesEnabled cannot be null.
10000328	400	GlobalAccountEnabled cannot be null.
10000329	400	PoliciesEnabled cannot be null.
10000330	400	SimOrderingEnabled cannot be null.
10000331	400	SmsAccess cannot be null.
10000332	400	SubscriptionChannel cannot be null.
10000333	400	VoiceAccess cannot be null.
10000334	400	SmsEnabled cannot be null.
10000335	400	VoiceEnabled cannot be null.
10000336	400	BlockBackwardSimState cannot be null.

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Error Code	HTTP Code	Error Message
10000337	400	DefaultSimState cannot be null.
10000338	400	DisplayDiagnosticWizard cannot be null.
10000339	400	DisplaySpotlight cannot be null.
10000340	400	FixedIpEnabled cannot be null.
10000341	400	TestReadyDataLimit cannot be null.
10000342	400	TestReadySmsLimit cannot be null.
10000343	400	TestReadyVoiceLimit cannot be null.
10000344	400	TestReadyCsdLimit cannot be null.
10000345	400	TestReadyDataState cannot be null.
10000346	400	FileExportMethod cannot be null.
10000347	400	EulaEnabled cannot be null.
10000348	400	ShowAcctSupport cannot be null.
10000349	400	ShowSpSupport cannot be null.
10000436	400	Shared can only be true.
10000437	400	Shared can not be used along with accountId.
10000438	404	Resource not found - Invalid username.
10000439	400	Unauthorized access to the user.
10000440	400	You cannot reset an API key for a user with access to the web interface.
10000441	400	Invalid RatePlan Version.
10000442	400	More than one RatePlan having the same name. Description: The rate plan name is not unique. Try using the rate plan version number along with the name to indicate a unique rate plan.

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Error Code	HTTP Code	Error Message
10000443	400	Invalid BillingCycleStart. Description: The billing cycle start date doesn't correspond to a valid billing cycle.
10000500	400	Service provider not found.
10000501	400	Could not create account.
10000504	400	Account already exists.
10000505	400	Invalid region Id.
10000515	400	Invalid BillingCycleStart.
10000516	400	Not allow to set BillingCycleStart.
10000517	400	The ipv4Address/ipv6Address is already in use, please assign a different one.
10000521	400	Invalid Country Code.
10000528	400	Partner Account creation not allowed.
10000530	400	Account Id cannot be null.
10000532	500	CUSTOM_CHARGE: Error creating a new custom charge.
10000533	404	Resource not found - invalid charge ID.
10000534	400	Custom Charge reference Id cannot be empty/null.
10000535	400	Custom Charge description cannot be empty/null.
10000536	400	Charge amount cannot be null.
10000537	400	Charge frequency cannot be null.
10000538	400	Empty Rate Plan Id.
10000539	400	Empty Device State.
10000540	404	Resource not found - Invalid operatorId.
10000541	400	Invalid Custom Charge Status.
10000542	400	Empty charge status.

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Error Code	HTTP Code	Error Message
10000543	400	Invalid charge ID.
10000545	400	Invalid charge frequency value.
10000546	400	Invalid Device State.
10000547	400	Unpermitted Charge Status.
10000548	400	Primary contact firstName and lastName are required.
10000549	400	Error deleting custom charge.
10000550	400	Invalid status transition. Description: You cannot delete a charge unless its status is NEW or INACTIVE.
10000551	400	AccountId mismatch for given chargeID. Description: The specified account doesn't contain the specified charge.
10000552	400	ReferenceId mismatch with given chargeId. Description: The reference ID/charge ID combination does not exist in the account.
10000553	400	The service provider doesn't contain the charge ID.
10000557	404	Resource not found - accounts for given service provider.
10000559	500	Unable to fetch custom charges.
10000562	400	Invalid OperatorId for the account provided.
10000563	400	Empty OperatorId.
10000567	404	Invalid Account Name.
10000568	400	Username already used. Please try a different one.
10000569	400	Invalid Authority.
10000570	400	Authority is required.
10000571	400	Account Name is required.

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Error Code	HTTP Code	Error Message
10000572	400	Service Provider Name is required.
10000574	400	AccessType is required.
10000575	400	Invalid Language value provided. For permissible values, please check the API documentation.
10000580	400	Customer details required.
10000581	400	Customer name is required.
10000582	400	Invalid customer name. Number of characters allowed 1 - 60.
10000583	400	Account Id is required.
10000584	400	Security answer missing.
10000585	400	Security question missing.
10000586	400	Invalid security question. Number of characters allowed 0 - 100.
10000587	400	Invalid security answer. Number of characters allowed 0 - 100.
10000588	400	Maximum 4 contacts can be associated with a customer.
10000589	400	Invalid contact name. Number of characters allowed 0 - 40.
10000590	400	Invalid contact title. Number of characters allowed 0 - 40.
10000591	400	Invalid contact phone. Number of characters allowed 0 - 25.
10000592	400	Invalid contact mobile. Number of characters allowed 0 - 40.
10000593	400	Invalid contact email. Number of characters allowed 0 - 320.
10000594	400	Invalid address line1. Number of characters allowed 0 - 40.
10000595	400	Invalid address line2. Number of characters allowed 0 - 40.
10000596	400	Invalid city. Number of characters allowed 0 - 40.

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Error Code	HTTP Code	Error Message
10000597	400	Invalid state/region. Number of characters allowed 0 - 40.
10000598	400	Invalid country. Number of characters allowed 0 - 100.
10000599	400	Invalid postal code. Number of characters allowed 0 - 15.
10000602	400	Duplicate customer name.
10000603	400	Invalid account. Account erasure request is in progress for the account.
10000604	400	Invalid Customer Id.
10000605	404	Customer not found with given Id.
10000700	400	Effective Date exceeds maximum - Effective Date must be within 365 days from today
10000800	202	Validation failed due to invalid target account or invalid target sim state.
10001550	400	You cannot get an API key for a user with access to the web interface.
10002117	400	modifiedTill date is earlier than modifiedSince date.
10002118	400	modifiedTill date is after the max allowed modifiedTill date.
10002119	400	modifiedSince is a future date.
10030000	400	Cannot activate a SIM which is already activated.
15000001	400	Start date should be paired with end date.
15000002	400	Either provide start and end date or billing cycle.
15000003	400	End date should be greater than start date.
15000004	400	Maximum duration between start date and end date is allowed 30 days.

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Error Code	HTTP Code	Error Message
15000005	400	Records count exceeded the maximum permissible limit in response(15k).
15000006	400	Provided login session is not valid.
15000007	400	Not a valid group by element.
15000008	400	Not a valid metric element.
15000009	400	Provide one element in metrics parameter.
15000010	400	Max limit for Top K is 100.
15000015	400	Invalid element provided in sort_by parameter.
15000020	400	Requested report type is not valid.
15000022	400	Requested field name is not valid.
15000025	400	Request does not belong to the user.
15000026	400	OperatorId and acctId are mandatory fields.
15000027	400	Requested accountId(s) in the filters section do not belong to the user.
15000028	400	Both operatorId and accountId can not be null or empty.
15000029	400	Incorrect filter value(s) provided for {fieldNames}.
15000030	400	Requested category is not valid.
15000031	400	Category is empty.
15000032	400	Requested metrics is not valid.
15000033	400	Internal system error.
15000034	400	Requested filter is not present in the fields.
15000035	400	Metric is empty.
15000036	400	GenerationEndDate cannot be before GenerationStartDate.
15000037	400	Mandatory Fields are missing for the report.

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Error Code	HTTP Code	Error Message
15000038	400	Past dates are not allowed. Please choose a valid date.
15000039	400	Invalid date format.
15000040	400	Invalid report format. Only CSV and DAT format are supported.
15000041	400	Invalid Frequency.
15000042	400	Please contact Cisco Support to enable SFTP.
15000043	400	Email Id(s) must be provided.
15000044	400	Invalid email provided for email notification.
15000045	400	DaysOfMonth must be specified for a monthly request.
15000046	400	DaysOfWeek must be specified for a weekly request.
15000047	400	Generation start date and end date are required for daily, weekly, and monthly requests.
15000048	400	Filtering data from other accounts is not permitted.
15000049	400	For the filter section allowed options for the operator are Range, <, >, =.
15000050	400	Only one metric can be selected at a time.
15000051	400	The requested Sub-Category is not valid.
15000052	400	Sub-Category is only applicable for Billing account level report.
15000053	400	'Subscription Charges' and 'Rating Zone ID' cannot be provided together.
15000054	400	'Rating Zone ID' cannot be provided with any Prepaid Term fields.
15000055	400	Requested billing cycle is not valid.

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Error Code	HTTP Code	Error Message
15000056	400	Sub-category must be provided for billing account level report.
15000057	400	Filter section value can not be empty.
15000058	400	Only one value is allowed when using the <, > or = operator.
15000059	400	The Range operator requires exactly two values.
15000060	400	Billing cycle is not allowed for this report type.
15000061	400	Billing cycle is empty.
15000062	400	Unit is not valid.
15000063	400	Unit is mandatory for the usage filter.
15000064	400	Start date and end date cannot be empty for a date range.
15000065	400	End Date can not be before the Start Date.
15000066	400	Report period is not allowed.
15000067	400	Only one-time frequency supports a date range and report period for the usage report.
15000068	400	DaysOfMonth and DaysOfWeek are only allowed for weekly or monthly reports.
15000069	400	Requested category, subcategory, or metrics configuration is invalid for Device report type.
15000070	400	GenerationDate can not be before EndDate.
15000071	400	Requested start and end dates must be within the last 6 months.
20000000	404	Requested resource not found.
20000001	404	Resource not found - Invalid ICCID.

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Error Code	HTTP Code	Error Message
20000002	404	Resource not found - Invalid smsMsgId.
20000005	404	Resource not found - Invalid Role.
20000003	404	Resource not found - Invalid accountId.
20000009	404	Resource not found - Invalid Package.
20000013	400	Account billing settings cannot be edited for service provider account.
20000014	400	Empty monthly devices minimum term (months).
20000016	400	Invalid activation proration.
20000015	400	Empty monthly device minimum.
20000017	400	Invalid renewal proration.
20000018	400	Invalid sim fee scale.
20000021	400	Invalid default activation fee scale.
20000020	400	Invalid default activation plan.
20000019	400	Sim fee should be between [0-9999999].
20000022	400	Default activation fee should be between [0-9999999].
20000023	400	Invalid enable retail billing.
20000025	400	Invalid prepaid renewal policy.
20000024	400	Invalid default timezone.
20000026	400	Invalid prepaid renewal rate plan.
20000029	400	Monthly device minimum should be between [0-999.99].
20000028	400	Monthly devices minimum term should be between [0-999].
20000027	400	Invalid activation grace period.
20000030	400	Invalid minimum activation term.

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Error Code	HTTP Code	Error Message
20000033	400	Invalid request to perform edit operation.
20000032	400	Billing commitment settings cannot be edited for Essential/Partner accounts.
20000031	400	Invalid request to view account billing settings.
20000034	400	Account billing settings cannot be edited for NBIOT account.
20000037	400	Operation cannot be performed as there are no rate plans available for this account.
20000036	400	Invalid request to assign shared rate plans to the account.
20000035	400	Shared rate plans can only be viewed by Advantage/Essential accounts.
20000038	400	Cannot perform operation as one or more shared rate plans not available for this account.
20000039	500	Error adding shared rate plan id mappings to the account.
20000040	400	Shared rate plans can only be added by Advantage/Essential accounts.
20000041	400	Taxable cannot be null.
20000042	400	Billable cannot be null.
20000043	400	ProrationRule cannot be null.
20000044	400	RenewalProration cannot be null.
20000045	400	ActivationPlan cannot be null.
20000046	400	RetailBilling cannot be null.
20000047	400	ImmediateRatePlanChange cannot be null.
20000049	400	PrepaidRenewalPolicy cannot be null.
20000048	400	EventPlanEnabled cannot be null.

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Error Code	HTTP Code	Error Message
20000050	400	PrepaidRenewalRatePlanId cannot be null.
20000051	400	BillActivationGracePeriod cannot be null.
20000052	400	MonthlyDevicesMinimumTerm cannot be null.
20000053	400	MonthlyDeviceMinimum cannot be null.
20000054	400	BillMonthlyDeviceMinimum cannot be null.
20000056	400	Commitments settings cannot be null.
20000057	400	Failed to update shared plans. Cannot remove the account default rate plan.
20000055	400	BillMinimumActivationTerm cannot be null.
20000058	400	Requested resource not available for Shared Rate plan. Description: You cannot use shared rate plans in this function.
30000001	500	Unknown server error.
30000002	500	Control Center failed to submit the message to the SMSC.
30000003	400	Invalid Usage Type.
50001225	400	Shipping Address cannot be null as it is not same as the billing address.
50001226	400	PPU Address cannot be null is it as not same as the shipping address.
50001224	400	Billing Contact cannot be null is it is not same as primary contact.
50001229	400	Currency code is invalid.
50001227	400	Service Provider name is invalid.
50001228	400	Account type is invalid.

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Error Code	HTTP Code	Error Message
50001234	400	CommPlan not allowed for Partner Account.
50001233	400	JSON validation error.
50001230	400	Language is invalid.
50001231	400	Sim state name is invalid.
50001235	400	Invalid AccountSegment.
50001236	400	Unsupported Account Type.
101000013	400	Invalid eSIM ID (EID)
101000017	404	eSIM ID (eid) not found.
101000022	400	Invalid message type.
102000002	400	Cannot update the EID for a device in the service provider inventory account.
102000001	400	If you specify the EID parameter, you must supply a value.
102000003	400	You do not have access to the EID attribute.

REST API/Role Matrix

Control Center provides different API access for different user roles. Take a look at the REST API/Role matrix for all the details. You can download the matrix from the Downloads page in the Knowledge Base.

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REST API Troubleshooting Checklist

You can quickly troubleshoot REST API issues and identify the main points of failure by answering the questions in the following checklist. If you answer *Yes*, continue to the next question, unless the instructions tell you otherwise. If you answer *No* to a question, you've found the problem. Follow the instructions for correcting the issue.

API Setup and Access

1. **Is the URL correct?**

   Make sure you're using the correct URL with the HTTPS protocol. You can find the URL in the Knowledge Base on the APIs > REST APIs > Getting Started page.

2. **Is the API key valid?**

   Each user has a unique key that you can find in the user profile (click the Show API key button to see it) or at the top of each REST API function page in the Knowledge Base. Users in the same account cannot share API keys.

   Be aware that API keys can expire, depending on the password policies associated with your account. Users with web interface access can reset the API key in their user profile. Otherwise, a service provider or account administrator must reset the API key.

3. **Is the authorization header valid?**

   Ensure that you followed the 3-step process for creating a valid authorization header: (1) concatenate the user name and API key (separated by a colon) (2) encrypt the resulting string using Base64 (3) set the authorization header value to "Basic" followed by the encoded string.

4. **Does the API user have feature access?**

   If the API user has access to the web interface, log into Control Center and try to access the same feature in the web interface. (Remember the Control Center password is different from the API key.) If you can make the feature work in the web interface, you can make it work via API.

   You may need to use multiple user roles as there is no single role that will guarantee access for all API function calls. Cisco recommends creating a dedicated user or group of users whose sole purpose is to execute API functions. For more information about user roles and access, see *REST API/Role Matrix on the previous page*.

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5. **Is the SSL certificate valid and included in your trusted store?**

   The REST APIs support the HTTPS protocol (not HTTP). The Cisco API server uses an SSL certificate signed by a top tier Certificate Authority (CA) to handle HTTPS access. We recommend that you include the root certificate in your trust store (this is included by default on many platforms) and use a CA-based chain of trust, rather than hard coding the Cisco certificate in your API code.

API Coding

6. **Are the request parameters valid?**

   Typically, the returned error codes will help you identify any problems with incorrect parameter values. Common problems include: improper date formats (see Date/Time Formats) and invalid values for device attributes.

7. **Did you verify input parameters and output data with the Try It feature?**

   If you're unsure about the request parameters or the returned data, try executing the API function from within the Knowledge Base using the Try It feature. This approach can help you isolate problem areas in a particular function.

8. **Is the API code able to process the error codes returned by Control Center**?

   When an error occurs, Control Center returns a JSON-formatted response that contains a human-readable error message and an error code. Ensure that your code can handle all the errors that may be returned. The API documentation provides a list of possible errors for each function.

9. **Does the API response coding follow best practices?**

   After checking your network connection to Control Center, review your receiver code and error handling. Ensure that your code allows for additional, unexpected items in the response. Cisco may add new return values to the APIs.

API Results and Performance

10. **Did the API call return the expected results?**

    You can use a web services testing tool to set up and run API function tests and test suites to verify variable values at different stages of the API execution. Also, keep in mind that the role of the API user determines the features and data that the API call can access and return.

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11. **Did the API call return data with the expected performance?**

    Follow these standards to improve Control Center API performance.

    - Limit the number of active calls to one at a time and avoid concurrent API processing.
    - Do not exceed the calls per second limit.
    - Page size is limited for certain APIs. Make sure your code handles pagination properly.
    - Align the timing of your API calls with the frequency of your service provider's record updates. For example, if your service provider updates their usage information once per 6 hours, there's no need to request the information via API more often than once per 6 hours.
    - Be aware that the API calls are not intended to provide real-time monitoring of service usage. For more immediate feedback when certain conditions occur, use automation rules and push APIs.

    You can poll API calls that return small amounts of data more frequently (once per minute, if necessary) without affecting the overall performance of Control Center. See *API Policies and Best Practices on page 9* for more details.

If you haven't been able to identify the cause of the API issue, contact Cisco Product Support.

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QoS on Demand

The Control Center QoS on Demand feature includes several network APIs. These industry-standard APIs allow you to apply service changes to a network on demand, ensuring that devices and applications maintain optimal performance and service quality at all times.

- Get Services for Account
- Activate QoS Profile for Device
- Get QoS Job Status
- Revoke QoS Profile Activation

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Get Services for Account

Description

Used with QoS on Demand , this API returns the QoS profile for an enterprise account.

This API requires OAuth 2.0 authentication. To use this authentication method, run the Get Access Token API to request an access token from Control Center. Once you receive the access token, you must create an HTTP Bearer authorization header containing the access token, and include it in the REST API call. For documentation on OAuth 2.0 authentication, contact your Cisco account team representative.

Resource URL

GET BaseURL/v{apiVersion}/marketplace/{serviceType}{relatedParty.Id}{Field}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
serviceType	Enter "*qod*" for this parameter.
relatedParty.Id	The ID of the account whose Marketplace services you want to retrieve.
Fields	Specify the fields to include on the API response, separated by commas. Possible values include: • *ServiceType* • *Name* • *Category* • *Feature* • *State*
tokenExpiryInMinutes	(Optional) The token validity timer, in minutes. Suggested values are *60* or *120*. If you leave this field blank, Control Center uses the default value of 1.

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Response Parameters

Return Value	Description
category	The category for the service. For example, "Network" is the category for the service containing QoS profiles.
serviceType	The identifier of the service type. For example, "qod" is the service type for the service containing QoS profiles.
name	The name of the service. For example, "Quality on Demand for Uplift" is the name for the service containing QoS profiles.
state	Indicates the status of the QoS Profile. For example, if the state is *active*, Control Center is currently using that QoS profile.
feature	The QoS profiles and their attributes.

Feature Parameters

Return Value	Description
name	The name of the QoS profile.
featureCharacteristic	Characteristics describing the QoS profile.
valueType	The data type of the characteristic value (string, number, etc.).
value	The value of the characteristic (e.g., duration, QCI, etc.).

Request Example

```
curl -X GET --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Bearer <YOUR-ACCESS-TOKEN>" -d "{
  }" "https://<YOUR-BASE-URL>/rws/api/
  v1/marketplace/service?fields=category,serviceType,name,state,feature&offset
  =0&limit=10&relatedParty.Id=131491803
```

Response Example

```
{
     "category":"Network",
     "serviceType": "qod",
     "name": "Quality on Demand for Uplift",
     "state": "active",
     "feature": [
        {
          "name": "qosprofile1",
          "featureCharacteristic": [
            {
             "name": "ProfileName",
```

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```
            "valueType": "string",
            "value": "Priority Scheduling 2 hours"
           },
           {
            "name": "QCI",
            "valueType": "number",
            "value": 7
           },
           {
            "name": "Duration",
            "valueType": "number",
            "value": 7200
           }
          ]
        }
     ]
}
```


Errors

Error Code	HTTP Code	Error Message
400	400	AccountId is mandatory.
400	400	Access Denied.
401	401	Invalid credentials.
500	500	Unknown server error.

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Get QoS Job Status

Description

Used with QoS on Demand , this API retrieves the status of a QoS profile provisioning request for a device.

Resource URL

GET BaseURL/v{apiVersion}/qod/{jobId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
jobId	The unique Job ID returned by the Activate QoS Profile for Device API during activation.

Response Parameters

Return Value	Description
jobId	The unique Job ID returned by the Activate QoS Profile for Device API during activation.
status	Indicates the current status of the provisioning request. Possible values are: • ***Activated***. The QoS profile was successfully activated. • ***Pending***. The request was received, and provisioning is in progress. • ***Failed***. The provisioning request failed for the device. • ***Unknown***. Control Center could not determine the status of the request.

Request Example

```
curl -X GET --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  }" "https://<YOUR-BASE-URL>/rws/api/v1/qod/919"
```

Response Example

```
[
  {
     "jobid": "919",
     "status": "ACTIVATION_IN_PROGRESS",
  },
  {
```

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```
    "jobid": "919",
    "status": "ACTIVATION_SUCCESS",
  }
]
```

Errors

Error Code	HTTP Code	Error Message
2310000	400	Invalid jobId - Value must be a number.
2310004	404	Job does not exist.
10000011	400	One or more required fields are missing.
10000024	400	Invalid apiVersion.
30000001	500	Unknown server error.

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Revoke QoS Profile Activation

Description

Used with QoS on Demand , this API deactivates a previously activated QoS profile for a device.

Resource URL

PATCH BaseURL/v{apiVersion}/qod/{jobId}

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
jobId	The unique Job ID returned by the Activate QoS Profile for Device API during activation.
status	Enter *Revoke* for this parameter.

Response Parameters

Return Value	Description
status	Indicates the current status of the revocation request. Possible values are: • ***Revocation in progress***. The revocation process has started and is ongoing. • ***Revoked***. The QoS profile activation has been successfully revoked.

Request Example

```
curl -X PATCH --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  }" "https://<YOUR-BASE-URL>/rws/api/v1/qod/919"
```

Response Example

```
[
  {
     "jobid": "919",
     "status": "REVOCATION_IN_PROGRESS",
  },
  {
     "jobid": "919",
     "status": "REVOCATION_SUCCESS",
  }
]
```

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Errors

Error Code	HTTP Code	Error Message
2310000	400	Invalid jobId - Value must be a number.
2310001	400	Invalid duration - Value must be between 15 minutes and 4 hours (in seconds).
2310005	400	Invalid update operation.
2310006	409	Job not qualified for revocation.
10000001	401	Invalid credentials. Description: Control Center uses this error message when the API credentials are invalid or when the IP address is not within the allowed range.
10000011	400	One or more required fields are missing.
10000024	400	Invalid apiVersion.
20000000	404	Requested resource not found.
20000001	404	Resource not found - Invalid ICCID.
20000009	404	Resource not found - Invalid Package.
30000001	500	Unknown server error.

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Activate QoS Profile for Device

Description

Used with QoS on Demand , this API activates a QoS profile for a device.

Resource URL

PUT BaseURL/v{apiVersion}/qod

Request Parameters

Parameter	Description
apiVersion	The version number for this API. The current version for all functions is 1.
iccid	The ICCID of the device.
qosProfile	The identifier for the QoS profile that you want to activate. To view a list of available QoS profiles, run the Get Services for Account API.
duration	(Optional) The duration for the QoS profile in seconds. When the duration expires, Control Center re-activates the previous QoS profile. If you do not provide a duration, Control Center uses the duration defined in the Marketplace package.
ipAddress	(Optional) IP address of the device. This field is optional, if you leave it blank, you must enter a value for the apn parameter.
apn	(Optional) If the device has multiple APNs, specify an APN. This field is optional, if you leave it blank, you must enter a value for the ipAddress parameter.

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Response Parameters

Return Value	Description
jobId	A unique identifier for tracking the QoS profile activation request.
status	Indicates the status of the provisioning request. Possible values are: • ***Success***. Provisioning completed successfully, and the request was applied to the device. • ***Pending***. The request was received, and provisioning is in progress. • ***Failed***. Provisioning request failed for the device. The description field provides a reason for the failure. • ***Revoke***. Removes the current QoS profile from the device.
description	(Optional) Provides additional details about the provisioning process or the reason for a failure. Example: Subscriber does not exist on the network.

Request Example

```
curl -X PUT --header "Content-Type: application/json" --header "Accept:
application/json" --header "Authorization: Basic <YOUR-ENCRYPTED-
CREDENTIALS>" -d "{
  }" "https://<YOUR-BASE-URL>/rws/api/v1/qod"
```

Response Example

```
{
     "jobid": "3453453",
     "status": "ACTIVATION_IN_PROGRESS"
}
```

Errors

Error Code	HTTP Code	Error Message
400300	500	System internal error.
1700102	400	Invalid IP address.
1800218	404	Invalid UMTS or LTE apn.
2310001	400	Invalid duration - Value must be between 15 minutes and 4 hours (in seconds).
2310002	500	Error while fetching entitlement.

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Error Code	HTTP Code	Error Message
2310003	409	A job is in progress.
2310007	409	SIM not active.
10000011	400	One or more required fields are missing.
10000021	400	Invalid iccid.
20000001	404	Resource not found - Invalid ICCID.
20000009	404	Resource not found - Invalid Package.
30000001	500	Unknown server error.

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Chapter 2

SOAP APIS

SOAP API support is included with all Control Center accounts, including trial accounts. This section provides an overview of the APIs, including the development process and coding standards.

- **Function Documentation**. For reference information about individual API functions, see the online API schema. Within the Knowledge Base, go to the APIs > SOAP APIs page and click the Go to API Schema button.
- **Path/License Key**. To access the SOAP API path and license key for your account, go to the APIs > SOAP APIs page in the Knowledge Base.

Essential/Lite Differences. For information about Essential and Lite account differences, see the *Control Center User Guide*.

Roles with Access

Service provider, account, and customer users can use the APIs. However, user access is limited by the following:

- **Access Type**. The user must have an Access Type of *API Only* or *Both API and UI* in order to access the APIs. You can find that information on the User Details page: Admin > Users > *User Name* link > Access Type.
- **Role**. If a user with UI access can perform the function within the Control Center web interface, then a user with the same role and API access will be able to complete the task via an API call. There is no single role that will guarantee access for all function calls. See the API/Role matrix in the Knowledge Base for a complete list of roles and the APIs they can use.

Typically, an enterprise will create user accounts with access to both the UI and the APIs for their engineering team during development. For

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auditing purposes, Cisco recommends that production code use a dedicated Control Center user whose sole purpose is to execute API functions. When you create the user, you can specify that the user has API-only access to Control Center.

Field Restrictions. While multiple roles may have access to the same API, not every role will have access to all the fields listed in the documentation. You can verify field access for a particular role by logging into the web interface as a user with the same role who also has web interface access and then looking for the field.

AccountAPIOnly Role. This legacy role has access to Control Center through the API, but not through the web interface. The purpose of this role was to provide restricted access to Control Center for service provider partners using a different platform. We recommend that current users take advantage of the Access Type attribute to create users with API-only access to Control Center rather than creating users with the AccountAPIOnly role.

IP Address Restrictions . If an enterprise implements IP Address Restrictions, an API user might not have API access if their IP address falls within a restricted range.

Single Sign-On Restriction. APIs are not available to users who access Control Center using the Single Sign-On (SSO) authentication process. SSO users are UI-only users by default.

Terminology

Terminals. Many API functions refer to a device as a “terminal.” The terms are interchangeable.

Network Access Config. This is a legacy Control Center term for a communication plan. We continue to use the term because several long-standing APIs use the term in their function names.

For information about other terms, refer to the Glossary in the main user guide.

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Getting Started

To get started with the SOAP APIs, read through the following information:

Topic	Description
SOAP API Development Process	Describes the steps involved in API development from setting up the environment to going live in production.
SOAP API Code Standards	Explains topics such as authentication, common function parameters, scheduling, and more.
WSDL Files	Describes all the WSDL files that are available.
Best Practices	Provides guidance for creating programs that ensure effective API performance for all users.
SOAP API Error Handling	Discusses the standard SOAP Fault element.

SOAP API Development Process

Here's the overall SOAP API development process:

Step 1: Set up development environment

First, you'll need to download the WSDL and XML schema files for the API. You can get these files from the Knowledge Base: APIs > SOAP APIs > Getting Started > WSDL Files. Then, follow the tutorial instructions for your preferred coding environment (Java, C#, Perl, PHP, or Ruby). You can use any language that supports SOAP web services.

Step 2: Locate your API license key

To use any API calls, you must supply a license key. You will find both the license key and the SOAP API path on the APIs > SOAP APIs page in the Knowledge Base.

You will need separate license keys for the sandbox environment and the production environment. When you request access to the sandbox environment, Cisco sends you the sandbox path and a license key.

Customer Access. Be aware that Control Center doesn't display the API license key to users with customer roles. A customer must request the key from their enterprise.

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Step 3: Create/test code

The online technical documentation provides code samples as well as detailed function descriptions to help you create your own code. When you're ready, you can test the code in the sandbox (if a sandbox account has been created for you) or in your production account. Be aware that the sandbox and production environments are completely separate. When you use the sandbox environment, there's no way you can damage production data with your test code.

Step 4: Go live

If you are using a sandbox account to test your code, you'll need to replace the credentials, license key, and URL in your calls when you're ready to put your code into production. Be aware that you cannot migrate any data from the sandbox environment to the production environment or vice versa.

In production environments, make sure you replace the sandbox URL with the production URL. Production URLs must use the full path URL, for example:

- **Sandbox SOAP URL**: https://jpotest.jasper.com/ws/service/echo
- **Production URL**: https://api.jasper.com/ws/service/echo

Sandbox Support. Be aware that Control Center's API test environment, JPOTest, is not governed by Cisco's service-level agreement. Cisco does not guarantee continuous support for the sandbox environment, but we will do our best to keep the environment available.

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SOAP API Authentication

Control Center requires three pieces of information for SOAP API authentication: user name, password or API key, and a license key.

- **User Name/Password**. All Control Center users have a user name, but some users might not have a password, depending on the account password policies. SOAP API calls must contain the user name and can use either the password or an API key for authentication.
- **API Key.** Every user with *API Only* or *Both API and UI* access has a unique API key. Users can use either an API key or a password for authentication. Users with web interface access can generate and view their own API key on the User Profile page. Users without web interface access must request an API key from an administrator user with web interface access. Be aware that you'll need different API keys for the production and sandbox environments.
- **License Key.** Every account has one license key that is shared by all users and required for SOAP API authentication. To get your license key, open the Knowledge Base and go to the APIs > SOAP APIs page. If you access different service providers using separate instances of Control Center, you'll need a separate license key for each service provider. For more information, see *Working with Multiple Service Providers on page 410*. Be aware that the license key for the sandbox environment is different from the license key for the production environment.

The SOAP API header specifies the user name and password (or API key). Use the following header for all function calls. Do not add any other XML elements.

```
<soapenv:Header>
  <wsse:Security soapenv:mustUnderstand="1" xmlns:wsse="http://docs.oasis-
open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd">
    <wsse:UsernameToken>
      <wsse:Username>?</wsse:Username>
      <wsse:Password Type="http://docs.oasis-open.org/wss/2004/01/oasis-
200401-wss-username-token-profile-1.0#PasswordText">?</wsse:Password>
    </wsse:UsernameToken>
  </wsse:Security>
</soapenv:Header>
```

The license key appears in the body of each SOAP call. Refer to the SOAP API Schema in the Knowledge Base for details.

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SOAP API Code Standards

This section provides some high level information about how the SOAP API calls work.

Standard	Description
SOAP request and response	Each API function maps to two SOAP functions, a request that sends the input to Control Center and a response from Control Center containing the requested data. This document treats each API pair as a single entity for the sake of discussion.
Role Access	See the *SOAP API/Role Matrix on page 430* for details about which functions each user role can perform.
URL	When you use an API function, you must send the request to a service provider-specific URL. To get your URL, open the Knowledge Base and go to the APIs > SOAP APIs page. Use the HTTPS protocol for data security and do not modify the path in any way. If your company uses multiple service providers, you'll have a unique SOAP URL for each of them (see *Working with Multiple Service Providers on page 410*). Be aware that the URL for the sandbox environment is different from the URL for the production environment.
Effective Date	Some APIs have an effective date attribute that allows you to schedule the task at a future date. All dates are in UTC and future dates cannot be more than 365 days in the future. For the task to take effect immediately, pass a null value for the effective date.
Build ID	Every API response contains the build ID of the current Control Center release. This value has a maximum length of 100 characters.
Message ID/ Correlation ID	If you anticipate sending multiple API requests within a short period of time, you can use message IDs to match the function request with the corresponding response. In the request, simply send a unique string (called a message ID). Control Center will include the same string (called a correlation ID) in the response. Even if you don't intend to use this parameter, you must supply a placeholder value (for example, *fred*) in every request.

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Standard	Description
Version Number	Both the API request and response contain a version number. In the API request, the version number refers to the version of the API that you're using. Because Control Center doesn't currently use this string, the value is optional. However, we recommend using "1.0" as a best practice. In the API response, the version number refers to the current release of Control Center, for example: 6.14. Both version numbers have a maximum length of 100 characters.
Input Array	For APIs that take an array as input, there is a limit of 50 input items. The API returns results for any items in the array that have data. An error occurs only when **all** of the items in the array are invalid. For example, GetSessionInfo allows you to retrieve session information for up to 50 different ICCIDs. If you supply an array of four ICCIDs where one of the ICCIDs is invalid, Control Center will return data for the three valid ICCIDs and no error for the fourth. You would receive an error only if all four ICCIDs were invalid.
Pagination	For APIs that typically return numerous results, Control Center supports two pagination parameters: pageNumber and totalPages. You request a specific page number and the function returns an array of records from that page and a count of the total pages in the result set. Once you know the total number of pages, we recommend you request each page in sequence until you receive the entire result set. The page size is limited to 50 records, a number that you cannot change. However, if you do not specify a page number, Control Center will return only the first 1000 records and a totalPages value of -1.
Extra Elements in the Response	Over time, Cisco may add new elements to an API. All API responses will continue to conform to the published XML schema, but it's important to design your code so unexpected elements in the API response won't cause a process failure.
SSL Certificates	The Cisco API server uses an SSL certificate signed by a top tier Certificate Authority (CA) to handle HTTPS access. We recommend that you include the root certificate in your trust store (this is included by default on many platforms) and use a CA-based chain of trust, rather than hard coding the Cisco certificate in your API code. Otherwise, you will need to update your API code if the Cisco certificate changes in the future.

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WSDL Files

The following table describes the WSDL files available for Control Center's API functions. To download the files, open the Knowledge Base from within Control Center and go to APIs > SOAP APIs > Getting Started > WSDL Files.

WSDL File	Description
JasperAPI.xsd	XML Schema for all Cisco web service calls.
Terminal.wsdl	WSDL for device-related calls such as GetTerminalDetails, EditTerminal, etc.
Billing.wsdl	WSDL for usage and invoice calls such as GetTerminalUsage, GetInvoice, etc.
Sms.wsdl	WSDL for SMS calls such as GetSMSDetails, SendSMS, etc.
Echo.wsdl	WSDL for basic testing of the Cisco web service (echoes a message back to you).
Account.wsdl	WSDL for account calls such as GetAccount, GetAccountIdByAcctName, etc.
NetworkAccess.wsdl	WSDL for communication plan calls such as GetNetworkAccessConfig, EditNetworkAccessConfig, etc.
Eventplan.wsdl	WSDL for event plan calls such as GetTerminalEvents, ActivateTerminalEvent, etc.
GlobalSimTransfer.wsdl	WSDL for Global SIM transfer calls such as TransferGlobalSim, GetGlobalSimTransferStatus etc.
Order.wsdl	WSDL for SIM order calls such as UpdateOrderStatus etc.
AddRoutableMSISDN.wsdl	WSDL to request a routable MSISDN.

Header Format. The Control Center WSDL files do not contain a SOAP API header. See *SOAP API Authentication on page 395* for the recommended header format.

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SOAP API Error Handling

When a SOAP fault occurs, Control Center returns a standard Fault element. See the W3C documentation for information about the standards. Control Center uses the fields as follows. A safe maximum length for these fields is 100 characters.

Fault Element Field	Description
faultCode	Standard value defined by the W3C, such as "SOAP-ENV:Server". It is always 100 characters or less.
faultString	Control Center error code, for example "200100". See *SOAP Error Messages on page 421* for details. This string is always 100 characters or less.
faultActor	Currently unused by Control Center.
detail	An XML structure containing the error details, intended for troubleshooting purposes. See an example below. This can have an arbitrary length depending on the error. It is optional for customers to process, so they can ignore the field (or truncate at a certain number of characters) as needed.

```
<detail>
<jws:requestId xmlns:jws="http://api.jasperwireless.com/ws/schema">
          AM6Sm6OiMPqsiF2T
          </jws:requestId>
<jws:error xmlns:jws="http://api.jasperwireless.com/ws/schema">
          No terminal usage found
          </jws:error>
<jws:exception xmlns:jws="http://api.jasperwireless.com/ws/schema">
          com.jasperwireless.ws.ApiException
          </jws:exception>
<jws:message xmlns:jws="http://api.jasperwireless.com/ws/schema">
          iccid=12345
          </jws:message>
</detail>
```

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SOAP API Functions

This section lists all the Control Center SOAP API functions by category. For detailed reference information about each function, look in the Knowledge Base: APIs > SOAP APIs > SOAP API Schema.

Devices

The device APIs provide access to detailed device information, including the current session. You can also change attribute values.

API Call	Description
AssignOrUpdateIPAddress	Changes the value of a given device IPv4 address. This function allows you to manage IPv4 addresses yourself without relying on the service provider. You can assign any IPv4 address you like, as long as it falls within a given range. In particular, you can reuse IP addresses that were associated with devices that are no longer in use. To assign IPv6 addresses, you must use the Edit Device Details REST API. This API is not yet supported for devices using a 5G SA network, even if they fall back to an LTE or GPRS network.
EditNetworkAccessConfig	Changes the communication plan (Network Access Config) associated with a given device.
EditTerminal	Changes the value of a single attribute for a given device. You can specify that the change take effect immediately or schedule it for some future date.
GetModifiedTerminals	Returns a list of ICCIDs for all the devices that have been modified since a given date and time. To get a list of all devices, simply omit the "since" parameter. For deployments with many devices, use pagination parameters and the "since" parameter to avoid time-outs. You might also consider using the Subscriber Delta FTP report.
GetNetworkAccessConfig	Returns the communication plan ID (Network Access Config ID) for one or more devices.
GetSessionInfo	Returns the current session information (IP address and session start time) for one or more devices. If the specified device is not in session, no information is returned.

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API Call	Description
GetTerminalDetails	Returns detailed information for each of the devices in a list.
GetTerminalLatestNetwork Registration	Returns the same information as GetTerminalLatestRegistration, but works for both SS7 and Diameter events. This API works with devices using either the GSM/GPRS or the LTE network. This API is not yet supported for devices using a 5G SA network, even if they fall back to an LTE or GPRS network.
GetTerminalLatest Registration	Returns information about a given device's most recent SS7 network registration, including the name of the registering service provider and the global title address of the network node. For devices using the LTE network, you must use GetTerminalLatestNetworkRegistration instead. This information can help you troubleshoot network problems by identifying when and where the device last successfully attached to the network. This API is not yet supported for devices using a 5G SA network, even if they fall back to a GPRS network.
GetTerminalsByIMSI	Returns a list of ICCIDs corresponding to a given set of device IMSIs.
GetTerminalsByMsisdn	Returns a list of ICCIDs corresponding to a given set of device phone numbers (MSISDNs).
GetTerminalsBySecureSimId	Returns a list of devices corresponding to a given Secure SIM ID.
SendCancelLocation	Forces a device off the network, causing it to re-register. Commonly used to correct network registration problems, this function is currently available in Control Center through Spotlight.

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Global SIMs

The Global SIM APIs allow you to manage over-the-air SIM swaps for devices that contain global SIMs. These APIs mirror many of the functions you can perform through the Control Center user interface. To identify a device, you must use the ICCID of the SIM under the primary service provider (the Primary ICCID).

API Call	Description
CancelGlobalSimTransfer	Cancels a global SIM transfer request for a given device.
GetGlobalSimTransferStatus	Returns the status of the swap process for a specified Global SIM.
TransferGlobalSim	Transfers a SIM from one service provider to another by changing the device IMSI. You must supply the ICCID of the SIM under the primary service provider, the name of the target service provider, the name of the global alliance, and the transfer type (standard or revert). If you supply a callback URL, Cisco sends a confirmation message when the process is complete.

Related API: Use the GetTerminalLatestRegistration API to help you locate a SIM before transferring it to a different service provider. This function returns information about a given device's most recent network registration, including the name of the registering service provider and the global title address of the network node.

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Messages (SMS)

The message APIs provide historical information about messages that a particular device has sent to or received from Control Center. You can also send messages to one or more devices.

API Call	Description
GetModifiedSMS	Returns a list of IDs for the messages that have been sent by a device to Control Center or received by a device from Control Center during a specified time period.
GetSMSDetails	Returns detailed information about one or more messages exchanged between the device and Control Center.
SendBulkSMS	Sends a message to one or more devices (by SIM ID).
SendBulkSMSToMsisdn	Sends a message to one or more phone numbers (by MSISDN).
SendSMS	Sends a message to a single device (by SIM ID).
SendSMSToMsisdn	Sends a message to a single phone number (MSISDN).

Message Text Encoding

The SendSMS API supports more than a half dozen types of SMS message encoding schemes. You specify the encoding scheme in the messageTextEncoding field. This composite field defines the encoding format for the message text itself (defaults to LITERAL) as well as the encoding scheme Control Center will use when submitting the message to the SMSC.

Use the information in the following table to determine which values to use.

- **Field Value.** The value used in the messageTextEncoding field in the SMS APIs.
- **SMS Encoding.** The encoding scheme used for the message text you supply in the API. Please ensure the value supplied in the messageText field uses the encoding scheme defined in the messageTextEncoding field.
- **Control Center Encoding.** Encoding scheme Control Center uses when submitting the message to the SMSC. The service provider's SMSC must be configured to support this scheme.

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Field Value	SMS Encoding	Control Center Encoding
LITERAL	LITERAL	Default encoding scheme that depends on your service provider's network configuration.
BASE64	BASE64	Default encoding scheme that depends on your service provider's network configuration.
LITERAL_IRA	LITERAL	ASCII
BASE64_IRA	BASE64	ASCII
LITERAL_BINARY	LITERAL	8 bit encoding
BASE64_BINARY	BASE64	8 bit encoding
LITERAL_GSM7	LITERAL	GSM 7 bit encoding
BASE64_GSM7	BASE64	GSM 7 bit encoding

Be aware that only LITERAL and BASE64 encoding apply to the GetSMS API.

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Rate Plans

This set of APIs allows you to manage the rate plans connected with a particular device. You can modify a device's standard rate plan, change the plans in the device's queue, and manage any events associated with the device. These APIs do not allow you to create or edit the rate plans themselves.

API Call	Description
ActivateTerminalEvent	Assigns an event rate plan to a device. The device must have an *Activated* SIM status.
DeleteTerminalEvent	Cancels a scheduled event for a particular device. You cannot cancel an event that's in progress.
EditTerminalRating	Adds a rate plan to the top of a device's queue. If a device is using a monthly rate plan and has an *Activated* SIM state, you can change the rate plan immediately using EditTerminal.
GetAvailableEvents	Returns a list of all the event rate plans available to a given device.
GetTerminalEvents	Returns a list of all events associated with a device, including historic events, current events, and future scheduled events.
GetTerminalRating	Returns the current base rate plan and all queued rate plans for a given device.
QueueTerminalRatePlan	Adds a rate plan to the end of a device's queue. To add a plan to the beginning of the queue, use EditTerminalRating.
RemoveRatePlanFromQueue	Removes the specified rate plan from a device's queue.

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Usage

You can examine the data, message, and voice usage for an individual device with the following APIs. For information about usage across all devices and rate plans during a billing cycle, refer to the invoice API.

API Call	Description
GetTerminalUsage	Returns the data usage amount for a device during a particular billing cycle. To access the device usage for the current month, use GetTerminalDetails and examine the MonthToDateUsage field.
GetTerminalUsageDataDetails	Returns information about all the data sessions that occurred during a particular billing cycle for a given device.
GetTerminalUsageSmsDetails	Returns information about all the messages sent or received by a device during a particular billing cycle.
GetTerminalUsageVoiceDetails	Returns information about all the voice calls received or placed by a device during a particular billing cycle.

Invoices

Control Center provides a single API to access the high level details of an invoice. For more information about individual device usage during a billing cycle, refer to the usage APIs.

API Call	Description
GetInvoice	Returns invoice data for a given account and billing cycle. This data includes the Details information that appears at the top of the Billing > Invoices > Invoice Details page, but not the information that appears in the tabs across the bottom of the page. For example, you can get: • Total subscription charges • Total usage charges for data and/or SMS messages in excess of included usage • Total billable usage volume by all billable SIMs • Activation charges, should any apply • Other charges such as network services fees You can also use the Usage APIs to access device usage information for each billing cycle.

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Users

The user API allows you to create new users.

API Call	Description
CreateUser	Creates a Control Center user with the given information.

Test

A single test API allows you to test your connection to Control Center during development.

API Call	Description
Echo	Simple test call that returns a value you send as input.

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SOAP API Use Cases

This section describes several use cases or design patterns that are common across all applications that access Control Center.

Analyzing Device Usage

Using the API set you can analyze device usage on a regular basis to determine if any devices are approaching or exceeding their included usage. If so, you can move the devices to a different rate plan with a higher usage allotment and a lower fee, as compared with the subscription and overage fees in the original rate plan. The following API functions give you slightly different views into device usage:

- **GetTerminalUsage** returns the data usage of a device during a particular billing cycle. The TotalDataVolume attribute tells you how much data the device used overall. The BillableDataVolume attribute tells you how much of that data was subject to overage charges.
- **GetTerminalUsageDataDetails** returns usage information by zone or by country. You can also examine the length of each data session. GetTerminalUsageSMSDetails and GetTerminalUsageVoiceDetails will provide the same information for SMS and voice service.
- **GetTerminalDetails** provides the MonthToDateUsage for the current billing cycle. The OverageLimitReached attribute will also tell you if the device has started incurring overage charges.

If you discover that a device has used more data than you would like, you can assign a different rate plan to the device using one of these two APIs:

- **EditTerminal**. If the device is currently using a monthly rate plan, use this function to change the rate plan.
- **EditTerminalRating**. If the device is currently using a prepaid plan, you can't change the plan until the prepaid plan expires. Then you can use the device queue to assign the next rate plan. If you can anticipate which rate plan will be most appropriate, you can set up the queue in advance or you can use EditTerminalRating to dynamically add a new rate plan to the queue. When the prepaid rate plan expires, Control Center will automatically assign the rate plan at the top of the queue to the device.

In this way you can cycle through all your devices and adjust their rate plans, if necessary.

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Automation Rules. For certain use cases, you might consider using an automation rule in Control Center (Automation > Rules > Actions menu > Create New button). These rules offer a variety of ways to monitor usage, manage subscriptions, and provision SIMs—without any programming.

Synchronizing Databases

Many enterprises maintain an internal database, separate from Control Center, with device attributes, usage, and message information. The API set gives you an easy way to synchronize your internal systems with Control Center. Here's the process:

For Device Attributes

1. Use **GetModifiedTerminals** to return a list of devices whose information has changed since a specified date.
2. Use **GetTerminalDetails** to return the detailed information for just the devices returned in step 1.
3. Update your internal database with the data returned in step 2.

For Messages

1. Use **GetModifiedSMS** to return a list of messages that have been sent or received during a certain time period.
2. Use **GetSMSDetails** to return the detailed information for just the messages returned in step 1.
3. Update your internal database with the data returned in step 2.

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Working with Multiple Service Providers

Many companies with connected devices contract with more than one service provider. As long as all these service providers use the Cisco platform, you can create a single application that can access and manage information in any service provider’s instance of Control Center. All you need to do is store log-in parameters for each service provider instance and then use the appropriate set of parameters when necessary.

Understanding Log-in Credentials

In Control Center every account, representing an enterprise, has one and only one service provider as a parent. Each account contains a collection of devices as well as a set of users who can view and manipulate these devices. When a user logs in, Control Center knows which account, service provider, and devices are associated with the user and limits the user's activities accordingly.

A global enterprise with a connected device deployment that spans multiple service providers will have multiple Control Center accounts and, thus, multiple user log-ins within the web application. Since web services use the same security infrastructure as the web application, each API request must include a unique set of user log-in credentials for the corresponding service provider.

For each service provider, you'll need the following information:

- **URL**. Each service provider has a unique URL for their instance of Control Center.
- **Account Credentials**. Your company will have a unique account with each service provider. For each account you must supply a username and password for a user with the appropriate privileges.
- **License Key**. You must use a different license key to access the data in each service provider's instance of Control Center.

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Code Sample

The code sample below shows an API request for the GetModifiedTerminals web service, with the credentials and custom information shown in red. You must send the SOAP request to the appropriate API URL for the service provider you want to access. For example: https://<OperatorApiHost>/ws/service/terminal

```
<soapenv:Envelope xmlns:sch="http://api.jasperwireless.com/ws/schema"
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
 <soapenv:Header>
  <wsse:Security soapenv:mustUnderstand="1" xmlns:wsse="http://docs.oasis-
open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd">
   <wsse:UsernameToken wsu:Id="UsernameToken-16847597"
xmlns:wsu="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-
utility-1.0.xsd">
    <wsse:Username>UserName</wsse:Username>
    <wsse:Password Type="http://docs.oasis-open.org/wss/2004/01/oasis-200401-
wss-username-token-profile-1.0#PasswordText">Password</wsse:Password>
   </wsse:UsernameToken>
  </wsse:Security>
 </soapenv:Header>
 <soapenv:Body>
  <sch:GetModifiedTerminalsRequest>
   <sch:messageId>?</sch:messageId>
   <sch:version>?</sch:version>
   <sch:licenseKey>xxxxxxx-xxxx-xxxx-xxxx-xxxx</sch:licenseKey>
   <sch:since>2009-09-01T00:00:00Z</sch:since>
  </sch:GetModifiedTerminalsRequest>
 </soapenv:Body>
</soapenv:Envelope>
```

Storing Log-in Credentials

Global enterprises can build Control Center API access from a single code base that works across service providers by storing API request parameters for each service provider in a table structure. The service provider table is accessible to the calling logic, which uses the information in the table to build the web service requests for each service provider and then processes the results. This data-driven approach is extensible and grows with you as you expand your IoT footprint without requiring additional code.

The following example shows a logical view of how a single customer (Acme Widgets) served by two service providers could design a single solution to:

1. Find all devices that have been updated since their last check (on 09/01/2009).
2. Retrieve current status details for each device with updates and apply those changes to their internal device tracking application.

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The following table contains information for each service provider serving Acme Widgets through the Control Center platform.

	Service Provider 1	Service Provider 2
URL	api.operator1.com	api.operator2.com
Username	acmeOp1User	acmeOp2User
Password	acmeOp1Password	acmeOp2Password
License Key	c0b244928-2bc9-57d6-1111	b0a877928-de19-75a67-2222

API Logic

For each service provider, the code must:

- Retrieve the URL, log-in credentials, and license key for the enterprise's account with the service provider.
- Construct the appropriate GetModifiedTerminalsRequest call using the information.
- Submit the GetModifiedTerminalsRequest to the service provider's API URL.
- Receive the resulting list of modified ICCIDs from GetModifiedTerminalsResponse.
- For each modified ICCID returned in GetModifiedTerminalsResponse, pass the ICCID to GetTerminalDetailsRequest and then store the returned device attributes in the local device tracking system.

If the enterprise expands into a new geography, the standard API logic above works without modification as long as you insert a new row into the credential table for the service provider serving the new region.

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Tutorials

This section contains instructions and tips for using various coding languages with the Control Center APIs.

Java

Follow these steps to begin using Java:

1. Install the Java(TM) Development Kit 1.5+.
2. Install Maven.
3. Download the javaSample.zip file from the Knowledge Base: Home > APIs > SOAP APIs > Tutorials > Java. Extract the files in any directory.
4. Open a command prompt at that directory. Type the following to install xmldsig.jar into your local Maven repository (use "/" instead of "\" if you're on Linux or Mac):

   mvn install:install-file -Dfile=lib\xmldsig.jar -DgroupId=javax.xml.crypto -DartifactId=xmldsig -Dversion=1.0 -Dpackaging=jar
5. Then type the following to compile and package the sample files:
   mvn clean package assembly:single
6. Type the following to call Echo API (use "/" instead of "\" if you're on Linux or Mac):
   java -cp target\Sample-1.0-jar-with-dependencies.jar com.jasperwireless.ws.client.sample.EchoClient <licenseKey>
7. Type the following to call the GetTerminalDetails API for the given ICCID (use "/" instead of "\" if you're on Linux or Mac):
   java -cp target\Sample-1.0-jar-with-dependencies.jar com.jasperwireless.ws.client.sample.GetTerminalDetailsClient <licenseKey> <username> <password> <iccid>
8. (Optional) If you need to pass binary text (for SMS messages, for example), use the following code for Base64 encoding/decoding.

Encode	`Base64 base64 = new Base64();`
	`byte[] encodedMsg = base64.encode`
	`(smsMessage.getMessageText().getBytes());`

Decode	`byte[] decodedMsg = base64.decode(msgTxt.getBytes());`
	`if (decodedMsg != null) {`
	`msgTxt = new String(decodedMsg);}`

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Notes

- Request/Response XML strings are printed to System.out for debugging purposes.
- The SAAJ library is used to handle SOAP calls.
- The XWSS library is used to secure the message with username/password token (which requires the file securityPolicy.xml).
- You can use an XML binding library (such as JAXB) to further simplify your development process.

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C#

Follow these steps to begin using C#:

1. Right click on the project in the "Solution Explorer" window and choose "Add Web Reference ..." In Visual Studio® 2008, right click on the project in the "Solution Explorer" window, choose "Add Service Reference ...", then click "Advanced...", then click "Add Web Reference..."
2. Input the correct URL for WSDL file, for example http://apitest.jasperwireless.com/ws/schema/Terminal.wsdl. See *WSDL Files on page 398* for the production URL.
3. Add the file SecurityHeader.cs into your project to override the generated proxy and add the XWSS security header to every request.
4. Go to the class TerminalService in the newly-created web reference (for example, Reference.cs). You can do this by going to Object Browser, searching for "TerminalService", and double-clicking on the TerminalService class.
5. Add the following line above the namespace declaration at the top of Reference.cs:
   ```
   using com.jaspersystems.api;
   ```
6. Add this public member to every service class in the auto generated Web Reference proxy, say in class TerminalService:
   ```
   public SecurityHeader securityHeader;
   ```
7. Add the following line to every method that will be called in the auto generated Web Reference proxy, say in TerminalService just above public GetModifiedTerminalsResponse GetModifiedTerminals(...):
   ```
   [System.Web.Services.Protocols.SoapHeaderAttribute("securityHeader")]
   ```
8. Follow the sample GetModifiedTerminals.cs to make the API call.

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9. (Optional) If you need to pass binary text (for SMS messages, for example), use the following code for Base64 encoding/decoding.

Encode	`byte[] inArray = new byte[msgTxt.Length];`
	`int x;`
	`for(x=0; x < msgTxt.Length; x++) {`
	`  inArray[x] = (byte)charArray[x];`
	`}`
	`string encodedString = Convert.ToBase64String`
	`(inArray,0,inArray.Length,Base64FormattingOptions.None);`

Decode	`byte[] outArray = Convert.FromBase64String(respMsg);`
	`char[] outCharArray =new char[outArray.Length];`
	`for (x = 0; x < outArray.Length; x++)`
	`{`
	`  outCharArray[x] = (char)outArray[x];`
	`}`
	`string decodedString = new string(outCharArray);`

Sample Files

You can download the sample files from the Knowledge Base: Home > APIs > SOAP APIs > Tutorials > C# > Sample Files.

- GetModifiedTerminals.cs
- SecurityHeader.cs

Troubleshooting

When there is a SOAP exception, Visual Studio displays the error code only. To see the detailed message, click on "View Detail...", expand SOAPException > Detail, and take a look at the InnerText field.

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Perl

Follow these steps to begin using Perl:

1. Install the following modules:
   - SOAP::Lite: simple SOAP library
     For example, if you're using ActivePerl on Windows, use the following command:
     ppm install http://theoryx5.uwinnipeg.ca/ppms/SOAP-Lite.ppd
   - Crypt::SSLeay: HTTPS support
     For example, if you're using ActivePerl on Windows, use the following command:
     ppm install http://theoryx5.uwinnipeg.ca/ppms/Crypt-SSLeay.pp
2. Download the sample files below.
3. Open the echo.pl sample file in a text editor and put in your own license key. Then type perl echo.pl to call the Echo API.
4. Open the getTerminalDetails.pl sample file in a text editor and put in your own license key, userName, password, and ICCIDs. Then type perl getTerminalDetails.pl to call the GetTerminalDetails API.

   Note: The SOAP::Lite debug messages are enabled in the sample files, so that you can see the request/response XML. Once you are comfortable with the calls, you may want to turn this off by replacing SOAP::Lite +trace => ['debug']; with SOAP::Lite;
5. (Optional) If you need to pass binary text (for SMS messages, for example), use the following code for Base64 encoding/decoding.

Encode	`use MIME::Base64;` `my $encoded = encode_base64('Hello World');`
Decode	`my $decoded = decode_base64($encoded);`

Sample Files

You can download the sample files from the Knowledge Base: Home > APIs > SOAP APIs > Tutorials > Perl > Sample Files.

- echo.pl
- getTerminalDetails.pl

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PHP

Follow these steps to begin using PHP:

1. Install the following libraries:
   - NuSOAP: simple SOAP library
   - PHP cURL: HTTPS support
2. Download the sample files below.
3. Open the echo.php sample file in a text editor and put in your own license key. To call the Echo API, type `php echo.php`. Be aware that nusoap 0.9.5 shows some warnings at runtime. These are harmless and can be ignored.
4. Open the getTerminalDetails.php sample file in a text editor and put in your own license key, userName, password, and ICCIDs. To call the GetTerminalDetails API, type `php getTerminalDetails.php`.
5. (Optional) If you need to pass binary text (for SMS messages, for example), use the following code for Base64 encoding/decoding.

Encode	`$encoded = base64_encode('Hello World');`
Decode	`$decoded = base64_decode($encoded);`

Sample Files

You can download the sample files from the Knowledge Base: Home > APIs > SOAP APIs > Tutorials > PHP > Sample Files.

- echo.php
- getTerminalDetails.php

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Ruby

Follow these steps to begin using Ruby:

1. Install the Savon SOAP client. Type gem install savon to install via Ruby Gems.
2. Download the sample files below.
3. Open the echo.rb sample file in a text editor and put in your own license key. To call the Echo API, type ruby echo.rb.
4. Open the getTerminalDetails.rb sample file in a text editor and put in your own license key, userName, password, and ICCIDs. To call the GetTerminalDetails API, type ruby getTerminalDetails.rb.
5. (Optional) If you need to pass binary text (for SMS messages, for example), use the following code for Base64 encoding/decoding.

Encode	`require 'base64'` `encoded = Base64.encode64('Hello World');`
Decode	`decoded = Base64.decode64(encoded);`

Sample Files

You can download the sample files from the Knowledge Base: Home > APIs > SOAP APIs > Tutorials > Ruby > Sample Files.

- echo.rb
- getTerminalDetails.rb

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XML Samples

The online API schema contains XML samples for some commonly used function calls:

- GetTerminalDetails
- GetTerminalUsage
- GetInvoice
- EditTerminal (change the SIM status)
- EditTerminal (change the rate plan)
- GetModifiedTerminals
- GetTerminalUsageDataDetails
- UpdateSecureSimCredentials

You can view the samples in the Knowledge Base: Home > APIs > SOAP APIs > XML Samples.

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SOAP Error Messages

Error Code	Message
General	
10000	Wildcard search is too general
10001	Maximum size exceeded
Devices	
100000	Device error
100100	Device not found. The ICCID doesn't exist in the caller's account. This type of error can occur when you try to use a sandbox account to access a production device.
100101	Invalid ICCID
100200	Invalid device change type
100300	Invalid edit terminal request
100400	Invalid device(s) specified
100401	Invalid device ID specified
100500	Offer not found
100501	Invalid offer
100510	Campaign not found
100520	Invalid rate plan
100800	Effective Date must be within 365 days from today
Billing	
200000	Billing error
200100	Invoice does not exist
200200	No device usage found
Admin	
300000	Admin error
300100	Invalid Qcode
300200	Invalid account name
300201	Account name not unique
300300	Invalid email

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Error Code	Message
300301	Email not unique
API	
400000	API error
400100	Invalid license key. Please use a valid API license key. Make sure you're using the correct license key for the current environment. Sometimes people mistakenly use their sandbox license key on the production site or vice versa.
400101	License has exceeded the rate limit for API calls.
400200	Security validation error. Your username or password is incorrect. Make sure you're using the right credentials for the current environment. Sometimes people mistakenly use their apitest credentials on the production site or vice versa.
400201	Access violation error. Please check your login; you may be trying to access a device that belongs to a different account or service provider. This error can happen if the caller has accounts with multiple service providers and they're using the login from the wrong service provider.
400300	System internal error
400400	Too many entries
400500	Invalid ICCID
SMS	
500000	SMS error
500100	From Date greater than To Date
500200	Too many entries
500300	SMS message not found
500400	Message text null
500500	Message text greater than 160 characters
500600	Send message failure
500700	Device not active
500800	Decoding error

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Error Code	Message
500900	Decoded message text null
Network Access Config (also called a Communication Plan)	
800000	Network Access Config not found
800100	Invalid edit Network Access Config request
Messaging	
900100	No active session
900101	Protocol not supported
900102	Invalid port
900103	Communication failed
900104	Communication timeout
Accounts	
1000100	Account not found
1000400	Too many accounts
1000500	Service provider not found
1000501	Could not create account
1000502	Internal Error - default trial communication plan not found for service provider
1000503	Internal Error - default trial rate plan not found for service provider
1000504	Account already exists
1000505	Invalid region ID
1000506	Default communication plan not found for service provider
1000507	Service Provider Account ID is not unique; multiple accounts found.
1000508	Cannot transfer device to this account
1000509	Cannot transfer device to the same account
1000510	Cannot transfer device when one of the accounts is an inventory account
Edit Device Rating	
1100100	Edit device rating error
1100200	Term end date must be greater than term start date

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Error Code	Message
1100300	Renewal rate plan is blank
1100400	Renewal rate plan is not valid for the device
1100500	Renewal mode is not valid
1100600	Access denied. Cannot edit term start and end dates
1100700	Failed to remove rate plan from queue
1110100	Add device renewal error
LinePayStatus	
1120100	Edit LinePayStatus error
1120101	Invalid LinePayStatus parameter
1120102	Invalid operation for SimStatus is Purged
1120104	No terminals associated with Service Provider Account ID
Event Plans	
1200100	Invalid event name
1200200	Event instance not found
1200300	Event instance started already
Connection Manager	
1300100	Configuration file not found
1300200	Failed to upload diagnostics data
1300300	Invalid sequence number
Users	
1400100	Try a different user name.
1400101	Customer not found
1400102	Role not found
1400103	Could not create user
1400104	Control Center uses this error message for an invalid password and an invalid IP address: • The password is invalid. • IP address restriction failure.
1400105	Invalid authentication token
1400106	Not authorized to create this role
1400107	Peered user information must be specified

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Error Code	Message
1400108	Account not peered
1400109	User not found
1400110	Could not delete user
1400111	Could not reset user password
Passwords	
1400133	Password should not contain user name
1400134	Password contains consecutively repeated alphanumeric characters that exceeds the maximum
1400135	Password contains alphanumeric sequence that exceeds the maximum
1401001	Illegal characters in password
1401002	Password length less than the required minimum
1401003	Password length more than the maximum
1401004	Contains fewer alphabetic characters than the required minimum
1401005	Contains more alphabetic characters than the maximum
1401006	Contains fewer numeric characters than the required minimum
1401007	Contains more numeric characters than the maximum
1401008	Contains fewer special characters than the required minimum
1401009	Contains more special characters than the maximum
1401010	Password matches one of the previous passwords
1401011	Contains fewer uppercase characters than the required minimum
Vault	
1500100	IMSI not found
1500200	Start Date is not before End Date
Rate Plans	
1600100	Invalid rate plan or not a wholesale rate plan
Communication Plans	
1700100	Invalid communication plan

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Error Code	Message
1700101	Invalid APN
1700102	Invalid IP address
Global SIM	
1800100	Unknown
1800101	Global SIM not found
1800102	Global partner not found
1800103	Invalid GlobalSimTransferId in the request
1800104	Invalid transfer request. Primary ICCID hasn't been transferred to any partner.
1800105	ICCID already has a global transfer pending.
1800106	ICCID doesn't have any pending Global Transfer request.
1800107	Global Alliance not found.
1800108	Target service provider doesn't exist.
1800109	Target Global SIM subscription information not found.
1800110	Account mapping not defined for Global SIM transfer.
1800111	Global SIM is in invalid state for Global SIM transfer.
1800112	Target subscription is already in use.
1800113	G&D IMSI swap failed for the Global SIM.
1800114	Cisco IMSI swap failed for the Global SIM.
1800115	Global SIM not live
1800116	Global SIM transfer can't be cancelled for the SIM at this stage.
1800117	Global SIM target subscription partially in use.
1800118	Global SIM target IMSI already in use.
1800119	Target SIM not present in expected account.
1800120	Subscription already associated with a chip.
1800121	No free slots found for chip.
1800122	Selected slot is not found for chip.
1800123	Selected slot is occupied for chip.
1800124	The provided login session is not valid.
1800125	The login session has expired, please re-authenticate.

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Error Code	Message
1800126	The user is not authorized to perform the operation.
1800127	The user could not be found.
1800128	The provided password is not valid.
1800129	The user is blocked.
1800130	The user is expired.
1800131	Too many failed login attempts, user is now blocked.
1800132	Invalid request, duplicate messageId used.
1800133	Global SIM has always roaming reimsi rule associated with it.
1800134	Target Global SIM subscription mapping cannot be created.
1800135	Global SIM transfer type not specified.
1800136	Global SIM has already been transferred.
1800138	No suitable target subscription found in the inventory pool of the target account.
1800139	SIM Profile with given External Profile ID does not exist.
1800140	Target SIM does not exist.
1800141	Source SIM cannot be prepared for Global SIM transfer.
1800142	Source SIM cannot be un-prepared on failed Global SIM transfer.
1800143	Global SIM Profile mapping does not exist.
1800144	Target Global SIM subscription mapping does not exist.
1800145	The SIM is not of a valid Global SIM type.
1800146	Target SIM cannot be prepared for reverse Global SIM transfer.
1800147	Target SIM cannot be un-prepared on failed reverse Global SIM transfer.
1800148	Read timeout while invoking transfer scheduling API.
1800149	Connection timeout while invoking transfer scheduling API.
1800150	Error while invoking transfer scheduling API.
1800160	Error in changing SIM State.
1800161	Target SIM cannot be created for Global SIM transfer.

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Error Code	Message
1800162	Source SIM cannot be deactivated for Global SIM transfer.
1800163	Target SIM cannot be activated for Global SIM transfer.
1800164	Global SIM is in invalid state for reverse Global SIM transfer.
1800165	Target SIM creation cannot be un-done on failed Global SIM transfer.
1800166	Global SIM transfer can't be resumed because it has already been scheduled for cancel.
1800167	Invalid cancel request. ICCID has no FAILED transfer associated with it.
1800168	G&D subscription cleanup failed for the Global SIM.
1800169	Internal configuration error in setting up the API invocation, please contact customer support.
1800170	The OTA Algorithm chosen for this partner is not supported. Please choose a different partner that supports G&D OTA algorithm.
1800171	Global SIM Transfer to Agency partners is not supported for now.
1800172	Global SIM target MSISDN assignment failed.
1800173	Can not get the acctBillSetting according to the ICCID.
1800174	Partner MSISDN Assignment API invocation limit reached.
1800175	Partner MSISDN Assignment API invocation failed.
1800176	G&D API authentication error: failed to verify password.
1800177	Could not update MSISDN on the Subscription, either IMSI is wrong or Subscription is locked due to Subscription Management activity.
1800178	Target Subscription is already associated with this chip.
1800179	Card Profile for the chip is not allowed by profile permit list associated with the subscription.
1800180	Could not reserve the virtual subscription for a slot of the chip.
1800181	There is already an ongoing transaction for the chip.
1800182	Unhandled exception in G&D server.
1800183	Either IMSI or CustomerId is wrong, or IMSI is not associated with a Chip

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Error Code	Message
1800184	Not possible to perform the operation on the subscription due to its wrong state.
1800185	Target virtual-subscription does not exist in G&D.
1800186	The target virtual-subscription either does not exist in G&D or cannot be imported from partner server.
1800187	The primary subscription does not exist in G&D.
1800188	The Account Custom Field or Service Provider account ID should not be null while making SolicitarSWAP request.
1800189	The request to the Target service provider's NMS for MSISDN assignment was not successful, and the swap has Failed.
1800190	The eUICC ID for the SIM is not found.
1800191	The request to the Target service provider's NMS for MSISDN assignment timed out, and the swap has Failed.
1800195	The G&D API invocation failed.
1800601	The SIM has a fixed IP Address.
1800211	The SIM is in an invalid state for InterSP Transfer.
1800215	There is no inventory account associate with the target SIM service provider.

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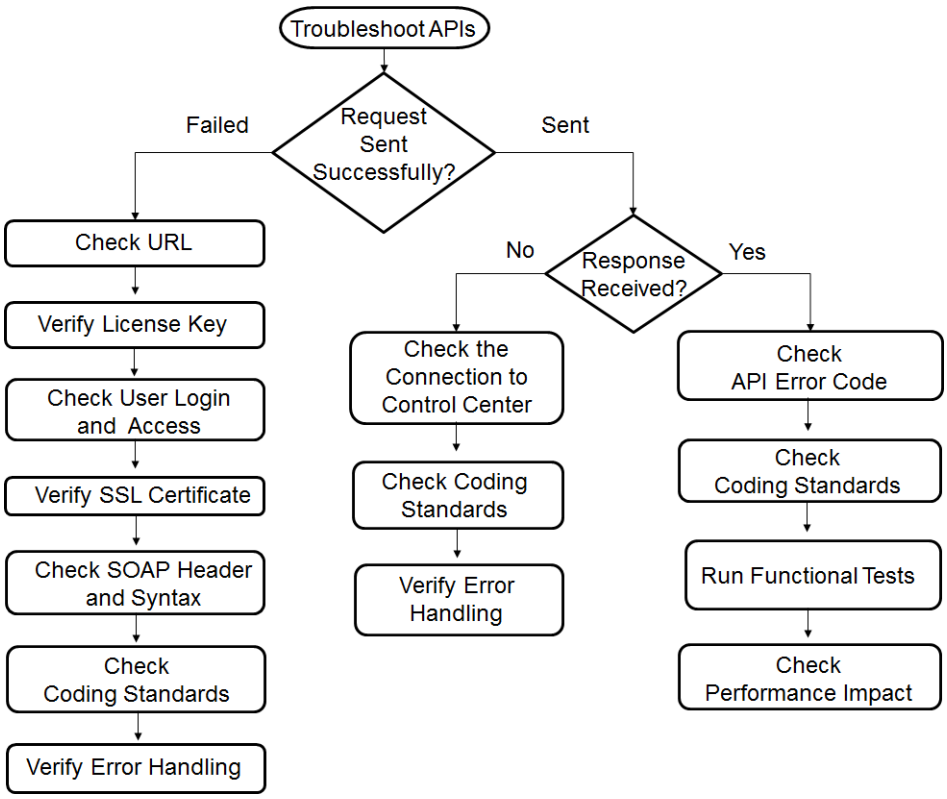
SOAP API/Role Matrix

Control Center provides different API access for different user roles. For details, take a look at the API/Role matrix. You can download the matrix from the Downloads page in the Knowledge Base. The SOAP APIs appear on the second page.

SOAP API Troubleshooting Checklist

You can quickly troubleshoot API issues and identify the main points of failure by answering the questions in the following checklist. If you answer *Yes*, continue to the next question, unless the instructions tell you otherwise. If you answer *No* to a question, you've found the problem. Follow the instructions for correcting the issue.

Sandbox APIs. When troubleshooting API issues, keep in mind that the sandbox APIs do not migrate to the production environment.



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API Setup and Access

1. **Is the URL correct?**

   Make sure you're using the correct URL with the HTTPS protocol. You can find the API server URL on the APIs > SOAP APIs page in the Knowledge Base. In production environments, make sure you replace the sandbox URL with the production URL. Production URLs must use the full path URL, for example:

   https://api.jasper.com/ws/service/echo

2. **Is the API license key valid?**

   Each account has one license key that you can find on the APIs > SOAP APIs page in the Knowledge Base. You will need to use a separate license key for the sandbox environment and the production environment. When you get access to the sandbox, you receive a license key. See *SOAP API Code Standards on page 396*.

3. **Does the API user have login access?**

   Each time you use an API function call, you must supply a user name for a user who is allowed to perform that function. You may need multiple user roles as there is no single role that will guarantee access for all API function calls. Cisco recommends creating a dedicated user or group of users whose sole purpose is to execute API functions. For more information about user roles and access, see *SOAP API/Role Matrix on the previous page*.

4. **Is the password/API key valid?**

   In addition to the user name and license key, every API call must have either a password or an API key for authentication purposes. Ensure that you're using the correct value.

5. **Does the API user have feature access?**

   If the API user has access to the web interface, log into Control Center and try to access the same feature in the web interface. (Remember the Control Center password is different from the API key.) If you can make the feature work in the web interface, you can make it work via API.

6. **Is the SSL certificate valid and included in your trusted store?**

   The Cisco API server uses an SSL certificate signed by a top tier Certificate Authority (CA) to handle HTTPS access. We recommend that you include the root certificate in your trust store (this is included by default on many platforms) and use a CA-based chain of trust, rather than hard coding the Cisco certificate in your API code. For more information see, *SOAP API Code Standards on page 396*.

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7. **Do the SOAP tags in the request specify valid definitions?**

   Verify that the SOAP envelope tag specifies the Cisco schema. Also, make sure the SOAP header tag contains the correct username and password definitions. The correct license key should be included in the SOAP body tag. A separate license key is used for the sandbox environment and the production environments. The body tag should also include any method calls and parameter values. For more examples of valid XML used in common API calls, see *XML Samples on page 420*.

```
<!-- Description of API call -->
<soapenv:Envelope xmlns:sch="http://api.jasperwireless.com/ws/schema"
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header>
    <wsse:Security soapenv:mustUnderstand="1" xmlns:wsse="http://docs.oasis-
open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd">
      <wsse:UsernameToken wsu:Id="UsernameToken-16847597"
xmlns:wsu="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-
utility-1.0.xsd">
        <wsse:Username>username</wsse:Username>
        <wsse:Password Type="http://docs.oasis-open.org/wss/2004/01/oasis-
200401-wss-username-token-profile-1.0#PasswordText">password</wsse:Password>
      </wsse:UsernameToken>
    </wsse:Security>
  </soapenv:Header>
  <soapenv:Body>
    <sch:API call>
      <sch:messageId>?</sch:messageId>
      <sch:version>?</sch:version>
      <sch:licenseKey>xxxxxxx-xxxx-xxxx-xxxx-xxxx</sch:licenseKey>
      <sch:iccid>device ID</sch:iccid>
    </sch:API call>
  </soapenv:Body>
</soapenv:Envelope>
```

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8. **Does the API request coding follow the code standards and best practices?**

   When coding with Cisco APIs, we recommend that you follow these standards:

   - Limit the number of active calls to one at a time and avoid concurrent API processing.
   - Do not exceed the calls per second limit.
   - Use pagination controls to limit the number of returned records.
   - Use UTC date and time formats. Pass a null value for a scheduled task to take effect immediately.
   - Specify a message ID in each request. Control Center includes the message ID as the correlation ID in the response. You can use these IDs to match the function requests with the corresponding responses.
   - Limit input arrays to 50 input items. Be aware that an error occurs only when every item in the array is invalid. For example, if you supply an array of four ICCIDs where one of the ICCIDs is invalid, Control Center will return data for the three valid ICCIDs and no error for the fourth.
   - Be aware that the version number in the request is optional and refers to the version of the API. The maximum length is 100 characters.

   For more details, see *SOAP API Code Standards on page 396* and *API Policies and Best Practices on page 9*. If you receive an error code related to the API request, see *SOAP Error Messages on page 421* for a complete list of error descriptions.

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9. **Does the API response coding follow the code standards and best practices?**

   After checking your network connection to Control Center, review your receiver code and error handling.

   - Design your code to allow for additional unexpected items in the response. Cisco may add additional return values to the APIs.
   - Be aware that the build ID in the response indicates the current Control Center build number. This value has a maximum length of 100 characters.
   - Be aware that the version number in the response indicates the current version of Control Center. The maximum length is 100 characters.
   - Be aware that Control Center includes the message ID as the correlation ID in the response. You can use these IDs to match the function requests with the corresponding responses.

   For more information see, *SOAP API Code Standards on page 396*. If you receive an error code related to the API response, see *SOAP Error Messages on page 421* for a complete list of error descriptions.

10. **Is the API code able to process standard SOAP fault elements from Control Center?**

    When a SOAP fault occurs, Control Center returns a detail element in the response body element that contains the standard fault elements (faultCode, faultString, and faultActor). The maximum length for these fields is 100 characters.

Fault Element Field	Description
faultCode	Standard value defined by the W3C, such as "SOAP-ENV:Server". It is always 100 characters or less.
faultString	Control Center error code, for example "200100". See *SOAP Error Messages on page 421* for details. This string is always 100 characters or less.
faultActor	Currently unused by Control Center.
detail	An XML structure containing the error details, intended for troubleshooting purposes. See an example below. This can have an arbitrary length depending on the error. It is optional for customers to process, so they can ignore the field (or truncate at a certain number of characters) as needed.

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```
<detail>
  <jws:requestId xmlns:jws="http://api.jasperwireless.com/ws/schema">
    AM6Sm6OiMPqsiF2T
  </jws:requestId>
  <jws:error xmlns:jws="http://api.jasperwireless.com/ws/schema">
    No terminal usage found
  </jws:error>
  <jws:exception xmlns:jws="http://api.jasperwireless.com/ws/schema">
    com.jasperwireless.ws.ApiException
  </jws:exception>
  <jws:message xmlns:jws="http://api.jasperwireless.com/ws/schema">
    iccid=12345
  </jws:message>
</detail>
```

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11. **Did the API call complete without an error code related to API access?**

    Control Center supports a wide variety of error codes to help clarify function call usage. The most common access-related errors are shown below.

Error Code	Description/Corrective Actions
100100	**Terminal not found.** The requested ICCID does not exist. Make sure the API call environment has access to the device.
400100	**Invalid license key.** Sometimes API users enter the sandbox license on the production site or vice versa. Verify the API license key.
400101	**License has exceeded the rate limit for API calls.** This error indicates that your account exceeded calls per second limit. Cisco may curtail API usage by returning error responses for any subsequent API calls within the measurement period (one second).
400200	**Security validation error.** This error commonly occurs when an API call uses the sandbox credentials on the production site or vice versa. Verify the correct username and password for the environment. For C# API calls, this error indicates an incorrect or missing SOAP security header. Add the SecurityHeader.cs file to your project and add SoapHeaderAttribute ("securityHeader") to every method that is called.
400201	**Access violation error.** This error typically occurs when the API user does not have access to the device or device attribute, such as a communication plan. Verify user access to the device or attribute. If the caller has accounts with multiple service providers, verify the login is from the right service provider. If the error persists, check the XML syntax with an XML validation tool.
1000504	**Error creating account <account_name>.** The entered account name already exists. Enter a different account name.

For a complete list of error descriptions, see *SOAP Error Messages on page 421*.

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12. **Did the API call return the expected results?**

    You can use a web services testing tool, such as SOAP UI, to set up and run API function tests and test suites to verify variable values at different stages of the API execution. Also, keep in mind that the role of the API user determines the features and data that the API call can access and return. If you receive an error code related to the returned data, see *SOAP Error Messages on page 421* for a complete list of error descriptions.

13. **Did the API call return data with the expected performance?**

    Follow these standards to improve Control Center API performance.

    - Limit the number of active calls to one at a time and avoid concurrent API processing.
    - Do not exceed the rate limit for API calls. Otherwise, you may receive error code 400101: License has exceeded the rate limit for API calls.
    - Align the timing of your API calls with the frequency of your service provider's record updates. For example, if your service provider updates their usage information once per 6 hours, there's no need to request the information via API more often than once per 6 hours.
    - Be aware that the API calls are not intended to provide real-time monitoring of service usage. For more immediate feedback when certain conditions occur, use automation rules and push APIs.

    You can poll API calls that return small amounts of data, such as GetModifiedTerminals, more frequently (once per minute, if necessary) without affecting the overall performance of Control Center. For more details about API performance recommendations and coding practices, see *SOAP API Code Standards on page 396*.

If you haven't been able to identify the cause of the API issue, contact Cisco Product Support.

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Chapter 3

PUSH APIS

Control Center also maintains a “push API” system that sends a programmatic notification to your application whenever a certain event occurs. For example, you might choose to receive notification when a device nears its data limit. Or, you might want to know if a device makes too many connections during a certain period of time.

To receive a notification, you must create an automation rule in Control Center and then configure it to send an API message. For this process to work, you need to set up a push receiver, or server, on your end to listen for messages, interpret the data, and then act on it appropriately.

While the majority of push APIs are associated with automation rules, Control Center also provides push APIs that allow service providers to automate SIM ordering and add-on package ordering. In addition, some REST APIs use push APIs to deliver asynchronous responses, also known as callbacks. See *Asynchronous Responses on page 45*.

Enabling Push APIs

Without Simplified Packaging, an enterprise must purchase an automation add-on package (Essential Automation or Advanced Automation) in order to use push APIs. For some push APIs, the enterprise must purchase another add-on package as well. For example, to use Location Based Services push API, an enterprise must also purchase the Location Based Services package.

Service providers always have access to the non-automation rule push APIs. Service providers also have access to the rule push APIs when creating rules for Advantage accounts, regardless of whether the accounts have purchased an automation package. However, service providers cannot create rules (or use rule push APIs) for Essential accounts unless the accounts purchase the required add-on packages. There are two exceptions to this restriction: service providers can create SIM Account Change and Available Static IP Address rules for any Essential account.

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Typical Event Sequence

Here is the typical sequence of events for push APIs:

Step 1: You Define an Automation Rule

First, you create an automation rule with the push API action. Using the rule creation pop-up (Automation > Rules >Actions menu > Create New button), find the rule that monitors the activity you're interested in. For example, you may want to receive notification when an activated device using a particular rate plan has failed to make a connection for more than 24 hours. Then, you instruct Control Center to send notification via the push API.

Depending on the URL you enter, Control Center will send the notification using either the HTTP or HTTPS protocol. For increased security, we recommend using HTTPS in the URL.

Step 2: Control Center Sends a Push API

When all the conditions of your automation rule are met, Control Center sends a message via push API to the server whose URL you specified in the rule settings. If a time-out occurs or your server returns a status code other than 200, Control Center will wait a few minutes and resend the message. Control Center will retry each notification a maximum of 20 times (with exponential back-off between attempts) before giving up. Be aware that the Cisco operations team may reduce the maximum number of retries during peak loads or error conditions in order to protect the network.

Step 3: You Verify the Signature (Optional)

You can add another level of security to the communication by implementing a Push API Shared Secret. In your account profile (Admin > Account Profile > Edit button), you can specify a string that Control Center will encrypt and include in the message. When you receive a message, you decrypt the signature and verify that it matches the original value. See *Implementing a Push Receiver on the facing page* for detailed instructions.

Step 4: You Process the Data

Then, you can parse the data in the message and take the appropriate action. In some cases that action might involve using the pull APIs to change values in the Control Center database. For the connectivity example, you might deactivate the device until a technician can diagnose the issue.

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Implementing a Push Receiver

To receive push API notifications, you must implement a simple receiver (for example, a Java servlet) that understands HTTPS or HTTP POST requests — depending on which protocol you specify in the URL.

Push API Notification Format

Each push API notification is an HTTPS/HTTP POST with the following URL-encoded form parameters:

Parameter	Description
eventId	A unique identifier for the event that remains the same even if the same event is re-sent multiple times due to network problems, server downtime, and so on. See *Retry Handling on the next page*.
eventType	The type of event, such as "CTD_USAGE" or "SESSION_START". See *Push API Reference on page 448* for the event type corresponding to each push API.
timestamp	The request time in UTC, such as: 2009-12-10T01:16:20.026Z.
signature	An HMAC-SHA1 hash of the timestamp using the API shared secret, which can be used to verify the identity of the caller. See *Signature Verification (Optional) on the next page*.
signature2	An HMAC-SHA256 hash of the timestamp using the API shared secret, which can be used to verify the identity of the caller. See *Signature Verification (Optional) on the next page*.

data	The XML content, specific to the event type. Refer to *Push API Reference on page 448* for the various push API data formats For example, here is the XML data content for a Session End event:
	 `<Session xmlns="http://api.jasperwireless.com/ws/schema">`
	 `<iccid>8901650500000002918</iccid>`
	 `<ipAddress>10.98.214.86</ipAddress>`
	 `<dateSessionStarted>2009-12-10T01:16:20.026Z</dateSessionStarted>`
	 `<dateSessionEnded>2009-12-10T01:16:25.026Z</dateSessionEnded>`
	 `</Session>`

For a notification sample, see *Sample Push API Message on page 443*.

REST API Callback Responses. Callback responses from REST APIs use the JSON format, not the XML format show above. For sample callbacks, refer to the documentation for a REST API that uses callbacks (for example, *Transfer Device to Account on page 259*).

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Signature Verification (Optional)

To verify that the request came from a trusted caller (such as Cisco), you can independently construct the signature using your Push API Shared Secret and confirm that it matches the signature in the request. You configure the Push API Shared Secret in your account profile (Admin > Account Profile > Edit button). Be aware that the Push API Shared Secret is a secret value, 1-100 characters in length, that you choose. It is different from the API license key.

To construct the signature, you take the bytes of the timestamp and apply a standard HMAC SHA2 hash using your shared secret as the key. See the sample code for an example. If the signatures don't match, you should not process the event. Instead, you should respond with the HTTP status code "403 Forbidden".

Data Processing

After parsing the notification, you can take the appropriate action in your system, such as inserting the event into a database. You might also call the Cisco APIs to take related actions on the device, such as changing the SIM state.

Once you have successfully processed the notification, respond with the HTTP status code "200 OK". If you return any other status code, Control Center will retry the notification.

Backward Compatibility. From time to time Cisco may make changes to the format of the push API notifications. Follow the best practices outlined in *Backward Compatibility on page 19* to ensure that your parsing code can handle any format changes.

Retry Handling

Although we expect your receiver to be up-and-running most of the time, Control Center provides an automatic retry mechanism to recover from temporary downtime or errors. If Control Center gets a time-out or a status code other than 200 when contacting your receiver, Control Center will wait a few minutes and then re-send the notification. Control Center will retry each notification a maximum of 20 times (with exponential back-off between attempts) before giving up.

After attempting the maximum number of tries, Control Center drops the message. If the failure state persists, Control Center will disable the rule. Control Center does not resend dropped messages; these messages are not recoverable. Be aware that the Cisco operations team may reduce the maximum number of retries during peak loads or error conditions in order to protect the network.

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SSL Certificate

In the automation rule interface, you can supply an HTTP or HTTPS URL as a destination for the push API. If you use an HTTPS URL, that URL must have an SSL certificate signed by one of the common trusted Certificate Authorities (CA). Ensure that all certificates in the chain are current and signed, and the host name matches the certificate. You may use a third party tool to validate the certificate setup.

Sample Code

The receiver can be written in any language that supports HTTP requests, such as Java, C#, or PHP. See the Java sample code in the pushApiJavaSample.zip file that you can download from the Knowledge Base (APIs > Push APIs > Implementing a Push Receiver > Sample Code). This sample code shows how to process push APIs returned by rules.

```
Received message from: [ip address]
timestamp=2010-01-07T01:20:55.685Z
eventId=SESSION_START-123
data=<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Session xmlns="http://api.jasperwireless.com/ws/schema">
<iccid>8901311242888845458</iccid>
<ipAddress>12.34.56.78</ipAddress>
<dateSessionStarted>2010-01-07T01:20:55.200Z</dateSessionStarted>
<dateSessionEnded>2010-01-07T01:20:55.200Z</dateSessionEnded>
</Session>
eventType=SESSION_START
signature=8DYYAlzX5TbzChTK/qpMWdi8flA=
===== Push event: session started for ICCID 12345 at 2010-01-07T01:20:55.200Z
```

REST API Callback Responses. Although the sample code does not show how to process JSON callback responses from REST APIs, you can adapt the code for this purpose.

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Sample Push API Message

Here is a sample push API message generated by the SIM Custom Field Change rule. All messages have similar headers. See the *Push API Reference on page 448* for the format of the data returned in each message.

```
Time: Wed, 01 Jul 15 04:00:04 -0700
Source ip: 218.107.13.5

Headers (Some may be inserted by server)
REQUEST_URI = /post.php
QUERY_STRING =
REQUEST_METHOD = POST
GATEWAY_INTERFACE = CGI/1.1
REMOTE_PORT = 45639
REMOTE_ADDR = 218.107.13.5
HTTP_CONNECTION = close
HTTP_HOST = posttestserver.com
CONTENT_TYPE = application/x-www-form-urlencoded>
CONTENT_LENGTH = 591
>UNIQUE_ID = VZPINNBx6hIAAGozNLkAAAAG
REQUEST_TIME_FLOAT = 1435748404.998
REQUEST_TIME = 1435748404

Post Parameters
key: 'eventId' value: 'SIM_CUSTOM_FIELD_CHANGE-151888917'
key: 'eventType' value: 'SIM_CUSTOM_FIELD_CHANGE'
key: 'timestamp' value: '2015-07-01T11:00:03.812Z'
key: 'signature' value: 'gBdsWXnCSllbu9uYktPY6bmu1RU='
key: 'signature2' value: '5F10RgcohjWBhlBP9h80w5i/jIiHJiv8eMsxRC+G+4I='
key: 'data' value: '<?xml version="1.0" encoding="UTF-8"
standalone="yes"?><SimFieldChange
xmlns="http://api.jasperwireless.com/ws/schema"><iccid>8986061569000000161</ic
cid><oldValue>testForPush1</oldValue><newValue>testForPush2</newValue><fieldNa
me>OPERATORCUSTOM1</fieldName></SimFieldChange>'
```

REST API Callback Responses. Callback responses from REST APIs use the JSON format, not the XML format show above. For sample callbacks, refer to the documentation for a REST API that uses callbacks (for example, *Transfer Device to Account on page 259*).

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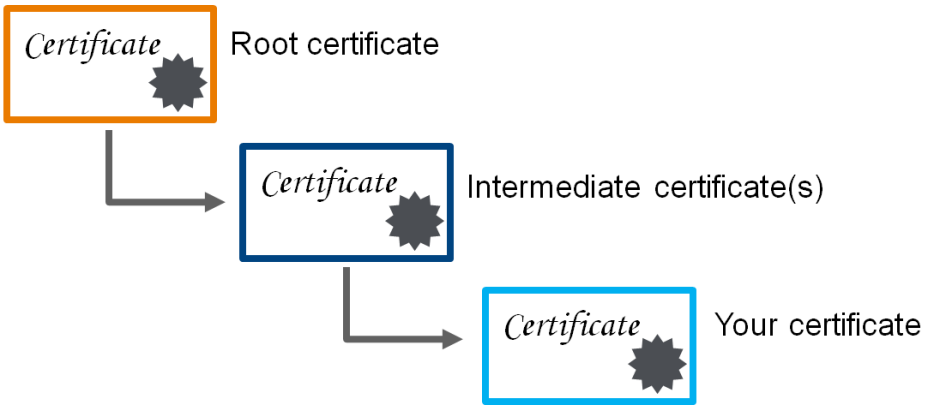
SSL Certificates

To secure the communication between Control Center and your application server, we recommend you use an HTTPS URL as the destination for the push API information we send. When you use HTTPS, you must purchase an SSL certificate and install it on your server. An SSL certificate encrypts the data connection and provides identification for your server so Control Center doesn't send sensitive data to the wrong location.

A certificate authority (CA) is an entity that issues SSL certificates and ensures that the owner of the certificate is who they say they are. Control Center accepts certificates from CAs in Cisco's trusted root store. When you purchase a certificate it must come from a trusted CA or the trusted CA must appear in the certificate's chain of trust.

A chain of trust refers to a series of certificates where one CA issues a certificate to another CA who in turn issues a certificate to another CA and so on. A chain of trust has a root certificate, multiple intermediate certificates, and the final certificate for the domain requiring certification. As long as the chain of trust associated with your SSL certificate contains one of the CAs in Cisco's trusted root store, Control Center will accept your SSL certificate.



Legacy CAs. Cisco will continue to support the CAs we have supported in the past (see *Legacy Trusted Certificate Authorities*). However, we encourage users to purchase new certificates from the Cisco-approved list.

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Tips for Purchasing SSL Certificates

- **Talk to the CA vendor.** Most CAs offer different types of SSL certificates with varying levels of security. Before purchasing a certificate from a CA on the list below, we recommend that you contact the CA's support team and ask them to provide the chain of trust for the certificate you plan to purchase. Verify that the chain of trust contains a CA on the trusted list.
- **Check the return policy**. Make sure you purchase a certificate from a CA with a reasonable return policy so you can get your money back if if you happen to purchase a certificate Control Center doesn't support. Many vendors offer a 15 or 30 day return policy.
- **Validate the certificate.** Test the certificate as soon as you receive it. We recommend installing the certificate on a different server from the one you're currently using so you don't disrupt a functioning environment. Create a test rule with a push API action and specify the location of the test server.
- **Keep certificates current.** SSL certificates expire. You'll need to renew the certificate on a regular basis to ensure the push APIs continue to function properly.

Cisco-Approved Certificate Authorities

For a list of Cisco-approved CAs, open the certificate authority spreadsheet in the Knowledge Base. You'll find it on the Downloads page and in this online topic (APIs > Push APIs > SSL Certificates > Cisco-Approved Certificate Authorities). The list is updated on a quarterly basis.

Legacy Trusted Certificate Authorities

See below for a list of legacy trusted certificate authorities. Control Center will continue to support these CAs. However, if you need to purchase a new certificate, we recommend you purchase one from the Cisco-approved list. The CAs below are shown alphabetically by alias.

Trusted Certificate Authorities
Alias: baltimorecodesigningca Owner DN: CN=Baltimore CyberTrust Code Signing Root, OU=CyberTrust, O=Baltimore, C=IE
Alias: baltimorecybertrustca Owner DN: CN=Baltimore CyberTrust Root, OU=CyberTrust, O=Baltimore, C=IE

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Trusted Certificate Authorities
Alias: entrust2048ca Owner DN: CN=Entrust.net Certification Authority (2048), OU=(c) 1999 Entrust.net Limited, OU=www.entrust.net/CPS_2048 incorp. by ref. (limits liab.), O=Entrust.net
Alias: entrustclientca Owner DN: CN=Entrust.net Client Certification Authority, OU=(c) 1999 Entrust.net Limited, OU=www.entrust.net/Client_CA_Info/CPS incorp. by ref. limits liab., O=Entrust.net, C=US
Alias: entrustglobalclientca Owner DN: CN=Entrust.net Client Certification Authority, OU=(c) 2000 Entrust.net Limited, OU=www.entrust.net/GCCA_CPS incorp. by ref. (limits liab.), O=Entrust.net
Alias: geotrustglobalca Owner DN: CN=GeoTrust Global CA, O=GeoTrust Inc., C=US
Alias: godaddyclass2ca Owner DN: OU=Go Daddy Class 2 Certification Authority, O="The Go Daddy Group, Inc.", C=US
Alias: soneraclass1ca Owner DN: CN=Sonera Class1 CA, O=Sonera, C=FI
Alias: soneraclass2ca Owner DN: CN=Sonera Class2 CA, O=Sonera, C=FI
Alias: starfieldclass2ca Owner DN: OU=Starfield Class 2 Certification Authority, O="Starfield Technologies, Inc.", C=US
Alias: verisignclass1ca Owner DN: OU=Class 1 Public Primary Certification Authority, O="VeriSign, Inc.", C=US

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Trusted Certificate Authorities
Alias: verisignclass2ca Owner DN: OU=Class 2 Public Primary Certification Authority, O="VeriSign, Inc.", C=US
Alias: verisignclass1g2ca Owner DN: OU=VeriSign Trust Network, OU="(c) 1998 VeriSign, Inc. - For authorized use only", OU=Class 1 Public Primary Certification Authority - G2, O="VeriSign, Inc.", C=US
Alias: verisignclass1g3ca Owner DN: CN=VeriSign Class 1 Public Primary Certification Authority - G3, OU="(c) 1999 VeriSign, Inc. - For authorized use only", OU=VeriSign Trust Network, O="VeriSign, Inc.", C=US
Alias: verisignclass2g2ca Owner DN: OU=VeriSign Trust Network, OU="(c) 1998 VeriSign, Inc. - For authorized use only", OU=Class 2 Public Primary Certification Authority - G2, O="VeriSign, Inc.", C=US
Alias: verisignclass2g3ca Owner DN: CN=VeriSign Class 2 Public Primary Certification Authority - G3, OU="(c) 1999 VeriSign, Inc. - For authorized use only", OU=VeriSign Trust Network, O="VeriSign, Inc.", C=US
Alias: verisignclass3g2ca Owner DN: OU=VeriSign Trust Network, OU="(c) 1998 VeriSign, Inc. - For authorized use only", OU=Class 3 Public Primary Certification Authority - G2, O="VeriSign, Inc.", C=US
Alias: verisignclass3g3ca Owner DN: CN=VeriSign Class 3 Public Primary Certification Authority - G3, OU="(c) 1999 VeriSign, Inc. - For authorized use only", OU=VeriSign Trust Network, O="VeriSign, Inc.", C=US

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Push API Reference

This section contains reference information about each of the push APIs, including the rules that call them, the format of the data contained in the message, and the event type. The parser should look for the event type to determine which rule caused the message.

The table below provides a quick summary of the rules and corresponding push APIs.

Rule	Push API
Available Static IP Address	AvailableStaticIpAddress
Change of Cell ID	LbsCellIdChange
Cycle To Date Data Usage	CtdUsage
Cycle To Date Data Usage in a Zone	CtdZUsage
Cycle To Date Voice Usage	CtdVoiceUsage
Cycle To Date Voice Usage in a Zone	CtdVoiceZUsage
IMEI Change	SimImeiChange
IMEI Permitlist Action	ImeiPermitlist
Monthly Pooled Data Usage	MonthlyDataUsage
Monthly Pooled SMS Usage	MonthlySmsUsage
Network Registration in a Zone	NetworkRegistrationInZone
No connection	NoConnection
Number of Session Connections (24 hours)	TooManyDailyConnection TooFewDailyConnection
Recent Data Usage (24 hours)	Past24HDataUsage
Recent SMS Usage	SmsUsage
Recent Voice Usage (24 hours)	Past24HVoiceUsage
Registration in a Zone	RegistrationInZone
Secure SIM	SecureSimAlert
Session End	Session
Session Start	Session
SIM Account Change	AccountTransfer
SIM Custom Field Change	SimFieldChange
SIM Data Limit	SimDataLimit

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Rule	Push API
SIM Event Plan	SimEventPlan
SIM Expiration	SimExpiration
SIM MSISDN Change	MsisdnChange
SIM Plan Completion	SimPlanComplete
SIM Rate Plan Change	SimRatePlanChange
SIM State Change	SimStateChange
SMS-MO Received	SmsMoReceived
Too many connections (Cycle to Date)	TooManyCtdConnection

Non-Rule Push APIs

Task	Push API
Service Provider-Designed Package Workflow	PremiumServiceOrderType
SIM Order Workflow	OrderCallback

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AccountTransfer

Description

Contains detailed information about a device transferred from one account to another.

Rule Name

SIM Account Change

Event Type

SIM_ACCOUNT_CHANGE

Source

```
<xs:complexType name="AccountTransferInfoType">
  <xs:annotation>
    <xs:documentation>
        Detailed information about Account Transfer
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="msisdn" type="xs:string" minOccurs="1"/>
    <xs:element name="imei" type="xs:string" minOccurs="1"/>
    <xs:element name="imsi" type="xs:string" minOccurs="1"/>
    <xs:element name="previousSimState" type="xs:string" minOccurs="1"/>
    <xs:element name="newSimState" type="xs:string" minOccurs="1"/>
    <xs:element name="previousAccountId" type="xs:long" minOccurs="1"/>
    <xs:element name="newAccountId" type="xs:long" minOccurs="1"/>
    <xs:element name="previousAccountName" type="xs:string" minOccurs="1"/>
    <xs:element name="newAccountName" type="xs:string" minOccurs="1"/>
    <xs:element name="dateChanged" type="xs:dateTime" minOccurs="1"/>
    <xs:element name="shippedDate" type="xs:dateTime" minOccurs="1"/>
    <xs:element name="deferActionStartDate" type="xs:dateTime"/>
    <xs:element name="deferActionPeriod" type="xs:int"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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AvailableStaticIpAddress

Description

Specifies the total number of fixed IP addresses in a pool and determines how many are still available for assignment.

Rule Name

Available Static IP Address

Event Type

AVAILABLE_STATIC_IP_ADDRESS

Source

```
<xs:complexType name="AvailableStaticIpInfoType">
  <xs:annotation>
    <xs:documentation>
        The detail information about Available Static IP action
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="apnName" type="xs:string" minOccurs="1"/>
    <xs:element name="totalNumberOfIpAddresses" type="xs:string"
minOccurs="1/>
    <xs:element name="numberOfAvailableIpAddresses" type="xs:string"
minOccurs="1"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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CtdUsage

Description

Contains cycle-to-date data usage information for the specified device (ICCID).

Rule Name

Cycle To Date Data Usage

Event Type

CTD_USAGE

Source

```
<xs:complexType name="CtdUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SIM CTD Usage.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="dataUsage" type="xs:string"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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CtdVoiceUsage

Description

Contains cycle-to-date voice usage information for the specified device (ICCID).

Rule Name

Cycle To Date Voice Usage

Event Type

CTD_VOICE_USAGE

Source

```
<xs:complexType name="CtdVoiceUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SIM CTD Voice Usage
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="ctdVoiceMoUsage" type="xs:string"/>
    <xs:element name="ctdVoiceMtUsage" type="xs:string"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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CtdVoiceZUsage

Description

Contains cycle-to-date voice usage information in a zone for the specified device (ICCID).

Rule Name

Cycle To Date Voice Usage in a Zone

Event Type

CTD_VOICE_ZUSAGE

Source

```
<xs:complexType name="CtdVoiceZUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SIM CTD Voice Usage in a zone.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="ctdVoiceMoUsage" type="xs:string"/>
    <xs:element name="ctdVoiceMtUsage" type="xs:string"/>
    <xs:element name="zoneName" type="xs:string"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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CtdZUsage

Description

Contains cycle-to-date data usage information in a zone for the specified device (ICCID).

Rule Name

Cycle To Date Data Usage in a Zone

Event Type

CTD_ZUSAGE

Source

```
<xs:complexType name="CtdZUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SIM CTD Usage in a zone.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="dataUsage" type="xs:string"/>
    <xs:element name="zoneName" type="xs:string"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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ImeiPermitlist

Description

Contains the IMEI value for a device that is not on the IMEI permit list. Also specifies whether Control Center blocked or allowed the data connection.

Rule Name

IMEI Permit List Action

Event Type

IMEI_PERMITLIST

Source

```
<xs:complexType name="ImeiPermitlistInfoType">
<xs:annotation>
    <xs:documentation>
     The detail information about Imei permit list action
    </xs:documentation>
</xs:annotation>
<xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="imsi" type="xs:string" minOccurs="1"/>
    <xs:element name="msisdn" type="xs:string" minOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="customerName" type="xs:string" maxOccurs="1"/>
    <xs:element name="imei" type="xs:string" minOccurs="1"/>>
    <xs:element name="permitlistAction" type="xs:string" minOccurs="1"/>
    <xs:element name="dateAction" type="xs:dateTime" maxOccurs="1"/>
    <xs:element name="simState" type="xs:string" maxOccurs="1"/>
    <xs:element name="endConsumerId" type="xs:string" maxOccurs="1"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
</xs:sequence>
<xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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LbsCellIdChange

Description

Provides location information for a device.

Rule Name

Change of Cell ID

Event Type

LBS_CELL_ID_CHANGE

Source

```
<xs:complexType name="LbsCellIdChangeInfoType">
  <xs:annotation>
    <xs:documentation>
        Location based service cell Id change Info for sim
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="msisdn" type="xs:string" minOccurs="0" maxOccurs="1"/>
    <xs:element name="date" type="xs:dateTime" minOccurs="1" maxOccurs="1"/>
    <xs:element name="newCellId" type="xs:long" minOccurs="1" maxOccurs="1"/>
    <xs:element name="newCellLAC" type="xs:int" minOccurs="1" maxOccurs="1"/>
    <xs:element name="newServingMCC" type="xs:int" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="newServingMNC" type="xs:int" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="newAccuracy" type="xs:int" minOccurs="0" maxOccurs="1"/>
    <xs:element name="newCity" type="xs:string" minOccurs="0" maxOccurs="1"/>
    <xs:element name="newState" type="xs:string" minOccurs="0" maxOccurs="1"/>
    <xs:element name="newCountry" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="oldCellId" type="xs:long" minOccurs="0" maxOccurs="1"/>
    <xs:element name="oldCellLAC" type="xs:int" minOccurs="0" maxOccurs="1"/>
    <xs:element name="oldServingMCC" type="xs:int" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="oldServingMNC" type="xs:int" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="oldAccuracy" type="xs:int" minOccurs="0" maxOccurs="1"/>
    <xs:element name="oldCity" type="xs:string" minOccurs="0" maxOccurs="1"/>
    <xs:element name="oldState" type="xs:string" minOccurs="0" maxOccurs="1"/>
    <xs:element name="oldCountry" type="xs:string" minOccurs="0"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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MonthlyDataUsage

Description

Contains information about monthly billable data usage for pooled rate plans.

Rule Name

Monthly Pooled Data Usage

Event Type

MONTHLY_DATA_USAGE_EXCEEDED

Source

```
<xs:complexType name="MonthlyDataUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information is about monthly billable data usage for pooled rate
plans
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="accountName" type="xs:string" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="zoneName" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="totalIncludedZoneUsage" type="xs:decimal" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="totalActualZoneUsage" type="xs:decimal" minOccurs="1"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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MonthlySmsUsage

Description

Contains information about monthly billable SMS usage for pooled rate plans.

Rule Name

Monthly Pooled SMS Usage

Event Type

MONTHLY_SMS_USAGE_EXCEEDED

Source

```
<xs:complexType name="MonthlySmsUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information is about monthly billable SMS usage for pooled rate
plans
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="accountName" type="xs:string" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="zoneName" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="smsUsageType" type="xs:string" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="totalIncludedZoneUsage" type="xs:decimal" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="totalActualZoneUsage" type="xs:decimal" minOccurs="1"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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MsisdnChange

Description

Contains information about a device with a MSISDN change.

Rule Name

SIM MSISDN Change

Event Type

MSISDN_CHANGE

Source

```
<xs:complexType name="MsisdnChangeInfoType">
  <xs:annotation>
    <xs:documentation>
        The detail information about msisdn change
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" maxOccurs="1"/>
    <xs:element name="imsi" type="xs:string" maxOccurs="1"/>
    <xs:element name="imei" type="xs:string" maxOccurs="1"/>
    <xs:element name="accountId" type="xs:long" maxOccurs="1"/>
    <xs:element name="simState" type="xs:string" maxOccurs="1"/>
    <xs:element name="customerName" type="xs:string" maxOccurs="1"/>
    <xs:element name="endConsumerId" type="xs:string" maxOccurs="1"/>
    <xs:element name="oldMsisdn" type="xs:string" maxOccurs="1"/>
    <xs:element name="newMsisdn" type="xs:string" maxOccurs="1"/>
    <xs:element name="accountName" type="xs:string" maxOccurs="1"/>
    <xs:element name="dateChanged" type="xs:dateTime" maxOccurs="1"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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NetworkRegistrationInZone

Description

Contains information about a device that registered in a given zone. If an SS7 event on a GSM/GPRS network triggers the rule, then the GT Address field will contain a value. If a Diameter event on an LTE network triggers the rule, then the Host Name field will contain a value. Any client-side program that processes the push notification must be able to handle these two different fields.

Network Type	Protocol	Push API Notification Field
GSM/GPRS	SS7	gtAddress
LTE	Diameter	hostName

5G SA Support. This feature is not available to devices using the 5G SA network.

Rule Name

Network Registration in a Zone

Event Type

NETWORK_REGISTER_IN_ZONE

Source

```
<xs:complexType name="NetworkRegistrationInZoneInfoType">
  <xs:annotation>
    <xs:documentation>
        Detailed information about SIM registration in a given zone
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="gtAddress" type="xs:string"  minOccurs="0"
maxOccurs="1"/>
    <xs:element name="zoneName" type="xs:string"/>
    <xs:element name="eventDate" type="xs:dateTime" minOccurs="1"/>
    <xs:element name="hostName" type="xs:string"  minOccurs="0"
maxOccurs="1"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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NoConnection

Description

Contains information about a device with no connections within the given period.

Rule Name

No connection

Event Type

SESSION_NO_CONNECTION

Source

```
<xs:complexType name="NoConnectionInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about a SIM has no connections within latest given
hours.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="evaluatingTime" type="xs:dateTime" minOccurs="1"/>
    <xs:element name="noConnectionThreshold" type="xs:long" minOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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OrderCallback

Description

Contains detailed information about the SIM order.

Rule Name

Not applicable

Event Type

ORDER_CALLBACK_RESPONSE

Source

```
<xs:complexType name="OrderResponseType">
  <xs:annotation>
    <xs:documentation>
      The detail information about the SIM Order
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
  <xs:annotation>
    <xs:documentation>
      The push notification should contain orderId, orderStatus, acctID, price
and quantity.
    </xs:documentation>
  </xs:annotation>
    <xs:element name="contactName" type="xs:string" minOccurs="0"/>
    <xs:element name="contactPhone" type="xs:string" minOccurs="0"/>
    <xs:element name="contactFax" type="xs:string" minOccurs="0"/>
    <xs:element name="email" type="xs:string" minOccurs="0"/>
    <xs:element name="orderId" type="xs:long" minOccurs="0"/>
    <xs:element name="orderStatus" type="xs:string" minOccurs="0"/>
    <xs:element name="orderDate" type="xs:dateTime" minOccurs="0"/>
    <xs:element name="acctName" type="xs:string" minOccurs="0"/>
    <xs:element name="po" type="xs:string" minOccurs="0"/>
    <xs:element name="implementationDate" type="xs:dateTime" minOccurs="0"/>
    <xs:element name="expedited" type="xs:string" minOccurs="0"/>
    <xs:element name="acctId" type="xs:long" minOccurs="0"/>
    <xs:element name="price" type="xs:decimal" minOccurs="0"/>
    <xs:element name="quantity" type="xs:int" minOccurs="0"/>
    <xs:element name="orderTotal" type="xs:string" minOccurs="0"/>
    <xs:element name="desc" type="xs:string" minOccurs="0"/>
    <xs:element name="opn" type="xs:string" minOccurs="0"/>
    <xs:element name="operatorAcctId" type="xs:string" minOccurs="0"/>
    <xs:element name="ratePlan" type="xs:string" minOccurs="0"/>
    <xs:element name="communicationPlan" type="xs:string" minOccurs="0"/>
    <xs:element name="shipStatus" type="xs:string" minOccurs="0"/>
    <xs:element name="shipAddr1" type="xs:string" minOccurs="0"/>
    <xs:element name="shipAddr2" type="xs:string" minOccurs="0"/>
    <xs:element name="city" type="xs:string" minOccurs="0"/>
    <xs:element name="stateRegion" type="xs:string" minOccurs="0"/>
    <xs:element name="country" type="xs:string" minOccurs="0"/>
    <xs:element name="postalCode" type="xs:string" minOccurs="0"/>
    <xs:element name="methodAndShippingCompany" type="xs:string"
minOccurs="0"/>
    <xs:element name="custAcctNo" type="xs:string" minOccurs="0"/>
```

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```
    <xs:element name="notes" type="xs:string" minOccurs="0"/>
    </xs:sequence>
</xs:complexType>
```

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Past24HDataUsage

Description

Contains the ICCID and data usage for a device whose usage has exceeded a specified amount within the past 24 hours.

Rule Name

Recent Data Usage (24 hours)

Event Type

PAST24H_DATA_USAGE_EXCEEDED

Source

```
<xs:complexType name="Past24HDataUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SIM Past 24 hours Data Usage
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="dataUsage" type="xs:string"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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Past24HVoiceUsage

Description

Contains the ICCID and voice usage (both MO and MT) for a device whose usage has exceeded a specified amount within the past 24 hours.

Rule Name

Recent Voice Usage (24 hours)

Event Type

PAST24H_VOICE_USAGE

Source

```
<xs:complexType name="Past24HVoiceUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SIM Past 24 hours Voice Usage
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="dailyVoiceMoUsage" type="xs:string"/>
    <xs:element name="dailyVoiceMtUsage" type="xs:string"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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PremiumServiceOrderType

Description

Contains the information about service provider-created Marketplace package offerings. Control Center sends a notification when an enterprise user orders a service provider-created package from Marketplace.

Rule Name

Not applicable

Event Type

MARKETPLACE_THIRDPARTY_ORDER_RESPONSE

Source

```
<xs:complexType name="PremiumServiceOrderType">
  <xs:annotation>
    <xs:documentation>
      It contains information about third-party provided Marketplace
      orders.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:annotation>
      <xs:documentation>
        The push notification should contain information for all
        Marketplace order push notifications for a third-party
        package lifecycle, which includes Activation, Cancellation,
        and Trial End notification.
      </xs:documentation>
    </xs:annotation>
    <xs:element name="alertType" type="xs:string">
      <xs:annotation>
        <xs:documentation>
          "NEW"       : Order Created Alert
          "CANCEL"    : Order Cancelled Alert
          "TRIALEND"  : Order Trial period end Alert
          "UNKNOWN"   : Unknown
        </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="orderId" type="xs:long" minOccurs="0"/>
    <xs:element name="acctId" type="xs:long" minOccurs="0"/>
    <xs:element name="operatorId" type="xs:long" minOccurs="0"/>
    <xs:element name="operatorAcctId" type="xs:long" minOccurs="0"/>
    <xs:element name="orderStatus" type="xs:string">
      <xs:annotation>
        <xs:documentation>
          "A"         : Active
          "C"         : Cancelled
          "P"         : Processing
          "T"         : In Trial
          "U"         : Unknown state
        </xs:documentation>
      </xs:annotation>
```

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```
    </xs:element>
    <xs:element name="acctName" type="xs:string" minOccurs="0"/>
    <xs:element name="operatorName" type="xs:string" minOccurs="0"/>
    <xs:element name="packageId" type="xs:long" minOccurs="0"/>
    <xs:element name="packageName" type="xs:string" minOccurs="0"/>
    <xs:element name="serviceType" type="xs:string" minOccurs="0"/>
    <xs:element name="orderedBy" type="xs:string" minOccurs="0"/>
    <xs:element name="acctPrimaryContactName" type="xs:string"
      minOccurs="0"/>
    <xs:element name="acctPrimaryContactEmail" type="xs:string"
      minOccurs="0"/>
    <xs:element name="quantity" type="xs:long" />
    <xs:element name="orderDate" type="xs:dateTime">
      <xs:annotation>
        <xs:documentation>
          Date and time when the Marketplace order was created for
          the account.
        </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="cancelledDate" type="xs:dateTime">
      <xs:annotation>
        <xs:documentation>
          Date and time when the Marketplace order was cancelled for
          the account.
        </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="trialEnabled" type="xs:boolean" minOccurs="1"
      maxOccurs="1"/>
    <xs:element name="trialStartDate" type="xs:dateTime">
      <xs:annotation>
        <xs:documentation>
          Date and time when the Marketplace order free trial starts
          for the account.
        </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="trialEndDate" type="xs:dateTime">
      <xs:annotation>
        <xs:documentation>
          Date and time when the Marketplace order free trial ends
          for the account.
        </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="billStartDate" type="xs:dateTime">
      <xs:annotation>
        <xs:documentation>
          Date and time of Account Billing start date.
        </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="billEndDate" type="xs:dateTime">
      <xs:annotation>
        <xs:documentation>
          Date and time of Account Billing end date.
        </xs:documentation>
      </xs:annotation>
```

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```
    </xs:element>
    <xs:element name="customFields" type="PrmSvcOrderCustomFieldsType"
      minOccurs="0" maxOccurs="5"/>
    <xs:element name="errorMsg" type="xs:string" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
```

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RegistrationInZone

Description

Contains information about a device that registered in a given zone. SS7 events in a GPRS network trigger the rule associated with this push API.

5G SA Support. This feature is not available to devices using the 5G SA network.

Rule Name

Registration in a Zone

Event Type

REGISTER_IN_ZONE

Source

```
<xs:complexType name="RegistrationInZoneInfoType">
  <xs:annotation>
    <xs:documentation>
        The detail information about SIM registered in a given zone
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="gtAddress" type="xs:string"/>
    <xs:element name="zoneName" type="xs:string"/>
    <xs:element name="eventDate" type="xs:dateTime" minOccurs="1"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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SecureSimAlert

Description

Contains the ICCID of a device with a Secure SIM credential mismatch.

Rule Name

Secure SIM

Event Type

SECURE_SIM

Source

```
<xs:complexType name="SecureSimAlertInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about Secure SIM credential mismatch.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="dateAdded" type="xs:dateTime"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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Session

Description

Contains session information for a particular device.

If the IP address uses the IPv4 format, it appears in the ipAddress field. If the IP address uses the IPv6 format, it appears in the ipv6Address field.

Rule Name

- Session End
- Session Start

Event Type

- SESSION_STOP
- SESSION_START

Source

```
<xs:complexType name="SessionInfoType">
  <xs:annotation>
    <xs:documentation>
        The detail information about SIM session.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string"/>
    <xs:element name="ipAddress" type="xs:string"/>
    <xs:element name="ipv6Address" type="xs:string" minOccurs="0"/>
    <xs:element name="dateSessionStarted" type="xs:dateTime" minOccurs="1">
      <xs:annotation>
        <xs:documentation>
                Date when the session started
            </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:element name="dateSessionEnded" type="xs:dateTime" nillable="true"
minOccurs="1">
      <xs:annotation>
        <xs:documentation>
                Date when the session ended (if applicable)
            </xs:documentation>
      </xs:annotation>
    </xs:element>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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SimDataLimit

Description

Contains information about a device on a prepaid rate plan that comes close to using its included usage.

Rule Name

SIM Data Limit

Event Type

DATA_LIMIT

Source

```
<xs:complexType name="SimDataLimitInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about a SIM on a prepaid plan comes close to using its
included usage.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="totalIncludedUsage" type="xs:decimal" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="totalActualUsage" type="xs:decimal" minOccurs="0"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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SimEventPlan

Description

Contains information about an event rate plan associated with a device. Control Center sends this notification when an event rate plan term begins or ends.

Return Value	Description
iccid	The ICCID of the device using the event rate plan.
eventName	The name of an event rate plan.
eventId	A unique identifier for this particular combination of an event rate plan and a device.
instanceEventId	A unique identifier for the current instance of this event rate plan.
changeType	Indicates whether the information is for the beginning of the term (Start) or the end of the term (End).
effectiveDate	The date and time when the event rate plan instance term begins. Date format: yyyy-MM-ddTHH:mm:ssZ.
effectiveEndDate	For the Start change type: The date and time when the event rate plan instance term is scheduled to end, based on the term length. For the End change type: The date and time when the event rate plan instance term actually ended. Date format: yyyy-MM-ddTHH:mm:ssZ.

Rule Name

SIM Event Plan

Event Type

SIM_EVENT_PLAN_NOTIFICATION

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Source

```
<xs:complexType name="SimEventPlanNotificationType">
  <xs:annotation>
    <xs:documentation>
        SIM Event Plan Notification
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="eventName" type="xs:string" minOccurs="1"/>
    <xs:element name="eventId" type="xs:long" minOccurs="1"/>
    <xs:element name="instanceEventId" type="xs:long" minOccurs="1"/>
    <xs:element name="changeType" type="xs:string" minOccurs="1"/>
    <xs:element name="effectiveDate" type="xs:dateTime" minOccurs="1"/>
    <xs:element name="effectiveEndDate" type="xs:dateTime" minOccurs="1"/>
  </xs:sequence>
  </xs:complexType>
```

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SimExpiration

Description

Contains information about a device on a prepaid rate plan with one week left in the subscription term.

Rule Name

SIM Expiration

Event Type

EXPIRATION

Source

```
<xs:complexType name="SimExpirationInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about a SIM on a prepaid plan comes close to its
expiration date.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="expirationDate" type="xs:date" minOccurs="0"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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SimFieldChange

Description

Contains information about a device with a change to a custom field.

Rule Name

SIM Custom Field Change

Event Type

SIM_CUSTOM_FIELD_CHANGE

Source

```
<xs:complexType name="SimFieldChangeType">
  <xs:annotation>
    <xs:documentation>
        The detail information about a change in a SIM field
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="oldValue" type="xs:string" minOccurs="1"/>
    <xs:element name="newValue" type="xs:string" minOccurs="1"/>
    <xs:element name="fieldName" type="xs:string" minOccurs="1"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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SimImeiChange

Description

Contains the current and previous IMEI values for a device.

Rule Name

IMEI Change

Event Type

IMEI_CHANGE

Source

```
<xs:complexType name="SimImeiChangeInfoType">
  <xs:annotation>
    <xs:documentation>
        The detail information about SIM Imei Change
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="previousImei" type="xs:string" minOccurs="1"/>
    <xs:element name="currentImei" type="xs:string" minOccurs="1"/>
    <xs:element name="dateChanged" type="xs:dateTime" minOccurs="1"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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SimPlanComplete

Description

Contains the rate plan name and expiration date for a device on a prepaid rate plan whose term has just expired.

Rule Name

SIM Plan Completion

Event Type

PREPAID_PLAN_COMPLETION

Source

```
<xs:complexType name="SimPlanCompleteInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about Sim's prepaid plan end.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="expirationDate" type="xs:date" minOccurs="0"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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SimRatePlanChange

Description

Contains the names of the old and new rate plans for a device whose rate plan has just changed.

Rule Name

SIM Rate Plan Change

Event Type

SIM_RATEPLAN_CHANGE

Source

```
<xs:complexType name="SimRatePlanChangeInfoType">
  <xs:annotation>
    <xs:documentation>
        The detail information about SIM Rate Plan Change
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="oldRatePlanName" type="xs:string" minOccurs="1"/>
    <xs:element name="newRatePlanName" type="xs:string" minOccurs="1"/>
    <xs:element name="dateChanged" type="xs:dateTime" minOccurs="1"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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SimStateChange

Description

For a device whose SIM state has changed, contains the prior and current SIM state values.

Rule Name

SIM State Change

Event Type

SIM_STATE_CHANGE

Source

```
<xs:complexType name="SimStateChangeInfoType">
  <xs:annotation>
    <xs:documentation>
        The detail information about SIM State Change
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="previousState" type="xs:string" minOccurs="1"/>
    <xs:element name="currentState" type="xs:string" minOccurs="1"/>
    <xs:element name="dateChanged" type="xs:dateTime" minOccurs="1"/>
    <xs:any processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:anyAttribute processContents="lax"/>
</xs:complexType>
```

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SmsMoReceived

Description

Contains information about a message sent by a device to a specific short code. The SMPP data coding values are:

- 0 — Default Alphabet
- 1 — IRA/IA5 (CCITT T.50)/ASCII (ANSI X3.4)
- 3 — Latin 1 (ISO-8859-1)
- 4 — Octet unspecified (8-bit binary); message content is encoded with Base64
- 8 — UCS2 (ISO/IEC-10646)

Rule Name

SMS-MO Received

Event Type

SMS_MO_RECEIVED

Source

```
<xs:complexType name="SmsMoReceivedInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SMS MO received event. When the smppDataCoding
is 4, message content is encoded with Base64.
        SMPP data coding:
        0 Default Alphabet
        1 IRA/IA5 (CCITT T.50)/ASCII (ANSI X3.4)
        3 Latin 1 (ISO-8859-1)
        4 Octet unspecified (8-bit binary)
        8 UCS2 (ISO/IEC-10646)
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="customerName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="msisdn" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="messageContent" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="smppDataCoding" type="xs:int" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="shortCode" type="xs:string" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="receivedTime" type="xs:dateTime" minOccurs="1"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

The timestamp format is: yyyy-MM-ddTHH:mm:ss.SSSZ. For example: 2017-03-10T00:40:02.846Z

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SmsUsage

Description

Contains SMS usage information for a device.

Rule Name

Recent SMS Usage

Event Type

SMS_USAGE

Source

```
<xs:complexType name="SmsUsageInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about SMS Usage.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1" maxOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="alertType" type="xs:string" minOccurs="1"
maxOccurs="1"/>
    <xs:element name="totalDailySmsUsage" type="xs:decimal" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="totalCtdSmsUsage" type="xs:decimal" minOccurs="1"
maxOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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TooFewDailyConnection

Description

Contains information about the number of daily sessions for a device that has connected less often than expected.

Rule Name

Number of Session Connections (24 hours)

Event Type

PAST24H_SESSION_USAGE_LESSTHAN

Source

```
<xs:complexType name="TooFewDailyConnectionInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about a SIM has fewer connections than expected in
trailing 24 hours.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="currentSessionUsage" type="xs:long" minOccurs="1"/>
    <xs:element name="sessionUsageThreshold" type="xs:long" minOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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TooManyCtdConnection

Description

Contains information about a device with more connections than expected in the current billing cycle.

Rule Name

Too Many Connections (Cycle to Date)

Event Type

CTD_SESSION_USAGE_EXCEEDED

Source

```
<xs:complexType name="TooManyCtdConnectionInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about a SIM has more connections than expected in the
current billing cycle.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="currentSessionUsage" type="xs:long" minOccurs="1"/>
    <xs:element name="sessionUsageThreshold" type="xs:long" minOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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TooManyDailyConnection

Description

Contains information about the number of daily sessions for a device that has connected more often than expected.

Rule Name

Number of Session Connections (24 hours)

Event Type

PAST24H_SESSION_USAGE_EXCEEDED

Source

```
<xs:complexType name="TooManyDailyConnectionInfoType">
  <xs:annotation>
    <xs:documentation>
        The information about a SIM has more connections than expected in
trailing 24 hours.
    </xs:documentation>
  </xs:annotation>
  <xs:sequence>
    <xs:element name="iccid" type="xs:string" minOccurs="1"/>
    <xs:element name="accountName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="ratePlanName" type="xs:string" minOccurs="0"
maxOccurs="1"/>
    <xs:element name="currentSessionUsage" type="xs:long" minOccurs="1"/>
    <xs:element name="sessionUsageThreshold" type="xs:long" minOccurs="1"/>
  </xs:sequence>
</xs:complexType>
```

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